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# Statistical Methods in Human Genetics

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Indranil Mukhopadhyay · Partha Pratim Majumder

# Statistical Methods in Human Genetics




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ISSN 2523-3114 ISSN 2523-3122 (electronic)
Indian Statistical Institute Series
ISBN 978-981-99-3219-1 ISBN 978-981-99-3220-7 (eBook)
https://doi.org/10.1007/978-981-99-3220-7

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*We dedicate this book to our mothers.*

# Preface

This book is primarily meant for the practicing biologist, particularly, the human geneticist. Human geneticists and other biologists use a variety of statistical methods to draw inferences from the results of their experiments. However, many scientists use statistical packages without really understanding the statistical methods. They are often unable to carry out customized statistical analyses demanded by the data, because the statistical package will not allow customisation.

In this book, we have tried to address these limitations in two ways. First, throughout the book, we have used data gleaned from a few types of experiments. For example, an experiment was performed to measure the expression of some genes. Such experiments are now very popular. Hundreds of papers are published every year to measure gene expression in various tissues and to compare differences in gene expression levels between tissues or in the same tissue under different environmental exposures or conditions. Expression differences are of paramount importance in the study of human disease. Study of gene expression differences can inform the genetic basis of pathophysiology.

By using data of a single type of experiment, we believe that the focus of the reader will be glued to the statistical methodologies presented, instead of having to grapple with multiple data types and experimental designs. However, we have also used other data sets to explain a few other statistical methods required beyond the realm of gene expression analysis. When our reader can focus on the statistical methodology more sharply, we believe that the learning will also be better, and applications will be more innovative. Throughout the book, we hand-hold readers to understand statistical methodologies which the readers also will find useful in analysing their data. We start with statistical techniques of data exploration, data summarisation, and visualisation, and slowly step into more intricate statistical methods, always focusing on applications to genetics. We have provided descriptions of statistical methods for the analysis of absolute and differential levels of gene expression when the properties of the population distribution of the level of gene expression are known. We also describe methods when the properties are unknown or do not conform to a 'standard' distribution. Additionally, in modern genetic data analysis, some specialised concepts and methods are used, that do not normally arise outside of the field of genetics. For

example, the concept of 'population stratification' arises in view of heterogeneity of ethnicities of humans and consequent differences in frequencies of some relevant genetic variables. In association analysis, ignoring these frequency differences can play havoc on inferences; handling these required some special methodologies. We explain some of these specialised statistical methods, as well; in particular, some that are important and contemporaneous.

Although human genomic data are often multivariate, it has not been possible for us to do full justice to multivariate statistical methods. However, we did not wish to ignore such methods altogether. We felt it important for us to introduce the reader to the problems of analysis when multiple variables are simultaneously considered for analysis and to describe some preliminary methods of multivariate statistics.

Concurrently with describing modern statistical methods, we also develop R programs, starting from the scratch (that is, introducing R), so that the reader can perform analysis of her/his own data as required, without having to mould the data to fit the requirements of a statistical package. R is free, and as a community-developed product is available in the public domain.

We hope that this book will become the standard text for biologists, those who carry out quantitative biological research, especially in human genetics.

Kolkata, India

Indranil Mukhopadhyay
Partha Pratim Majumder

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**Partha Pratim Majumder** is the founder of the National Institute of Biomedical Genomics, Kalyani, West Bengal, India. He is currently a National Science Chair of the Government of India. He concurrently holds academic positions in many national institutes. He serves on the governing boards and executive committees of the Human Genome Organisation, Human Cell Atlas, and International Common Disease Alliance. Earlier, he has served as Member of governance of the International Genetic Epidemiology Society, the Indian Society of Human Genetics, the Indian Society for Medical Statistics, and the American Society of Human Genetics, among others. He is an elected Fellow of all science academies of India, the International Statistical Institute, and The World Academy of Sciences. He has served as the President of the Indian Academy of Sciences. His research interests are in genetic epidemiology, human biomedical genomics, human genome diversity and evolution, and statistical genetics.

He has been awarded the G. N. Ramachandran Gold Medal (2021) by the CSIR; Barclay Memorial Medal (2020) by The Asiatic Society; Sir P. C. Ray Memorial Medal (2020) by the University of Calcutta; Golden Jubilee Commemoration Medal (2018) by the Indian National Science Academy; Centenary Medal of Excellence (2014) by the School of Tropical Medicine; TWAS Prize in Biology (2009) of The World Academy of Sciences, Trieste; G. D. Birla Award for Scientific Research

(2002) by the K. K. Birla Foundation, New Delhi and Kolkata; the Om Prakash Bhasin Award in Biotechnology (2001) by the Om Prakash Bhasin Foundation, New Delhi; the Ranbaxy Research Award in Applied Medical Sciences (2000) by the Ranbaxy Science Foundation, New Delhi; and the New Millennium Science Medal (2000) by the Indian Science Congress Association and the Council for Scientific and Industrial Research (CSIR), the Government of India.

# Chapter 1
# Introduction to Analysis of Human Genetic Data




## 1.1 Need for Genetic Data Analysis

Biological data present a multitude of challenges in terms of presentation and analysis. The most interesting feature is that data may come from diverse but related different domains including botany, zoology, fishery, genetics, genomics, neuroscience, anthropology, etc. Types of data differ; often a data set comes along with some unique characteristics of its own. However, each data set has a common goal: how to analyse the data properly and scientifically so that we are able to extract important information that is not apparent by a cursory examination of the data. It seems imperative to extract and exploit the full information contained in the data to come out with a nice story that explains and throws insight into the actual problem for which the data have been generated. Naturally, careful study of the data should lead to further more important questions about the problem concerned. Sometimes, it may lead to more interesting questions that are not intended to be studied initially, but that would shed light on important features unknown at least at the beginning of the experiment or the data analysis exercise. Whenever we have data, usually we have a tendency to jump on it and calculate some statistical measures as analysis of that data. However, it should start not with calculations but by asking questions, as many as you can. We need to understand the background of the experiment and the hypothesis and/or objective that triggers this experiment. Understanding the whole scenario is essential to appreciate the problem at hand. Thus, first, we should ask questions to dissect it, to get a holistic view as well as specific features of data and then only can start the appropriate analysis.

So, it is essential to identify the unique features of the data and frame the scientific problems in terms of several questions that would lead to detailed analysis. But the analysis completely depends on the relevance of the questions to the problem attempted at being solved, availability of the data, reliability, and quality of the data generated and stored, and knowledge of statistical analytic techniques. Data management and storage should also claim an important place in data analytics, especially in today's world of data science. Moreover, during the process of generation of data

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I. Mukhopadhyay and P. P. Majumder, *Statistical Methods in Human Genetics*,
Indian Statistical Institute Series, https://doi.org/10.1007/978-981-99-3220-7_1

some data might be missing, fully or partially; however, these data should not be discarded as we can get some information rather than no information from this type of partially missing data. However, statistical analysis of data with missingness is more complicated but should be undertaken, although advanced statistical techniques are required for this purpose.

On the other hand, in order to carry out an appropriate protocol for data analysis, one needs to know the basic structure and nature of the data, other important features that might be a distinguishable character for this particular data set, the way it has been generated and stored, etc. Thus, a complete understanding of the whole experiment and data generation process is essential to proceed further with the data analysis and eventual conclusion relating to the scientific questions asked.

Thus, the most important issue is to learn how to understand the problem clearly so that we can generate relevant questions relating to the data. Note that these questions might be of general nature, but care should be taken to identify the special questions that are only relevant to the data set considered for the analysis. This problem formulation stage should be done with utmost care and caution. Entire downstream analysis depends on this very crucial stage. Inadequate understanding of the data and the objective of its analysis may lead to erroneous results, which when interpreted, may not show any parity to the actual problem being investigated; the entire exercise would become futile. Hence, after the analysis, we need to interpret the results in relation to the actual problem and questions asked at the beginning of this exercise. If we see that interpretation of results really seems meaningful in relation to the problem, we can say that statistical analysis has been done appropriately. If we are not satisfied, we have to revisit the entire exercise from the beginning. At the end, we need to tell a story based on our data analysis.

Human genetic data analysis, the main focus of this book, is no more different than analysis of other types of data. However, because of advances in technologies, the diversity of human genetic data is large. Some popular data sets are voluminous and present major challenges in data handling and manipulation, in preparation for statistical analysis. It is interesting to note that Mendel's laws on heredity were formulated on the basis of careful statistical analysis of data collected from carefully designed experiments. During that period, statistical analysis had not flourished, in fact, statistics was not really a recognised discipline. However, it was the genius of Mendel that triggered the first statistical data analysis in genetics leading to fundamental laws of genetics being derived.

Statistics, as many people think, is not really a bunch of complicated mathematical methods with a fearsome look full of complicated symbols and notations. Rather it is a beautiful garden where flowers and green leaves play with gentle breeze full of ideas and thoughts. We should have an open mind to embrace new ideas, try to be patient and understand the scientific problem, and spend some time only for thinking about the intricate and interesting features of our data and experiment along with scientific question, and the solution will provide more light than we have ever thought.

In this book, we explain admixture of general purpose statistical methods and some specific methods applicable to human genetic data. We have kept the use of notations and symbols to a minimal level and almost omitted any detailed mathemat-

ical derivations. However, we have included derivations of a few important results in the appendix. We start with a few typical data sets and explain the types of questions that can be generated from the data. Later, we extend the data set and naturally also the question set, in order to get clearer answers to the questions. We consider mainly data sets that consist of gene expression data and genetic data with genotypes and other variables. Each chapter starts with a set of questions and the entire chapter is then devoted to a discussion of statistical methodology that can be used to address those questions and will eventually lead to the appropriate analysis. As a chapter progresses, more intricate sets of questions are considered along with their solutions through statistical analysis.

## 1.2 Nature of Analysis

Choice of techniques of statistical analysis of genetic data depends on the nature of data. We must have a clear understanding about the nature of the data and the scientific questions that underlie the generation of the data set; otherwise, the entire exercise of data analysis may lead to misleading results. Note that some typical features of the data may pose considerable challenges that may not even be solved through standard or common statistical methods. A common tendency in such situation is to bypass or ignore this special feature and do a quick statistical analysis using common methods. However, this would end up with very weak or even misleading conclusions that may not be replicable in the future. Although a set of data may have potential and be sufficiently informative to results producing valid inferences, we may lose that only due to our ignorance or reluctance to understand the data carefully and address the challenges posed by the data. Let us discuss this with one example.

To check whether a gene has any effect on a disease, we may collect data from tissues harvested from affected organs (for example, from the liver of a liver cancer patient and the patient's blood) from the same patient and measure their gene expression values. It is not unnatural that while processing the biological samples, some data might not be available due to degradation of sample quality or technological artefacts. Suppose that finally gene expression data for both normal and tumour samples are available for 20 individuals whereas for 4 individuals data for normal tissues are lost or unavailable. One may opt for two methods of analysis of such data.

**Option 1** Analyse the data only based on 20 paired samples for which full data is available and ignore the remaining 4 samples.

**Option 2** Consider the data of normal tissues for 20 individuals and tumour samples of 24 individuals. Then do a statistical test that should be applied to two independent samples.

However, both the above options for analysis are wrong, or at least inappropriate. Why? Note that since normal as well as tumour tissues are collected from the same individual, the data on expression levels of genes are expected to be biologically related (and also statistically). The second option is untenable since it does not use the

information on the possible relationship between levels of gene expression between tumour and normal samples obtained from the same patients. In fact, this is a wrong method of statistical testing. On the other hand, in the first option, we are discarding information that are in four tumour samples although we have spent money and time to generate the data. Thus, neither of the above tests should be applied in this case. If one does so, this is only due to ignorance of statistical knowledge or reluctance to exploit the full information that is available in the entire sample. However, it is not very difficult to analyse such data, although it would not be a standard one. So we should be honest, open-minded, and enthusiastic to learn how to use statistics with its full power to come up with the best possible analysis given the intricate nature and challenges posed by the data.

It is crucial to understand the data. The very first step in the central dogma of genetics is related to DNA sequence being transcribed to mRNA. The most common variant at one nucleotide position that we use is single nucleotide polymorphism (SNP). Note that a SNP has exactly two alleles giving rise to three genotypes, one of which is observed in an individual. Let two alleles at a bi-allelic locus be $A$ and $T$ and hence the possible genotypes be $AA$, $AT$, or $TT$. Note that here we consider $AT$ and $TA$ genotypes as a single class $AT$. So genotype data have no numerical value, but only three categories. For the sake of analytical convenience, we may assign numbers 0, 1, and 2 respectively, but care should be taken while interpreting the result. The main reason being one can assign any numbers to these three categories and the final result should not be dependent on what numbers we assign to genotypes.

This categorial variable can be converted to a meaningful variable that takes only a few discrete values. If we consider the number of minor alleles in a genotype at bi-allelic locus, it can take values '0', '1', and '2' depending on the number of minor allele in the genotype. In the above example, if $T$ is the minor allele, the genotypes $AA$, $AT$, and $TT$ would produce 0, 1, and 2 values respectively. Thus, it becomes a discrete variable. However, whether we work with discrete variable or the original categorical variable depends on the problem that we are trying to address.

So genotype data are categorical, having three categories. However, sometimes during the data collection, data for a few other variables may be collected naturally. For example, when we collect blood or tumour tissue from a person, we may collect data on her/his age, gender, smoking or food habit, etc. These auxiliary variables, also known as 'covariates', contain important information that should be included in the analysis to get a better insight into the problem that is attempted to be solved using the biospecimens.

In another scenario, to study whether a gene is somehow affecting a disease, we generate gene expression data. These data are continuous in the sense that these can take any value, up to a certain place of decimal, within a given interval. Suppose a particular set of gene expression data reveals that the expression level of a gene varies between 0 and 10.95 (for instance). The analysis will differ completely from that used for genotype data. Moreover, it also depends on whether normal and tumour samples are collected from the same individual or different individuals.

The above examples are only a few in the bigger set of data types generated through experiments. Thus, data might be discrete, categorical, continuous as well

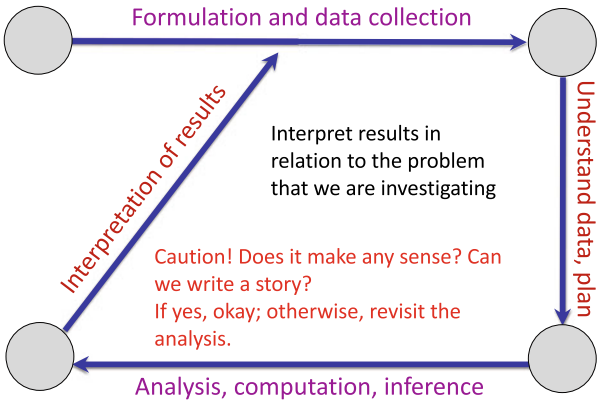



**Fig. 1.1** A schematic of data analysis protocol

as a mixture of all these. We should take utmost care to understand the nature of data, spend sometime to find statistical methods that should be most appropriate, try to use the maximum possible information contained in the sample, at the end interpret the results based on domain knowledge and see whether it sounds meaningful with respect to the actual problem that initiated this analysis. (Fig. 1.1) We shall take our decisions on this experiment to the next step, may be based on information available from other sources; then only we will be able to advance ourselves towards deciphering the genetic architecture of the disease or the characteristic.

## 1.3 R: A Versatile Tool for Genetic Data Analysis

Most current data sets are too voluminous to be amenable to manual analysis; using a computer has become inevitable. Broadly, we may use

(1) programming language and write programs of several statistical measures and mathematical functions, run our own program to carry out the analysis, or
(2) a software package, pull required functions from the package, and use them to perform the analysis.

Both approaches have their own drawbacks as well as advantages. Software can be used when we do routine calculations. It is usually menu-driven, user-friendly and one has to remember the way a function is to be used. This part is not very difficult. But when we want to do novel analysis based on some novel scientific questions and ideas, software usage is limited because we have to find appropriate functions and use them one by one, each time going to the menu, pulling that function, exercising it, and finally combining the result in the same order. In most cases, menus or functions are used by clicking the mouse on the appropriate tab.

On the other hand, writing ones own code has a tremendous advantage. We can always customise our code according to our needs and questions asked, whether novel or standard or a combination of both. But in order to do that, we need to learn how to write programs using a particular computing language like C, C++, FORTRAN, etc. This requires extensive training.

To bridge the gap between the two, the computing world offers something which can be treated as a combination of the two approaches. Among many efforts, Roy Ihaca and Robert Gentleman of Auckland University, New Zealand, developed a programming language, named it as **R**, and gave it to the world absolutely free. Usually, this is known as "R programming language and environment".

R is very user-friendly; a little bit of training in programming can make anyone comfortable to use it. Data entry or reading data from files can be done very easily using a command consisting of only a few words. Basic operations like addition, multiplication, etc., and many simple mathematical functions like logarithm, exponential, etc., can be easily done with commands similar to the operation name itself. Multitude of simple and complex statistical measures, statistical testing procedures, fitting regression lines, etc., are done using standard R commands. Moreover, it is convenient to merge a few R commands to do more sophisticated data analysis. We can also save the results in an easy-to-understand format and produce nice graphs leading to a better understanding of data and its analysis. The recent introduction of R studio and tidyverse has made our lives more simple and provides us with tools for extensive analysis of data.

This book is written with a focus to do our own calculation using R. A brief exposure to R is given in Appendix A. Although it is given in Appendix, spending some time towards learning R might be very effective and comforting while doing data analysis.

## 1.4 Some Remarks

The remainder of the book deals with statistical methods to analyse genetic data with the help of R only. In each analysis, we provide the R command and the output generated. All graphs used in this book are produced only using R. This will give an idea about the beauty and power of R language in generating useful graphs and diagrams and to do any kind of statistical analysis. The discussion is motivated by the data itself through generation of adequate questions or queries, analysing them with the logic of the method chosen, and its execution using R along with the interpretation of the results. This would provide a holistic approach to data analysis using statistical intuition, subject knowledge, and computational methodology.

## Exercise

1.1 Write three advantages of using R in statistical analysis of data.
1.2 Is there any issue in using standard software for statistical analysis? If so what are they?
1.3 The main steps in statistical data analysis are (1) actual numerical computation, (2) understanding the problem, (3) our expectation about solving a few relevant questions, (4) inference through statistical analysis, and (5) interpretation of results. Order these steps from the starting step of data analysis to the end. If any step is missing, what is that step and where does it come in the sequence of data analysis steps?
1.4 Is it wise to always convert one type of data to another? Justify your answer with examples.
1.5 Explain with an example how you can convert a continuous data into a categorical data. Is it wise to do that? Justify your answer.
1.6 As described in Sect. 1.2, give another concrete example where we may do common mistake during data analysis. Try to suggest what should be the appropriate method that you can think of? You need not explain or develop any new statistical method, but an idea about the solution would be fine.
1.7 Suppose we have collected data on 50 individuals on four different variables. Missing observations are artefacts for many mechanisms, some known, and some might even be unknown. During the analysis, it is observed that some observations are missing. What do you do if

(a) information of two individuals are missing completely for all four variables.
(b) information for three individuals are missing for only two variables.
(c) information for three individuals are missing for only one variable, however, for one individual, although all information is available but we have enough reason to be suspicious of the fact that data on one variable might be wrong.

1.8 Suppose a person gives some data to a statistician. The statistician immediately calculates a few measures and do a quick analysis without asking any question. Do you support the act of that statistician? Justify your answer.
1.9 "Statistical methods are always generic, not problem specific"—A scientist makes an interpretation of this statement as if one knows some statistical methods, s/he can do data analysis for any data set whatsoever. What are your views about this? Explain clearly, may be with relevant examples.
1.10 "In most cases we can categorise or discretise continuous data into categorical or discrete variable. Hence it is enough to know statistical methods that deal with categorical and discrete variables." Do you agree with this? Justify your answer with some examples.

# Chapter 2
# Basic Understanding of Single Gene Expression Data




Various physiological processes of a living organism are controlled by proteins. Information on production of proteins is encoded in the DNA of each cell of an organism. The information in the DNA is transcribed into RNA and then translated into proteins. This is known as the central dogma of biology.

In a cell, not all genes are transcribed. Subsets of genes that are transcribed provide a cell with its identity, cell-type. Cells are specialised and activities carried out within a cell depends on RNA molecules produced within cell. Transcriptome is the collection of RNA molecules transcribed within a cell. Technology tried to address the difficulty of characterising the transcriptome of a single cell in the following way. Usually a large set of cells from a tissue is collected and the transcriptome of this bulk of cells is characterised. Even in a single tissue, characteristics of the bulk transcriptome - that is, which genes have been transcribed and at what levels—are highly variable across individuals, which may be beyond differences in age, gender, life-exposures and ethnicity. Characteristics of the transcriptome can also indicate the state of health or disease of the tissue in an individual.

The transcriptome is complex, because there are different types of RNA, called RNA species, not all of are messenger RNA (mRNA) which are directly involved in the production of proteins. Messenger RNA (mRNA) is called so because it encodes proteins via the genetic code. Genetic code is actually DNA information used to produce amino acids, which are the building blocks of a protein. Therefore, mRNA is the most widely studied RNA molecule.

These studies which were conducted using methods that were low-throughput and involved considerable human intervention evolved over time. Initially studies were limited to measuring single transcripts using methods known as northern blotting or quantitative polymerase chain reaction. Rapidly, technologies and methods were developed to quantify genome-wide gene expression levels called transcriptomics. Carried out by hybridization-based methods using microarray technologies producing high-throughput data at a modest cost (Schena et al., 1995), these experiments and the resulting data were fraught with many errors and artefacts (Casneuf et al., 2007; Shendure and Ji 2008). To resolve these problems, sequence-based methods

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I. Mukhopadhyay and P. P. Majumder, *Statistical Methods in Human Genetics*,
Indian Statistical Institute Series, https://doi.org/10.1007/978-981-99-3220-7_2

were developed to directly determine transcriptome sequences. However, many limitations remained.

Introduction of high-throughput massively-parallel sequencing technology revolutionised the field of transcriptomics. RNA analysis now rapidly done by sequencing of cDNA (Wang et al., 2009) called RNA-sequencing (RNA-seq) provided a deep and quantitative view of gene expression, alternative splicing, and allele-specific expression. Typically, an RNA-seq experiment involves three steps:

- isolating RNA from cells collected from a tissue.
- converting RNA to cDNA sequencing library. This is a critical step in which the desired RNA molecules (mRNA but not ribosomal RNA) are selected.
- mRNAs are then converted to cDNA by using reverse transcriptase, fragmented, and adaptors required for sequencing are ligated to the fragments; this creates the sequencing library using next-generation sequencing (NGS) platform.

A whole new world of accurate and massive data ushered in for comparing changes that result from different physiological conditions or in various disease states. The RNA-seq data provides an accurate, high-resolution view of the global transcriptional landscape in platforms like NGS and formats like FASTQ.

Thus, gene expression data provide an indication of the activity of genes that can be used for many genetic and genomic studies. Analysis of gene expression data has become extremely important during the last few decades.

But before analysing the data we need to understand the data as clearly as possible. Here Table 2.1 represents a set of observations corresponding to the expression of a single gene for a set of individuals. Although data are collected for a number of cases and controls, to start with we only consider the control data. Later on we shall discuss analysis of gene expression data for both cases and controls and also for multiple genes.

In a specific study, after the execution of various primary steps of data analysis (alignment, assembly, estimation of expression level), RNA-seq data of a study involving $n$ individuals can be summarised in the form of a data matrix $((x_{ij}))$, where $x_{ij}$ denotes the expression level of gene $i$ $(i = 1, 2, \ldots, p)$ where $p$ is the total number of genes in a human genome in the $j$-th individual $(j = 1, 2, \ldots, n)$. In the context of a disease study, instead of the absolute expression level of the $i$-th gene in the $j$-th patient, the differential expression level of the gene compared to

**Table 2.1** Gene expression values for a single gene

| | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2.67 | 0.74 | 4.60 | 1.95 | 1.64 | 2.57 | 1.69 | 0.23 | 2.94 | 2.30 | 3.32 | 3.42 | 4.00 | 2.66 | 9.21 |
| 1.40 | 4.91 | 1.45 | 1.99 | 6.96 | 2.41 | 4.57 | 1.98 | 4.06 | 4.64 | 2.90 | 1.66 | 4.97 | 4.03 | 9.01 |
| 3.58 | 2.81 | 0.02 | 4.29 | 2.60 | 6.67 | 4.00 | 1.24 | 3.72 | 3.32 | 5.75 | 3.99 | 4.07 | 2.81 | 0.01 |
| 2.16 | 3.38 | 4.21 | 0.79 | 1.08 | 3.53 | 4.35 | 3.45 | 2.95 | 1.61 | 3.36 | 0.15 | 2.04 | 2.78 | 3.46 |
| 1.56 | 0.57 | 3.46 | 2.37 | 5.36 | 1.80 | 3.15 | 1.86 | 4.04 | 2.77 | 2.51 | 3.47 | 3.25 | 2.42 | 9.98 |
| 1.16 | 0.71 | 1.70 | 4.49 | 6.10 | 2.39 | 4.27 | 0.88 | 2.86 | 3.62 | 4.22 | 2.52 | 2.75 | 1.44 | 2.31 |

either a matched unaffected individual or a tissue that is unaffected by the disease collected from the same patient and this is presented as data matrix $((x_{ij}))$.

However, for the sake of simplicity, in Table 2.1, we consider only a set of observations $\{x_1, x_2, \ldots, x_n\}$ as the gene expression values of a particular gene corresponding to $n$ different individuals. Note that we assume that these $n$ individuals are selected randomly, i.e. without any preference to anybody, from a larger group of individuals. Gene expression values for this larger group, is commonly known as 'population' under study and the study is conducted through the sample of $n$ expression values corresponding to $n$ randomly chosen individuals.

Note that a population is a large group of individuals or items and we are interested in studying this population and deciphering hidden patterns. However, usually it is too large to collect information on all items. So we draw a random sample of smaller size, which would be manageable for analysis, and study these sample values to infer about the population. It is important that we must ensure that sample represents the salient features of the population. Usually a random sample is a good one, which is a collection of items or individual values selected without attaching any preference to any member of the population. This forms the basis of data analysis.

However, in some situations we may not have a random sample in the truest sense. Certain ascertainment criteria are considered while collecting such sample. In such a case, we should be very careful to consider the effect of such criteria and adequate correction is necessary while analysing the data. Without this correction, analysis might be biased and there is a high chance that we miss some important features. Sometimes sample generation mechanism has some subtleties that might get ignored during routine analysis leading to wrong or vague conclusions. Thus deep understanding of data and the mechanism of its generation is essential to exploit the full information contained in the sample.

## 2.1 Generating Questions

Just looking at the data would not reveal its inherent pattern or information. However, based on domain knowledge and common intuition we can think of several questions that would be helpful in understanding the data as well as extracting salient features of the data. Research begins with questions. And the solutions or attempts to answer these questions are essential to understand the data, its nature and inherent pattern, the truth it may reveal, and for further investigation. So we begin with a few questions about the data. Although here we refer to the gene expression data, this should be a common theme before actually starting any analysis of any kind of data.

Here data means statistical data; the observations corresponding to different units that are subject to some random factors, some known, some may be unknown, but will always contain some degree of variation among them. Variation is the most important characteristic for any statistical data; without it we cannot do any statistical analysis.

With respect to the data presented in Table 2.1, a few questions that can be deemed important are:

(1) What is the general nature of the data, if we look at it very naively? Can we represent the data using a diagram to reveal some general patterns?
(2) Are all observations in a reasonable range or some observations show unusual behaviour compared to the rest of the data?
(3) Can we have an idea about the gene expression on an average, that can be considered as a representative value? Can we devise any measure that depicts this idea?
(4) Can we have any idea about the spread of the observations? Can we devise any measure to present this idea?
(5) Can we compare this so called average value with any previously known knowledge, may be based on literature survey or other studies?
(6) Can we infer about the same for any measure that represents the overall scatteredness of the data?
(7) A very important technical question might be: Can we think that observations come from a normal distribution? Or does there exist a random variable that follows normal distribution, from which these samples are drawn? This question is for a technical reason, but has great impact on the methods of data analysis. In fact the answer to this question will guide the entire downstream analysis of data.
(8) When we have two samples, for cases and controls, can we compare the general features between these two groups? How valid is that comparison?
(9) If we have data on multiple genes for a group of individuals, can we think of a subgroup of genes that might act in a similar fashion?

Many other questions can be asked, but for the time being we shall restrict ourselves only to these and explore the methods of addressing them in the best possible way.

## 2.2 Visualising Data

Experiments generate data i.e. raw data, as it is termed. They are usually, but not always, numerical observations. Looking at hundreds (or less) of such numerical figures only add more complexity in understanding. Hence first we have to present the raw data so that it might reveal some information, patterns, insight, even if at a crude level. This presentation should give some confidence that at least we should be able to get something out of it that relates to our primary question(s). Note that these primary questions or objectives need to be sorted out at the very beginning, during the planning of the experiment that would generate the data.

So, presentation of data is very important. It would guide our analysis plan. And visualisation of data, if possible, should be the most effective and attractive way of presentation. However, visual presentation should be as simple as possible, but at the same time should reveal salient features of the data. Hence, it is expected that every part of the diagram presenting the data should have some meaning of its own; there should not be any part left redundant. This visualisation provides a feeling of

the data, which is the first and perhaps the most important step in data analysis. A number of visual representation techniques are available; we discuss here two most popular and informative way of presenting this kind of data (Table 2.1).

Note that gene expression data are continuous in nature, meaning that it can take any value within a particular interval. This interval may be large or small, but the variable i.e. gene expression in this case, can take any value in that interval. This kind of variable that can take any value within a certain range, is known as "continuous variable". A random sample for this variable will constitute frequency data. It is so called because within a small interval we may have more than one, in fact a large or moderate number of observations. Thus this type of data is known as "frequency data".

Naturally larger the data size, more will be the information contained in the data set, better insight can be obtained about the data and the domain knowledge related to it. However, due to several resource constraints, like time of collecting samples and associated cost, it may not always be possible to go beyond a reasonable size of data, sometimes restricting even to a small sample size. Thus appropriate statistical analysis can only throw light to draw inferences about the question asked and it should start with some exploratory analysis, may be in terms of naive diagrammatic representation of the data or some elementary analysis.

Two most common diagrams used frequently are **histogram** and **boxplot**. Before discussing how to draw these diagrams, we need to understand the general feature of frequency data; then only we can develop diagrams for its representation.

### *2.2.1 Frequency Data*

A value in a data set may occur more than once or a number of observations in a data set may lie within an interval. Such types of data, where an observation may occur more than once, are called "frequency data". A continuous variable is a quantity or entity that can take any possible values within a certain interval. Some examples of continuous variables are: level of mRNA expression, viral load in 1 mL of blood, length of a leaf, weight of fish caught, etc. Naturally this type of variable can take any real number within a range or interval thus retaining the notion of continuity in a common sense and hence is called continuous variable. In some situations, especially when the underlying variable is continuous, each observation may occur only once, but a few observations are very close to each other. This type of data is also an example of frequency data. In Table 2.1 data on level of gene expression can take any value within a certain range, 0.01 to 9.98. This indicates that the data set has minimum value of 0.01 and maximum value of 9.98. All other values in this data set lie within these two values. There is no real number within this interval 0.01 to 9.98 that this variable cannot potentially take, although in reality, it takes a few numbers because the sample size is small, the measuring instruments can measure only up to second place of decimal; there are other experimental artefacts also.

But how do we find the minimum and maximum values of the data set? That's easy with R. This is the first time in this book where we are going to use R. A quick look at the Appendix would help in working with the actual data. First we would need to input the data so that we can read it within R. For instance, we can store the entire data in a variable named '*x*', (say). *min(x)* and *max(x)* functions calculate minimum and maximum of the data stored in *x* respectively.

```
> x<- c(2.67, 0.74,4.60,1.95,1.64,2.57,1.69,0.23,2.94,2.30,3.32,9.21,1.40,4.91,
1.45,1.99,6.96,2.41,4.57,1.98,4.06,4.64,2.90,9.01,3.58,2.81,0.02,4.29,2.60,6.67,
4.00,1.24, 3.72,3.32,5.75,0.01,2.16,3.38,4.21,0.79,1.08,3.53,4.35,3.45,2.95,1.61,
3.36,3.46,1.56,0.57, 3.46,2.37,5.36,1.8,3.15,1.86,4.04,2.77,2.51,9.98,1.16,0.71,
1.70,4.49,6.10,2.39,4.27,0.88, 2.86,3.62,4.22,2.31)      ## input data and store in 'x'
> min(x)                    ## finding minimum of values stored in 'x'
[1] 0.01
> max(x)                    ## finding maximum of values stored in 'x'
[1] 9.98
```

Once we have an idea about the possible values of the variable in the sample or the range of variables in which the data are expected to lie, we should proceed with the analysis part. A visual representation would be the first step that might guide the analysis because it gives an impression (idea) of the distribution of observations in the given sample (minimum to maximum).

### *2.2.2 Histogram*

The most commonly used diagram to represent data corresponding to a continuous variable is known as "histogram". Histogram provides information and an idea about the 'distribution' of observations, its shape and many other features. Now, 'distribution' means how the data (sample observations) are distributed over different small intervals (non-overlapping) within the range of data. To do this, we first divide the range ($maximum - minimum$) into a number of intervals, called class intervals. The smallest and the highest values of a class are known as lower class limit and upper class limit respectively for that class. We calculate the frequency of each class i.e. number of observations belonging to each class and hence a frequency distribution is obtained. Here we distribute the entire data over several classes thus constructed. However, these classes may be disjoint, meaning that although for the given data set we can accommodate all the values into classes, but theoretically some values, being real numbers, may fall between the upper limit of a class-interval and the lower limit of the next class-interval. To get rid of this problem and make it theoretically stronger in covering the whole range of values, we change the lower limit and upper limit of a class by lower class boundary and upper class boundary respectively.

Let $k$ be the number of classes, $f_i$ be the the frequency of $i$-th class, and $L_i$ and $U_i$ be respectively the lower and upper class limits of the $i$-th class ($i = 1, \ldots, k$). Then

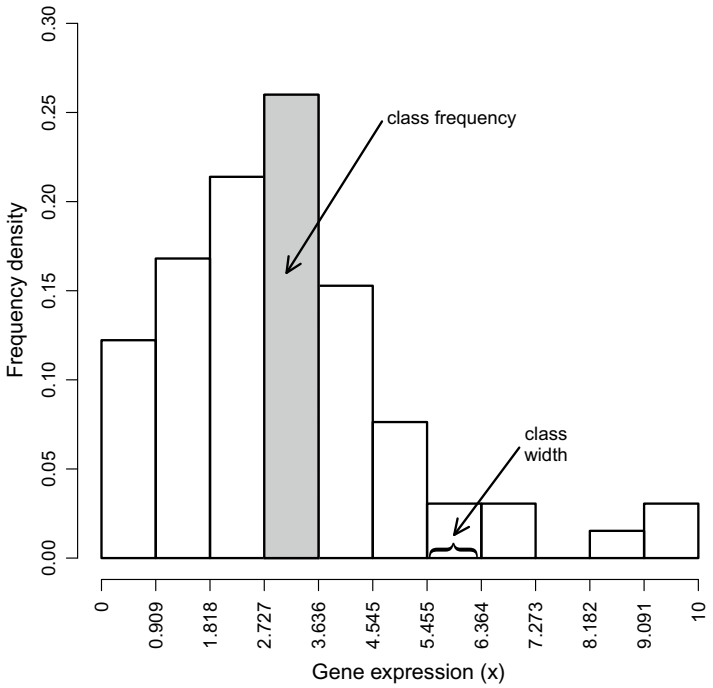



**Fig. 2.1** Histogram of single gene expression data for 90 samples

the lower (LCB) and upper (UCB) class boundaries of the $i$-th class are given by respectively, $LCB_i = L_i - (L_i - U_{i-1})/2$ and $UCB_i = U_i + (L_{i+1} - U_i)/2$. Let $w$ be the class-width of each class defined as $UCB - LCB$, assuming that the class widths are same for all classes. However, this assumption can be relaxed. We count the number of observations in each class, known as the class frequency. We also calculate frequency density as the ratio of class frequency to class width ($f_i/w$), for each class. This gives an idea about the density of values in a class. We then plot the class boundaries on the horizontal axis and draw a rectangle with height being equal to the frequency density and breadth as class width. This diagram is known as histogram (Fig. 2.1). Clearly, the area of each rectangle is equal to the class frequency and the total area under the histogram is the total frequency, because,

$$\text{Area of } i\text{-th class} = \text{frequency density} \times \text{class width} = \frac{f_i}{w} \times w = f_i = \text{ class frequency of } i\text{-th class}$$
$$\text{and total area under histogram } = f_1 + f_2 + \cdots + f_k = n = \text{ total frequency.}$$

Now let's try to understand this histogram in a very simple way. No complicated mathematics, only a careful look at the histogram while keeping in mind about the nature and context of our data set.

It is intuitively clear from the Fig. 2.1 that the class [2.727, 3.636] has the highest frequency and the frequencies in the classes to the left of this class decreases gradually whereas that to right decreases relatively sharply. Thus there is a concentration of observations around [2.727, 3.636] and also a long tail exists to the right spreading up to 10. Also some values to the right end seems little strange compared to the general nature of the data set. Thus it creates a suspicion whether observations in classes lying to the extreme right end are natural to obtain from the experiment that generates the data.

Another interesting feature can be observed if we study the histogram in depth. Suppose we draw a histogram using relative frequency density in the vertical axis. Relative frequency density of a class is defined as the ratio of relative frequency (=class frequency/total frequency) to the class width of the corresponding class. Since without any loss of generality we can assume that class widths for all classes are same, only one class width value would suffice for the entire calculation. Naturally the area of each rectangle represents the relative frequency of that class. Area under the histogram is total relative frequency, which is always 1, irrespective of the total frequency, because

$$\text{Area of } i\text{-th class} = \text{relative frequency density} \times \text{class width} = \frac{f_i}{nw} \times w = \frac{f_i}{n} = \text{ relative class frequency of } i\text{-th class}$$

$$\text{and total area under histogram } = \frac{f_1}{n} + \frac{f_2}{n} + \cdots + \frac{f_k}{n} = 1 = \text{ total relative frequency (since } f_1 + \cdots + f_k = n).$$

Drawing a histogram using relative frequency has immense advantage in downstream analysis (as we shall see later) mainly due to the fact that the total area of the histogram is always 1 whatever be the value of $n$. The value 1 is interpreted as the total relative frequency (or total normalised frequency). This gives us a nice opportunity to compare two frequency distributions, even when the total frequencies are not same for the two data sets.

Now, if we were to increase the number of observations i.e. total frequency and at the same time reduce the class width, we will get more and more dense rectangles but width being smaller and smaller, area of each rectangle would be smaller; the total area would always remain equal to 1 in each case, irrespective of the number of classes as well as total number of observations. Hence in the limiting case, when the total frequency is extremely large and the class width is extremely small, we get a smooth curve as the histogram. This is known as **frequency curve**. This frequency curve is very important in statistics and forms the basis of the underlying probability structure for the frequency distribution of a continuous variable. This curve can be modelled in a probabilistic framework and hence easy to study extensively. So histogram, which is based on real data, can be viewed as a basic tool to advance towards a theoretical probability structure to which we can assign sample observations as realisations of random experiments. This transition from real data to theoretical sophistication is a beautiful journey in statistics and may bring flexibility, comfort, and mathematical justification while doing data analysis.

Note that by constructing a frequency distribution of a continuous variable we lose individual values taken by the variable and this information cannot be retrieved unless we again go back to the original raw data, which is a humongous task. However, this sacrifice in losing some information is intended for analysis of data and answering the general questions laid down in Sect. 2.1. Moreover, as in most cases, for a judiciously chosen class width, since the observations lie densely within a class, losing individual values will not affect much to get an overall idea about the distribution of sample observations.

R code for drawing histogram is simply 'hist()'. But this histogram is drawn using frequency as the height of each rectangle rendering difficulty in interpretation of area of each rectangle and the entire area under the histogram. Moreover, it uses a default choice of the number of classes.

Now, the question is: what is the ideal number of class intervals? Ideally we should organise data so that the number of class intervals is neither too large nor too small. Otherwise the distribution would either be too smooth or too jagged, i.e. it would lose its latency. An empirical formula for the number of class intervals is an integer value closest to square root of the number of observations.

```
> br <- round(seq(0,10,length=12,3)      ## creating user-defined class boundaries
> hist(x,breaks=br, freq=F)              ## draws histogram with relative frequency density
```

R code for drawing a histogram is given in the box above. First we split the range of observations into 11 classes. 'br' gives the boundary points of the consecutive classes. Note that 'length=12' would be the number of boundaries which is one more than the number of classes. 'hist' gives the R-code for histogram; 'freq=F' is required when we draw histogram using relative frequency density (always recommended). It gives us Fig. 2.1 except the labels inside the graph and grey shaded region (for this we need to write little more sophisticated R code). Figure 2.2 presents three histograms with varying number of classes; 10 is closest value to the square-root of $n$ in this case. So the choice of $k$ as an approximation to $\sqrt{n}$ works! However, the user is free to choose their own number of classes while drawing a histogram.

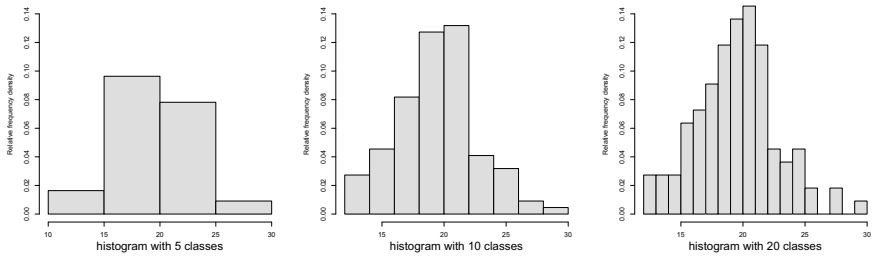



**Fig. 2.2** Histogram with different number of classes

Clearly, a little bit of difference in the number of classes in histogram for the same data will not matter much because this is a random sample and it represents general features of the population only approximately.

Note that on R console, each line starts with a '>' sign. This sign will appear automatically when you press enter. R Code for drawing histogram is given in Appendix B.

### *2.2.3 Ogive*

The frequency distribution of a continuous variable like the gene expression values can also be represented through another very useful diagram, known as '**ogive**'. This diagram represents the 'cumulative frequency', less-than type and/or greater-than type for the variable concerned. Cumulative frequency of less-than (greater-than) type of a class is defined as the number of observations that are less (greater) than or equal to the upper (lower) class boundary of that class.

We plot less-than type cumulative frequency for a given class corresponding to its upper class boundary and join the consecutive points by straight lines. The graph thus obtained is called an **ogive**. Ogive also emphasizes the fact that in constructing or representing a frequency distribution of a continuous variable we assume that the class frequency is distributed uniformly throughout the class, which in fact, is the fundamental assumption behind a continuous frequency distribution. In a similar fashion, we can plot the cumulative frequency greater-than type corresponding to the lower class boundary of that class and join the consecutive points by straight lines.

The first ogive looks increasing in nature while the second one is decreasing as we go along higher values of class boundaries. So these two ogives must intersect at a point. An interesting interpretation can be given to the point of intersection of these two ogives thus obtained. Let this point be denoted by $O$. Drop a perpendicular from $O$ which cuts horizontal axis at $M$ and the corresponding value is $F_M$ at the vertical axis. Clearly the number of observations more than $M$ along the less-than type ogive is same as the number of observations below $M$ along the greater-than type ogive. Naturally there are 50% observations that are less than $M$ as well as greater than $M$ and hence $F_M = 45$ as total frequency is 90. This point $M$ is called median of the given data set. We shall discuss some interesting properties of median later.

At this point, we expect that readers have some preliminary knowledge of R programming or they should read Appendix A at the end of this book, which contains a brief tutorial on R. The R code for obtaining ogives as in Fig. 2.3 based on the data in Table 2.1, is given in Appendix B.

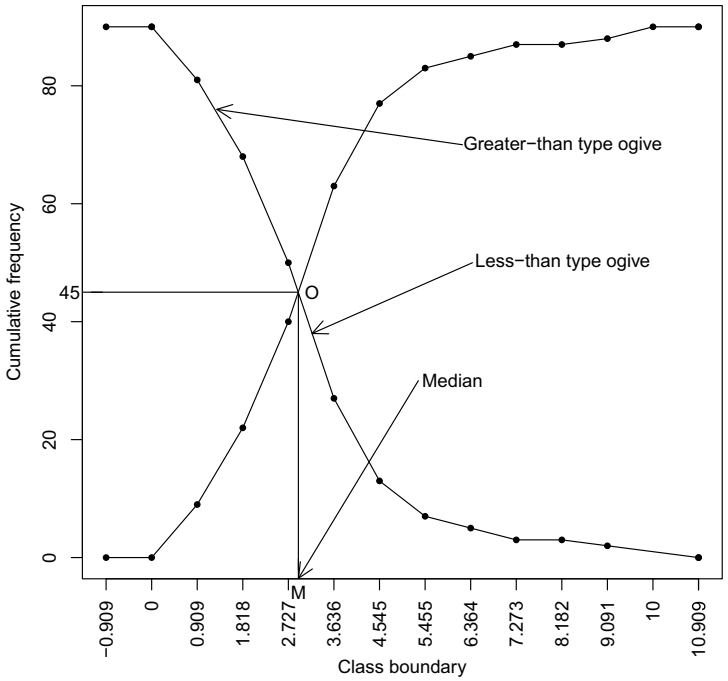



**Fig. 2.3** Ogive of single gene expression data for 90 samples

### *2.2.4 Boxplot*

A natural tendency of data is to cluster around some particular value(s) as well as to cover a range of values. These two features are probably most important in statistics to understand the general characteristics of the data. Before discussing some measures to represent these features, we would like to discuss a nice visual representation provided by a plot, known as '**boxplot**'. This plot provides summary measures of the data in one single diagram and can also be used to compare two or more distributions with respect to these characteristics.

A boxplot consists of a rectangle with a thick horizontal bar representing the median of the frequency distribution. The lower and upper horizontal lines of the middle rectangle represent first and third quartiles respectively. The two tails stretching towards the top and bottom are known as whiskers. These whiskers indicate how far the data extends on either side of the first quartile and third quartile. One can draw custom-made boxplots by controlling the whiskers to cover a substantial number of observations. There are some points that are scattered along the lines of whiskers but far beyond it. These points are known as 'outliers'. However, these outliers can be controlled by the user so that one can identify some important outliers that should not be deleted while doing the analysis. Boxplot is a nice tool to detect outliers present in

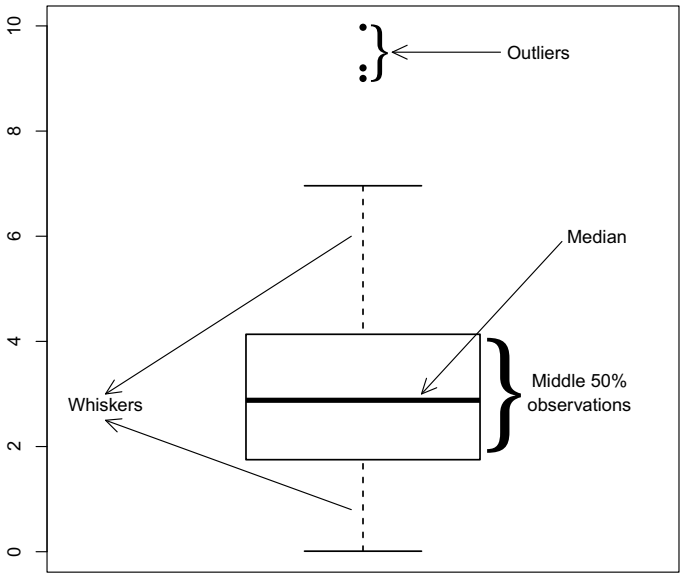



**Fig. 2.4** Boxplot of single gene expression data for 90 samples

the data. Figure 2.4 represents the boxplot of a single gene expression data as given in Table 2.1.

The R command for drawing a boxplot is simply 'boxplot(b)', where 'b' contains the data list for Table 2.1.

```
> x<- c(2.67, 0.74,4.60,1.95,1.64,2.57,1.69,0.23,2.94,2.30,3.32,9.21,1.40,
4.91,1.45,1.99,6.96,2.41,4.57,1.98,4.06,4.64,2.90,9.01,3.58,2.81,0.02,4.29,
2.60,6.67,4.00,1.24,3.72,3.32,5.75,0.01,2.16,3.38,4.21,0.79,1.08,3.53,4.35,
3.45,2.95,1.61,3.36,3.46,1.56,0.57,3.46,2.37,5.36,1.8,3.15,1.86,4.04,2.77,2.51,
9.98,1.16,0.71,1.70,4.49,6.10,2.39,4.27,0.88,2.86,3.62,4.22,2.31)     ## input
                                                                  data and store in 'x'
> boxplot(x)                    ## draws boxplot using values stored in 'x'
```

However, the diagram for boxplot in Fig. 2.4 is drawn using a series of little more sophisticated commands as given in Appendix B.

The boxplot in Fig. 2.4 shows that the distribution is symmetric about 2.8 (approximately) and has almost equal spread with respect to this value. The upper whisker is longer than the lower whisker indicating that the data are more scattered towards the right side of median than the left. It also indicates the presence of outliers which are marked as solid points along the line of whiskers on the upper side. This means that there are a few (in this case 3) values of gene expressions that are much deviated than the general data cloud and need special attention.

The experimenter who generated these data should check how these *unusual* data points or outliers, as they are commonly called, are generated in order to identify

possible errors associated with these points. In most situations we would just delete these outliers and work with rest of the data as the number of outliers is less and hence we assume that it should not affect the general nature of the data. Moreover, these data points may be due to artefacts of the experiment. However, special attention should be given to identify the actual reason for the presence of such data. If it is justified that the presence of these outliers is a natural phenomenon, then we should not omit them; rather special care should be taken to extract inference from such data.

## 2.3 Summary Measures

While visual representation of data is very useful in understanding the nature of the data, these do not provide succinct quantitative descriptions of the data. Therefore, a small number of quantitative and objective measures that capture the nature of the data are required. These can explain the nature and trend of data, decipher the hidden pattern that is not evident immediately and provide future guidelines about the scientific experiments as a whole. However, ideas behind these measures would come from the diagrammatic representations of the data. Diagrams provide an overall presentation of data, some hints about salient features, and an indication of some hidden information that needs to be unearthed and explored. So, developing a few quantitative measures should start with adequate visual representation(s) of data.

### *2.3.1 Measures of Central Tendency*

A careful look at a histogram reveals that area of some rectangles in the middle portion is much more than those in the tail regions. This clearly indicates the abundance of observations around a value in the middle region of the entire range of observations. Interestingly this is a common scenario in almost every data set. Thus we can say that for any data set, there is a tendency of the observations to cluster around some central value(s). Now our task is to develop some appropriate measures that can capture this feature of the data.

Consider again two more histograms (Fig. 2.5) or frequency curves (limiting forms of histograms). These histograms are drawn by increasing total frequency and decreasing class widths to have a feel about the structure of the data. All histograms and frequency curves reveal that the data have a high concentration around some central value, which in these case (Fig. 2.1) is about 2.9 (approximately). To apprehend this tendency, we define some measures, known as measures of central tendency, because they are expected to provide a quantification of the tendency of data to cluster around some central value(s). It is clear that both histograms reveal more or

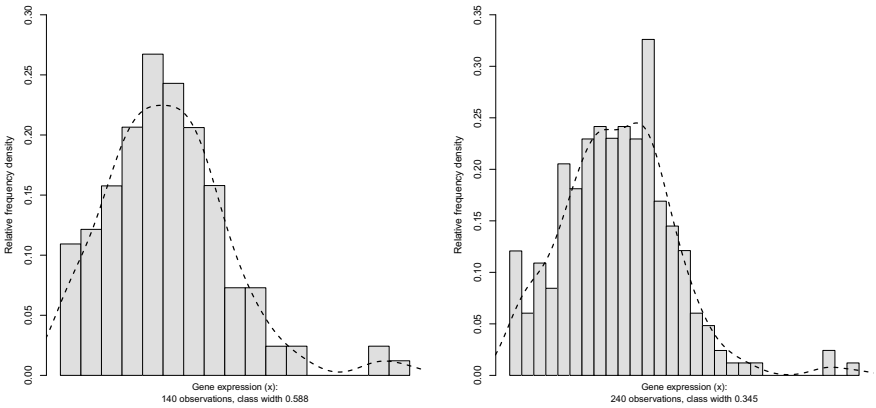



**Fig. 2.5** Histograms of expression data with increasing sample size and decreasing class widths. Frequency curves are drawn using dotted lines

less the same overall picture about the data and hence population. Thus a moderate number of appropriately chosen sample, more specifically a random sample, would be enough to reveal the main features.

### Arithmetic mean

10 children in my class were nice throughout the week. I gave 4.8, 4.5, 5.6, 5.7, 4.4, 4, 4.9, 5, 5.8, 5.3 dollars respectively (5.3$ means 5$ and 3 cents) to 10 children so that they could buy candies during the weekend. Later, next Monday the children were although happy but a little upset because not all of them got the same amount from me. I realised my mistake. Actually the amounts I gave to the ten children were not same. I had met each child at a different period and had randomly handed them some amount. So the question is: if I were to give the same amount, how much money would each child get?

The answer is easy. Total amount is 50$ that needs to be distributed among 10 children. So each child is supposed to get 5$. This is what is called arithmetic mean. In other words, if we distribute the total amount equally among a number of children (or individuals/units), each one would receive an amount that is equal to the arithmetic mean.

Thus a simple and most commonly used measure of central tendency is known as 'arithmetic mean' or simply 'mean', defined as $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$ where $x_1, x_2, \ldots, x_n$ are $n$ observations corresponding to $n$ units sampled from the population. By applying this formula we get the mean gene expression value to be 3.1005 for the data given in Table 2.1. For a frequency distribution in class intervals, mainly for a continuous distribution, the arithmetic mean is calculated assuming that the class frequency is distributed uniformly around the mid-point of each class in that interval. Hence the

formula is given by: $\bar{x} = \frac{1}{N}\sum_{i=1}^{n} x_i f_i$ where $N$ is the total frequency and $f_i$ is the frequency of the $i$-th class for which the mid-point is $x_i$, for $i = 1, \ldots, n$.

Since nowadays, we calculate everything using computer programming or by a software, there is no need to calculate mean using the second formula. We can easily calculate the mean for a set of observations, however large or small. Another practical tip: in a real data set sometimes a few observations are missing and are recorded as 'NA'. But R can tackle it directly; calculating 'mean' while eliminating those 'NA' values.

```
> mean(x)                    ## calculates mean of values stored in 'x'
> mean(x,na.omit=T)                  ## if data set contains 'NA'
```

Although mean is intuitively an appealing measure of central tendency, it suffers from some drawbacks. In a set of experimental data, some values are observed to be very different from most other values. These few (might be one or two) observations can pull the arithmetic mean towards them thus distorting the position of mean around which majority of the values lie. The mean, therefore, is highly affected by such extreme observations, known as 'outliers'. In this data set, it is clear from the histogram and especially from the boxplot that there are few outliers that may affect the arithmetic mean. So in this situation mean might give misleading value as a measure of central tendency and should not be used in such situations.

**Median**

To avoid the impact of outliers in measuring central tendency, another measure known as 'median' is preferred in many situations. The median is the middle-most observation when the data are arranged in increasing or decreasing order. Suppose $n$ is odd, i.e. $n = 2m + 1$ for some integer $m$. If $x_1, x_2, \ldots, x_n$ are observations already arranged in increasing order, the median is defined as $\tilde{x} = x_{m+1}$. If $n$ is even, i.e. $n = 2m$ for some integer $m$, then any value between the two middle-most observations can be taken as median. However, for the sake of simplicity, we can define median as the mean of the two middle-most observations, i.e. $\tilde{x} = (x_m + x_{m+1})/2$. Clearly, median is not much affected by the presence of extreme values. The median gene expression value is 2.835. The R command for calculating median is '$median(x)$' (note how simple the commands are to calculate mean or median!).

Suppose that we have been given a frequency distribution for a continuous variable instead of raw data points. In such a situation, we need to use a formula to calculate median. This formula is derived using the basic assumption that the class frequency is distributed uniformly around the mid-value (class mark) of each class interval. This is given by: $\tilde{x} = x_l + (N/2 - F_l)w/f_0$ where $x_l$ is the lower class boundary of the class for which cumulative frequency (less than type) is less than or equal to $N/2$, $N$ is the total frequency, $w$ is the class width, $F_l$ is the cumulative frequency up to $x_l$, and $f_0$ is the frequency of the median class. Here we assume that the class

widths are the same for all classes in the frequency distribution constructed based on the given data.

However, like mean, we usually calculate median using computer programming or software and hence the formula for calculating median using grouped frequency data is not required at all.

```
> median(x)          ## calculates median of values stored in 'x'
> median(x,na.omit=T)                    ## if data contains 'NA'
```

Clearly mean is much more than median for the given data set. It indicates presence of outliers that is also confirmed by boxplot of the data. Hence, depending on our objective, we have to decide the appropriate measure of central tendency for further downstream analysis.

### *2.3.2 Measures of Dispersion*

From Figs. 2.1 and 2.4 it is clear that although data are clustered around central values like 3.1005 or 2.835, observations are spanned over a range of values as seen from the span of the histogram or boxplot. Note that outliers may belong towards either or both ends of histogram or boxplot; one end shows more stretched compared to other end. Thus we also need to study the nature of dispersion or scatteredness present in the data along with the mean. Note that if a data set has no dispersion, all values would be same and hence there is no need to do any statistical analysis whatsoever. In fact the study of dispersion is most important in Statistics, but taking careful attention to central tendency at the same time. This simple philosophy triggers many branches in statistics and rigorous research is still continuing to explore several aspects and features that germinate from it. To develop some measures of the degree of dispersion in a data set, we need to understand its meaning and interpretation along with the inherent nature of data.

Central tendency is always observed for any kind of data. But there is another inherent characteristic without which no statistical analysis is possible or meaningful. I realised it with a real life experience, a scary incident indeed! It happened to me long back. While trekking in the mountains we came across a beautiful small river, the width of the river was not so wide. Our leader said "Cross the river. It's not very deep." Being very excited to cross a river bare feet, I walked into the river boldly (and of course carelessly)! I had the confidence that the river was not too deep; there was no question of drowning, or falling into the cold water. But I did tumble, fall, get soaked and wet in that freezing water. Why? I ignored what is known as '**dispersion**'. Our leader intended to mean that the average depth is not too much. That was indeed true, but at some places it was deeper than rest of the places while in some places it was very shallow. Mistakenly I fell into the deeper region, tumbled and got

drenched in the freezing water! Because there was dispersion. So dispersion or how data are scattered is extremely important and is the main reason of statistical analysis.

**Mean deviation**

If there is no dispersion in the data, all observations would have been same and this value is a representative of the entire data sets, a measure of central tendency. Because there is dispersion, values are different. The concept is so simple! Hence dispersion is always observed and should be studied around a measure of central tendency. We can take the difference of each observation from any chosen measure of central tendency, say $A$. Thus, we consider $x_i - A$ for all $i = 1, \ldots, n$ and take their mean. However some of these differences are positive and some are negative, hence the sum might be approximately or exactly equal to 0, which is counter intuitive. Moreover, we are only interested in measuring the extent of dispersion without considering whether an observed value is lower or higher than the measure of central tendency i.e., ignoring the sign of differences. We need to remember that we are interested in knowing how far an observation is from a central tendency measure, not the direction (positive or negative). So, instead of just the difference, we can take absolute value of each difference or deviation from an appropriate measure of central tendency. An average or mean of such deviations can be taken as a nice measure of dispersion present in the data. This is known as 'mean deviation about $A$' and is given as:

$$MD_A = \frac{1}{n}\sum_{i=1}^{n}|x_i - A|$$

where $A$ is an appropriately chosen measure of central tendency.

It can be shown that $MD_A$ is minimum if we take $A = \tilde{x}$, i.e. median, which is also a measure least affected by outliers. Thus we define another good measure of deviation, somehow less affected by outliers, as mean deviation about median,

$$MD_{\tilde{x}} = \frac{1}{n}\sum_{i=1}^{n}|x_i - \tilde{x}|.$$

Similarly we can define mean deviation about mean as

$$MD_{\bar{x}} = \frac{1}{n}\sum_{i=1}^{n}|x_i - \bar{x}|.$$

Note that it always true that $MD_{\tilde{x}} \leq MD_{\bar{x}}$ for any data set. For the gene expression data (Table 2.1), $MD_{\tilde{x}} = 1.358$ and $MD_{\bar{x}} = 1.341$. R code to calculate mean deviation about mean and median is given below.

```
> md.median=mean(abs(x - median(x)))          ## mean deviation about median
> md.median                                   ## showing output on the screen
> round(md.median,3)          ## if you want the output up to three decimal places
> md.mean=round(mean(abs(x - mean(x))),3)  ## mean deviation about mean up
                                                to three decimal places
```

**Standard deviation**

Although $MD_{\bar{x}}$ has an advantage that it is somewhat less affected by outliers, its major disadvantage is that it is not really easy to handle mathematically for further analysis. Working with mean deviation sometimes may be difficult. To make our lives simpler, we consider squaring the values of these differences (instead of absolute deviations) and take their mean. However, there is a problem. While squaring the differences, we have also squared the unit in which observations are measured. An easy solution to get rid of this problem is to just take the square-root of the mean thus obtained. Hence we get a measure of dispersion with respect to the chosen average $A$, as given by,

$$SD_A = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - A)^2}.$$

This is known as root mean square deviation about $A$. Since for different choices of $A$, we get different values and it can be very large for some choices of $A$, it is preferable to take the value of $A$ for which this will be minimum. It can be shown mathematically that $SD_A$ would be minimum if we take $A = \bar{x}$. Hence, we define standard deviation as a measure of dispersion as

$$s_x = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2}.$$

The standard deviation of gene expression values for the data in Table 2.1 is found to be 1.865 using the R command for standard deviation as '$sd(x)$'. Remember that this command calculates the standard deviation taking $n - 1$ in the denominator instead of $n$. So a simple multiplication factor would give $s_x$ as we defined just now.

```
> sd(x)                       ## denominator being n-1
> sd(x)*sqrt((n-1)/n)         ## denominator being n, as defined
```

Note that if the sample size is moderately large, it does not matter whether we take $n$ or $n - 1$ in the denominator; both results would be approximately same. Standard

deviation as obtained in R using 'sd()' is 1.8649 and that using divisor $n = 90$ as 1.8545. Note that $MD_{\tilde{x}} = 1.341$, which is naturally less than $s_x$ as it is less affected by outliers. It can also be proved mathematically that for any data set, $MD_{\tilde{x}} \leq s_x$. However, standard deviation is the most popular and widely used measure of dispersion.

### Quartile deviation

Based on the above discussion, we need to device another measure of dispersion that is least affected or not affected at all by outliers. It is clear that this measure should be constructed using median or measures like median that is not affected by outliers. To define such a measure, that is least affected by outliers, we first define quartile for a distribution. We define $Q_1$, and $Q_3$ as 1st, and 3rd quartile respectively, as the value of the variable such that the number of observations less than that value is 25% and 75% respectively. Similarly we can define $Q_2$ as the value below which there are 50% observations. Clearly $Q_2$ is the median and the interval $(Q_1, Q_3)$ contains middle 50% observations. Naturally any measure based on $Q_1$ and $Q_3$ would be free from the effect of outliers. Longer this interval, more disperse the data would be. Thus we define a new measure of dispersion based on quartiles only, known as **quartile deviation**, as:

$$QD = \frac{Q_3 - Q_1}{2}$$

For the data set in Table 2.1, $QD$ is found to be 1.07.

Note that although this measure of dispersion is completely robust to the presence of outliers, it depends only on the middle 50% observations and ignores 50% observations that lie towards the ends or 'tails' of the distribution. Naturally this incorporates serious loss of information from the data as a price to remove the effect of outliers completely. This might be one reason for not using quartile deviation very frequently as a measure of dispersion. Sometimes $Q_3 - Q_1$ is known as **interquartile range**.

```
> QD= (quantile(x,0.75)-quantile(x,0.25))/2      ## calculates and stores
                                                    quartile deviation
> QD                                  ## showing quartile deviation on the screen
```

Till now we have discussed the intuitive approach to address the question (1)–(4) as mentioned in Sect. 2.1 and the corresponding mathematical formula based on statistical intuitive argument. Recalling these questions, a careful look at histograms and/or boxplots reveals the general features of data, apply our thought process and the development of measures of central tendency and dispersion is followed naturally.

## 2.4 Points to Remember

We have discussed a few visual tools for representation of data for a single variable and some basic summary measures to understand the characteristics of data. The data we consider here is gene expression data for a single gene for a number of individuals randomly selected from a population. We must emphasise that these visual tools as well as summary measures are equally important and applicable to understand the salient features of any data set corresponding to a continuous variable. This should be the starting point of statistical analysis of the data set arising out of any random experiment that gives continuous frequency data. We should always give serious attention to the presence and effect of outliers as in the later sections we see how important and sometimes distorting effects it can cast in the data analysis.

Now one very simple but trivial question is whether one should or should not analyse the data. Suppose an experiment generates values all equal to 2.57; we cannot statistically analyse that data. The only reason is absence of any variation and we doubt whether the data has been generated out of a random experiment. Such data are not amenable to statistical data analysis. If values in the data are different (may not be all values but mostly) and there is a prior knowledge about data generation process that ensures introduction of randomness, we must do statistical analysis in the best possible way. Anything between these two extreme scenarios should be guided by the jurisprudence of the experimenter, nature of data, knowledge about the data generation process, and most importantly the hypothesis and objective of the experiment.

## Exercise

2.1 Based on some ideas and queries, a scientist performed an experiment that gives data on expression values of 12 genes from 102 individuals. Can you guess what should be the questions that triggered this experiment? Try to come up with as many questions as possible.

2.2 In Question 2.1, after generating the data it is observed that data from all genes from all individuals were not obtained. Can you give some idea about the data structure and relevant questions that might be absent previously?

2.3 Consider the following data set.

**data set 1:**

3.5, 3.7, 5.7, 4.2, 5.4, 2.9, 2.3, 8.6, 3.4, 3.1, 4.5, 7.8, 6.9, 6.2, 2.6, 6.0, 5.0, 10.3, 9.9, 11.9, 7.3, 3.0, 3.9, 2.0, 6.7, 4.0, 5.7, 1.4, 6.0, 2.4, 3.7, 9.2, 3.8, 4.5, 9.9, 6.3, 2.0, 5.2, 4.0, 1.4, 2.0, 4.4, 7.9, 1.6, 12.6, 3.9, 4.0, 3.3, 4.7, 2.9, 1.3, 4.5, 3.7, 4.9, 2.2, 2.4, 2.5, 1.7, 3.7, 3.3

(a) What is the type of the data?

(b) If it is continuous, draw a histogram with 6 classes and also draw a boxplot. If not, don't go for Questions 2.3 (c) and justify your answer.

(c) Draw the above histogram and boxplot in one graph. [hint: use the R command 'par(mfrow=c(1,2))']
(d) What is the percentage of observations that is less than 5.2?
(e) What is the percentage of observations that is more 3.7?
(f) What is the percentage of observations that are greater than or equal to 4.5?
(g) What percentage of observations lies between 3.84 and 6.12?
(h) Calculate appropriate measures of central tendency and dispersion.
(i) Comment on your findings.
(j) Calculate mean eliminating upper 10% and lower 10% observations.

2.4 Summary measures are defined mathematically so that we can calculate them using a set of observations. However, interpretation is necessary. Interpret arithmetic mean, median and standard deviation in simple language. Use some examples to explain your interpretation.

2.5 The hemoglobin level (g/dl) of a sample of 50 men are given below.
17.0, 17.7, 15.9, 15.2, 16.2, 17.1, 15.7, 17.3, 13.5, 16.3, 14.6, 15.8, 15.3, 16.4, 13.7, 16.2, 16.4, 16.1, 17.0, 15.9, 14.0, 16.2, 16.4, 14.9, 17.8, 16.1, 15.5, 18.3, 15.8, 16.7, 15.9, 15.3, 13.9, 16.8, 15.9, 16.3, 17.4, 15.0, 17.5, 16.1, 14.2, 16.1, 15.7 15.1, 17.4, 16.5, 14.4, 16.3, 17.3, 15.8
Calculate Mean, Median, Range, Variance, Standard deviation and Quartile deviation.

2.6 In reference to Question 2.5, let $x_i$ denote the hemoglobin level (g/dl) of $i$-th individual; $i = 1, 2, \ldots, 50$. Define a new variable as $y_i = x_i - \bar{x}$, $i = 1, 2, \ldots, 50$. Calculate Mean and Variance of $y$.

2.7 Here is a frequency distribution of the Apgar scores for 100 low birthweight infants. Find the Mean and SD of Apgar scores.

| Argar Score | Frequency |
|---|---|
| 0 | 6 |
| 1 | 1 |
| 2 | 3 |
| 3 | 4 |
| 4 | 5 |
| 5 | 10 |
| 6 | 11 |
| 7 | 23 |
| 8 | 24 |
| 9 | 23 |
| Total | 100 |

2.8 In data set 1 (Question 2.3), suppose we remove the observation 12.6 from the data set and calculate new mean and new standard deviation based on remaining observations. It is seen that both new mean and new standard deviation are smaller than the original mean and original standard deviation. However, if

we remove the observation 4.7 from data set 1, we see that both new mean and new standard deviation are larger than the original mean and original standard deviation. Can you explain these situations?

2.9 Suppose there are 100 observations in a data set. We can calculate arithmetic mean $\bar{x}$, all three quartiles i.e., $Q_1$, $Q_2$(=median), $Q_3$ and standard deviation $s$. Now replace top 25% observations by $Q_3$ and the bottom 25% observations by $Q_1$. What would be the effect of such operation on the summary measures mentioned above. Justify your answer clearly.

2.10 Consider the following two data sets.

**data set 2:**

10.01, 14.18, 9.95, 11.71, 12.26, 9.86, 13.33, 10.70, 7.46, 11.19, 8.07, 6.23, 7.17, 7.77, 9.95, 12.16, 10.55, 10.03, 14.11, 9.84, 9.36, 11.18, 9.61, 9.99, 10.98, 8.90, 11.09, 10.42, 11.28, 12.85, 10.10, 10.39, 8.99, 10.41, 9.31, 9.11, 10.62, 11.18, 10.35, 6.02, 4.06, 10.54, 8.95, 5.58, 10.40, 9.65, 12.45, 12.15, 9.66, 13.30, 8.12, 11.73, 11.01, 13.53, 10.80, 11.14, 11.60, 10.39, 9.39, 8.38, 11.16, 13.00, 9.72, 6.38, 8.83, 8.33, 9.51, 5.35, 9.04, 13.35, 9.26, 11.56, 10.09, 8.62, 9.75, 10.11, 10.09, 10.37, 8.84, 6.32

**data set 3**

10.68, 10.92, 12.27, 17.62, 15.55, 12.04, 12.91, 11.20, 13.55, 11.72, 16.27, 16.88, 15.52, 17.14, 14.16, 15.28, 18.92, 13.32, 19.80, 15.24, 12.57, 17.13, 13.27, 17.37, 11.51, 14.10, 15.29, 11.56, 13.43, 12.27, 13.90, 13.67, 12.49, 13.46, 15.40, 13.30, 15.86, 14.10, 12.28, 12.54, 16.97, 14.06, 14.75, 17.78, 15.34, 9.52, 15.91, 19.21, 12.72, 16.85, 18.64, 15.90, 12.35, 13.76, 15.78, 18.18, 9.63, 18.68, 17.11, 12.69, 11.15, 16.57, 11.81, 15.71, 17.29, 13.69, 16.46, 13.82, 9.18, 11.24, 20.09, 14.52, 13.93, 11.85, 15.34, 17.12, 12.92, 12.46, 13.46, 14.15, 14.97, 9.57, 16.12, 21.06, 15.37, 15.02, 16.44, 14.43, 19.24, 19.51

(a) Present these two data sets diagrammatically, preferably in a single graph, so that we can compare them. Write down your observations.
(b) Using the commands for Fig. 2.2 or other commands, can you add colours to different parts of the histogram drawn for data set 2, change the labels for x-axis and y-axis, change the legends and shade area of each rectangle?
(c) Draw two boxplots for data set 2 and data set 3 side by side on the same graph, and distinguish them by colours as well as legends.
(d) Analyse the data based on appropriate measures of central tendency and dispersion. Analysis means not only to look at the features of each data set, but a comparison between two data sets is essential.
(e) Comment on your findings.
(f) Calculate third quartile for both data sets and comment on the nature of distribution.
(g) Calculate first, second and third quartiles for both data sets and comment on the nature of distribution considering all these calculated measures.
(h) Identify at which values (at least approximately) the peak for frequency curve would occur for both data sets? Comment on your findings.

2.11 Can you explain intuitively why mean deviation about median is less than the mean deviation about mean?

2.12 Define a measure of central tendency ignoring top 10% and bottom 10% of the observations. Write an R code to calculate such a measure.

2.13 Draw a histogram for data set 3 in Question 2.10 using only 9 classes of equal width. Change the colour of inside and border of the histogram.

2.14 Draw Fig. 2.3 with less-than type ogive in red colour and greater-than type ogive in blue colour. Also try to make the code little shorter.

2.15 In drawing ogive, why do we join two consecutive points by a straight line? "In an ogive, the first and last line segments are always parallel to the horizontal axis" - justify this statement.

2.16 Draw two ogives using data set 3 as in Question 2.10. Calculate median from the ogive.

2.17 From the ogive drawn based data set 3 in Question 2.10, calculate percentage of observations that are

(a) less than 15,
(b) greater than 17, and
(c) between 12 and 18.

Do the same from the histogram also and check whether your answers match.

2.18 For the above three data sets, we want to study the dispersion.

(a) Calculate quartile deviation for each data set and comment on your findings.
(b) Do you think your comments remain more or less same if you compare their dispersions using standard deviations? Justify your answer.

2.19 Suppose we calculate means $\bar{x}_1$ and $\bar{x}_2$ based on two data sets of sizes $n_1$ and $n_2$ respectively. If we merge the two data sets, what would be the arithmetic mean ($\bar{x}$, say) of the combined group in terms of known quantities? Show that $\bar{x}$ lies between $\bar{x}_1$ and $\bar{x}_2$. Extending this problem to $k$ groups with means $\bar{x}_1, \ldots, \bar{x}_k$ with sample sizes for the groups as $n_1, \ldots, n_k$ respectively, find the expression of the arithmetic mean when all groups are merged together. Show that,

$$\min_{i=1,\ldots,k} \{\bar{x}_1, \ldots, \bar{x}_k\} \leq \bar{x} \leq \max_{i=1,\ldots,k} \{\bar{x}_1, \ldots, \bar{x}_k\}$$

2.20 Based on a set of 20 observations, an experimenter calculates arithmetic mean as 5.68. However, after a few days, while checking the entire process, it becomes clear that by mistake the experimenter has taken a value 12.4 while calculating the mean. Since this value is wrongly taken, so a correction is necessary. Moreover, at this stage the raw data are not available any more. The experimenter wants to recalculate the mean just by removing this wrong observation. How can he calculate the mean removing 12.4 and using only the available measures (not the data)? What would be the new corrected mean?

2.21 In Question 2.20, suppose it becomes known that the experimenter uses 12 instead of 2.4. How does he modify his calculation? What would be the corrected value of mean?

2.22 The weight (Kg.) of 10 individuals are recorded as: 52, 62, 54, 82, 49, 68, 62, 34, 97, 72. After scrutiny, it is found the lowest and the highest weights are wrongly reported. The true highest is 82 Kgs and the second highest weight is 79 Kgs. If the average weight is 67 Kgs, what is the true lowest weight?

2.23 Let $z_i = x_i + y_i, i = 1, 2, \ldots, 7$. Show that the mean of $z$ is the sum of mean of $x$ and mean of $y$ using the data. Also calculate the variance of $z$. Is it equal to the sum of $var(x)$ and $var(y)$?

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| $x$ | 21 | 71 | 32 | 78 | 56 | 73 | 74 |
| $y$ | 82 | 172 | 102 | 167 | 134 | 160 | 155 |

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# Chapter 3
# Basic Probability Theory and Inference




Most people have an apprehension that probability theory is a very difficult area of mathematics or statistics. However, if we don't go into the details and intricacies of probability theory, it is very intuitive and easy to understand. In fact, probability theory forms the core of real data analysis. It is indeed vast. But to begin with, a simple understanding of probability theory is not at all difficult and can be framed or constructed from real-life phenomena. People always wonder: (1) why do we need probability theory? (2) does it have anything to do with a real-life situation or experiment?

Well, the concept of probability is probably at the core of any experiment that generates data. To understand its concept clearly, we need to know a few important ideas, which are common in daily life.

## 3.1 Random Experiment

Probability is there in our everyday life. We make probabilistic statements every now and then, knowingly or unknowingly. When you go out, your mother tells you to carry an umbrella because it might rain after a little while. Looking at the sky, she predicts not the rainfall but the possibility of raining. A cloudy sky might lead her to say that there is a high chance of rain today; so you should carry an umbrella.

From these statements, two facts are obvious: (1) she predicts based on the condition of the sky, how it was yesterday and/or how it looks today and (2) her prediction is not with certainty, rather there is some amount of uncertainty associated with her statement. This uncertainty is nothing but the backbone of probability theory that provides a measure of the quantification of uncertainty about an event whose outcome may not be always predicted accurately, but can be guessed.

Usually, whatever be the nature of data, they are subject to some random causes that give rise to fluctuations present in the data. We call such an experiment 'random

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I. Mukhopadhyay and P. P. Majumder, *Statistical Methods in Human Genetics*,
Indian Statistical Institute Series, https://doi.org/10.1007/978-981-99-3220-7_3

experiment'. Precisely, a random experiment is an experiment whose outcome cannot be predicted beforehand with certainty, but we might know all possible outcomes that this experiment would generate.

The most classical example of a random experiment is the 'coin tossing experiment'. If we toss a coin, we don't know its outcome; however, we know that it must be either 'head' or 'tail'. If we throw a die, we know that the upper face will show one of the numbers from the set $\{1, 2, 3, 4, 5, 6\}$; but we cannot say with certainty which one will appear. This uncertainty is due to the inherent randomness of the experiment, in this case observing the number on the upper face when we throw a die. Similarly choosing a person at random (i.e. without attaching any preference to anybody) from a group of persons having three categories of blood pressure, viz., low, normal, and high blood pressure, is another example of a random experiment. We know that a randomly selected person will have either low or normal or high blood pressure (all possible cases), but we would be able to know the actual status of blood pressure only when a person is selected and her/his blood pressure level is measured. These are some examples of a random experiment.

## 3.2 Event

An 'event' may be defined as an outcome or a set of outcomes or a combination of outcomes of a random experiment. Consider a random experiment where we toss a coin two times and note the outcome. Let $H$ and $T$ denote 'head' and 'tail' respectively. So, $\{HH\}$ denotes two heads, $\{HT\}$ denotes head in the first toss and tail in the second toss, etc. These events are called 'elementary events'. In this case, all possible elementary events are:

$$\{HH\}, \{HT\}, \{TH\}, \{TT\}$$

However, we can think of another event $A$ as '*getting at least one head*'. So, here $A = \{HH, HT, TH\}$. This event, which is a combination of elementary events is sometimes known as a 'composite event'. Whatever be the name, its treatment in probability theory is basically same; the only advantage of an elementary event is that probability of elementary events are easy to calculate and probability of composite events can be derived using probabilities of some elementary events that constitute the composite event.

Consider a die throwing experiment. When we throw a die, '*getting number 4*' is an elementary event. But '*getting odd number*' means observing either 1 or 3 or 5, i.e. it consists of three elementary events and is a composite event. Similarly, '*getting an odd number which is less than 5*' is an example of a composite event. In the case of each of these events, we don't know what we get if we throw a die. Uncertainty again! But we understand that the degree of uncertainty varies depending on events.

Now that we have an idea about random experiments and associated events, we should try to define probability of an event, be it elementary or composite.

## 3.3 Idea and Definition of Probability

It is now clear that there is some amount of uncertainty in almost all real-life phenomena or experiments. We need quantification of this uncertainty as the degree of uncertainty is not same in all cases. Probability is basically how we can quantify the uncertainty in an experiment, or rather the uncertainty to observe an event. Before starting a game, the referee tosses a coin in the air and the captain of one team shouts 'head'. The choice of 'head or 'tail' may depend on her/his prior belief, but it is expected that she/he has an equal chance of winning or losing the toss; otherwise, it would be unfair. Assume that the coin is a fair coin. So, we say that chance of getting a 'head' in a single toss is 50%, i.e. probability of winning (or losing) is $\frac{1}{2}$. 'head' can occur in only one way out of two possible ways as the coin has only two faces marked as 'head' and 'tail'. We can interpret this in another way. If we toss 10 times, we expect to get an equal number of heads and equal number of tails approximately (we can never be sure!). Considering this, the classical definition of probability of an event $A$ (getting 'head' in a toss is an event) is given by,

$$P(A) = \frac{\text{number of outcomes favourable to the event } A}{\text{total number of events}}$$

Note that here we assume all outcomes are equally likely to occur.

However, this classical definition has a few drawbacks. Without going into the details of it, we can discuss this with an example. Suppose that in my wrist watch the big hand strikes at 12. What is the probability that it will stop functioning before it strikes 6?

Looking carefully it reveals that the big hand may stop at any moment between two 12s; that means the number of total cases is really infinite. But the number of favourable cases is also infinity because there are enumerable points on the wristwatch between 12 and 6 where it can stop. So, probability of the desired event cannot be calculated, if we stick to the classical definition.

But anyone can give the correct answer as $\frac{1}{2}$! It is easy to see that the totality of cases is the entire circumference whereas favourable situation is the half circumference (Fig. 3.1), because the grey curve represents the favourable space, whereas the total circumference is the total possible space. Clearly, the length of the grey curve is exactly half of the total circumference; hence, the probability would be $\frac{1}{2}$.

Thus, classical definition is not a general one; one can give a more precise definition of probability. The formal definition of probability requires some mathematical concepts, which we state without details.

Let $\left(\Omega, \mathscr{A}, P\right)$ be a probability space where $\Omega$ is the sample space, a collection of all outcomes, $\mathscr{A}$ is the $\sigma$-field generated by the class of subsets of $\Omega$, and $P(.)$ is a set function satisfying the following conditions:

(1) $P(A) \geq 0$ for any $A \in \mathscr{A}$,
(2) $P(\Omega) = 1$, and

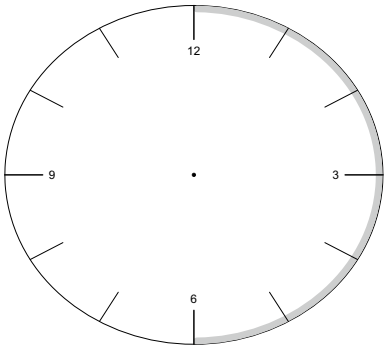



**Fig. 3.1** Clock showing how the probability that the big hand might stop between 12 and 6. Grey part is the favourable region whereas the entire circumference is the total possible region where the big hand might stop

(3) If $A_1, \ldots, A_k$ be mutually exclusive events such that $A_i \in \mathscr{A}$, $\bigcup_{i=1}^{k} A_i \in \mathscr{A}$, then $P\left(\cup_{i=1}^{k} A_i\right) = \sum_{i=1}^{k} P(A_i)$.

**A few examples**

**Ex-3.1** A coin is tossed two times. The possible elementary events are: $\{HH\}$, $\{HT\}$, $\{TH\}$, $\{TT\}$. Assuming the coin is unbiased, i.e. getting a head is as likely as getting a tail, we might expect to observe $\{HH\}$ only once if we repeat the experiment four times. So,

$$P(\{HH\}) = \frac{1}{4}.$$

Let $A$ be the event of getting at least one $H$. Here, occurrence of any one of $\{HH\}$, $\{HT\}$, and $\{TH\}$ would indicate the occurrence of the event $A$ and there are altogether four such elementary events when we toss the coin two times. Hence,

$$P(A) = \frac{3}{4}.$$

**Ex-3.2** A group consists of 26 unaffected and 3 affected people. Select one person at random. What is the probability that the selected person is affected?
Number of events favourable to our desired event is 3 because selection of any one of three affected people will satisfy our criterion. On the other hand, the total number of selections is 29 and any one of 29 people may be selected at random. Hence,

$$P(\text{selecting one affected people}) = \frac{3}{29}.$$

## 3.4 Some Facts

Now we state a few facts, or results (without proof) related to the probability theory. These are very useful to proceed further. Although we are not giving any proof of these results, however, we try to explain them through diagrams so that a clear idea would emerge. A few of them come directly from the definition of probability. But, first, we define a few operations using sets (or events) as follows:

**Complement of a set:** $A^c$, which indicates non-occurrence of $A$. Let $A$ be an event that an odd number appears when a die is thrown. So, $A^c =$ { numbers that are not odd } $= \{2, 4, 6\}$.

**Union of sets:** $\{A \cup B\}$, i.e. $\{A$ or $B\}$ indicates that either $A$ occurs or $B$ occurs or both occur. In die throwing experiment with one die, if $A =$ { number appears is odd } $= \{1, 3, 5\}$, $B =$ { multiple of 3 appears } $= \{3, 6\}$, then $A \cup B$ means any one of the numbers in the set $\{1, 3, 5, 6\}$ would appear.

**Intersection of sets:** $\{A \cap B\}$, i.e. $\{A$ and $B\}$ indicates that both $A$ and $B$ occur. In the previous example, $A \cap B = \{3\}$, because only 3 satisfies both $A$ and $B$.

**Mutually exclusive sets:** Two sets $A$ and $B$ are mutually exclusive or disjoint sets if $A \cap B = \phi$, where $\phi$ denotes a null set. In throwing a die, let $A =$ { an odd number appears } $= \{1, 3, 5\}$ and $B =$ { an even number appears } $= \{2, 4, 6\}$. Therefore, $A \cap B = \phi$, i.e. null set; hence $A$ and $B$ are mutually exclusive sets or disjoint sets.

Figure 3.2 presents a diagrammatic representation of these four types of set operations. Note that any set operation when applied on a set or a number of sets, will give another set.

Now we state a few facts.

(1) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
(2) For any $A$ and $B$ such that $A \subset B$, $P(A) \leq P(B)$.
(3) $0 \leq P(A) \leq 1$.
(4) $P(A^c) = 1 - P(A)$.
(5) $P(\phi) = 0$ where $\phi$ is a null set.
(6) If $A$ and $B$ are mutually exclusive events, then $P(A \cap B) = 0$.

Mathematical proofs along with intuitive 'proofs' are given in Appendix C.

### *A Few Examples*

We now give a few examples that use these facts. If we can define an event as a set, it is very easy to calculate probabilities of several events (sets). First, we try to find out probabilities of simple or elementary events. Then using them and the above-stated facts, we deduce probability of a little more complicated event that can be a combination of elementary events.

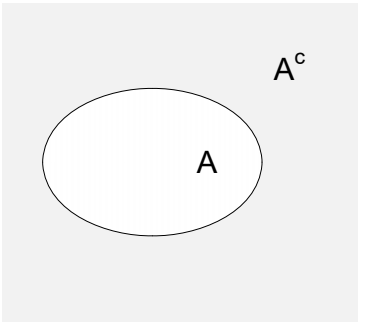



Fig. 3.2A : Shaded region is complement of set A

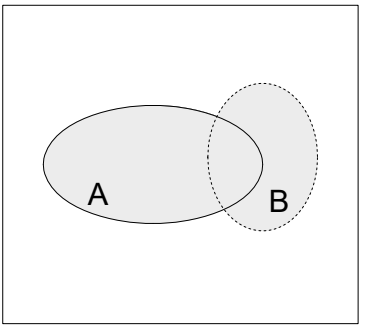



Fig. 3.2B: Shaded region is union of two sets

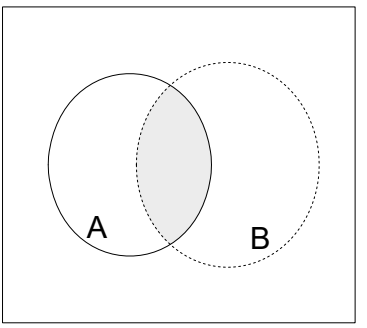



Fig. 3.2C: Shaded region is intersection of two sets

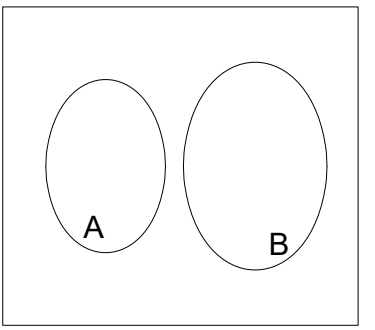



Fig. 3.2D: Mutually exclusive sets

**Fig. 3.2** A few set operations

**Ex-3.3** A die is thrown. Let $A = \{$ number appears is odd $\}$, $B = \{$ number appears is a multiple of 3$\}$. Clearly, $A = \{1, 3, 5\}$, $B = \{3, 6\}$ so that $(A \cap B) = \{3\}$. Now,

$$P(A) = \frac{\text{number of cases favourable to } \{1, 3, 6\}}{\text{total number of cases}} = \frac{3}{6}; \text{ similarly, } P(B) = \frac{2}{6},$$

$$P(\text{number appears is odd and a multiple of 3}) = P(A \cap B) = \frac{1}{6},$$

$\therefore$ $P(\text{number appears is a multiple of 3 or an odd number})$

$$= P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{3}{6} + \frac{2}{6} - \frac{1}{6} = \frac{4}{6} = \frac{2}{3}.$$

**Ex-3.4** Let $X =$ genotype of a randomly selected person at a bi-allelic locus with alleles $D$ and $d$. Define $A = \{DD\}$, $B = \{Dd\}$, and $C = \{dd\}$. Also it is

known that $P(A) = 0.3$, $P(B) = 0.2$, and $P(C) = 0.5$. Then,

$$P(X \text{ is homozygous}) = P(A \text{ or } C) = P(A \cup C) = P(A) + P(C) - P(A \cap C) = 0.3 + 0.5 - 0 = 0.8$$

Note that $A$ and $C$ cannot occur simultaneously because a person cannot have two genotypes $DD$ and $dd$ at the same locus; hence, $P(A \cap C) = 0$.

**Ex-3.5** A coin is tossed three times. Let $A = \{\text{at least one head appears}\}$. Noting all possible outcomes when a coin is tossed three times, we have $A = \Big\{\{HHH\}, \{HHT\}, \{THH\}, \{HTH\}, \{HTT\}, \{THT\}, \{TTH\}\Big\}$.
Clearly $P(A) = \frac{7}{8}$. However, it is sometimes difficult to count all such cases favourable to $A$, especially when the number of tosses is large. So we use complementation rule to evaluate $P(A)$ very easily.
Note that $A^c = \{\text{ no head appears}\} = \{TTT\}$ and naturally $P(A^c) = \frac{1}{8}$. Therefore, we have,

$$P(A) = 1 - P(A^c) = 1 - \frac{1}{8} = \frac{7}{8} \qquad \text{[matches with our direct calculation!!!]}$$

**Ex-3.6** In Example 3.4, let $F = \{\text{ genotype contains at least one } D \text{ allele }\}$. Clearly, $F = C^c$, where, as given in the previous Example 3.4, $C = \{\text{ genotype contains no } D \text{ allele }\} = \{dd\}$.
So, $P(F) = P(C^c) = 1 - P(C) = 1 - 0.5 = 0.5$.

Before proceeding further, we introduce another concept named 'random variable' that is related to probability theory. The idea of a random variable and subsequent treatment help in calculation of probability of many composite events very easily.

## 3.5 Random Variable

Let us consider the classical coin tossing experiment. Suppose we toss a fair coin three times and we are interested to evaluate probabilities of several events related to this experiment. But, first let's write down all possible outcomes and the corresponding probabilities (Table 3.1).

Now, we define a quantity $X$ as

$$X : \text{ number of heads in three tosses of a fair coin}$$

Clearly, $X$ takes the values 0 or 1 or 2 depending on the outcome. For example, if $\omega = \{HHH\}$, $X$ takes the value 3, i.e. $X(\omega) = 3$; similarly, if $\omega = \{HTT\}$, $X$

**Table 3.1** Outcomes and corresponding probabilities when a fair (unbiased) coin is tossed three times

| Event ($\omega$) | HHH | HHT | HTH | HTT | THH | THT | TTH | TTT |
|---|---|---|---|---|---|---|---|---|
| Probability | $\frac{1}{8}$ | $\frac{1}{8}$ | $\frac{1}{8}$ | $\frac{1}{8}$ | $\frac{1}{8}$ | $\frac{1}{8}$ | $\frac{1}{8}$ | $\frac{1}{8}$ |

**Table 3.2** Outcomes, random variable realisations and corresponding probabilities when a coin is tossed three times

| $\omega$ | HHH | HHT | HTH | HTT | THH | THT | TTH | TTT |
|---|---|---|---|---|---|---|---|---|
| Probability | $\frac{1}{8}$ | $\frac{1}{8}$ | $\frac{1}{8}$ | $\frac{1}{8}$ | $\frac{1}{8}$ | $\frac{1}{8}$ | $\frac{1}{8}$ | $\frac{1}{8}$ |
| $X$ | 3 | 2 | 2 | 1 | 2 | 1 | 1 | 0 |

becomes 1, i.e. $X(\omega) = 1$ and so on. We can present the possible values that $X$ takes depending on $\omega$ in Table 3.2.

By introducing $X$ in this way, we may have some advantages. We can describe any event, elementary or composite in terms of value(s) of $X$ and calculate the probability corresponding to it. For example,

$$P(\text{exactly 2 heads}) = P(HHT, HTH, THH) = \frac{3}{8} = \boldsymbol{P(X = 2)}$$

$$P(\text{at least 2 heads}) = P(HHH, HHT, HTH, THH) = \frac{4}{8} = \boldsymbol{P(X \geq 2)}$$

This '$X$' is known as 'random variable'; it varies, i.e. takes different values depending on different outcomes, and they are random in the sense that outcomes are random, i.e. cannot be predicted beforehand.

But the major advantage is that, sometimes, we can evaluate these probabilities without even looking at (or listing) all possible outcomes of an experiment! Looks surprising, but, yes, it's true. We shall discuss it later. Rather, now we shall consider a few examples that will use the rule of probability theory (discussed in Sect. 3.4).

**Ex-3.7** A die is thrown. Let us define a few events and evaluate their probability explaining each event in terms of a random variable $X$ (say), as given below. Define a random variable $X$ as the number appearing on the upper face of the die when it is thrown once.

$$A = \{\text{number appearing is more than 1}\} = (X > 1) = \{2, 3, 4, 5, 6\}$$
$$B = \{\text{number appearing is an odd number}\} = (X \text{ is odd}) = \{1, 3, 5\},$$
$$C = \{\text{number appearing is more than 4}\} = (X > 4) = \{5, 6\},$$
$$D = \{\text{number appearing is more than 5}\} = (X > 5) = \{6\}$$
$$P(A) = P(X > 1) = 1 - P(X = 1) = 1 - \frac{1}{6} = \frac{5}{6}$$
$$P(B \cap D) = P(\{X = \{1,\ 3,\ 5\}\}\ \textit{and}\ \{X > 5\}) = P(\phi) = 0,$$
[because no number satisfies both $B$ and $D$ simultaneously]
$$P(B \cup D) = P(B) + P(D) - P(B \cap D) = P(\{X = \{1,\ 3,\ 5\}\}) +$$
$$P(\{X > 5\}) = \frac{3}{6} + \frac{1}{6} = \frac{2}{3}$$
$$P(B \cup C) = P(B) + P(C) - P(B \cap C) = P(X = \ odd) + P(X > 4)$$
$$- P(X = 5) = \frac{1}{2} + \frac{2}{6} - \frac{1}{6} = \frac{2}{3}$$

We come back to probability theory again and introduce another idea, 'conditional probability'.

## 3.6 Conditional Probability

The name suggests that there should be 'something', may be an event which presents a condition. Probability of an event that we plan to calculate is related somehow to this condition. Since we always evaluate probability of an event, this conditioning should be with respect to another event. Let us discuss this with the help of an example.

Two boys Harry and Biltu are playing with a die with six faces numbered $\{1, 2, \ldots, 6\}$. Harry has already thrown the die and knows that the outcome is an odd number, i.e. either of 1 or 3 or 5. He asks Biltu "Hey, the number appeared is odd. Now can you tell me the probability that it may be a number greater than 1?"

Well, the answer is simple! Given that the number appeared is odd and there are three such numbers, only possible numbers that are greater than 1 are 3 or 5. Hence, Biltu immediately replies, it's easy! It is $\frac{2}{3}$.

Using notation of events, define two events $A$ and $B$ associated with the same random experiment. We are interested in evaluating $P(A$ given that $B$ has occurred). We denote this by $P(A|B)$ and is defined as:

$$P(A \cap B) = P(B)P(A|B) \text{ i.e. } P(A|B) = \frac{P(A \cap B)}{P(B)} \text{ provided } P(B) > 0.$$

Figure 3.3 presents the idea behind conditional probability. A portion of $A$ lies within $B$. While calculating conditional probability of $A$ given that $B$ has occurred

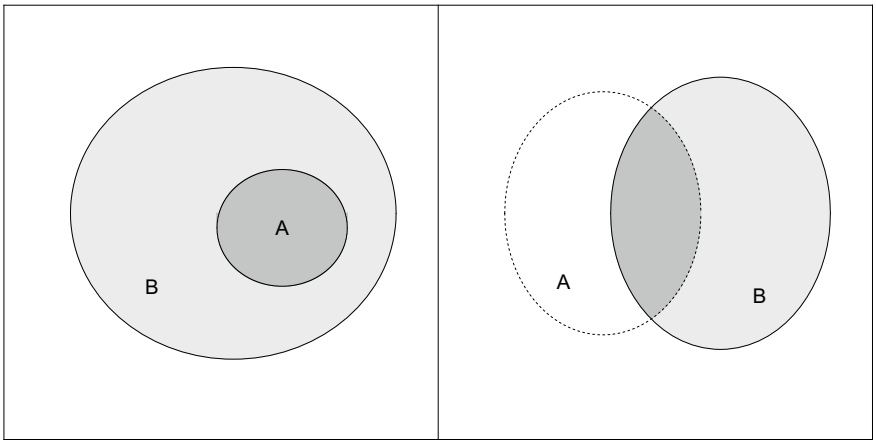



**Fig. 3.3** Conditional probability diagram: dark-shaded region indicates the required event, whereas lighter-shaded region indicates the conditioning event

already, we focus only on the dark-shaded region, i.e. portion of $A$ within $B$ and it is compared with respect to totality of $B$.

In the context of the above example, denote $A = (X > 1)$ and $B = (X \text{ is odd})$ and we want to evaluate $P(A|B)$. Using the definition conditional probability and noting that $P(A \cap B) = \{3, 5\}$, the probability that Harry asks for can be deduced as

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{2/6}{3/6} = \frac{2}{3}.$$

It is exactly the same answer that Biltu got by direct calculation. So depending on the situation, we can evaluate conditional probability either directly or using the formal definition. Both must give the same result.

**Ex-3.8** It is seen that the number of individuals having genotypes $AA$, $Aa$, and $aa$ at a bi-allelic locus are 30, 123, and 365 respectively. However, the number of affected individuals in these three groups respectively are 1, 4, and 29. Given that a person's genotype is $aa$, what is probability the person is affected?

The number of persons with genotype $aa$ is 365, out of which 29 are affected. So, the required probability is $\frac{29}{365}$, by direct calculation. Now, we calculate this using the formal definition. First define $B =$ { person with genotype $aa$} and $C =$ { person is affected}. Hence,

$$P(B \cap C) = P(\text{ a person with genotype } aa \text{ is affected}) = \frac{29}{518}$$
$$\text{and } P(C) = P(\text{ a person has genotype } aa) = \frac{365}{518}.$$
$$\therefore P(B|C) = \frac{P(B \cap C)}{P(C)} = \frac{29/518}{365/518} = \frac{29}{365}.$$

**Ex-3.9** Probability that a person will suffer from a disease if the person also carries a particular mutation is 0.7. The disease prevalence is 0.08. What is the probability that an affected individual carries the mutation?

Let $M = \{$ person carries mutation $\}, D = \{$ person suffers from the disease $\}$.
Here, it is given that $P(D|M) = 0.7$ and $P(D) = 0.08$.
Hence, $P(\text{ an affected person carries the mutation }) = P(D \cap M) = P(D|M)P(M) = 0.7 \times 0.08 = 0.056$.

Although from the above example, it seems that direct evaluation is easy, in many cases, the use of formal definition helps us a lot in understanding and evaluating probabilities in many complicated problems (to be seen later)!

### *3.6.1 Breaking a Probability Down by Conditioning*

In real life, an event may occur with many other events. In Example 3.8, selection of affected people is related with three groups with different genotypes. We cannot select an affected person directly; we have to select an affected person from any of three groups. In other words, the probability of selecting an affected person varies between genotype groups. Sometimes, it would be very useful if we can express an event (or set) in terms of other sets. Naturally, that motivates us to write the probability of the desired set in terms of probabilities of other sets, if possible. However, the probabilities of other sets should be evaluated easily; otherwise, this exercise has no value.

Look at Fig. 3.4. Part of $A$ occurs with $B$ and the rest of $A$ occurs with $B^c$. It is clear that $P(A)$ should be in terms of other probabilities, more specifically in terms of several conditional probabilities. We try to evaluate $P(A)$ using other probabilities.

$$\begin{aligned} P(A) &= P(\{A \text{ and } B\} \text{ or } \{A \text{ and } B^c\}) \\ &= P(\{A \cap B\} \cup \{A \cap B^c\}) \\ &= P(A \cap B) + P(A \cap B^c) \\ &\qquad [\text{since } A \cap B \text{ and } A \cap B^c \text{ are mutually exclusive events}] \\ &= P(B)P(A|B) + P(B^c)P(A|B^c) \\ &\qquad [\text{using definition of conditional probability}] \end{aligned}$$

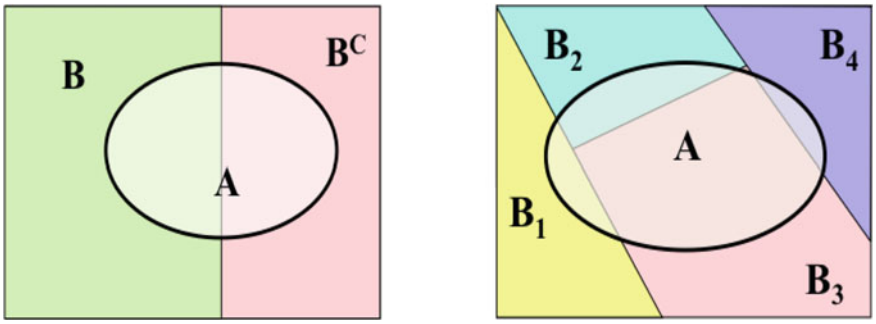



**Fig. 3.4** An event $A$ can be written in terms of two events and more than two events

Now suppose that we try to extend this idea when $A$ occurs with more than two events, say $B_1, B_2, \ldots, B_k$ with $k \geq 1$. Again looking at Fig. 3.4 and proceeding similarly, we have,

$$\begin{aligned} P(A) &= P\big[(A \cap B_1) \cup (A \cap B_2) \cup \cdots \cup (A \cap B_k)\big] \\ &= P(A \cap B_1) + P(A \cap B_2) + \cdots + P(A \cap B_k) \\ &= P(B_1)P(A|B_1) + P(B_2)P(A|B_2) + \cdots + P(B_k)P(A|B_k) = \sum_{i=1}^{k} P(B_i)P(A|B_i) \end{aligned}$$

**Ex-3.10** Suppose in a population at a bi-allelic locus with two alleles $A$ and $a$, the genotype probabilities (commonly known as genotype frequencies) for genotypes $AA$, $Aa$, and $aa$ are $p_{AA}$, $p_{Aa}$, and $p_{aa}$ respectively. Also, suppose the conditional probability of being affected for a given genotype $g$ is $f_g$, where $g = AA, Aa, aa$. What is the disease prevalence?

Here, $D = \{\text{disease}\}$, $f_{AA} = P(D|AA)$, $f_{Aa} = P(D|Aa)$, $f_{aa} = P(D|aa)$, $p_{AA} = P(AA)$, $p_{Aa} = P(Aa)$, and $p_{aa} = P(aa)$. Hence, the disease prevalence or the probability of disease becomes

$$\begin{aligned} P(D) &= P\big[(D \cap AA) \cup (D \cap Aa) \cup (D \cap aa)\big] \\ &= P(AA)P(D|AA) + P(Aa)P(D|Aa) + P(aa)P(D|aa) = \\ &\quad p_{AA}f_{AA} + p_{Aa}f_{Aa} + p_{aa}f_{aa} \end{aligned}$$

**Ex-3.11** There are two urns. Urn 1 contains 4 white and 3 black balls, while Urn 2 contains 5 white and 6 black balls. One ball is drawn at random and we don't know from which urn it is drawn. What is the probability that the ball is white?

First, define a few events that will facilitate the understanding and hence computation of the required event. Let $A = \{\text{ball drawn is white}\}$, $B_1 = \{\text{Urn 1 is selected}\}$, and $B_2 = \{\text{Urn 2 is selected}\}$.
Since we don't know from which urn the ball is finally drawn, we have to consider all possibilities, i.e. simultaneous occurrence of $A$ with both $B_1$ and $B_2$. Since we can first select any urn randomly, we have

$$P(B_1) = P(B_2) = \frac{1}{2}.$$

Again, by the given conditions, we have

$$P(A|B_1) = \frac{4}{7},\ P(A|B_2) = \frac{5}{11},$$
$$\text{and hence, } P(A) = P\big[(A \cap B_1) \cup (A \cap B_2)\big] = P(B_1)P(A|B_1) + P(B_2)P(A|B_2) = \frac{1}{2}.\frac{4}{7} + \frac{1}{2}.\frac{5}{11} = \frac{79}{154}$$

### 3.6.2 *Bayes' Theorem*

Continuing with Example 3.8, one can ask a very pertinent question: if we know the disease prevalence and genotype frequencies in a population, can we find the probability that a person has genotype $AA$ if a person is known to be affected?

This type of question has a deep interpretation. Here, we try to find the probability of a *cause* when *an event has occurred* by looking at *prior probabilities* of several causes and occurrence of the current event under each such cause. This question gave rise to one of the most celebrated theorems in probability theory, the 'Bayes' Theorem', postulated and proved by Reverend Bayes.

We use the same set-up as described in Sect. 3.6.1 for both two events as well as $k$ events.

**Theorem 1** $P(B|A) = \frac{P(B)P(A|B)}{P(B)P(A|B)+P(B^c)P(A|B^c)}$

***Proof***

$$P(B|A) = \frac{P(B \cap A)}{P(A)} = \frac{P(B \cap A)}{P\big[(A \cap B) \cup (A \cap B^c)\big]} = \frac{P(B \cap A)}{P(A \cap B) + P(A \cap B^c)}$$
$$= \frac{P(B)P(A|B)}{P(B)P(A|B) + P(B^c)P(A|B^c)}$$

For more than two conditioning events $B_1, B_2, \ldots, B_k$, we have Bayes' theorem as:

**Theorem 2** $P(B_1|A) = \frac{P(B_1)P(A|B_1)}{\sum_{i=1}^{k} P(B_i)P(A|B_i)}$

Proof follows exactly in the same way as in Theorem 1. Thus, through Bayes' theorem, we get the *posterior probability* of $B_i$ in terms of *prior probabilities* $P(B_i),\ i = 1, 2, \ldots, k$ and the conditional probabilities $P(A|B_i),\ i = 1, 2, \ldots, k$.

**Ex-3.12** In Example 3.10, what is the probability that genotype of an affected person is $AA$?

In the same set-up on Example 3.10, we have

$$P(AA|D) = \frac{P(AA)P(D|AA)}{P(AA)P(D|AA) + P(Aa)P(D|Aa) + P(aa)P(D|aa)} = \frac{p_{AA} f_{AA}}{p_{AA} f_{AA} + p_{Aa} f_{Aa} + p_{aa} f_{aa}}$$

**Ex-3.13** Two different coins, coin 1 is unbiased and coin 2 is biased with $P(Head|\text{ coin 2}) = 0.7$. David selects one coin at random and toss it once. The coin shows '$Tail$'. What is the probability that the unbiased coin, i.e. coin 1 is used?

This is a direct application of Bayes' theorem. However, we don't need to guess whether we apply Bayes' theorem or not. It will reveal itself if we start writing down several events associated with this problem and evaluate the probability of the required event.
Let $A_1 = \{\text{ Coin 1 is used}\}$, $A_2 = \{\text{ Coin 2 is used}\}$, and $B = \{\text{ Tail appears}\}$. So, we have $P(A_1) = P(A_2) = 0.5$, since a coin is selected at random. Also $P(B|A_1) = 0.5$ and $P(B|A_2) = 1 - 0.7 = 0.3$. Hence,

$$P(A_1|B) = \frac{P(A_1 \cap B)}{P(B)} = \frac{P(A_1)P(B|A_1)}{P\left[(A_1 \cap B) \cup (A_2 \cap B)\right]} = \frac{P(A_1)P(B|A_1)}{P(A_1)P(B|A_1) + P(A_2)P(B|A_2)} = \frac{0.5 \times 0.5}{0.5 \times 0.5 + 0.5 \times 0.3} = \frac{0.5}{0.5 + 0.3} = \frac{5}{8}$$

**Ex-3.14** In connection with Example 3.11, suppose we know that the ball drawn is white. We want to know the probability that it comes from Urn 2.

Again direct application of Bayes' theorem yields the result. We have

$$P(B_2|A) = \frac{P(B_2)P(A|B_2)}{P(B_1)P(A|B_1) + P(B_2)P(A|B_2)} = \frac{\frac{1}{2} \cdot \frac{5}{11}}{\frac{1}{2} \cdot \frac{4}{7} + \frac{1}{2} \cdot \frac{5}{11}} = \frac{35}{79}$$

## 3.7 Independence

Sometimes, it may so happen that the conditioning event has nothing to do with the event under consideration. That means the conditional probability of $A$ given the event $B$ is same as just $P(A)$, i.e. $P(A|B) = P(A)$. In such a situation, we can say that $B$ has no influence on $A$ and hence the events $A$ and $B$ are 'independent'. Formally, we can say that $A$ and $B$ are independent if (assuming $P(A) > 0$ or $P(B) > 0$),

$$P(A \cap B) = P(A).P(B)$$

This is equivalent to $P(A|B) = P(A)$ because, if $A$ and $B$ are independent,

$$P(A \cap B) = P(B).P(A|B) = P(A).P(B) \text{ if } P(A|B) = P(A)$$
$$\text{Similarly, } P(A \cap B) = P(A).P(B) \implies P(B)P(A|B) = P(A).P(B) \text{ i.e. } P(A|B) = P(A)$$

**Ex-3.15** $X$ = outcome of a die roll, $Y$ = outcome of another die roll. Any event involving $X$ and any event involving $Y$ are independent.

**Ex-3.16** $E_1$ = genotype of an individual is $AA$, $E_2$ = genotype of that individual's father is $AA$. Clearly, $E_1$ and $E_2$ are not independent because father always transmits one allele to his offspring.

**Ex-3.17** $F_1$ = an individual gets '$a$' allele from his mother, $F_2$ = the individual gets an '$A$' allele from his father. Here, $F_1$ and $F_2$ are independent events because mother transmits her allele to offspring independent of whatever allele is transmitted by father. Implicitly we assume that random mating occurred.

**Ex-3.18** Suppose a coin is tossed 10 times. Probability of getting a head in a single trial (i.e. single toss) is $p$, where $0 < p < 1$. What is the probability that each of first 3 tosses will give head and each of the last seven tosses will give tail?

Clearly, $P(\text{ first toss gives head }) = p$. $P(\text{ second toss gives head })$ is again $p$, because the outcome of second toss is not in any way dependent on the outcome of the first toss.
So, Probability of getting head in first two tosses $= p \times p = p^2$, since the two events are independent. Hence, proceeding this way, we have the required probability as $p^3(1-p)^7$.

## 3.8 Discrete Probability Distributions

Let's go back to random variable. Referring to Table 3.2, we have further questions: although we understand that random variable might ease our calculation and description of probabilities of several events, but can it offer any real advantage? Table 3.2 shows that we have to look at the probabilities of each outcome in the table and evaluate probability of an event in terms of the random variable $X$! This cannot be a real advantage because we are depending on all outcomes to evaluate the probability of an event. This becomes a nearly impossible (or impossible) task if the number of tosses becomes large. For 10 tosses of a coin, the total number of outcomes is 1024; very difficult to list them! Fortunately, there is a much simpler way to handle these kinds of situation. We can write a function, a mathematical function that will immediately provide the probability that $X$ takes a particular value. Let us explain this using an example.

Consider the coin tossing experiment where we toss an unbiased coin three times. Clearly, each trial, i.e. throwing a coin once, has two outcomes, 'Head' or 'Tail', may be denoted in general by 'Success' or 'Failure' respectively. You may also designate a 'Tail' by 'Success'. It depends on the problem and how one decides to frame it. Since we are using the same coin and tossing is done under identical conditions, we may assume that the probability of getting 'Success' remains same from trial to trial. Moreover, since the outcome of a toss does not depend on any previous tosses, we assume that trials are independent. These kinds of trials are known as 'Bernoulli trials'. Under these assumptions, we define a random variable $X$ as the number of 'heads' in three trials (i.e. tosses). Then the probability that $X$ takes a value $x$, i.e. $P(X = x)$ is given by a function

$$P(X = x) = f(x) = \binom{3}{x}\left(\frac{1}{2}\right)^{x}\left(\frac{1}{2}\right)^{3-x}, \quad x = 0, 1, \ldots, 3$$

Using different values of $x = 0, 1, 2, 3$, we have the respective probabilities as

$$P(X = 0) = \left(\frac{1}{2}\right)^{3} = \frac{1}{8}, \quad P(X = 1) = 3\left(\frac{1}{2}\right)^{1}\left(\frac{1}{2}\right)^{2} = \frac{3}{8},$$
$$P(X = 2) = 3\left(\frac{1}{2}\right)^{2}\left(\frac{1}{2}\right)^{1} = \frac{3}{8}, \quad P(X = 3) = \left(\frac{1}{2}\right)^{3} = \frac{1}{8}$$

which matches with that given in Table 3.2.

In general, if we throw the coin $n$ times, i.e. $n$ Bernoulli trials, and probability of 'Success' in a single trial is $p (0 < p < 1)$, we have

$$P(X = x) = f(x) = \binom{n}{x} p^{x} q^{n-x}, \quad x = 0, 1, \ldots, n; \quad q = 1 - p, 0 < p < 1 \tag{3.1}$$

The evaluation of the expression in (3.1) is easy. Following Example 3.18, the probability that the first $x$ tosses yield heads (successes) and next $n - x$ trials yield

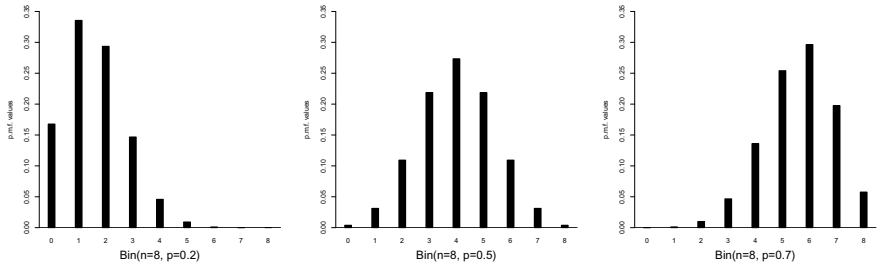



**Fig. 3.5** p.m.f. of Binomial distribution with $n = 8$ and different values of $p$

tails (failures) is $p^x(1-p)^{n-x}$. Now we relax our idea. The $x$ successes can occur in any $x$ positions out of $n$ trials in $\binom{n}{x}$ ways. Hence, the required probability of observing $x$ successes in $n$ tosses becomes same as given in (3.1), with $q = 1 - p$.

Note that the above function $f(x)$ has two important properties:

(1) $f(x) \geq 0$ for all $x$.
(2) $\sum_x f(x) = 1$.

This kind of function $f(x)$ corresponding to a random variable $X$ (say) satisfying the above two properties is known as a 'probability mass function (p.m.f.)' and gives the probability of different events (described in terms of $X$) very easily directly through $f(x)$.

The particular function in (3.1) is known as p.m.f. of a Binomial random variable; we say that $X$ follows Binomial distribution with parameters $n$ and $p$ and denote this by $X \sim Bin(n, p)$.

To get an idea how the probabilities vary depending on the values of $X$, we present p.m.f. by means of a diagram, known as 'Column diagram'. This simply plots a vertical bar whose height represents the probability $P(X = x)$ for all possible values of $x$ (Fig. 3.5).

From Fig. 3.5 it seems that binomial probabilities are symmetrical when $p = 0.5$. However, it has a tail towards right and left respectively for $p = 0.2$ and $p = 0.7$.

## 3.9 Expectation and Variance

We had introduced a term 'parameter' in Sect. 3.8. In Binomial distribution, there are two parameters, $n$ and $p$. A parameter characterises a distribution and we need to know the value of a parameter in order to know about the probability distribution. In fact, if we know the value of all parameters in a probability distribution, we would be able to know everything about that distribution. Moreover, some important features of a probability distribution are always in terms of parameters. Analogous to arithmetic mean, a measure of central tendency, here also we can conceptualise similar idea with respect to a probability distribution. This is called 'Expectation'.

### 3.9.1 Expectation

The expectation of a random variable is the long-term average (mean) outcome. With reference to the example of tossing a fair coin three times, i.e. for a $Bin(3, \frac{1}{2})$ distribution we can think that the number of 'heads' is 0, 1, 2, and 3 with probabilities $\frac{1}{8}$, $\frac{3}{8}$, $\frac{3}{8}$, and $\frac{1}{8}$ respectively. Although we are not sure which outcome we are going to observe while conducting the experiment, however, we can expect to see the value taken by $X$. Also, note that chance of observing 1 is three times that for 0 and so on. Hence, expected value of $X$, i.e. expectation of $X$, denoted by $E(X)$ becomes

$$E(X) = 0.\frac{1}{8} + 1.\frac{3}{8} + 2.\frac{3}{8} + 3.\frac{1}{8} = \frac{3}{2} = 1.5$$

This can be looked as a weighted arithmetic mean of possible values that a discrete random variable can take with weights as the corresponding probabilities.

Note that $E(X) = \frac{3}{2}$, i.e. a fraction, although in real life we never see fractional value taken by $X$ when $X \sim Bin(3, \frac{1}{2})$. However, this is what we can expect; it may not necessarily be an integer even when the random variable under consideration takes only integer values. Expectation, as the term indicates, gives the value that we would think or expect to get even when we just know the distribution of the random variable under consideration. This value, may or may not be realised when the experiment is actually performed; however, we can expect to get a realisation close to the 'expectation of the variable', i.e. $E(X)$.

It can be shown that when $X \sim Bin(n, p)$, we have $E(X) = np$. This expression can be obtained from an intuitive point of view. The probability of observing a 'success' in a single trial is $p$, i.e. $p$ is the number of success in one trial. Hence, in $n$ trials, the number of successes should be $np$ (just use unitary method!).

**Ex-3.19** A die is thrown and $X$ denotes the number appearing on the upper face of the die. What is the expectation of $X$?
Here $X$ takes the values $1, 2, \ldots, 6$, each with probability $\frac{1}{6}$. So the p.m.f. of $X$ is:

$$f(x) = P(X = x) = \frac{1}{6}, \quad x = 1, 2, \ldots, 6$$

Hence $E(X) = 1.\frac{1}{6} + 2.\frac{1}{6} + \cdots + 6.\frac{1}{6} = \frac{7}{2} = 3.5$.

**Ex-3.20** A person is selected from a group of 28 individuals out of which 5 are affected by a disease. What is the expected number of affected persons in the sample?

Here $X$ is the number of affected persons in a sample of size 1 (only one person is selected). So, $X$ takes values either 0 or 1. Clearly, $X \sim Bin(1, \frac{5}{28})$ and so $E(X) = 1 \times \frac{5}{28} = \frac{5}{28}$.

### 3.9.2 Variance

Similar to the idea of expectation, we can think of variance of a random variable as the long-term spread (variation) of its outcomes. With reference to the previous example, i.e. for a $Bin(3, \frac{1}{2})$ distribution, we can think that the number of 'heads' is different if we repeat the experiment and there is an inherent probability associated with each observation. Since the outcomes vary randomly but within the set of values $\{0, 1, 2, 3\}$, we can think of a measure of variability analogous to variance as a measure of dispersion. Hence, variance of $X$, denoted by $V(X)$ becomes

$$V(X) = E\big(X - E(X)\big)^2 = (0 - 1.5)^2.\frac{1}{8} + (1 - 1.5)^2.\frac{3}{8} + (2 - 1.5)^2.\frac{3}{8} + (3 - 1.5)^2.\frac{1}{8} = \frac{3}{4} = 0.75$$

Since expectation for a $Bin(3, \frac{1}{2})$ random variable is 1.5, we expect to see 1.5 but there is a variation of 0.75 or a standard deviation $\sqrt{0.75} = 0.866$ associated with this.

It can be shown that when $X \sim Bin(n, p)$, we have $V(X) = npq$.

**Ex-3.21** In Example 3.19, what is $V(X)$?
Here, $X$ takes the values $1, 2, \ldots, 6$, each with probability $\frac{1}{6}$ and $E(X) = 3.5$. Hence,

$$E(X) = (1 - 3.5)^2.\frac{1}{6} + (2 - 3.5)^2.\frac{1}{6} + \cdots + (6 - 3.5)^2.\frac{1}{6} = 2.917.$$

### 3.9.3 A Few More Discrete Distributions

There are some frequently occurring discrete random variables with corresponding probability distributions that might be deemed appropriate in explaining real-life situations or phenomena. Among them, the most popular ones are Poisson and negative binomial random variables. If $X$ follows a negative binomial distribution, it indicates the number of failures before the $r$th success. So in this case, there is no bound to the number of trials; it continues until we get $r$ successes. We present the p.m.f., $E(X)$ and $V(X)$ for these distributions (along with Binomial distribution) in Table 3.3.

**Table 3.3** Probability mass functions of three discrete distributions: Binomial, Poisson, and negative binomial

| Distribution | $f(x) = P(X = x)$ | $x$ | Parameter(s) | $E(X)$ | $V(X)$ |
|---|---|---|---|---|---|
| Binomial distribution | $\binom{n}{x} p^x q^{n-x}$ where $q = 1 - p, 0 < p < 1$ | $x = 0, 1, \ldots, n$ | $n, p$ | $np$ | $npq$ |
| Poisson distribution | $\frac{e^{-\lambda}\lambda^x}{x!}$, $\lambda > 0$ | $x = 0, 1, 2, \ldots$ | $\lambda$ | $\lambda$ | $\lambda$ |
| Negative binomial distribution | $\binom{x+r-1}{r-1} p^r q^x$ where $q = 1 - p, 0 < p < 1$ | $x = 0, 1, 2, \ldots$ | $p$ | $rq/p$ | $rq/p^2$ |

## 3.10 Continuous Probability Distributions

We have discussed a few variables like number of heads in three tosses of a coin, number of *A* alleles in the genotype of an individual at a particular locus, etc. However, as stated earlier, all these variables can take only isolated or specified values depending on the random experiment. Consider another variable, say, height of a student. Now the value of height varies from one student to another (in a particular population) but it can take any value within a certain range. Suppose if the population is kindergarten students in a city, we can think that the values of height for different students might vary between 2 feet and 4 feet. But it can be any value (in decimals or fraction) within this range. This kind of variable is called 'continuous random variable'. Another example related to genetics might be gene expression values. It is a continuous variable and can take any value within its own range. Note that although we may get a set of distinct values up to a few places after decimal, however, it is possible to get a more accurate value if we use more sophisticated instrument for measuring. Thus, in this sense, it is continuous as there is always a possibility of observing any value, which is not the case for a discrete variable.

But how do we attach probability to such a continuous random variable? Recall the discussion in Sect. 2.2.2 and Fig. 2.1. We know that a histogram provides a graphical presentation of frequency distribution of a continuous variable $X$ (say). Careful understanding of histogram and Fig. 2.1 reveal some interesting facts. If we draw histogram using relative frequency density on the vertical axis, the total area under the histogram is one. The total area always remains same, i.e. its value is 1 always, whatever be the number of observations or the number of classes. Now letting class width ($\omega$) of each class very small and at the same time increase the number of observations ($n$), histogram would approach to a smooth curve having this property (Fig. 2.5). This limiting form of histogram, i.e. 'frequency curve', is obtained (theoretically) when $\omega \to 0$ and $n \to \infty$. Let's denote this frequency curve by $f(x)$ where $x$ represents the value of the variable of interest; in our case, it may be gene expression value. Clearly, $f(x)$ has two important properties that are quite apparent: (1) $f(x) \geq 0$ for all $x$, and (2) total area under $f(x)$ over its entire range is 1.

Now we define a continuous random variable $X$ that can take any value within a feasible range (i.e. a set of values) depending on the random experiment. Examples of such random variables may be height, weight, HDL level, total cholesterol, gene expression values, etc. Any realisation of $X$ based on a random experiment is denoted by $x$ and the limiting form of the histogram is denoted by $f(x)$. We may say that $f(x)$ is a probability density function (p.d.f.) of $X$ and it always satisfies the two properties as given below.

1. $f(x) \geq 0$ for all $x$, and
2. $\int_a^b f(x)dx = 1$, where $a$ and $b$ are minimum and maximum possible values that $x$ can take.

This function, $f(x)$ is called a 'probability density function' (p.d.f.) of $x$ because at a point $c$, $f(c)$ represents the limiting relative frequency density (in a limiting histogram), and hence can be considered as 'probability density' at $c$. The probability that $X$ lies between any two values $c$ and $d$ (say) with $c < d$ is given by

$$P(c \leq X \leq d) = \int_c^d f(x)dx.$$

So probability is interpreted as area under the curve $f(x)$ between two values just as area under histogram represents relative frequency between two values.

Clearly, probability at a particular point is zero as area on a point is always zero. So, we have

$$P(X = x) = 0 \text{ for all } x$$
$$\implies P(c \leq X \leq d) = P(c < X \leq d) = P(c \leq X < d) = P(c < X < d) = \int_c^d f(x)dx \text{ for } c < d,$$

because this is the area under the curve $f(x)$ between $c$ and $d$. Proceeding similarly as in discrete case, we can define expectation and variance, denoted by $\mu$ and $\sigma^2$ respectively as

$$\mu = E(X) = \int_a^b xf(x)dx, \quad \sigma^2 = V(X) = \int_a^b (x-\mu)^2 f(x)dx$$

### *3.10.1 Normal Distribution*

In statistical analysis, the most common and widely used distribution for a continuous variable is known as 'normal' distribution. Suppose that a random variable $X$ has a normal distribution. Then the p.d.f. of $X$ is

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad -\infty < x < \infty, -\infty < \mu < \infty, \sigma^2 > 0.$$

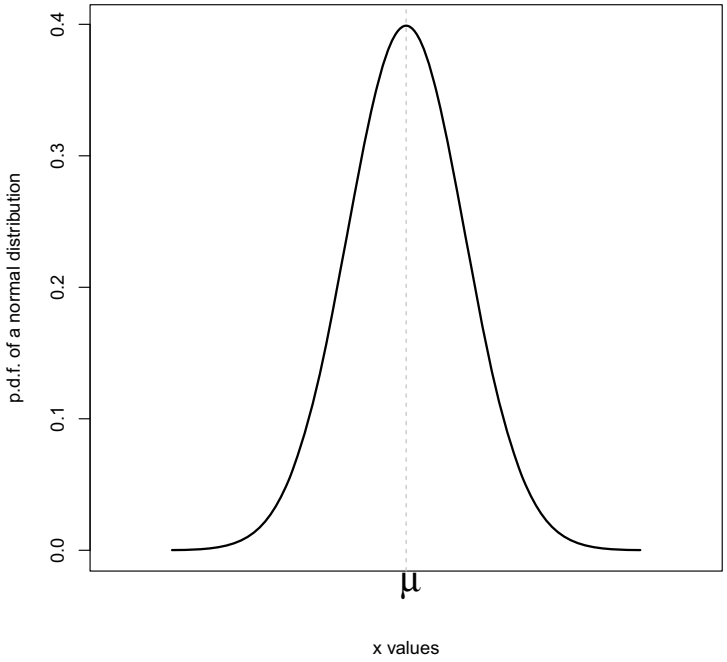



**Fig. 3.6** p.d.f. of a normal distribution with mean at $\mu$

If a continuous random variable $X$ follows a normal distribution with mean $\mu$ and variance $\sigma^2$, we denote this by $X \sim N(\mu, \sigma^2)$. If we look at the graph of normal distribution (Fig. 3.6), immediately some nice features emerge. Its p.d.f. is symmetric about $\mu$, two tails are falling down on both sides smoothly and, at a regular manner, it has a peak at mean value indicating that mean and median are same, most of the values, i.e. around 99.73% values will lie between $\mu - 3\sigma$ and $\mu + 3\sigma$ whatever be the values of $\mu$ and $\sigma$, etc. It can be shown that $E(X) = \mu$ and $V(X) = \sigma^2$.

**Calculation of normal probability**

Sometimes, it is essential to calculate probability of a certain event based on a normal random variable. Let us explain this using an example.

**Ex-3.22** Suppose gene expression values for a gene are known to follow a normal distribution with mean 1 and variance 0.2. We might be interested in evaluating probabilities of different events like:

22.1 What is the probability that the gene expression value might be less than or equal to 0.82?
22.2 What is the probability that the gene expression value might be greater than 3.2?
22.3 What is the probability that for a randomly chosen person, the gene expression value will lie between 0.6 to 1.3?

Well, to answer these questions, we would need to know how to calculate $P(a \leq X \leq b)$ for any $a < b$ when $X \sim N(\mu, \sigma^2)$. This requires evaluation of the integral $\int_a^b f(x)dx$ where $f(x)$ is the p.d.f. of a normal distribution. Unfortunately, there is no closed form by which we can immediately calculate this probability for given values of $a$ and $b$. But fortunately, we know R language and we can use R to calculate probability of any event when $X \sim N(\mu, \sigma^2)$. The standard R code is available for evaluating one-sided probability, i.e. for $P(X \leq x)$ or $P(X > x)$. Note that it is immaterial to consider $<$ or $\leq$ as explained earlier. Let's try to calculate the above probabilities.

The R code for calculating probability of $X \leq x$ is given by

```
> pnorm(x,mean=μ, sd=σ)
```

Here, 'norm' stands for 'normal' distribution, mean, and sd are numerical values of mean and standard deviation of $X$ respectively, to be given as input inside the function 'pnorm( )'. Using this function, our command for **22.1** would be

```
> pnorm(0.82,mean=1, sd=sqrt(0.2))          ## P(X ≤ 0.82|X ~ N(1, 0.2))
```

which gives the value 0.3436609; 'sqrt(a)' indicates the 'square root of $a$'. If we want the value up to four places after decimal, we can modify the command as

```
> round(pnorm(0.82,mean=1, sd=sqrt(0.2)),4)
```

This gives the value $P(X \leq 0.82|X \sim N(1, 0.2)) = 0.3437$. Note that 'round(a, 4)' function rounds the value of '$a$' up to four places after decimal.

**(22.2)** R code for calculating probability that gene expression value might be greater than 3.2 is given by

```
> pnorm(3.2,mean=1, sd=sqrt(0.2),lower.tail=F)
```

which gives the value $4.341614e-07$ means the value is 0.0000004341614. To get the right-sided probability, we have to write 'lower.tail=F' inside the 'pnorm( )' function; default is for calculation of left-sided probability. Hence, this gives the value $P(X > 3.2|X \sim N(1, 0.2)) = 4.3416 \times 10^{-07}$.

**(22.3)** To calculate $P(0.6 \leq X \leq 1.3)$, we first calculate $P(X \leq 1.3)$ and subtract $P(X \leq 0.6)$ from it. Since $P(0.6 \leq X \leq 1.3) = P(X \leq 1.3) - P(X < 0.6)$, we have the R code as

```
> pnorm(1.3, mean=1, sd=sqrt(0.2),lower.tail=T) - pnorm(0.6, mean=1,
sd=sqrt(0.2),lower.tail=T)
```

and the value is 0.5632858. Note that here we have written "lower.tail=T" inside the 'pnorm' function; but it is not necessary as long as we calculate the left-sided probability. However, to avoid confusion, it's better to explicitly write "lower.tail=T" or "lower.tail=F" depending on whether we want left-sided or right-sided probability respectively.

Again, if we want the value up to three places of decimal, the R code should be

```
> round(pnorm(1.3, mean=1, sd=sqrt(0.2),lower.tail=T) - pnorm(0.6,
mean=1, sd=sqrt(0.2),lower.tail=T), 3)
```

which gives the value 0.563.

### *3.10.2 Few Other Distributions*

We now present a few more distributions, mainly pictorially. Although 'Normal distribution' is the most popular and widely applicable probability distribution of a continuous random variable, some other distributions may occur sometimes in many real-life situations. However, there is no hard and fast rule to identify the appropriate distribution, which the underlying variable would follow. We can draw a histogram or boxplot and looking at the shape and features of the diagram, we might be able to guess a probability distribution. After this step, we usually estimate the parameters of the distribution and fit that distribution to the data. Next step is to check or justify whether the fit is good. There are some statistical techniques to evaluate the goodness of fit of the probability distribution to the dataset. Sometimes, based on prior knowledge we may assume the probability distribution. It is important to note that larger the sample size, better will be the justification about the goodness of fit. Although there are many such probability distributions, we consider three more probability distributions, which are common and have inetresting features.

$$\begin{aligned}
&\textit{Normal distribution}: \quad X \sim N(\mu, \sigma^2): \quad f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\frac{(x-\mu)}{\sigma}^2}, \ -\infty < x < \infty \\
&\textit{Laplace distribution}: X \sim \textit{Laplace}(\mu, \sigma): \quad f(x) = \frac{1}{2\sigma} e^{-\frac{|x-\mu|}{\sigma}}, \ -\infty < x < \infty \\
&\textit{Gamma distribution}: \ X \sim \textit{Gamma}(\alpha, \beta): \quad f(x) = \frac{\alpha^\beta}{\Gamma(\beta)} e^{-\alpha x} x^{\beta-1}, \ x > 0 \\
&\textit{Cauchy distribution}: \ X \sim \textit{Cauchy}(\mu, \sigma): \quad f(x) = \frac{\sigma^2}{\pi} \frac{1}{\sigma^2 + (x-\mu)^2}, \ -\infty < x < \infty
\end{aligned}$$

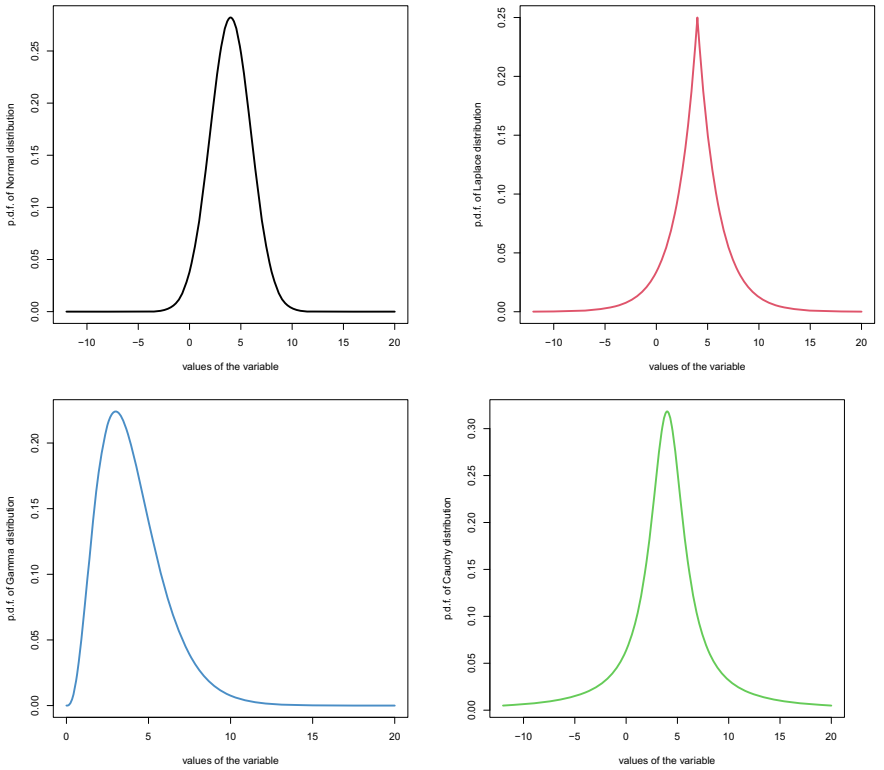



**Fig. 3.7** p.d.f.s for four distributions

There are specific reasons behind choosing these distributions. Figure 3.7 presents the p.d.f.s of these four probability distributions for some specific values of respective parameters. We use the same location and dispersion parameters for all four distributions so as to have a clear idea when all four p.d.f.s are drawn on same plot, i.e. superimposing all four graphs. Clearly, p.d.f. of a normal distribution is symmetric about $\mu$ (here $\mu = 5$) with the peak gradually and smoothly falling on both sides in a symmetric manner. Thus, the curve is symmetric about $\mu$. We choose the Laplace distribution because like normal distribution it is also symmetric about mean but it has a high kurtosis, i.e. a fatter tail on both sides compared to normal distribution. We select the gamma distribution because it is positively skewed having a longer tail towards the right side of the mean and realisations start from 0. Cauchy distribution is chosen because it has much fatter tails and although symmetric about median, its mean and variance do not exist. This distribution has a special role in statistics and is used to demonstrate some interesting facts which would need a detailed explanation.

A few interesting observations are immediately visible from Fig. 3.7. Normal, Laplace and Cauchy distributions are symmetric about 5, i.e. if we fold around the value 5, the right side and left side will fall on top of each other. But gamma distri-

bution is not symmetric; it is tilted to the left having a peak at a value less than 5. But peak occurs for other distributions exactly at 5. Laplace distribution has a very sharp peak at 5. Careful attention also reveals that beyond -10 to the left and beyond 20 to the right, a few values may still be observed. Normal distribution has almost no value, whereas there is substantial contribution at these regions for Laplace and Cauchy distributions. Hence, these two distributions are characteristically different from normal distribution. Gamma distribution has a long tail towards the right indicating the existence of some large values. These facts, although very simple to observe, play a very important role in any data analysis. Sometimes, another distribution is used in some situations, known as 'Exponential distribution'. This distribution can be viewed as a special case of gamma distribution, with $\beta = 1$. Thus, the p.d.f. of an exponential distribution is with parameter $\alpha$ :

$$f(x) = \frac{1}{\alpha} e^{-x/\alpha}, \quad x > 0, \alpha > 0.$$

It can be shown that for an exponential distribution with parameter $\alpha$, mean is $\alpha$, and variance is $\alpha^2$.

### *3.10.3 Important Results Related to Probability Distributions*

On many occasions, we need to standardise the data, i.e. subtract mean from every observation and divide it by its standard deviation. For normal distribution, even after standardisation, the variable follows a normal distribution with mean 0 and variance 1; but this is not true for all distributions. Moreover, if $X$ and $Y$ are two independent random variables both following normal distributions, then any linear function of $X$ and $Y$ also follows a normal distribution. Some very interesting results are available for normal distribution. These results have immense implications and use in data analysis. We state a few results below, without proof.

**Theorem 3** *If* $X \sim N(\mu, \sigma^2)$, *then* $Z = \frac{X-\mu}{\sigma} \sim N(0, 1)$.

**Theorem 4** *Suppose* $X \sim N(\mu_1, \sigma_1^2)$, $Y \sim N(\mu_2, \sigma_2^2)$, *and X and Y are independently distributed. Then for any constants (non-random, real numbers)* $a, b$, *and c,* $a + bX + cY \sim N(a + b\mu_1 + c\mu_2, b^2\sigma_1^2 + c^2\sigma_2^2)$.

Now, let's draw a random sample from $X$. Let it be $X_1, X_2, \ldots, X_n$, where $n$ is the sample size. During developing statistical testing procedures for some problems, we shall see that a few quantities are of utmost important. These quantities are functions of $X_1, X_2, \ldots, X_n$. Note that $X_1, X_2, \ldots, X_n$ constitute a random sample from $X$. Random sample means, first, we draw an item from the population and measure the value of the variable $X$; then we draw another one but it has nothing to do with the first observation, and so on. Thus, observations correspond to random variables that are copies of the original variable $X$ and hence they are *identically* distributed. Moreover, since these identically distributed random variables $X_1, X_2, \ldots, X_n$ constitute

a random sample, they are independent of each other. These two facts lead to some nice results. We present these results as theorems (without proof).

**Theorem 5** *Let $X_1, X_2, \ldots, X_n$ be independently distributed random variables with $E(X_i) = \mu_i$ and $V(X_i) = \sigma_i^2$ for all $i = 1, 2, \ldots, n$. Then, we have*

$$E\Big(\sum_{i=1}^{n} X_i\Big) = \sum_{i=1}^{n} \mu_i, \text{ and } V\Big(\sum_{i=1}^{n} X_i\Big) = \sum_{i=1}^{n} \sigma_i^2.$$

**Corollary 1** *If $\mu_i = \mu$ for all $i = 1, \ldots, n$, we have, $E\Big(\sum_{i=1}^{n} X_i\Big) = n\mu$. Similarly, if $\sigma_i^2 = \sigma^2$ for all $i = 1, \ldots, n$, we have $V\Big(\sum_{i=1}^{n} X_i\Big) = n\sigma^2$.*

**Theorem 6** *Define $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$, as the sample mean. Clearly, $\bar{X}$ is also a random variable. Then $\bar{X}_i \sim N(\mu, \sigma^2/n)$ if $X_i \sim N(\mu, \sigma^2)$ i.i.d. for $i = 1, 2, \ldots, n$.*

**Remarks:** Theorem 5 is true for discrete as well as continuous variables as long as expectation and variance exist for all variables.

Theorem 5 can be envisaged intuitively also. To see this easily, assume without any loss of generality that all variables are positive. Now if we add two variables, the values corresponding to the sum of variables become larger than the individual variables. Hence it is expected that expectation of the sum of variables would be large compared to that for individual variables. Similarly, variance becomes large as larger values would show more variability in the data obtained from sum of variables. Hence, these results become intuitively more natural.

**Theorem 7** *Let $X_1, \ldots, X_n$ are independently and identically distributed (i.i.d.) random variables each following a normal distribution with mean 0 and variance 1, i.e. $X_i \sim N(0, 1)$, $i = 1, \ldots, n$ and $X_1, \ldots, X_n$ are independent. Then $\sum_{i=1}^{n} X_i^2 \sim \chi_n^2$.*

So, we get a new distribution, called $\chi_n^2$ distribution, pronounced as 'chi-square distribution'. As per definition, if $Y$ follows a chi-square distribution with $n$ **degrees of freedom**, or notationally $Y \sim \chi_n^2$, the p.d.f. of $Y$ is given by

$$f(y) = \frac{\left(\frac{1}{2}\right)^{n/2}}{\Gamma(n/2)} e^{-y/2} y^{\frac{n}{2}-1}, \quad y > 0,$$

where $\Gamma(m)$ is a standard 'Gamma' function defined as $\Gamma(m) = \int_0^\infty e^{-x} x^{m-1} dx$. Although it is not essential to memorise the p.d.f. of $Y$, but we should have a clear idea about what is called 'degrees of freedom'.

**Degrees of freedom:** Drosophila, a fruit fly is flying freely in a room. A room is identified by length, width, and height, along $x$-axis, $y$-axis together $xy$-plane (floor) and $z$-axis (in the direction of floor to ceiling vertically). The position of the fly at any point inside the room is determined by these three axes. If we drop any axis, we cannot identify its position. So the degrees of freedom of the fly is three. Now assume that for some reason (restriction), the fly was to fall into a cup of water, soaking its wings, rendering it unable to fly, but capable of crawling on the floor. Now, its position can be determined only by $x$- and $y$-axes; we don't need the $z$-axis, which is no more active due to its wet wings, i.e. one restriction. Hence, in this case, the degrees of freedom reduce to 2.

Thus, we can think the degrees of freedom as the number of independent variables involved in the expression. Without any restriction, the position of the fly is a function of three independent axes. But with one restriction degrees of freedom is reduced to 2, i.e. one less than three. For a set of $n$ independent variables, degree of freedom is $n$. If we put 1 restriction, it becomes $n-1$. In general, if we put $k$ $(1 < k < n)$ independent linear restrictions, the degrees of freedom come down to $n-k$.

Now that, we have normal and $\chi^2_n$ distribution at our hands, we have a very important result, as stated below.

**Theorem 8** *Define* $s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$, *as the sample variance. Clearly,* $s^2$ *is also a random variable. Then*

*(1)* $\frac{(n-1)s^2}{\sigma^2} = \sum_{i=1}^{n}(X_i - \bar{X})^2/\sigma^2 \sim \chi^2_{n-1}$, *and*

*(2)* $\bar{X}$ *and* $s^2$ *are independently distributed.*

Note that although $s^2$ involves $n$ independent variables $X_1, \ldots, X_n$, but they are subject to one linear restriction, i.e. $\sum_{i=1}^{n}(X_i - \bar{X}) = 0$. Hence, the degrees of freedom associated with $\frac{(n-1)s^2}{\sigma^2}$ reduces to $n-1$. The second part of Theorem 8 is very important and it can be exploited to develop some statistical tools in data analytics. Independence of sample mean and sample variance is a typical characteristic for normal distribution that makes our life simple in dealing with many data analysis problems in genetics and other areas.

Two more distributions that we would need to know are $t$ and $F$ distribution. We mention them without the explicit expression for their probability density functions. These distributions involve two independent variables.

**Theorem 9** *Suppose* $X \sim N(0,1)$, $Y \sim \chi^2_n$, *and* $X$ *and* $Y$ *are independently distributed. Then,* $\frac{X}{\sqrt{Y/n}} \sim t_n$, *where* $t_n$ *is known as* $t$ *distribution with* $n$ *degrees of freedom.*

**Theorem 10** *Suppose* $X \sim \chi^2_m$, $Y \sim \chi^2_n$, *and* $X$ *and* $Y$ *are independently distributed. Then,* $\frac{X/m}{Y/n} \sim F_{m,n}$, *where* $F_{m,n}$ *is known as* $F$ *distribution with degrees of freedom* $m$ *and* $n$.

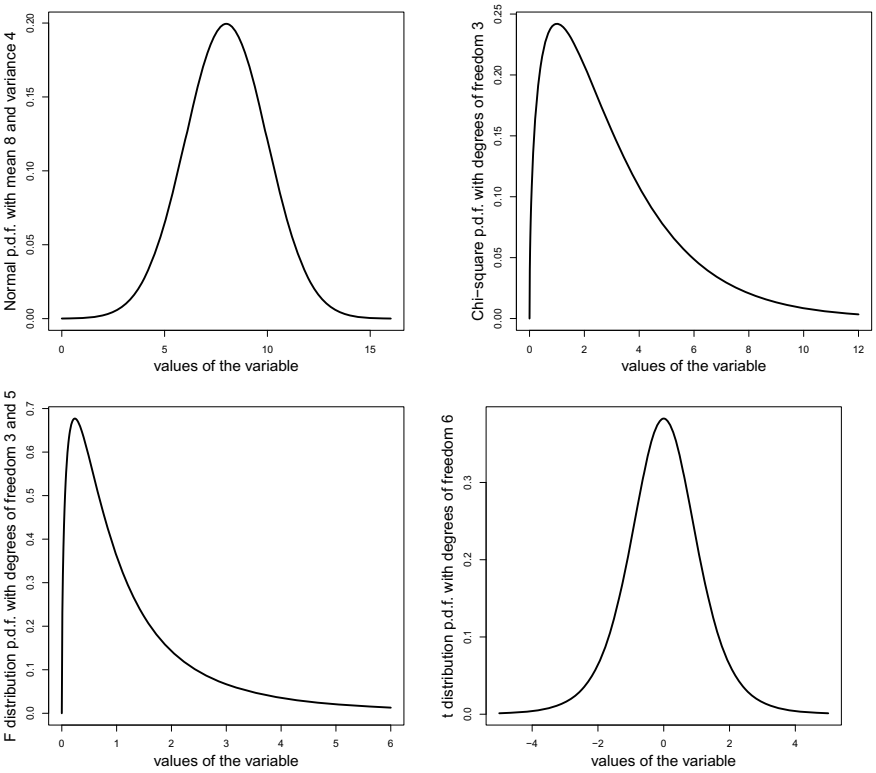



**Fig. 3.8** p.d.f.s for four probability distributions: Normal Chi-square, F, and t

The following theorem deals with two independent $\chi^2$ variables. It states that if the two chi-square variables are independent, their sum also follows chi-square distribution with degrees of freedom being the sum of degrees of freedom of the respective variables.

**Theorem 11** *Suppose* $X \sim \chi^2_m$, $Y \sim \chi^2_n$ *and* $X$ *and* $Y$ *are independently distributed. Then,* $X + Y \sim \chi^2_{m+n}$

Although we did not give the exact form of the p.d.f of $t$ or $F$ distribution, Fig. 3.8 presents graphs of their density functions. It is clear from the graphs, that $t$-distribution is symmetric about 0 but have much fatter tails on either side compared to normal distribution whereas $F$ distribution and $\chi^2$ distribution are not symmetric. In fact, $F$ distribution has a longer tail towards the right side and its peak lies on the left side of its range. Naturally, mean, variance, and other properties of $\chi^2_n$, $F_{m,n}$, and $t_n$ distributions depend on the degrees of freedom. Height of the peak also depends on degrees of freedom.

With this knowledge, we can now proceed to develop a few testing methods required to address several problems relating to gene expression data and other continuous variables that occur in practice in any discipline.

## Exercise

3.1 Two dice are thrown. Naturally, we observe two numbers appearing on the upper faces of two dice. What is the probability that

(a) sum of these two numbers is 7,
(b) product of these two numbers is 12, and
(c) maximum of these two numbers is 5?

3.2 Two dice are thrown. What is the probability that sum of the numbers appearing on the upper faces of two dice is 7 when it is known that

(a) one die shows 4,
(b) at least one die shows 4,
(c) the difference of two numbers is 3, and
(d) minimum of two numbers is 1?

3.3 A group consists of 56 individuals of which 17 are affected by a disease. One person is selected at random. What is the probability that the person is affected?

3.4 A group consists of 72 individuals of which 18 are affected by a disease. Five persons are selected at random. What is the probability that

(a) two are affected,
(b) at least one is affected, and
(c) two are affected when it is known that at least one is affected?

3.5 A fair coin is tossed 4 times. What is the probability that we can get

(a) three consecutive heads once,
(b) two consecutive heads at least once,
(c) three consecutive tails once, and
(d) all tosses yield tails?

3.6 Explain why there is a mistake in each of the following statements:

(a) The probability that David will pass the yearly examination is 0.72 and the probability that he will not pass is $-0.28$.
(b) The probability that it will be raining tomorrow is 0.82 and the probability that it will not be raining tomorrow is 0.32.

(c) The probability that last year's champion in a football tournament will win its first match in this year's tournament is 0.76, the probability that it will tie the game is 0.09, and the probability that it will win or tie is 0.92.
(d) The probabilities that a person will make 0, 1, 2, 3, or 4 spelling mistakes while typing a page are 0.13, 0.24, 0.38, 0.16, and 0.19 respectively.
(e) The probabilities that a student passes in English literature and mathematics are 0.46 and 0.39 respectively, whereas probability passing in either one of these two subjects is 0.87.

3.7 The probability that the annual maintenance of a new PCR machine will be rated as very easy, easy, average, difficult, or very difficult are, respectively 0.07, 0.3, 0.36, 0.16, and 0.11. Find the probabilitues that annual maintenance of the machine will be rated as:

(a) average or better;
(b) average or worse;
(c) difficult or very difficult;
(d) neither very difficult nor very easy.

3.8 From a pack of cards, suppose three cards are drawn at random. What is the probability that

(a) all three are aces?
(b) none of these three is an ace?
(c) two are black and one is red?
(d) each of the three cards belongs to different suits?
(e) each of the three cards belongs to different denominations?
(f) all three cards are neither black nor aces?
(g) all of them are spades when it is known that they belong to any black suit?
(h) If the cards are drawn randomly one by one and first two cards are red, what is the probability that the third card drawn will also be red?

3.9 A box contains 5 red and 5 white balls while another box contains 7 red and 4 white balls. A box is selected at random and one ball is drawn from that box. What is the probability that selected ball is white?

3.10 In problem 3.9, suppose it is seen that the ball drawn is white, but we don't know from which box it was drawn. Find the probability that it was selected from the second box.

3.11 A population consists of three different ethnic groups A, B and C (say) in 2 : 3 : 5 proportions. Proportion of persons suffering from a rare disease in these three groups is 0.02, 0.01, and 0.005 respectively. What is disease prevalence in that population? Suppose one person is selected and is found to affected. What is the probability that the person belong to ethnic group C?

3.12 Mental stress may be related to high blood pressure. Probability that a person gets stressed is 0.3. Probability of observing high blood pressure when a person is stressed is estimated as 0.7 whereas the same probability is only

0.1 when the person is not stressed. Suppose probability that the indicator of blood pressure shows high is 0.1. What is the probability that the person was under stress when blood pressure was measured as high?

3.13 Proportion of individuals suffering from cardiovascular disease is 0.12. A random sample of 24 individuals is taken from that population. What is the probability that

(a) exactly 4 of have cardiovascular disease?
(b) at least 2 of them have cardiovascular disease?
(c) 3 individuals are suffering from the disease when it is known that 2 have cardiovascular disease?
(d) 3 individuals are suffering from the disease when it is known that at least 2 have cardiovascular disease?
(e) the number of individuals suffering from the disease exceeds the average number affected by the disease?

3.14 Suppose HDL level is assumed to be normal with mean 50 and variance 6. Suppose you measure HDL of a person chosen at random from the population. What is the probability that the HDL for that person

(a) is more than 55?
(b) is less than 40?
(c) lies between 35 and 45?

3.15 Suppose we select 20 individuals at random from the population whose LDL follows a normal distribution with mean 120 and variance 10. What is the probability that

(a) exact 4 individuals are having LDL more than 150?
(b) at least 4 individuals have LDL more than 150?
(c) at most 4 individuals LDL more than 150?

How many of them are expected to have LDL level (a) more than 140, and (b) lies between 130 to 140?

3.16 Suppose I change the numbers on the faces of a die. Instead of $\{1, 2, 3, 4, 5, 6\}$, I printed the numbers as $\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \frac{1}{6}\}$. Find the expectation and variance of the number of upper face(s) if

(a) the die is thrown only once,
(b) the die is thrown twice, and
(c) the die is thrown 10 times.

Can you get the answers of 3.16(b) and 3.16(c) without evaluating the probabilities, just by using 3.16(a)? If so, explain how and show that you arrive at the same answers. If not, explain why you cannot do?

3.17 Out of a group of 50 persons, 12 have some cardiovascular problems, 8 has hypertension and 3 have both. A person is selected at random. What is the probability that the person has

(a) both the problems?
(b) either hypertension or cardiovascular disease?
(c) neither hypertension nor cardiovascular problems?
(d) has cardiovascular problem but not hypertension?

3.18 In a family father has genotype $Aa$ and mother has genotype $Aa$ at a particular bi-allelic locus. Probabilities that a person with genotypes $AA$, $Aa$, $aa$ has a disease that is somehow caused by the genotype at this locus are 0.9, 0.4 and 0.001 respectively. What is the probability that a newborn baby might have the disease? What is the expected number of children having the disease if the couple has three children? You can assume random mating and allele frequency of allele $A$ is 0.1.

3.19 Twenty-four people had a blood test and the results are shown below.
A, B, B, AB, AB, B, O, O, AB, O, B, A, AB, A, O, O, AB, B, O, A, AB, O, B, A

(a) Construct a frequency distribution for the data.
(b) If a person is selected randomly from the group of twenty-four people, what is the probability that his/her blood type is not O?

3.20 Suppose chance of surviving from a disease is 0.95 if the person has no co-morbidity. However, the chance reduces to 0.62 and 0.34 if the person has cardiovascular problem and both cardiovascular issue and hypertension. In the population, it is known that prevalence of either one of these two is 0.12 whereas that for only cardiovascular issue is 0.08 and that for hypertension only is 0.06. What is the chance of survival if a person is affected by the disease? Suppose a person is selected from the population and is found to be affected by the disease. What is the chance that he is also suffering from hypertension?

3.21 A company has 12 statisticians, 9 mathematicians, and 23 computer scientists. The manager, who is a management person, wants to build a team consisting of 6 people for a project on data science. What is the probability that the team consists of

(a) 3 statisticians, 2 mathematicians and 9 computer scientists?
(b) no mathematician at all?

After selecting a team of 3 statisticians, 2 mathematicians and 9 computer scientists for the first project, the manager wants to gather another team for a second project. For this second project what is the probability that team consists of 5 statisticians, 3 mathematicians, and 8 computer scientists? Two persons are selected at random from the entire group of scientists. What is the probability that both of them are computer scientists and are idle at that moment?

3.22 For any two events $A$ and $B$, if $A$ and $B$ are independent events, show that $A^c$ and $B^c$ are also independent. Give an example.

3.23 Sometimes, to get answers to a sensitive question, a typical method, known as **randomised response technique** is used. Suppose we want to know probabilities of some events relating to smokers among school students. We construct 40 cards and write 'I smoke at least once a week' on 15 cards and 'I do not smoke at least once a week' on rest of the cards. Note that the number 15 is arbitrary. A sample of 294 students is selected randomly. Now we ask each student in the sample to select randomly one of the 40 cards and respond 'yes' or 'no' without disclosing the question. Here, there is no way that the experimenter knows what is written on the card chosen by each student. Selection of card is made with replacement. Let $A$ be the event that a student gives 'yes' and $S$ denote that a randomly selected student smokes at least once a week.

(a) Establish a relationship between $P(A)$ and $P(S)$.
(b) If 90 students answered 'yes' out of 294 students, an estimate of $P(A)$ can be given as $\frac{90}{294}$. Find an estimate of $P(S)$ using the result in part (a).

3.24 Suppose lengths of sides that are perpendicular to each other in a right angled triangle follow two independent normal distributions, each with mean 0 and variance 1. Length of each side is measured from its midpoint; hence it can be negative when measured to the left of the midpoint. What is the distribution of the square of the hypotenuse? If we consider 100 such triangles, how many you can expect to have hypotenuse with a length of more than 1.25?

3.25 Suppose height ($X$) of a group of college students follows a normal distribution with mean 167 cm and variance 15 cm$^2$. What is the probability that a student's height is (a) 172 cm, (b) 165 cm, and (c) either 170 cm or 171 cm or 172 cm? Given reasons in favour of your answers. Also calculate using R function $P(X > 172)$ and $P(X \leq 162)$. You will see that these two probabilities are same! Justify with reasons.

3.26 Suppose a sampling scheme suggests to continue sampling unless we get 4 affected individuals from a population. Assuming that the probability of an individual to be affected is 0.1, what is the probability that

(a) at least 4 selections are required?
(b) more than 8 selections are required?
(c) 7th selection will produce 4th affected person?
(d) 5 more selections are required when it is known that the first 5 selections did not give any affected individual?
(e) 5 more selections are required when it is known that the first 5 selections give only 2 affected individuals?

3.27 A special case of negative binomial distribution can be thought of when a random variable indicates the number of failures before the first success. What is the p.m.f. of $X$? Also find $E(X)$ and $V(X)$.

3.28 Suppose I toss a coin to select one out of two individuals. Probability of getting head in a single trial by this coin is known to be $\frac{3}{4}$. Since the coin is biased, I am unable to select one person unbiasedly, i.e. with probability $\frac{1}{2}$. So I devised

a scheme. The coin would be tossed two times and if $\{HH\}$ or $\{TT\}$ appears, no selection is made and tossing is continued. Person 1 is selected if $\{HT\}$ appears, and person 2 is selected if $\{TH\}$ appears.

(a) Show that under this scheme, the probability of selecting person 1 is $\frac{1}{2}$.
(b) What is the probability that person 1 will be selected on 8th toss?
(c) Since there is a possibility that the tossing might continue without any limit, what is the probability that person 2 will not at all be selected?
(d) What is the probability that a person will be selected on 10th toss of the coin?

3.29 There is a screening test for prostate cancer that looks at the level of PSA (prostate-specific antigen) in the blood. There are a number of reasons besides prostate cancer that a man can have elevated PSA levels. In addition, many types of prostate cancer develop so slowly that they are never a problem. Unfortunately, there is currently no test to distinguish the different types and using the test is controversial because it is hard to quantify the accuracy rates and the harm done by false positives. For this problem, we'll call a positive test a true positive if it catches a dangerous type of prostate cancer. We'll assume the following numbers:
Rate of prostate cancer among men over 50 is 0.0005,
True positive rate for the test is 0.9,
False positive rate for the test is 0.01. Let $T$ be the event a man has a positive test and let $D$ be the event a man has a dangerous type of the disease. Find $P(D|T)$.

3.30 Plot the p.d.f. of exponential distributions with parameters $\alpha = 1$ and $\alpha = 3$ on the same plot using R functions. Comment on your findings.

3.31 Plot the p.d.f. of a $\chi^2$ distribution with varying degrees of freedom, on the same graph. Comment on your findings.

# Chapter 4
# Analysis of Single Gene Expression Data




By now, we have some idea on how we should look at the data sets at the very outset. In order to get some feel about the data we need to explore it through some visual representation tools (histogram, boxplot, etc.) and a few suitable summary measures (mean, standard deviation, etc.) that may be guided by relevant graphs. These reveal some characteristics about the data set as given in Table 2.1. Now we disclose one important feature about the data given in Table 2.1.

The values in Table 2.1 are not the actual values; but they are log-transformed. We take the logarithm with base 2 for the raw data to get values in Table 2.1. This triggers a natural question: why have we taken transformed values instead of original values? Moreover, out of millions of transformations (theoretically), why do we suddenly prefer to use logarithm transformation? So we need to explain the choice of transformation and why this is regarded as the 'best' for this data set.

Knowledge on data generation in the case of microarray data already indicates that a log transformation might help managing the gene expression data. But is there any way to develop or choose a transformation objectively looking at the data?

Before discussing the transformation we demonstrate an exploratory tool in order to determine whether a set of observed values follows a particular distribution. Here by distribution, we mean, strictly speaking, a probability distribution of a random variable from which the data are supposed to be drawn (randomly). We have already discussed the notion of random variable and its corresponding probability distribution. This will help us to develop an exploratory visual tool that might guide us in identifying the distribution.

## 4.1 Q-Q Plot

'Quantile-Quantile plot' or commonly known as 'Q-Q plot' is a nice visual tool that explores the degree to which a particular theoretical distribution fits to a set of

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I. Mukhopadhyay and P. P. Majumder, *Statistical Methods in Human Genetics*,
Indian Statistical Institute Series, https://doi.org/10.1007/978-981-99-3220-7_4

numerical observations. The name suggests that this technique heavily depends on quantiles of the distribution. We define $p$-th sample quantile ($z_p$) as an observed value such the proportion of the number of observations that are less than or equal to $z_p$ is $p$ for $0 < p < 1$. Analogously $\zeta_p$ is said to be the $p$-th population quantile if $P(X \leq \zeta_p) = p$. Note that for $p = 0.5$, $\zeta_p$ is the population median while its sample analogue is the median obtained from the actual sample data. First, we explain the idea of Q-Q plot using a normal distribution and then its extension towards identifying any distribution will be immediate.

Suppose a random variable $X$ follows a normal distribution with mean 5 and variance 4. However, Theorem 3 establishes that we can easily standardise $N(\mu, \sigma^2)$ variable to a standard normal variable. Hence, without any loss of generality, we can work assuming that $X \sim N(0, 1)$. We then calculate population quantiles ($\zeta_p$) for several values of $p$. On the other hand, based on sample observations drawn from the same population, we also calculate several $z_p$'s, the sample quantiles for different values of $p$. Naturally, if the observations come from the same normal distribution, the sample quantiles should be very close to their corresponding population quantiles. If we plot population quantiles in the horizontal axis and the corresponding sample quantiles in the vertical axis, we can expect that all points should ideally lie on a straight line passing through the origin making an angle of $45^o$ with the horizontal axis. Any significant deviation of the points from the line would indicate that the distribution differs from the normal distribution. Larger the deviation, higher would be the distance of the sample observation from the normal distribution. To calculate sample quantiles, we must standardise each sample value, i.e. instead of considering $\{x_i,\ i = 1, \ldots, n\}$, we should consider $\{y_i = \frac{x_i - \bar{x}}{s},\ i = 1, \ldots, n\}$. Plot these sample quantiles corresponding to population quantiles calculated using a standard normal distribution. Let's demonstrate this idea first using a sample drawn from a normal distribution.

We draw 50 observations, drawn from $N(0, 1)$ distribution and calculate sample quantiles and population quantiles for the same values of $p$ (Table 4.1). Using the information as in Table 4.1, we can plot corresponding to population quantiles. Also, we draw a straight line that passes through the point $(0, 0)$ having an angle $45^o$ with the horizontal axis. This is what is known as 'Q-Q plot' (Fig. 4.1).

Clearly, the points are very close to the dotted line, which guides us to declare that data come from a normal distribution. Note that points may not necessarily lie on the dotted line, but they are close enough, which is a strong indication in favour of our decision. Although we can work with actual values, for normality checking, we can work with standardised values and compare to a standard normal distribution. The R code for from drawing Fig. 4.1 is given below.

**Table 4.1** Data and calculation of sample and population quantiles from N(0,1) distribution

| 100p% | $\zeta_p$ | $z_p$ |
|---|---|---|
| 10% | −1.28 | −1.29 |
| 20% | −0.84 | −0.67 |
| 30% | −0.52 | −0.45 |
| 40% | −0.25 | −0.26 |
| 50% | 0.00 | −0.04 |
| 60% | 0.25 | 0.21 |
| 70% | 0.52 | 0.52 |
| 80% | 0.84 | 0.86 |
| 90% | 1.28 | 1.34 |

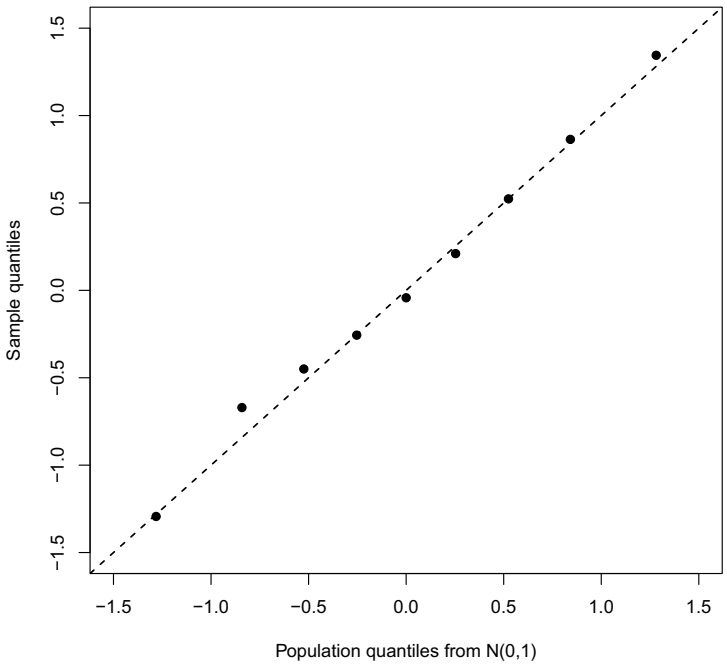



**Fig. 4.1** Q-Q plot to check whether standardised data of Table 4.1 come from a standard normal distribution

```
> x<- c(9.31, 6.25, 9.04, 8.72, 9.75, 9.02, 9.71, 10.24, 11.39, 12.49, 11.03, 7.38,
9.67, 11.43, 12.11, 5.38, 9.69, 7.98, 9.43, 11.27, 10.85, 9.28, 9.18, 11.07, 12.21, 12.36,
9.62, 9.73, 7.60, 6.57, 10.10, 8.70, 9.98, 10.59, 11.12, 10.42, 9.42, 12.91, 6.35, 10.17,
8.43, 8.86, 10.70, 10.72, 12.68, 10.61, 7.14, 8.66, 11.63, 11.53, 8.95, 7.48, 8.92, 9.63,
13.04, 8.44, 10.15, 9.23, 7.75, 8.96, 11.67, 10.66, 9.56, 8.60, 11.28, 9.28, 7.60, 9.68,
14.09, 7.03)                                   ## data input
> y<- (x-mean(x))/sd(x)                        ## standardises data
> zp<- quantile(y,probs = seq(0, 1, 0.1))      ## sample quantiles for
                                                 p = 0.1, 0.2, ..., 0.9
> zetap<- qnorm(seq(0, 1, 0.1))                ## population quantiles from N(0,1) for
                                                 p = 0.1, 0.2, ..., 0.9
> plot(zetap,zp,xlim=c(-1.5,1.5),ylim=c(-1.5,1.5),pch=19,xlab="Population quan-
tiles from N(0,1)",
    ylab="Sample quantiles")                   ## plot z_p versus ζ_p
> abline(0,1,lwd=1.5,lty=2)                    ## draws line through (0, 0) having 45^o angle
                                                 with horizontal axis
```

Table 4.1 and Fig. 4.1 demonstrate the closeness of data points from normal distribution. However, it is worth exploring the pattern of sample quantiles based on data from distribution other than normal through Q-Q plot. First, we generate 100 observations from each of four distributions viz. Normal, Laplace, Gamma, and Cauchy distributions. Calculate sample quantiles for different values of $p(0 < p < 1)$ after standardising the observations, i.e. if $\{x_i,\ i = 1, \ldots, n\}$ are the observations, we calculate sample quantiles for $y_i = (x_i - \bar{x})/s_x$, $i = 1, \ldots, n$. Plot these sample quantiles corresponding to population quantiles calculated on the basis of a standard normal distribution (Fig. 4.2).

The purple dotted line corresponds to the ideal fit to a normal distribution. We see that black points are closest to the purple line, whereas others are deviating from it, with red points showing the most deviation. Red points corresponds to Cauchy distribution which is known to have a very fat tail and thus far from a normal distribution. Thus, Q-Q plot gives us a nice visual tool to check whether sample observations in a data set came from a normal distribution. If the points fall very close to the line drawn through the origin and subtends an angle $45^o$ to the right of horizontal axis (i.e. x-axis), we can say that the data came from a normal distribution.

Note that using Q-Q plot, we can check whether the data come from any particular distribution. Assume a probability distribution, calculates the population quantiles based on this probability distribution, and plot them against the corresponding sample quantiles calculated on the basis of the given data. Here also if we see that points are very close to the line drawn through origin with an angle $45^o$ with the horizontal axis, we can say that the given sample observations have come from the chosen probability distribution. Closer the curve (for a large number of sample quantiles, it looks like a curve) to the line, stronger would be the evidence that data came from the assumed distribution.

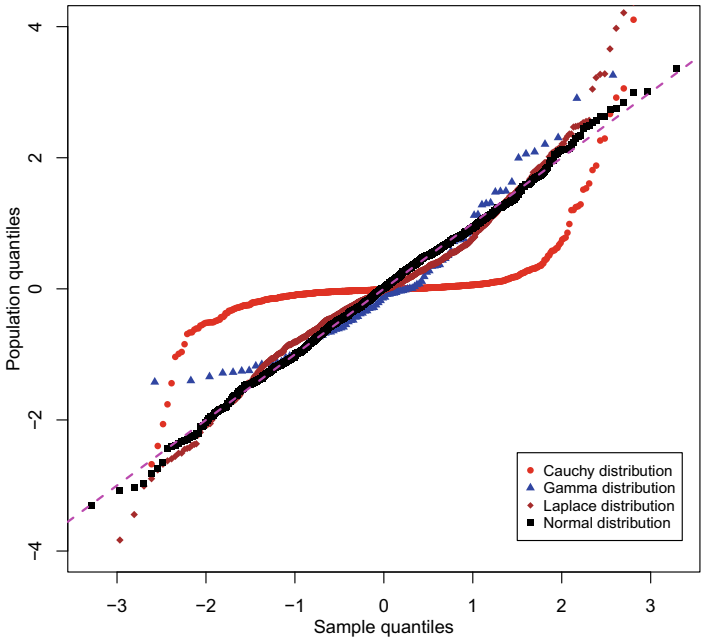



**Fig. 4.2** Q-Q plots for some distributions

```
> qqplot(x,y)          # 'x' contains random sample from the distribution, 'y'
                         contains data
```

The R code for drawing a Q-Q plot is very simple: '$qqplot()$'. Use the command '$?qqplot()$' in R terminal and a full help page will open. Use this function, play around it to explore how fascinating plots can be drawn, try to get the Fig. 4.1 as given here, and customise it using your own data.

## 4.2 Transformation of Variables

Usually, in the paradigm of gene expression data analysis, if $x$ is a value or realisation of the random variable under consideration, sometimes we start working with $log_2(x)$ instead of $x$. However, this is not a random transformation choice.

If we look at histogram or boxplot (Fig. 4.3) of the raw observations, we see an abundant presence of so-called 'outliers'; but it is not wise to think that actual experiment results in so many outliers. Moreover, the actual outliers are lost in the crowd and a histogram or boxplot fails to reveal any features of the data. Actually, the

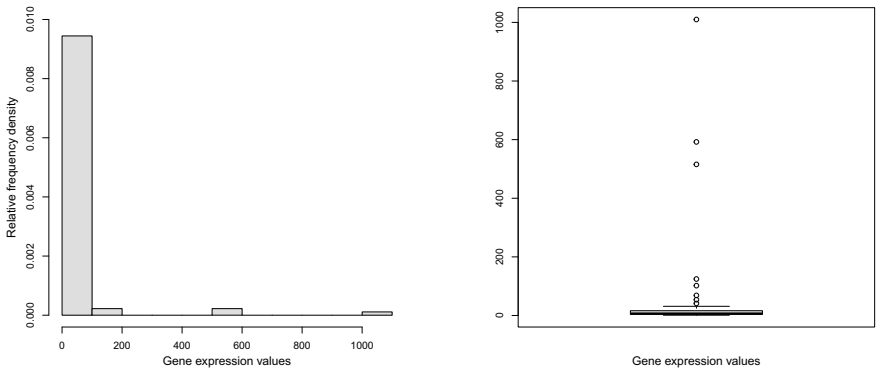



**Fig. 4.3** Histogram and boxplot of gene expression data for original observations

tremendous variation present in the data is due to artefacts of the actual experiment that generates the gene expression data.

Clearly, we cannot get adequate information from the graphs. It is also clear that except a few, all observations are clustered in a small interval (with respect to the scale of measurement), whereas there are two or three clusters with very large values. Boxplot also fails to reveal some idea as the so-called outliers are distributed over a long range towards the right of the mean and other values that constitute majority of the data are concentrated in a very narrow region. The coverage of most values is too narrow to reveal any information about the data. Moreover, it is unrealistic to think that most of the data in reality show very small dispersion as we know that microarray experiment or other gene expression data generating experiments always induce lots of artefacts and hence scatteredness to the data. So there is a need to filter them out or stabilise the observations, at least to some extent, and to dampen the affect of the large outliers and make them compatible with the data. The most effective way to do this is to transform the data. Thus, transformation of data before doing any analysis attracts serious attention among scientists.

There is also another reason in favour of transforming raw data. We know that if we are sure that the data arising out of a continuous variable follows a normal distribution, doing statistical analysis would be much easy. There are lots of statistical tests and methods applicable to data only when it follows a normal distribution. Thus, normal distribution plays a crucial role in analysis. If from a simple Q-Q plot, we see that the data are nowhere near a normal distribution and we can validate this idea using a statistical test (to be discussed later), we are not in a position to use the myriads of available methods at our disposal assuming that the underlying distribution as normal. However, it is exciting to know that in many situations, appropriately transformed values follow a normal distribution while the original values do not. So there is a sound technical reason to consider the possibility of undertaking such transformation before starting any analysis of data.

However, the type of transformation should be determined very carefully, preferably in an objective manner. Box and Cox (1964) proposed a rule of transformation. Suppose $y$ is a variable that takes values $y_1, y_2, \ldots, y_n$. We want to transform these values so that the transformed values should very closely follow a normal distribution. Let $y^{(\lambda)}$ be the transformed variable from the original variable $y$ through a parameter $\lambda$. We have to find a $\lambda$ such that the transformed variables would be:

$$y^{(\lambda)} = \begin{cases} \frac{y^{\lambda}-1}{\lambda} & \text{if } \lambda \neq 0 \\ \log y & \text{if } \lambda = 0 \end{cases}$$

Estimation of $\lambda$ is difficult and demands lot of calculation and assumption. Rather we choose different values of $\lambda$ and draw a 'Q-Q plot' to check whether it would be a good fit to a normal distribution. There might be enumerable choices of $\lambda$ that we can try to find the appropriate transformation, but a little bit of brain storming based on the general nature of the data would reduce this burden to a great extent. Clearly, the values of raw gene expression data may be very large with high variation. So the appropriate value of $\lambda$ cannot be very large. We draw Q-Q plots for $\lambda = 2, 0.9, 0, -0.9, -2$ and see which is closest to the normal distribution. Note that we calculate population quantiles assuming normal distribution while drawing such a Q-Q plot. Naturally, a set of values coming from a normal distribution would always be very close to a straight line passing through the points $(0, 0)$ and $(1, 1)$, i.e. a line passing through origin making an angle $45^o$ with the horizontal axis. In Fig. 4.4, purple line is passing through the black points (stars) in the most representative way. Black points are drawn for $\lambda = 0$, i.e. they are log-transformed values. Thus, it establishes that $\lambda = 0$ is the most appropriate choice and hence the transformation that makes the given raw data closest to a normal distribution is logarithm transformed values.

Application of transformation of variables is extremely important in statistical analysis, especially when the distribution of the variable considered is not a normal distribution and sometimes it is very difficult to know the distribution even approximately. In such situations transforming a variable would make the data more manageable for analysis. Moreover, it is seen that in many cases it is possible to convert original distribution to a normal distribution, at least approximately for the given data set. Once we establish that the transformed data follows normal distribution to a certain degree of confidence, a whole new world of statistical analysis methods would open in front of us and many important information can be extracted from the given data, which at the beginning looks very complicated to analyse.

However, it is important to note that any information obtained after the analysis of data is based on the transformed data and not on the raw data. Once we get some information or inference from the transformed data, we must interpret it in terms of the original variable, although sometimes it is difficult to do so. So from now on the analysis of the data will be done using log-transfomed observations as given in Table 2.1. For the rest of the book, we always mean log-transformed values as gene expression data, unless otherwise mentioned.

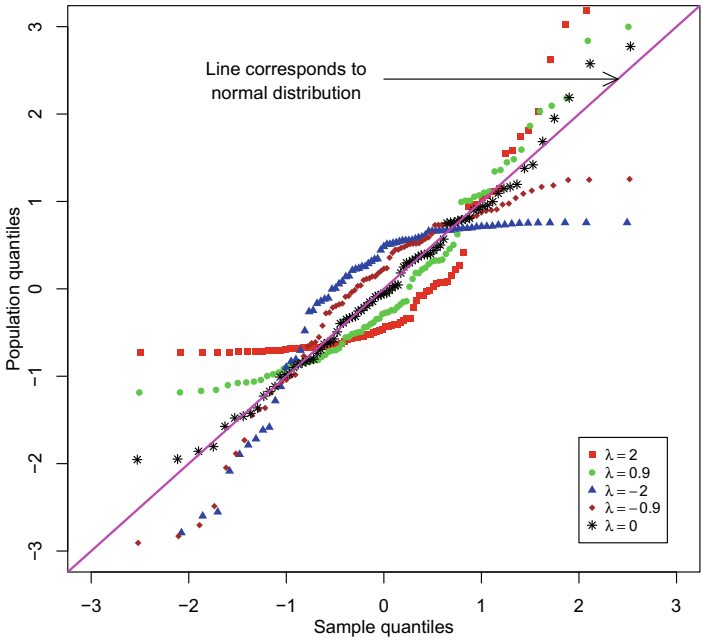



**Fig. 4.4** Q-Q plots showing that logarithm transformation is the best one

## 4.3 A Few Testing Problems

Questions 5–8 in Sect. 2.1 are related to testing some facts about the variable considered, which in our case is gene expression data. We address different types of testing problems, how they can be solved mainly from intuitive consideration, and how best we can interpret the results. But first, we quickly discuss some basic concepts and criteria that a testing procedure should satisfy in order to get meaningful and reliable results. Note that testing of hypothesis covers a major part of data analysis. It is seen sometimes that for the statistical problem under consideration, existing testing methods might fail to give confidence to a statistician because of the intricate nature of the data and/or the hypothesis or objective of the problem. In that case, we might have to develop new method of testing only specific to that problem. So it is imperative to understand the basic ideas behind testing of such statistical hypothesis.

### *4.3.1 Basics of Testing of Hypothesis*

We used to play cricket matches, everyday, during the afternoon after school hours. We made two teams and these two teams would play a series of games. Madhu, the captain of Team A observed that the other captain Shaun always called ‘head’

during the toss to determine which team would bat first. Batting first seemed to be more exciting. Moreover, Shaun used the same coin for the toss every day. Madhu observed that Shaun won the toss every time but once during the last six occasions. So he suspected that the coin Shaun uses was in all probability a biased one, i.e. the chance of getting 'head' was more than getting a 'tail'. Now the question is, how does one check that the coin is fair (unbiased), i.e. the chance (probability) of getting a 'head' is same as getting a 'tail' in a single toss.

Well, a little bit of thinking gives us a very simple strategy. Just toss the coin a fixed number of times, say 20. We expect to see around 10 heads (may be slightly more or slightly less). However, if the number of heads is very large, say more than 17, we may say that the coin is not fair. It shows more chance of getting a head than a tail. This is nothing but statistical testing of a hypothesis.

In the above example, we test a hypothesis, i.e. the coin is fair or unbiased against a suspicion that the coin is not fair. The first one, the hypothesis that we want to check is called 'null hypothesis' (denoted by $H_0$) and the other one, which we consider in case of a false null hypothesis, is called 'alternative hypothesis' (denoted by $H_1$). The strategy that determines whether null hypothesis is true or false is called the testing methodology or strategy.

A scientific hypothesis is usually generated based on available prior information (in most cases partial information). Often a pilot study is conducted to generate specific hypotheses. The hypothesis that is to be tested is called the null hypothesis. A decision on the null hypothesis is taken on the basis of observations collected as a result of a random experiment from the population under study. Usually, one has an alternative hypothesis in mind as well, which is automatically accepted (for most practical situations or further downstream analysis), if the null hypothesis is rejected. So based on a random sample of a fixed size, we can either reject the null hypothesis or accept it or can say that we need some more evidence to come to a firm decision. In this section, we shall discuss, in brief, the principles and methodology of testing of hypothesis.

For any null hypothesis, usually denoted by $H_0$, any one of the four possible situations can arise:

(1) Reject $H_0$ when $H_0$ is false: a correct decision.
(2) Reject $H_0$ when $H_0$ is true: an incorrect decision.
(3) Do not reject $H_0$ when $H_0$ is false: an incorrect decision.
(4) Do not reject $H_0$ when $H_0$ is true: a correct decision.

The second and third decisions are called Type I error and Type II error, respectively. In ancient Indian stories, there are a few instances of these errors. In 'Abhigyan Shakuntalam', King Dushyanta married Shakuntala when he met her in hermitage while hunting in a jungle. He gave her his royal ring to serve as her identity. However, while going to meet him, she lost the ring in the river. When she reached the palace and met him, the king failed to recognise her. Thus, Dushayanta committed a Type I error as he rejected Shakuntala as his wife when, in fact, she was his true wife.

In Mahabharta, Dronacharya was fighting for the Kauravs. During the war, an elephant named 'Aswathama', his son's namesake was killed. Yudhishthira on advice

of Krishna, went to Dronacharya and pronounced that 'Aswathama' was dead and added that it was an elephant in a low murmur. Dronacharya, on listening the first part of ever truthful Yudhishthira's sentence, presumed that his son was dead, and stopped fighting. Thus, Droncharya could be said to have committed a Type II error, i.e. accepting a statement when it was not true.

However, we should devise our testing method to avoid frequently committing these two types of errors. In other words, in good decision-making, the probabilities of these types of errors should be very low. However, for a fixed sample size, we cannot minimise the probabilities of both of these types of errors. Heuristically, this fact can be argued in a very simple way. Note that Type I error is related to rejection of $H_0$, whereas Type II error is concerned with not rejecting $H_0$. These two types of decisions are complementary to each other. Hence, if we decrease (or increase) the probability of one type of errors, it would automatically increase (or decrease) the probability of other type of error. So we cannot minimise the probabilities of both types of errors at the same time using a fixed sample.

On the other hand, we cannot afford to develop a testing procedure that would ensure a high probability of one type of error, be it Type I or Type II. One simple and effective way to get rid of this problem is to fix the probability of Type I error at a very low level, say 0.05 or 0.01, known as 'level of significance' and minimise the probability of type II error. We can also fix the probability of Type II error and minimise Type I error. Whether we choose the first option or the second one, depends on the problem. Given a statistical testing problem, first identify which error is more serious. Keep the probability of that error at a very low level and minimise the probability of the other error. This will ensure that by adopting this testing method, we are not committing a serious mistake in any way. Usually, it is seen that in most cases type I error is more serious than type II error.

We define the statistical Power of a test as:

$$Power = 1 - P(Type\ II\ error)$$

and develop our testing method by maximising the power and keeping probability of Type I error fixed at a prespecified level of significance.

Now we use a statistic (defined as a function of sample observations) that would be used for testing the null hypothesis and also for estimating the population parameter. Note that $H_0$ carries some information about the parameter. So based on data on a set of samples, we first calculate the value of the test statistic. Then, we calculate the probability of obtaining the observed value of statistic or higher (lower) assuming that $H_0$ is true. This probability value is known as the p-value. The p-value has a very important role in getting inference through testing of hypothesis; however, it should be interpreted carefully.

Another very important but often not focused upon is the design by which we draw the sample observations. In almost all testing problems we draw the samples randomly, i.e. following a random mechanism or design, when we collect sample observations from the population of interest. As an example, suppose we want to test whether systolic blood pressure of students of class XI in a particular school

is 120. For this, we may go to that class and select a number of students, say 12, just by looking at their roll numbers or sitting arrangement. This looks like a random sample because we do not know the names of any students. However, this is not really a random sample, because the mechanism by which the students are drawn is not random. Instead, a meaningful design would be as follows. We write the roll number of each student on a piece of pear, fold it and put it in a jar. After shuffling the jar we pick up one paper, unfold it, note down the roll number, measure the blood pressure of the student corresponding to that roll number, and return the paper in the jar. Repeat this procedure until we have picked 12 papers following the same procedure. Finally, we get 12 observations of blood pressure that constitute our sample. This, in fact, is a random sample, because there is a clear random mechanism or experiment at the back of the selection process. Note that we may not return the paper after each draw; even in that case it remains a random sample, although the design is slightly different.

Thus, in a testing paradigm, we always work with a sample whose observations are drawn randomly. This clearly indicates that drawing an observation or its value has nothing to do with the values of other observations. Thus, the selection mechanism induces randomness or *independence* among the observations. In other words, sample observations (or the identical random copies of original variable) are independent of each other. This independence plays a very important role in the development of different testing procedures based on different hypotheses. Henceforth, unless otherwise mentioned, we assume that samples are independently distributed or simply, we work with random or independent samples in all hypothesis testing problems.

### *4.3.2 Interpretation of p-value*

Associated with any testing problem, a widely known and used quantity is p-value. Basically, p-value gives the strength of our decision regarding $H_0$. If the p-value is very small, we may be highly confident in rejecting $H_0$. On the other hand, large p-value indicates that the evidence is too weak to reject $H_0$. Usually, but not always, it is compared to the level of significance, which is taken to be very small. Sometimes people involving in such analysis consider a rule of thumb to compare p-value to a small value, usually 0.05 or 0.01. But in such cases, serious confusion may arise if not interpreted correctly.

Suppose the p-value corresponding to a test statistic for a testing problem is 0.00003. We can immediately reject $H_0$ as the p-value is extremely small, compared to 0.05. However, a p-value of 0.000000023 still provides stronger evidence in favour of rejecting $H_0$ than the previous one. On the other hand, if it is 0.1543, we can say that it is too large compared to 0.05 and hence we do not reject $H_0$.

Now consider two situations; one produces a p-value 0.048 and the other gives 0.052. Considering the level of significance as 0.05, thumb rule suggests that we reject $H_0$ in the first case and do not reject $H_0$ in the second case. Strictly speaking, these conclusions are not very meaningful. In fact, a p-value of 0.048 shows weak

evidence in favour of rejection of $H_0$. On the other hand, if we do not reject $H_0$ in the second case only because it is slightly more than 0.05, we will make serious mistake. Note that 0.048 and 0.052 fall on either side of 0.05 with an absolute difference 0.002. This difference may occur due to a change in a single observation or a random fluctuation only in one observation. In other words, very minor changes in a single observation changes the decision, which should not be appropriate if we just use our common sense. Thus, in such situations, when p-value is reasonably close to 0.05, we can say that there exists a marginal significance of the effect of some factors that distorts the value specified by $H_0$. Instead of taking a very firm decision about $H_0$ at this stage, we need to investigate it more carefully either using more samples or doing an in-depth study of the entire phenomenon or understanding the study design that generates the data or by some other process. We should also check the assumptions under which the test is carried out and look for other powerful tests from the statistics literature, if the situation so demands.

Looking at the definition as well as the cited example it is clear that lower the p-value, stronger is the evidence against $H_0$.

We will now discuss some examples of testing methods that are commonly used. Careful attention should be given to the assumptions under which the tests are derived, the actual methodology of the test, and how the resulting statistic is used to make the final inference.

### *4.3.3 Test for Mean: Single Sample Problem*

In the context of the gene expression data as given in Table 2.1, suppose the experimenter who generated this data set is interested to see whether the mean of expression values is around 4. Let $X$ be the random variable denoting gene expression of a gene under consideration. The observed data set is assumed to be a random sample from $X$. Then the null and alternative hypotheses are:

$$H_0 : \mu = \mu_0 \text{ against } H_1 : \mu \neq \mu_0, \text{ where } \mu_0 = 4, \text{ in this case.}$$

Since we have no prior information about what might be the true value of mean if $H_0$ is rejected, i.e whether it is more or less than $\mu_0$, we use the two-sided alternative hypothesis.

First, we assume that $X$ follows a normal distribution but unfortunately the standard deviation is unknown. This assumption of normal distribution must be validated by either a statistical test or by a strong belief based on prior knowledge about the nature of gene expression values for the gene. Q-Q plot might give an idea whether the data follow a normal distribution. However, a statistical test for testing whether a normal distribution is applicable will be discussed later.

Here, our hypothesis is related to the population mean $\mu$. So we should start by calculating the sample mean $\bar{x}$ based on the observed values. Then we check whether the sample mean is either too small or too large compared to population mean $\mu_0$,

under null hypothesis. Alternatively, we can look at the difference of $\mu_0$ from $\bar{x}$, i.e. the quantity $\bar{x} - \mu_0$. Any difference should be judged with respect to the scale of measurement. And we know that standard deviation is a good scaling measure. Hence, since this difference is in regard to its standard deviation ($\sigma/\sqrt{n}$), we divide the difference by $\sigma/\sqrt{n}$ and thus get our test statistic as:

$$\tau = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}.$$

It can be easily shown (Theorem 3) that $\tau$ follows $N(0, 1)$, a normal distribution with mean 0 and variance 1, under $H_0$. So we reject $H_0$ if observed $\tau$ is too large or the p-value corresponding to $\tau$ is very small compared to $\alpha (0 < \alpha < 1)$, the chosen level of significance. Throughout this book, we take $\alpha$ to be equal to 0.05, unless otherwise mentioned. However, the above testing procedure is based on the assumption that the standard deviation is known. But in reality, this assumption is rarely valid as it is difficult to know the value of $\sigma$ apriori. Thus, it is safe to assume that the standard deviation is unknown and in that case we cannot calculate the value of the test statistic, even under $H_0$, because we don't know the value of $\sigma$.

Since $\sigma$ is unknown, we replace $\sigma^2$ by its unbiased estimator $s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$ and get the test statistic as:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}},$$

which follows a $t$ distribution with $n - 1$ degrees of freedom, under $H_0$. Thus, we reject $H_0$ if the observed value of $t$ is either too small or too large or the p-value corresponding to the value of $t$ is very small compared to $\alpha$. This statistical procedure to test whether mean is equal to a specific value when standard deviation is unknown and the parent distribution is normal, is known as the famous **one-sample t-test**. This is based on $t$-distribution, which was first derived by a great statistician Student (his real name was William Sealy Gosset) in 1908. Hence, this test is sometimes known as Student's $t$-test.

If we look at the histogram of the observations, in Fig. 2.1, it may not be very clear that 4 is widely different from the calculated mean 3.1005 as there is a relatively long tail towards the right side. That is the reason we need to do statistical tests to ensure any strong evidence in favour of or against our belief.

Based on the observations in Table 2.1, we see that $t = -4.5755$ and degrees of freedom is 89. Hence, the corresponding p-value is 0.000015. Since the p-value is extremely small compared to 0.05, we can straightway reject $H_0$ concluding that, on the basis of the given data, there is significant evidence to believe that the mean gene expression value is different from 4.

```
> t.test(x, mu=4, alternative="two.sided")      ## single sample t-test with two sided
                                                   alternative hypothesis
```

The output on the screen contains the value of the test statistic and the p-value. This p-value can be compared to the chosen level of significance leading to the decision based on the given data. Note that if we have prior information that the mean might be greater than 4, we should take alternative hypothesis as $H_1 : \mu > 4$ and in the R code, replace 'two.sided' by 'greater' to get the result.

Advanced readers may ask why the test statistic follows a $t$-distribution. We explain this as a theorem below.

**Theorem 12** *Let* $X_i \sim N(\mu, \sigma^2)$, $i = 1, \ldots, n$ *i.i.d. Then* $\frac{\bar{X}-\mu}{s/\sqrt{n}} \sim t_{n-1}$.

***Proof*** By Theorem 6 and Theorem 8, $\bar{X} \sim N(\mu, \sigma^2/n)$, $\frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{n-1}$ and $\bar{X}$ and $s^2$ are independently distributed. It follows immediately using Theorems 6–8 that $\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0, 1)$, $\frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{n-1}$ and $\bar{X}$ and $s^2$ are independently distributed.

Take, $X = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$, and $Y = \frac{(n-1)s^2}{\sigma^2}$. Using Theorem 9 we have,

$$\frac{\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}}{\sqrt{\frac{(n-1)s^2}{\sigma^2}}} = \frac{\bar{X}-\mu}{s/\sqrt{n}} \sim t_{n-1}$$

**Corollary 2** *When* $H_0$ *is true, i.e. under* $H_0$, $\mu = \mu_0$, *putting* $\mu = \mu_0$, *we have,*

$$\frac{\bar{X}-\mu_0}{s/\sqrt{n}} \sim t_{n-1}.$$

### *4.3.4 Wilcoxon Single Sample Test*

Single sample $t$-test provides a method for testing whether the population mean is equal to a specified value when observations are randomly collected from a normal distribution. However, sometimes data are collected randomly but from a distribution which is not a normal distribution. In such situations, we cannot do the above $t$-test as the test statistic does not follow a $t$-distribution, even under $H_0$. In this connection, it is worthwhile to discuss a statistical test for mean or more generally for an appropriate test for location parameter when the random variable does not follow a normal distribution.

Since we assume here that the random variable describing the population does not follow a normal distribution, we cannot guarantee the existence of population mean. For example, if the variable follows Cauchy distribution, it's mean does not exist. Since we don't know the population distribution, it would be safe to assume that mean may or may nor exist. We should work with the most general situation. Hence, in such cases, we do not assume the existence of mean. However, it is obvious that median always exists and it is a very good measure of central tendency. Hence, in a paradigm where we don't have the liberty to assume normal distribution, we

consider median as our parameter of interest as a measure of central tendency for the population.

Suppose there is a strong evidence or prior knowledge to believe that the variable does not follow a normal distribution. In such a case we can do another test for location parameter, known as **Wilcoxon rank sum test** for a single sample. This test is based on the sum of ranks of absolute deviation of the values from the value of location parameter as specified by $H_0$. This kind of test does not use any assumption about the distribution of the random variable considered, except that it has a continuous distribution, i.e. the variable is an absolutely continuous random variable.

Let $\tilde{\mu}$ be the population median for $X$. So our hypotheses of interest are,

$$H_0 : \tilde{\mu} = \tilde{\mu}_0 \text{ against } H_1 : \tilde{\mu} \neq \tilde{\mu}_0,$$

where $\tilde{\mu}_0$ is value of median specified by $H_0$. Let $R_i$ be the rank of $d_i$ where $d_i = |x_i - \tilde{\mu}_0|,\ i = 1, \ldots, n$. We consider a test statistic as the sum of ranks for which $x_i - \tilde{\mu}_0$ is positive and calculate the corresponding p-value. Thus, our test statistic becomes,

$$W = \sum_{i=1}^{n} rank\{d_i : x_i > \tilde{\mu}_0\}.$$

For the data in Table 2.1, suppose we take $\tilde{\mu}_0$ as 3.5 and the Wilcoxon test statistic is found to be 1273 with p-value 0.00184. We naturally reject $H_0$ concluding that, on the basis of given data, there is a strong evidence to believe against the assumed median value 3.5 at 5% level of significance, even when we do not make or validate any assumption about the nature of the distribution of the variable. R code for Wilcoxon rank sum test would provide all information about this test including the p-value.

```
> wilcox.test(x, mu=4, alternative="two.sided")      ## single sample Wilcoxon
                                                        rank sum test with
                                                     ## two sided alternative
                                                        hypothesis
```

Note that nowhere in this test we are using any fact or property of a normal distribution or any other distribution. These kinds of tests are not based on any assumption of parent distribution and hence do not depend on any parameter of the actual distribution. Hence, these are called 'non-parametric' tests contrary to other 'parametric' tests like $t$-test, etc. that are dependent on parameters of the assumed distribution.

### 4.3.5 Test for Variance: Single Sample Problem

Sometimes we might be interested to see whether the variance of the distribution from which the samples are drawn is equal to a specific value. Here we assume that the parent distribution of the random variable is normal. Moreover, if the value of the variance is known based on prior knowledge, this should be validated based on given observations. Suppose here we want to test

$$H_0 : \sigma^2 = \sigma_0^2 \text{ against } H_1 : \sigma^2 > \sigma_0^2.$$

where $\sigma_0^2$ is the value of the variance specified by $H_0$. We assume that based on prior information, we are only interested to see whether true variance exceeds a certain value. It is believed that if the true variance is less than or equal to a certain value, the data seems to be stable to a certain permissible degree. Based on prior information, suppose we can assume that $\sigma_0^2 = 3$.

We start with comparing the sample variance $s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$ to $\sigma_0^2$ and consider the ratio $s^2/\sigma_0^2$ as our test statistic. We reject $H_0$ if observed value of $s^2/\sigma_0^2$ is too large compared to 1. It is known that when $H_0$ is true, by Theorem 8, $(n-1)s^2/\sigma_0^2$, a multiple of the ratio of sample variance to population variance (under $H_0$), follows a $\chi^2$ distribution with $n-1$ degrees of freedom. Since the sample size is 90, the degrees of freedom here is 89. The p-value corresponding to this statistic is obtained as 0.1445 and hence we have no reason to reject $H_0$ concluding that the true variance of gene expression values can be taken as 3 (at level of significance 0.05).

Suppose we don't know the function which does this test in R, we can easily write our own code.

```
> n<- length(x)                          ## number of observations
> test.statistic<- (n − 1) ∗ var(x)/(sigma ∗ sigma)        ## sigma=σ0
> pvalue <- pchisq(test.statistic,n-1,lower.tail=F)      ## calculates p-value
> pvalue                                 ## prints p-value on R terminal


> pchisq((length(x)-1)*var(x)/3,length(x)-1,lower.tail=F)   ## all previous
                                         ##commands in one line to calculate p-value
```

The function '$pchisq()$' calculates the probability that a $\chi^2$ variable is greater than the observed value of the test statistic, which is obtained by '$(length(x) - 1) * var(x)/3$' with degrees of freedom '$length(x) - 1$' for a right-sided test

('$lower.tail = F$'), where '$length(x)$' is the sample size. The output is only the p-value, which is 0.1445.

### 4.3.6 Test for Equality of Two Means

Consider the data given in Table 4.2. Gene expression values are given for two different groups of people. Two groups may be cases or controls, i.e. affected and unaffected individuals with respect to a particular disease, or two ethnic groups, etc. The two groups are considered independent in the sense that observations in one group are not at all related or affected by observations in the other group. Moreover, observations in each group are random samples from the respective populations.

We want to check whether the mean expression levels are same in the two distributions from which the two independent samples of sizes $n_1(= 30)$ and $n_2(= 24)$ are drawn. So our hypotheses of interest are:

$$H_0 : \mu_1 = \mu_2 \text{ against } H_1 : \mu_1 \neq \mu_2.$$

We assume that both distributions are normal, i.e. if $X$ and $Y$ are the two variables, we assume that $X \sim N(\mu_1, \sigma_1^2)$ and $Y \sim N(\mu_2, \sigma_2^2)$ and $X$ and $Y$ are independently distributed. To do an exact test, we have to assume that $\sigma_1^2 = \sigma_2^2$ although the common value is not known. There is a statistical test that can be used to validate the assumption about the variances. Under these assumptions, we consider the difference $\bar{X} - \bar{Y}$ and compare it to $\mu_1 - \mu_2$, which is 0 under $H_0$. However, since this difference is in regard to its standard deviation, which is unknown, we consider a test statistic under $H_0$ as:

$$t = \frac{\bar{x} - \bar{y}}{s\sqrt{(\frac{1}{n_1} + \frac{1}{n_2})}}, \text{ where } s^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}.$$

where $s_1^2 = \frac{1}{n_1 - 1}\sum_{i=1}^{n_1}(x_i - \bar{x})^2$ and $s_2^2 = \frac{1}{n_2 - 1}\sum_{i=1}^{n_2}(y_i - \bar{y})^2$.

**Table 4.2** Gene expression values for two independent samples

| First sample | | | | | Second sample | | | |
|---|---|---|---|---|---|---|---|---|
| 2.67 | 0.74 | 4.60 | 1.95 | 1.64 | 3.39 | 2.63 | 3.63 | 3.94 |
| 1.40 | 4.91 | 1.45 | 1.99 | 6.96 | 6.28 | 3.70 | 5.54 | 6.54 |
| 3.58 | 2.81 | 0.02 | 4.29 | 2.60 | 5.24 | 2.19 | 5.96 | 6.24 |
| 2.16 | 3.38 | 4.21 | 0.79 | 1.08 | 6.04 | 5.15 | 4.26 | 3.50 |
| 1.56 | 0.57 | 3.46 | 2.37 | 5.36 | 4.41 | 3.10 | 4.11 | 4.97 |
| 1.16 | 0.71 | 1.70 | 4.49 | 6.10 | 6.01 | 2.65 | 5.45 | 7.00 |

It can be shown that under $H_0$, the test statistic will follow a $t$-distribution with $n_1 + n_2 - 2$ degrees of freedom. Hence, we can calculate the p-value corresponding to it, which is 0.000044. So we reject $H_0$ concluding that based on the data in Table 4.2, there exists strong evidence to believe that the average gene expression values are different in the two populations.

```
> x<- c(2.67, 0.74, 4.60, 1.95, 1.64, 1.40, 4.91, 1.45, 1.99, 6.96, 3.58, 2.81, 0.02,
4.29, 2.60, 2.16, 3.38, 4.21, 0.79, 1.08, 1.56, 0.57, 3.46, 2.37, 5.36, 1.16, 0.71, 1.70,
4.49, 6.10)          ## first sample data input
> y<- c(3.39, 2.63, 3.63, 3.94, 6.28, 3.70, 5.54, 6.54, 5.24, 2.19, 5.96, 6.24, 6.04,
5.15, 4.26, 3.50, 4.41, 3.10, 4.11, 4.97, 6.01, 2.65, 5.45, 7.00)
                                        ## second sample data input
> t.test(x, y, alternative = "two.sided", var.equal=T)          ## two sample t-test
```

Note that we must mention that '$var.equal = T$' within the R function; otherwise it will do a test that will be valid only when the variances are unequal and unknown.

Output of $t.test$ gives the test statistic value as $-4.4602$ with corresponding p-value $4.406 \times 10^{-5}$. Since the p-value is very small compared to the chosen level of significance 0.05, we reject the null hypothesis at level of significance 0.05 concluding that, in the light of the given data, it seems that the population means are not equal to each other.

For advanced readers, we show how the test statistic follows $t$-distribution in the following theorem.

**Theorem 13** *Let $X_1, \ldots, X_{n_1}$ be i.i.d. random samples from $X \sim N(\mu_1, \sigma^2)$, $Y_1, \ldots, Y_{n_2}$ be another i.i.d. random samples from $Y \sim N(\mu_2, \sigma^2)$. Assume that the two samples are also independent of each other. Then,*

$$\frac{\bar{X} - \bar{Y} - (\mu_1 - \mu_2)}{s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{n_1+n_2-2}.$$

***Proof*** Clearly, $\bar{X} \sim N(\mu_1, \sigma^2/n_1)$, $\bar{Y} \sim N(\mu_2, \sigma^2/n_2)$, and $\bar{X}_1$ and $\bar{X}_2$ are independently distributed. So by Theorem 4, $\bar{X} - \bar{Y} \sim N(\mu_1 - \mu_2, \sigma^2(\frac{1}{n_1} + \frac{1}{n_2}))$.

Again, by Theorem 8 $\frac{(n_1-1)s_1^2}{\sigma^2} \sim \chi^2_{n_1-1}$, $\frac{(n_2-1)s_2^2}{\sigma^2} \sim \chi^2_{n_2-1}$ and they are independent. Hence, by Theorem 11,

$$\frac{(n_1-1)s_1^2}{\sigma^2} + \frac{(n_2-1)s_2^2}{\sigma^2} \sim \chi^2_{n_1+n_2-2}$$

$$\text{i.e. } \frac{(n_1+n_2-2)s^2}{\sigma^2} \sim \chi^2_{n_1+n_2-2} \qquad \text{where } s^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}$$

Note that sample means are independent as they come from two independent distributions. Moreover, sample mean and sample variance are also independently distributed

for normal distributions. Hence, it is clear that $\bar{X} - \bar{Y}$ and $s^2$ are independently distributed random variables. Hence, by Theorem 9,

$$\frac{\frac{\bar{X}-\bar{Y}-(\mu_1-\mu_2)}{\sqrt{\sigma^2\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}}}{\sqrt{\frac{(n_1+n_1-2)s^2}{\sigma^2}/(n_1+n_{2-2})}} \sim t_{n_1+n_2-2} \qquad \text{i.e.} \qquad \frac{\bar{X}-\bar{Y}-(\mu_1-\mu_2)}{s\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} \sim t_{n_1+n_2-2}$$

**Corollary 3** *Suppose $H_0$ is true, i.e. under $H_0$, $\mu_1 = \mu_2$. Hence, putting $\mu_1 = \mu_2$, we have, under $H_0$,*

$$\frac{\bar{X}-\bar{Y}}{s\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} \sim t_{n_1+n_2-2}.$$

### *4.3.7 Test for Equality of Two Variances*

In the two-sample t-test, along with normality assumption, we have also assumed that population variances are equal. This may or may not be true. Before doing a two-sample t-test, we have to check whether this assumption is valid. To validate our assumption of whether the two variances are equal, we have to develop a statistical test for $H_0 : \sigma_1^2 = \sigma_2^2$ against $H_0 : \sigma_1^2 \neq \sigma_2^2$.

Under the assumptions of normality, we take the ratio of sample variances, i.e. $s_1^2/s_2^2$ as our test statistic. If $H_0$ is true, the ratio of population variances is one. Hence, under $H_0$, we can expect that corresponding ratio of sample variances should be very close to 1. So, we reject $H_0$ if the observed value of $s_1^2/s_2^2$ is either too large or too small compared to 1. It can be shown that when $H_0$ is true, this statistic will follow an $F$-distribution with degrees of freedom $(n_1 - 1, n_2 - 1)$, i.e.

$$\frac{s_1^2}{s_2^2} \sim F_{n-1-1,n_2-1} \text{ under } H_0.$$

Hence, we can easily calculate p-value corresponding to the value of the test statistic using the $F$ distribution with degrees of freedom $n_1 - 1$ and $n_2 - 1$. Here the value of the test statistic is 1.637 with the corresponding p-value as 0.1144. Thus, based on the given data we can believe that our assumption about the equality of variances is correct. Hence, the population variances are equal for two distributions and we can do a standard two-sample $t$-test to see whether the average gene expressions for the two populations are same, based on the data given in Table 4.2.

```
> pf(var(x)/var(y), length(x)-1, length(y)-1, lower.tail=F)        ## calculates
                                                                   p-value using F distribution
```

For advanced readers, we derive the distribution of the above test statistic under $H_0$.

**Theorem 14** *Under the above set-up,* $\frac{s_1^2}{s_2^2} \sim F_{n_1-1,n_2-1}$ *under* $H_0$.

***Proof*** Note that $\frac{(n_1-1)s_1^2}{\sigma_1^2} \sim \chi^2_{n_1-1}$ and $\frac{(n_2-1)s_2^2}{\sigma_2^2} \sim \chi^2_{n_2-1}$. Moreover, $s_1^2$ and $s_2^2$ are independently distributed. Then by Theorem 10,

$$\frac{\frac{(n_1-1)s_1^2}{\sigma_1^2}/(n_1-1)}{\frac{(n_2-1)s_2^2}{\sigma_2^2}/(n_2-1)} \sim F_{n_1-1,n_2-1} \quad \text{i.e.} \quad \frac{s_1^2}{s_2^2}.\frac{\sigma_2^2}{\sigma_1^2} \sim F_{n_1-1,n_2-1}.$$

Hence, under $H_0 : \sigma_1^2 = \sigma_2^2$, we have $\frac{s_1^2}{s_2^2} \sim F_{n_1-1,n_2-1}$.

### 4.3.8 Wilcoxon Two-Sample Test for Locations

The two-sample $t$-test can be done only when the distributions are normal and variances are equal but unknown. But in many real-life situations data from either one or both populations do not follow normal distribution. Naturally, $t$-test in such a situation is not recommended and may mislead the entire conclusion. In such cases, there is a nice but simple test based on comparing the ranks of the observations from two groups, known as Wilcoxon two-sample test for equality of location parameters. Following similar logic as in Wilcoxon single sample test, we are going to develop a non-parametric test where no assumption is made about the distributions except that they are absolutely continuous. We consider median instead of mean as the location parameter.

Like Wilcoxon single sample test, here we consider the null hypothesis as $H_0 : \tilde{\mu}_1 = \tilde{\mu}_2$, i.e. whether the medians of two distributions are equal. Since we are not allowed to assume any distribution of the random variables considered, we have to construct a test using ranks of the observations. Thus, our hypotheses of interest become,

$$H_0 : \tilde{\mu}_1 = \tilde{\mu}_2 \text{ against } H_0 : \tilde{\mu}_1 \neq \tilde{\mu}_2.$$

Note that the decision about $H_0$ is taken using a test statistic and its distribution under $H_0$. First assume that $H_0$ is true. This indicates that two samples are drawn from the same distribution and so we have a random sample of size $n_1 + n_2$. Rank all these observations. Consider the sum of ranks corresponding to the observations that come from the second distribution. If $R_i$ is rank of the $i$-th observation from the second sample, we can construct the test statistic as:

$$S = \sum_{i=1}^{n_2} R_i.$$

Naturally, if $\tilde{\mu}_2 > \tilde{\mu}_1$, the distribution for the second sample is shifted to the right of that for the first sample, although a little overlap might be there. Consequently, $S$ would be very large. On the other hand, if $\tilde{\mu}_2 < \tilde{\mu}_1$, $S$ would be very small. This if $H_0$ is true, $S$ will neither be small nor large. So we reject $H_0$ if $S$ is either too large or too small. We can calculate p-value directly using the R function.

If we ignore that observations in Table 4.2 follow normal distributions, we can do Wilcoxon two-sample test to test $H_0 : \tilde{\mu}_1 = \tilde{\mu}_2$ against $H_1 : \tilde{\mu}_1 \neq \tilde{\mu}_2$; the p-value is obtained as 0.00006 which indicates that medians of the two groups are significantly different. The most important point to remember is that this test can be done when there is no assumption about the distribution concerned.

Sometimes, many people also do Mann-Whitney test for testing the same hypothesis. Note that although the test statistic is somewhat different than Wilcoxon test, basically one is obtained from the other through a linear relation. Hence, the p-value obtained on the basis of Mann-Whitney test is exactly same as that obtained through Wilcoxon test. Here nowhere have we assumed any specific probability distribution of $X$ and $Y$.

```
> wilcox.test(x, y, alternative = "two.sided")
```

First assume that our chosen level of significance is 0.05. Based on the data set (Table 4.2), the value of test statistic is 138, whereas the corresponding p-value is 0.0000601. The very small value of p-value compared to the level of significance indicates that on the basis of the given data, there is enough evidence to conclude that the medians and hence the central tendencies of two data sets are significantly different.

### *4.3.9 Test for Normality*

The above discussion establishes one important fact that basic testing methods depend very much on the assumption whether the data conforms to a normal distribution, especially when the data are of continuous type. We have seen that test for location (mean or median) can be done using t-test if the data are from normal distribution; otherwise, Wilcoxon test is appropriate. Hence, the appropriate test for a given problem depends on our knowledge of the parent population, i.e. distribution from which the data are supposed to come. To this end, although Q-Q plot can provide a nice visual tool to understand whether the variable follows a normal distribution, this is not a statistical test, in the true sense. However, there are some statistical tests available for this purpose. Among them, two are the most popular and we discus them in detail.

#### 4.3.9.1 Kolmogorov-Smirnov Test

Before discussing any test for normality, we have to understand one important fact. Normal distribution is entirely specified by its mean and variance, denoted respectively by $\mu$ and $\sigma^2$. These are called parameters, that characterise a probability distribution describing the population under consideration. For an exponential distribution, whose p.d.f. is given by $f(x) = \frac{1}{\theta}e^{-x/\theta}$, $x > 0$, the only parameter is $\theta$ and it is enough to know everything about this distribution if we know the value of $\theta$.

However, if we want to know from which distribution the data are coming, we cannot think of any particular distribution at the outset. That means we neither have any knowledge about the possible distribution nor any idea about the parameters nor even the number of parameters that it should have. Sometimes even population mean or variance may not exist for a typical distribution; however, any quantile-based measures of central tendency or dispersion always exist. For example, mean does not exist for a Cauchy distribution, but median does. On the other hand, although we know whether the variable under consideration is discrete or continuous, it would be too ambitious to think of the existence of p.d.f (p.m.f.) in case the variable is continuous (discrete). There are some instances where neither p.d.f. nor p.m.f. exists although the distribution function always exists. Thus, a distribution function seems to be a promising quantity that could have been used to develop a test to identify whether the data corresponding to random variable come from a normal distribution or from a specific distribution. This is also a non-parametric test as we are not assuming any p.d.f. (or p.m.f.) of the random variable from which data are generated.

Thus, to meet our objective, we can generalise the question as how to develop a test, more specifically a non-parametric test, to see whether the two distributions are same for two random variables. One is the specific distribution that we think to be true, whereas the other distribution is the actual distribution from which we have the actual data. Since we don't know the actual distribution, we have no idea about its p.d.f. But the fact that probability distribution function always exists enables us to develop a statistical test based on an analogous concept of distribution function that can be deduced only from the data. This quantity is known as the 'empirical distribution function'. We compare it to the known or our objective distribution. This idea is the basis of the famous Kolmogorov-Smirnov (KS) test.

Suppose that $\{X_1, \ldots, X_n\}$ is a random sample from the population whose distribution is given by $F(x)$. The empirical distribution function $\hat{F}_n$ is based on the given observations. We define $\hat{F}_n(x)$ for each $x$ as:

$$\hat{F}_n(x) = \frac{\text{number of observations } \leq x}{n}$$

If we want to test that the random sample is from a normal distribution, we can frame this problem as a testing of hypothesis problem. Here,

$$H_0 : X \sim \text{Normal distribution against } X \nsim \text{Normal distribution.}$$

Let $\Phi(x)$ denote the distribution function, i.e. probability distribution of a standard normal distribution (i.e. a normal distribution with mean 0 and variance 1). To test the above $H_0$, Kolmogorov and Smirnov proposed the test statistic, based on empirical distribution function, as:

$$D_n = \sup_x |\hat{F}_n(x) - \Phi(x)|.$$

The test statistic $D_n$ measures the maximum difference between the empirical distribution and standard normal distribution. Larger the value of $D_n$, less likely would $H_0$ be true.

Note that in most situations the sample mean is not very close to 0, neither the sample variance is close to 1. So first we standardise the sample values by subtracting sample mean from each observation and divide them by the sample standard deviation. We then apply Kolmogorov-Smirnov (KS) test to the standardised data. The p-value corresponding to this test is obtained either by using asymptotic distribution of $D_n$ under $H_0$ or by exact calculation for small samples.

```
> ks.test((x-mean(x))/sd(x), "pnorm")        ## Kolmogorov-Smirnov test
                                               for checking normality
```

Here, it is observed that the value of KS statistic for the data considered in Table 2.1 is 0.0959 with corresponding p-value 0.3796. Since the p-value is very large compared to the chosen level of significance 0.05, we can safely conclude that the data have been drawn from a normal distribution. Here we have taken a two-sided alternative as there is no prior information about stochastic ordering of the distribution with respect to a standard normal distribution. 'pnorm' in R code indicates that we are trying to see whether the data conforms to a normal distribution.

#### 4.3.9.2 Shapiro-Wilk Test

Another interesting test for normality was proposed by Shapiro and Wilk (1965) based on the results and principles of linear model theory in statistics. Before discussing this test, we first introduce the concept of order statistics. If we arrange the observations in increasing order, we can assume that there exists a random variable $Y_1 = min\{X_1, \ldots, X_n\}$; i.e. the realisation of $Y_1$ is the minimum of the set of observations. Similarly, we can define $Y_n$ as the maximum 'order statistic' as $Y_n = \max\{X_1, \ldots, X_n\}$. In general, we have $Y_1, Y_2, \ldots, Y_n$ as the set of order statistics corresponding to the random sample $\{X_1, X_2, \ldots, X_n\}$. Shapiro-Wilk (SW) test is described based on these order statistics.

It compares the sample variance for the ordered observations to its corresponding population-level variance. Let $m' = \{m_1, \ldots, m_n\}$ be the vector of expected values of a standard normal order statistics and $V = ((v_{ij}))$ be the corresponding variance-covariance matrix. Then the test statistic is proposed as:

$$W = \frac{(\sum_{i=1}^{n} a_i y_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2},$$

where $a' = (a_1, \ldots, a_n) = \frac{m'V^{-1}}{(m'V^{-1}V^{-1}m)^{\frac{1}{2}}}$.

As we know that for normal distribution, measures of skewness ($\sqrt{\beta_1}$) and kurtosis ($\beta_2$) are 0 and 3 respectively indicating that it is a symmetric distribution and has moderate kurtosis (i.e. 'mesokurtic'). So one can have a fairly good idea about the distribution just by looking at the value of $\sqrt{\beta_1}$ and $\beta_2$. Extensive empirical sampling studies have shown that the value of $W$ for non-null distributions usually tends to shift to the left of that for the null case. Note that here our null hypothesis of interest is that the data conforms to a normal distribution. It is also interesting to note that $W$ lies between $na_1^2/(n-1)$ and 1.

Moreover, the probability density function of $W$ for $n = 3$ is given by $f(w) = \frac{3}{\pi}(1-w)^{-\frac{1}{2}}W^{-\frac{1}{2}}$, $\frac{3}{4} \leq w \leq 1$. Thus, we can directly use $f(w)$ to calculate the p-value for $n = 3$. However, for $n > 3$, no compact form of the distribution is available even under $H_0$. Thus, we have to calculate the p-value approximately. Based on extensive simulations, however, it has been observed that this p-value, although approximate, is fairly accurate and can be used for all practical purposes.

Applying this test to the data, we see sometimes that the p-value is very small and we have no option but to conclude that the data do not follow a normal distribution. However, when we apply Kolmogorov-Smirnov test, we get a larger p-value that directs us to accept that the data follow a normal distribution. This apparent contradiction is actually due to the presence of outliers. If we remove the outliers and apply Shapiro-Wilk test to the data given in Table 2.1, we would get the p-value as 0.5074, thus matching our inference from KS test. So the KS test is more robust to the presence of outliers, whereas SW-test is more sensitive to it. The reason is that the KS-test is based on empirical distance, whereas SW-test is based on moments of higher order. Naturally, moments are highly affected by the outliers, thus making SW-test more sensitive to outliers compared to KS-test. Hence, we recommend applying both these tests for checking normality and accordingly taking a decision. This also shows the importance of studying outliers rather than ignoring them.

```
> shapiro.test((x-mean(x))/sd(x))      ## Shapiro-Wilk test for checking normality
```

## 4.4 Points to Remember

Not only in genetics or genomics, but for any statistical testing problem, we need to remember a few facts. Knowing what problem we are going to solve, we should try to understand the nature of the scientific problem and the objective of the experimenter. This could be done by asking many questions. A rigorous discussion might reveal a

few questions that demand statistical analysis. We must ensure at this point whether we are required to do any statistical testing of hypothesis. If so, first frame and define the null hypothesis and alternative hypothesis appropriately. Considerable time and effort should be devoted to do this; otherwise, the entire downstream analysis might not be correct and lead to misleading conclusion. Note that we must decide on both null and alternative hypotheses, keeping in mind the problem and its objective. Only then do we proceed to do sample collection and eventual testing of hypothesis. Framing or changing any hypothesis after data collection is not permitted; that might amount to manipulating the data and the results.

Once we are sure about the hypothesis to be tested, we must check whether the data show enough evidence that it comes from a normal distribution. If it follows a normal distribution, we may proceed to carry out our hypothesis testing assuming normal distribution. We must be careful about the assumptions on variance when testing for mean(s). Especially, in case of two-sample problems, we must check whether the variances are equal. If they are same, then only we can apply the two-sample $t$-test. On the other hand, if the assumption of normal distribution is not valid based on the given data, we cannot do a $t$-test; rather we should proceed to do non-parametric tests like Wilcoxon test. Thus, knowledge about the parent probability distribution is extremely important; it triggers and controls the entire analysis protocol.

For checking assumption of normal distribution, we should not forget to standardise the data before applying Kolmogorov-Smirnov test or Shapiro-Wilk test. However, for Shapiro-Wilk test, it is not mandatory to standardise the values because this test is specific to checking normal distribution only.

Finally, interpretation of p-value should be done cautiously and carefully. A simple comparison of p-value with respect to the chosen level of significance is not recommended. We need to understand the problem, how serious the type I error rate is, etc., to come to a conclusion based on p-value. We should fix the maximum permissible type I error rate (i.e. level of significance) before performing the statistical test, even before the data collection process or experiment. It should not be changed or adjusted after the test or looking at the p-value. Such a thing would amount to ignorance of the problem or unethical manipulation of results or both.

## Exercise

4.1 Consider the data set 1 as given in Exercise-2.3.

(a) Draw a Q-Q plot to check whether the data come from a normal distribution. Ensure your observation by a statistical test.
(b) Perform an appropriate test to check whether we can say that the population mean is around 3.9. Write your conclusion properly.

4.2 Consider the data set:
16.40, 13.14, 11.34, 18.80, 13.07, 8.12, 11.72, 10.53, 11.06, 6.06, 8.27, 12.05, 13.16, 8.87, 12.38, 13.44, 15.72, 16.22, 13.64, 10.44, 11.95, 7.69, 12.57, 9.13, 16.70, 13.35, 11.50, 10.09, 14.93, 15.47, 8.19, 7.24, 10.52, 12.14, 14.69, 8.84, 13.95

(a) Check whether the data come from a normal distribution.
(b) Draw a Q-Q plot and comment.
(c) Discuss in detail the statistical test that you want to use for testing $H_0 : \mu = 14$ against $H_1 : \mu > 14$. Write your conclusion.
(d) Calculate power of the test assuming values of $\mu$ as 12, 12.2, 12.4, 12.6, ..., 14.8, 15. Draw a graph showing powers against these $\mu$ values and comment. This graph is known as a 'power curve'.
(e) Do a statistical test for $H_0 : \sigma^2 = 4$ against $\sigma^2 \neq 4$.

4.3 Two independent samples from populations that are normally distributed produced the following statistics: for sample 1 the sample size was 25, the sample mean was 34.2 and the sample standard deviation was 12.6. For sample 2, the sample size was 27, the sample mean was 49.1 and the sample standard deviation was 19.4. Assume that population variances are equal. Carry out a statistical test for the null hypothesis that the two population means are equal, in favour of the two-sided alternative?

4.4 Consider data set 2 and data set 3 as in Exercise-2.10.

(a) Test whether the variances in the two data sets are equal.
(b) Do a statistical test to see whether each data set comes from normal distribution.
(c) Perform a test to check equality of means after doing appropriate preliminary analysis.
(d) Write your comments based on your analysis.

4.5 For the data set:
4.46, 3.54, 4.02, 4.60, 4.09, 4.32, 3.98, 4.48, 4.44, 4.43, 4.28, 3.75, 4.58, 4.49, 3.61, 4.59, 3.99, 4.27, 3.98, 3.94, 4.83, 3.08, 4.56, 4.43, 4.00, 4.59, 4.76, 4.51, 3.69, 4.72, 4.12, 4.50, 3.80, 4.30, 4.82, 4.18, 4.54, 4.39, 4.02, 4.70, 3.89, 3.79, 4.45, 3.18, 4.44, 4.45, 4.26, 4.02, 4.25, 3.82
Check whether the above data follows a normal distribution. If not, find a suitable $\lambda$ in the Box-Cox transformation so that the transformed variable follows a normal distribution. Compare mean and variance of both original observations and transformed observations. Comment on your findings.

4.6 For the data set given in problem 4.5, suppose we want to do an appropriate test for location.

(a) What statistical test you would adopt? Justify.
(b) Suppose the experimenter wants to do the same type of hypothesis based on original observations. What should be the most appropriate test in this case and why?

(c) Write a short report on the conclusions obtained in (a) and (b).

4.7 Consider the set of observations:
10.56, 8.58, 10.91, 9.72, 8.21, 8.58, 10.98, 10.36, 9.17, 12.11, 7.73, 14.01, 12.50, 9.21, 10.29, 9.45, 8.98, 8.54, 9.62, 10.60, 12.87, 6.34, 13.12, 8.26, 15.09, 7.66, 10.65, 14.72, 10.30, 7.42

(a) What are the values of first- and fourth-order statistics?
(b) What are the values of $F_n(x)$ at $x = 8.58, 12.50, 9.62$, and $14.72$? Draw a graph of $F_n(x)$ against different given values of $x$. While drawing this graph, write R code of your own to first calculate values of $F_n(x)$ for all possible values of $x$ and then draw an appropriate graph.
(c) Calculate $\Phi(x)$ values assuming $\mu = \bar{x}$ and $\sigma^2 = s^2$ for all given values of $x$. On the same graph obtained in 4.7(b), draw the graph of $\Phi(x)$ using a different colour and line type. Comment on your findings based on the two graphs thus obtained.
(d) To check whether the data follow a normal distribution, you apply both Kolmogorov-Smirnov test and Shapiro-Wilk test using R codes 'ks.test(x)' and 'shapiro.test(x)'. Note down the p-values for KS-test and SW-test. Do you see anything contradictory? If so, where is the fallacy? Explain and solve the fallacy. What is the right conclusion?
(e) Now test whether the location parameter may be taken as 10.43.

4.8 Choose a level of significance as 0.05. You are testing whether the data conform to a normal distribution. Based on a given data set, you get very small p-value for SW-test and relatively large p-value for KS-test. Naturally, you reject $H_0$ based on SW-test while you are unable to reject $H_0$ on the basis of KS-test. This seems an apparent contradiction. What might be the reasons and how do you solve it?

4.9 Interpretation of p-value is extremely important in testing statistical hypothesis. Assume that the level of significance for the test is 0.01. How do you interpret if the p-value is (a) 0.2415, (b) 0.03, (c) 0.008, (d) 0.0012, and (e) 0.0000023?

4.10 A data set contains the following observations:
3.52, 5.10, 4.34, 5.22, 3.55, 4.22, 4.89, 3.89, 4.29, 5.77, 5.74, 4.15, 4.78, 4.39, 4.80, 2.87, 1.86, 4.09, 5.04, 2.21, 2.22, 3.20, 4.63, 4.40, 4.20, 3.57, 4.82, 5.67, 6.22, 4.45, 5.95, 5.52, 4.10, 3.39, 4.45, 6.27, 4.46, 5.15, 4.16, 4.14, 2.51, 2.04, 2.74, 4.51, 2.81, 4.98, 4.53, 3.35, 3.99, 3.73, 4.82, 3.53, 3.90, 5.21, 3.42, 4.03, 4.00, 4.68, 4.11, 3.07
Answer the following questions, preferably using R.

(a) Draw a histogram and write down the main features that are immediately clear from the histogram.
(b) Arrange the observations in increasing order and subtract sample mean from each of them. Let us denote the transformed value as $y$. Thus, we get $y_1, y_2, \ldots, y_n$, where $n$ is the number of observations in the given data set.

(c) Calculate empirical distribution function at each point, i.e. at each value of $y$, calculate

$$\hat{F}(y) = \frac{\text{number of observations } \leq y}{n}.$$

(d) Calculate $F(y) = P(Y \leq y)$ for all such $y$ values when $Y \sim N(0, 1)$, i.e. using 'pnorm()' function of R.
(e) Plot $\hat{F}(y)$ and $F(y)$ against $y$ on the same graph and interpret your result.
(f) Again subtract sample median from ordered observation of $x$ values when they are arranged in increasing order. Let these values be $z_1, z_2, \ldots, z_n$.
(g) Calculate empirical distribuion function $\hat{F}(z)$ based on $z_1, z_2, \ldots, z_n$ as done in (c).
(h) Calculate $F(z) = P(Z \leq z)$ for all such $z$ values if $Z \sim Cauchy(0, 1)$, i.e. using 'pcauchy()' function of R.
(i) Plot $\hat{F}(z)$ and $F(z)$ against $z$ on the same graph and interpret your result.
(j) What is your overall conclusion?

4.11 Suppose a single sample t-test based on $n_1$ and $n_2$ observations produces p-values $P_1$ and $P_2$ respectively, In both cases, sample means are the same and so are the sample variances. Alternative hypothesis is greater than the type in both tests. What is the relation between $P_1$ and $P_2$ if $n_1 < n_2$?

4.12 Let type I error rate for testing $H_0 : \theta = 4$ against $H_1 : \theta < 4$ be 0.05. Power of the test at $\theta = 2$ is 0.69. How does the power get affected if the type I error rate is reduced to 0.01? If the test statistic ($T$) follows an exponential distribution with parameter $\theta$, what is the type I error rate if we reject $H_0$ when $T < 2$?

4.13 An experiment was conducted by students at a university in the fall of 1994 to explore the nature of the relationship between a person's heart rate and stepping up and down on steps of various heights. One of the objectives of this study was to see whether the height of steps matters. Students were randomly assigned to two groups: Low-steps, where the height of steps was 5.75 inches (this group was coded as '0'), and High-steps, where the height of steps was 11.5 inches (this group was coded as '1'). Student performed the exercise (stepping up and down) for three minutes, after which their heart rates were measured. The investigators had hypothesised that the average heart rate would be different between the two groups. A two-sample t-test was used to analyze the data. Answer the following questions:

(a) Explain why we use a two-sample t-test and not a z-test or a paired t-test.
(b) Is this a one-sided or two-sided test?
(c) The sample mean, SD of heart rates for low steps and high steps are 85, 7 and 97, 22 respectively. 20 and 25 students are assigned to the low and high steps group respectively. Test the investigator's hypothesis at 5% level of significance.

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# Chapter 5
# Analysis of Gene Expression Data in a Dependent Set-up




Gene expression data provides us with a big picture of the activity of the genes under study. However, it is of great interest to see whether the same gene behaves in the same way in tumour tissue as well as normal tissue. Or two genes in the same or different pathways behave similarly. Especially in studying cancer, it is expected that at least for some genes, expression patterns vary in different types of tissues or cells from different regions. In other words, a few genes might not show any significant differential signature in different tissues, while others might. One gene might act in synergy with another gene and show some significant activity or change in expression level. Naturally, this demands a thorough understanding and detailed study to get a clear picture of the gene expression pattern.

Multitude of genes exist in any living organism. Human genome alone consists of around 20000–30000 genes. They usually act in a dependent manner. It is expected that there are different types of interplay among genes, some act together in a pathway, some act based on other types of interactions, sometimes based on epigenetic triggers and other processes, etc. Various biological mechanisms and modalities play an important role at almost every stage, some of them being known, but probably mostly unknown. Thus, it is not a wise idea to study the expression of a gene individually; rather in order to get a more vivid picture, we need to study the expression data of two or more genes together.

This chapter is devoted to studying two genes together or the same gene in two different tissues. In this context, the first task should be to identify what are the possible queries that can be addressed through appropriate statistical analysis of the above phenomena. Needless to mention that as we move from simple to complex data, i.e. single gene to two or multiple genes, the statistical analysis might also move from simple to a more complicated type. This is necessary in deciphering the true picture of the phenotype and gene expression scenario. It is clear that data structure becomes more complex than what has been discussed already and demands careful attention, rigorous study, and new concepts and methods that are mandatory for undertaking appropriate analysis. However, the development of such statistical analysis methods emerges from simple intuition, the natural flow of domain knowledge and the nature of the problem.

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I. Mukhopadhyay and P. P. Majumder, *Statistical Methods in Human Genetics*,
Indian Statistical Institute Series, https://doi.org/10.1007/978-981-99-3220-7_5

## 5.1 Understanding the Data

Modern technology provides us an easy way to generate gene expression data. The data may come from two different tissues, viz, normal and tumour from the same individual. Since we get observations corresponding to the same gene but from two different regions of the same individual, it is natural to think that observations are correlated. Similarly, the expression data for two genes from a group of individuals may also be correlated as both genes belong to the same individual. However, two genes may belong to the same pathway or different pathways, same chromosome or different chromosomes. In other words, two or more genes may be co-regulated, i.e. action of one gene is triggered or influenced by another gene, not necessarily in the same direction. In fact, we can generalise this by considering the entire gene expression profile of thousands of genes from each individual thus giving rise to a huge amount of data. We will discuss its analysis later. For the moment, we concentrate on the first two types of scenarios based on two genes.

Whatever be the situation, it is imperative to understand that the nature of the data before starting the analysis. Table 5.1 presents data for a single gene expression from two different regions or tissues for a group of individuals whereas Table 5.2 gives gene expression data for two genes in a number of individuals. We can also think of a more general data set where data are generated for two groups of individuals, viz., cases and controls. For the moment, we assume that the data are given only for controls.

First and the most important step in the analysis of data should be to critically understand the data. This will reveal important features of the data. While exploring data we might be aware of the information that is not available at this moment. It is also important to know what is known and what is not known. Lack of a few particular features is not the weakness of the data. But it demands an appropriate data analysis protocol. The best way to proceed is to ask as many questions about the data as possible; then try to address them, objectively and through statistical tools. At the end, we expect to get a broader picture with an emphasis on special features of the data that may lead to further scientific questions.

## 5.2 Generating Questions

It is always advisable to generate important questions before starting any analysis when data are presented. These questions should be generated using common sense coupled with domain knowledge and expertise. Moreover, we need to understand the problem and the hypothesis or the objective of the data generation experiment. For the data presented in Tables 5.1 and 5.2, some questions might be as follows.

(1) For normal-tumour pair of samples, can we compare the activity of the gene on the basis of their expression data using any suitable diagram(s)?

**Table 5.1** Gene expression data for normal and tumour samples from same individuals

| Normal | Tumour | Normal | Tumour | Normal | Tumour | Normal | Tumour | Normal | Tumour | Normal | Tumour |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 3.26 | 4.53 | 2.85 | 5.13 | 4.81 | 3.71 | 4.3 | 3.45 | 1.62 | 4.66 | 3.75 | 1.31 |
| 2.58 | 1.28 | 0.95 | 3.92 | 3.8 | 4.9 | 3.77 | 1.94 | 1.16 | 3.19 | 3.94 | 6.56 |
| 0.85 | 3.56 | 0.56 | 5.79 | 2.11 | 3.04 | 2.63 | 1.43 | 4.15 | 4.57 | 2.31 | 4.2 |
| 1.96 | 2.08 | 2.94 | 0.93 | 3.86 | 7.91 | 7.2 | 6.02 | 2.54 | 4.79 | 3.64 | 6.18 |
| 5.12 | 1.71 | 2.29 | 0.8 | 0.35 | 2.66 | 2.85 | 2.59 | 2.95 | 3.41 | 6.85 | 7.87 |
| 3.58 | 2.08 | 3.44 | 6.21 | 2.13 | 2.01 | 2.39 | 4.39 | 3.94 | 2.98 | 4.84 | 5.05 |
| 3.61 | 3.58 | 0.25 | 2.06 | 4.74 | 4.8 | 4.16 | 4.81 | 4.7 | 4.39 | 1.41 | 2.71 |
| 2.84 | 2.29 | 1.74 | 3.66 | 6.71 | 7.15 | 2.22 | 2.12 | 4.61 | 4.64 | 5.49 | 9.15 |
| 4.29 | 0.17 | 2.27 | 8.24 | 4.73 | 4.26 | 0.53 | 4.13 | 3.24 | 2.42 | 1.89 | 9.34 |
| 3.68 | 1.72 | 3.24 | 6.76 | 0.41 | 4.33 | 5 | 3.45 | 1.34 | 4.41 | 3.42 | 1.35 |

**Table 5.2** Gene expression data for two genes from same individuals

| Gene1 | Gene2 | Gene1 | Gene2 | Gene1 | Gene2 | Gene1 | Gene2 | Gene1 | Gene2 | Gene1 | Gene2 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0.54 | 2.27 | 1.29 | 1.52 | 2.78 | 0.89 | 2.94 | 4.12 | 4.83 | 5.41 | 2.75 | 1.41 |
| 4.67 | 6.61 | 4.23 | 4.25 | 1.63 | 0.31 | 3.12 | 3.91 | 1.42 | 5.34 | 0.1 | 5.44 |
| 5.86 | 2.91 | 0.15 | 2.38 | −0.06 | 5.97 | 5.17 | 4.83 | 2.49 | 2.71 | 1.81 | 4.95 |
| 1.98 | 4.83 | 0.77 | 0.27 | 2.55 | 4 | 8.93 | 2.94 | 1.78 | 3.12 | 1.84 | 6.99 |
| 0.31 | 2.32 | 2.59 | 1.57 | 2.25 | 3.41 | 2 | 4.57 | 4.15 | 3.98 | 1.98 | 1.99 |
| 4.59 | 4.53 | 5.22 | 7.1 | 0.69 | 4.64 | 3.81 | 4.32 | 4.08 | 3.14 | 4.15 | 10.12 |
| 2.84 | 4.33 | 4.93 | 1.12 | 0.01 | 4.75 | 3.29 | 3.41 | 0.86 | 1.78 | 2.96 | 3.32 |
| 1.96 | 4.09 | 1.73 | 5.83 | 2.52 | 2.34 | 4.36 | 3.61 | 4.6 | 7.47 | 0.01 | 1.65 |
| 2.69 | 2.93 | 3.31 | 7.14 | 0.85 | 5.49 | 4.95 | 4 | 2.39 | 3.44 | 3.39 | 7.04 |
| 3.74 | 7.48 | 4.55 | 3.99 | 6.08 | 4.49 | 0.69 | 4.25 | −0.28 | 1.39 | 1.94 | 6.92 |

(2) Is there any observations that can be considered unreasonable or special with respect to the general nature of the data?
(3) Is there any special reason to study data for normal-tumour pair together instead of studying them individually?
(4) Can we have any idea of the general nature of the data with respect to central tendency, dispersion, or any other feature?
(5) Can we compare the expression level averages or their measures of dispersion?
(6) Can we think of the same set of questions (1)–(4) for expression of two genes as given in Table 5.2?
(7) Is there any reason to believe that the activity of Gene 1 has no relation with that of Gene 2? Is it possible to validate the belief using some statistical procedures or more objectively?
(8) Can we think that the observations come from a normal distribution? Or do there exist normal random variables from which these samples are drawn? This

question is for a technical reason, but has a great impact on the methods of data analysis.

(9) Can we somehow guess or predict the expression of Gene 2 when we have information on Gene 1? If so, how reliable would be our prediction?

Other questions can be asked for, but we concentrate to address these queries. Once this is done, the reader will get an idea how to address other relevant questions based on the given data set.

## 5.3 Visually Inspecting the Data

Only visualisation itself sometimes gives a lot of information about the general nature of the data that can be used as a base knowledge to do further analysis. Here also we consider only log-transformed values, i.e. the data given in Tables 5.1 and 5.2 are already log-transformed. This transformation dampens the effect of outliers to some extent at least and provides a rough idea about the underlying probability distribution of the variable. Choice of such a transformation is decided objectively; usually, we consider Box-Cox transformation.

Each of these data sets is a set of frequency data for a continuous variable. First, we can study histograms and boxplots for gene expression in normal and tumour tissues separately. Similarly for Table 5.2, we can think of same diagrammatic representation for the two genes separately. We draw histograms using relative frequency density to make an adequate comparison because the area under the histogram would be equal to one. This histogram would provide a sense of probability distribution. Also note that here we first study histograms for each gene although observations are given in pairs for each individual. Later on, we move on to discussing presentation of two genes together. We start with the simplest possible diagrams.

### *5.3.1 Histogram and Boxplot*

First, we draw two histograms using relative frequency density for direct comparison of two frequency distributions. Here, we should have a pair of observations for each unit or individual, i.e. we collect data for normal and tumour samples from the same individual or data for expression of two genes from each individual. However, due to many reasons, a few observations might sometimes be missing for one variable, while a few for other variables. This unwanted but unavoidable situation leads to unequal sample sizes for the two variables (normal and tumour or corresponding to two genes). So, it is recommended that we draw histograms using relative frequency density so that we can compare two samples visually through histograms.

If we look at the histograms (Fig. 5.1) carefully, we observe that the gene expression values for normal samples fall in the range bewteen 0 and 8, whereas the range

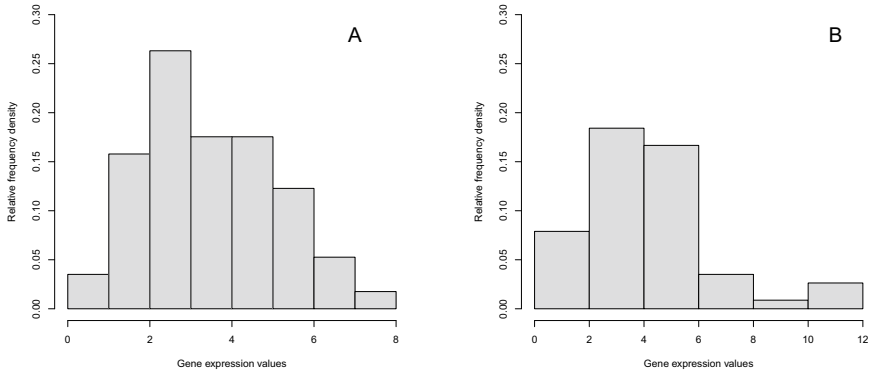



**Fig. 5.1** Histograms of gene expression data for **A** normal samples, and **B** tumour samples

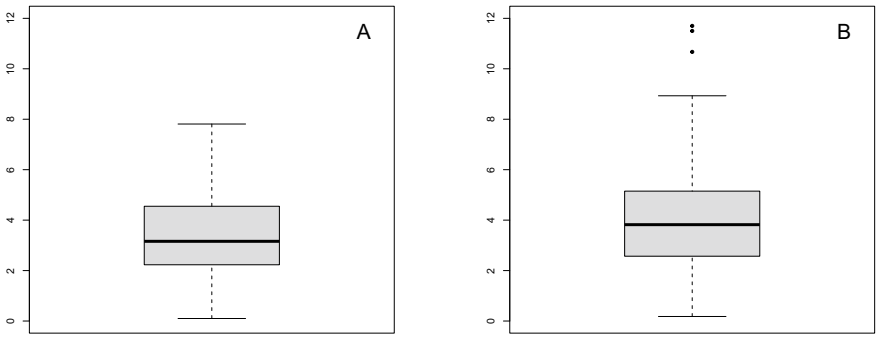



**Fig. 5.2** Boxplots of gene expression data for **A** normal samples, and **B** tumour samples

for tumour samples is 0–12. However, whether the overall gene expression level is significantly large in tumour compared to normal tissue is an important question. This question should be addressed objectively through an appropriate statistical method of testing, which we shall discuss later in this chapter. The histogram reveals that relative frequency spectrum has a peak in the middle (slightly left) and it falls gradually on either side of the peak for both samples. However, it is not clear only from histograms whether the distributions are very close to normal distributions. Moreover, although gene expression values for tumour samples extend more towards the right, it fails to provide any information about whether any or some observations are unusually large. That means histogram sometimes cannot give a clear idea whether there are few outliers in the data. For this, we have to draw boxplots.

A comparison of two boxplots (Fig. 5.2) reveals that the gene expression values for tumour samples have more dispersion compared to normal samples and cover a wider range of values. They are more variable. More variability means more movement. We need to interpret that in the context of the given problem. Median for tumour samples is slightly larger than that for normal samples, but whether this difference is

significant needs more sophisticated statistical analysis. Moreover, a few outliers are also present in tumour sample data. Thus, along with histogram, sometimes boxplots are necessary to get some more information, especially for the detection of outliers.

### 5.3.2 *Finding Relationship Between Genes*

However, these diagrams can only give an idea about the general nature of the data, about its location, extent of spread, presence of outliers, shape of the distribution, etc., separately for each gene or under each condition. But it fails to throw any light on the need of studying them together since we have considered one data set at a time without considering what happens to the other. It completely ignores the movement of one data with respect to another or any kind of relation between the two. Note that, for each individual (say $i$), we can always have data $(x_i, y_i)$ for expressions of two genes or for expression of the same gene in two types of tissues.

Let's have a look at a very simple plot that requires only common sense, not any complicated statistical or mathematical idea. From Table 5.1, just plot the points $(x_i, y_i)$, i.e. gene expression value in tumour $(y_i)$ against that in normal tissue $(x_i)$ in a graph as given Fig. 5.3.

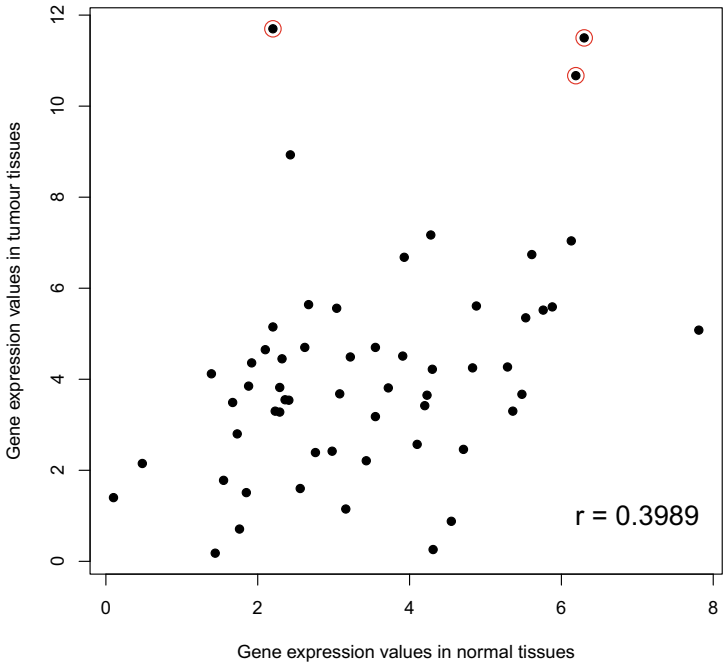



**Fig. 5.3** Scatter diagram of gene expression in normal-tumour samples

The general pattern of the data cloud is to move from left the bottom corner to the top right corner with a substantial amount of spread. This clearly shows that expression in tumour tissue increases with the expression in normal tissue. Thus, we are left with no option but to study these two genes together; otherwise, we risk losing a lot of information on how one acts with respect to the other or the influence of one variable on the other. This diagram, obtained just by plotting the points in a two-dimensional rectangular coordinate system, is known as '**scatter diagram**'.

Let us now study this scatter diagram. This is a very simple but extremely important diagram involving two variables, having immense significance in statistical analysis. Many interesting and informative features can be identified throughout this diagram. Clearly, the scatter diagram shows that the expression of gene in tumour tissues has a tendency to increase with that in normal tissues. This pattern will be more evident if we ignore the so-called 'outliers' (red encircled points in Fig. 5.3).

A scatter diagram represents the movement of a variable with respect to another variable. This type of tendency may be more or less prominent depending on data for different pairs of variables. Naturally, one may encounter several other patterns as described through a few other scatter diagrams in Fig. 5.4. We may have an increasing tendency of one variable with the other (Fig. 5.4A) meaning that as $x$ value increases, corresponding $y$ value also increases. Similarly, Fig. 5.4B presents a decreasing tendency. On the other hand, there might exist a stagnant relation meaning that one variable behaves completely independently or indifferently of whether another increases or decreases (Fig. 5.4C). Another interesting type of relation may also be revealed depending on the data. Figure 5.4D presents a typical scenario where, as the value of one variable increases, the value of another variable first decreases (increases) and after a certain value starts increasing (decreasing). Figure 5.4D presents such a scenario. Clearly, these types of relations indicate the presence of what is called 'linear relation' or 'near linear relation'. If one variable behaves in a non-linear fashion with the movement of the other variable, it also indicates no or very weak linear relation; however, a relationship is there!

After studying these diagrams, we are sure that there is a very important concept that needs to be studied thoroughly. If we look at the scatter diagrams (Fig. 5.4) carefully, we see that Fig. 5.4C represents no relation, whereas Fig. 5.4D indicates the existence of a non-linear, more specifically, a quadratic relationship between two variables $x$ and $y$. Moreover, patterns in Fig. 5.4A and that in Fig. 5.4B are similar as per as the trend is concerned (one increasing and another decreasing), but it is clear that points are more dispersed in Fig. 5.4B. So only the pattern or existence of a relationship is not enough; strength or extent of this relationship is also important and we should not ignore that.

So, we need to devise a quantity that will measure this relationship between two variables; rather the degree of linear relationship between two variables. Hence, we define a quantity, known as '**correlation coefficient**', denoted by $r_{xy}$ or $r_{yx}$ or simply $r$, when there is no confusion about the variables under study. The formula for correlation coefficient is given as

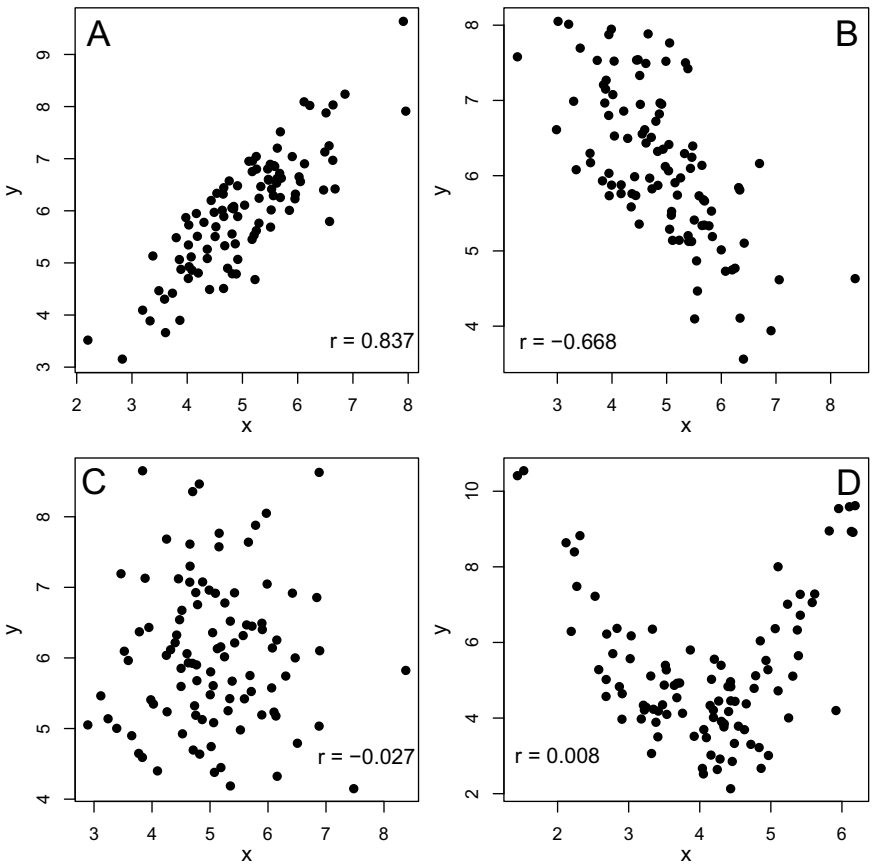



**Fig. 5.4** Scatter diagrams: **A** positive linear relation, **B** negative linear relation, **C** no relation, **D** non-linear relation

$$r = \frac{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2}.\sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \bar{y})^2}} = \frac{cov(x, y)}{s_x s_y}$$

where $cov(x, y) = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$ is known as 'covariance', an important measure to be used later in statistical analysis. Note that correlation coefficient is just the scaled covariance, the standard deviations of $x$ and $y$ are being used as scales.

Because of this scaling and using deviation from mean of the raw values instead of considering the raw values directly, $r$ enjoys a nice property that it always lies between $-1$ to $+1$, i.e. $-1 \leq r \leq +1$. This result has a huge significance; whatever

be the sets of values of two variables, it doesn't matter whether they are large or small or a mix of two, correlation coefficient cannot lie beyond these limits. If we go back to the paired data for which scatter diagrams were drawn, we see that the value of correlation coefficients are 0.837 and −0.668 for Figs. 5.4A, B respectively. These values clearly indicate the nature and degree of linear relationship. Figure 5.4A revealed more linearity that Fig. 5.4B in terms of explaining linear relationships. On the other hand, correlation coefficients are very close to 0 for both Figs. 5.4C, D, although the first one shows no relationship, whereas the second one indicates a strong presence of non-linear relationship. This justifies the notion that correlation coefficient represents the '*degree of linear relationship*' and *not any relationship* between two variables. Also note that the value of $r$ is 0.3989 for data in Fig. 5.3, whereas it is 0.837 for Fig. 5.4A. This is a clear indication that higher the value of $r$, the two variables are more strongly linearly related. This in turn explains the more apparent scatteredness in Fig. 5.3.

This discussion triggers another possibility that there might exist some non-linear relations for which $r$ would be approximately or exactly 0. In such a situation, we can use other measures like '**correlation index**' or '**correlation ratio**' to determine the strength of the non-linear relationship. However, we are not doing a detail discussion about these measures as we rarely need them in practice.

### *5.3.3 Regression*

When two variables are linearly related, it is natural to think whether the knowledge of the value of one variable can be used to guess the value of the other variable. The stronger the relationship, better would be the guess. It is also natural to have an idea about the extent of linear relationship between two variables in the sense that how well we can predict one variable using the other or whether we can predict at all. A scatter diagram supported with evidence through the value of correlation coefficient value would guide us in deciding whether we can predict one variable using the other. This prediction should be done through a mathematical or statistical model. Here, by model, we mean that two variables are related by a mathematical equation (at least approximately). Any pattern other than completely scattered points might be indicative of doing such an exercise of prediction or explaining the relationship between two variables through a model.

For this, we can try to fit a very simple equation, i.e. a straight line through the scattered points in an optimum way so that this line would represent the phenomenon depicted through $(x, y)$-points, in a 'best' possible way. All points on the $(x, y)$ plane need not lie on a straight line, because there are usually intrinsic errors in measurement or other factors that influence the observations, even if the two variables are highly linearly related. However, we may expect that points should be close to the fitted straight line. Closer the points, stronger would be the relationship revealed through the fitted equation.

If $y_i$ be the observed value and $Y_i$ is the so-called 'true value' corresponding to the $i$-th unit, then $y_i - Y_i$ is the difference that is due to other factors, including errors in measurement. This error component is an essential part of any model. To make it more clear, we can say that a mathematical model has two parts: (1) a part that can be explained through evidence (in this case data) and (2) another part that remains unexplained however rigorous the model might be. Since we don't know the actual model representing the relationship between two variables, we will never know it, however large data sets we might collect. So, we are only interested in reducing the error part of the model, that too based on our data (available evidence); it can never be reduced to zero or even if so, we will never know.

Clearly, larger the difference between the true (unknown) value and the observed value, larger would be the contribution of error part. Thus we are interested in the distance between observed and true values and hence we can take $|y_i - Y_i|$ or $(y_i - Y_i)^2$ as a measure of such distance. Now an average or sum of these distances for all observations can give us an idea of the overall deviation from the true values. The principle in finding the line of best fit, the linear regression equation, passing through a scatter of points is to minimise this 'average distance' and estimate the unknown quantities that are involved in the linear equation. Figure 5.5 provides a detailed pictorial representation of this scenario.

Let $y = \alpha + \beta x$ be the true equation between $x$ (independent variable or explanatory variable) and $y$ (response or dependent variable or study variable) and so for the $i$-th unit, we have the model:

$$y_i = \alpha + \beta x_i + e_i; \ \ i = 1, \ldots, n$$

where $e_i$ is the error or the amount of deviation from the true value associated with $y_i$. Naturally, $e_i$ is unknown and so we have to minimise some summary measures of $e_i$s to get the estimates of $\alpha$ and $\beta$. So, we minimise $\sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - Y_i)^2$ and get the estimates of $\alpha$ and $\beta$, denoted by $\hat{\alpha}$ and $\hat{\beta}$ respectively, as follows:

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x} \ \text{ and } \ \hat{\beta} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \frac{cov(x, y)}{var(x)}$$

The fitted regression equation using the estimates of $\alpha$ and $\beta$ becomes:

$$Y = \hat{\alpha} + \hat{\beta} x.$$

Figure 5.5 shows the best possible linear regression of $y$ on $x$ in the sense that the purple line minimises the error sum of squares for all values. However, a careful look reveals some more interesting facts. Although we claim that this would be the best fit (in least square sense), there are more points below the purple line than above

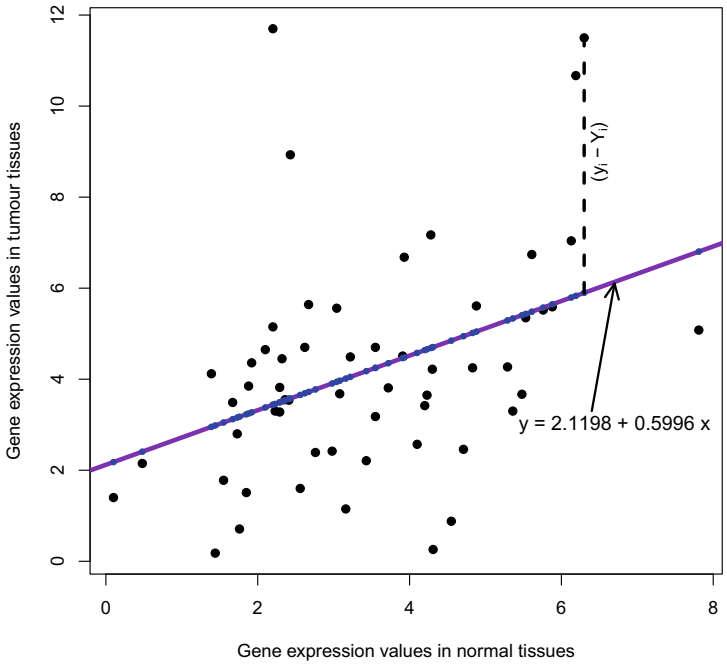



**Fig. 5.5** Least squares regression line to predict gene expression tumour using normal

it (Fig. 5.5). The reason is the presence of outliers (seen from boxplot Fig. 5.2B). So, we do the same exercise of removing the outliers and obtain Fig. 5.6. Here, the purple line is the least square regression that is obtained using all observations, whereas the red dotted line represents the same without outliers.

Thus Fig. 5.6 reveals the impact of outliers. It is clear from this analysis that only a few outliers can shift the regression line. In many biological data, the impact of outliers is sometimes very significant; mere presence of only a few such values can make a substantial change in the regression lines. It may give a completely misleading picture if we routinely or blindly do regression analysis using a data set without any exploratory analysis of data. Thus, we should always be cautious; first try to see and understand the data points very carefully, may be using some diagrammatic representation, and then take actions accordingly. There are other types of effects of outliers; but in each situation, the decision can be altered by its presence. It is to be noted that the number as well as the values of outliers play a very important role in statistical analysis.

If we want to fit a linear regression of $y$ on $x$, we can use R code.

```
> lm(y ~ x)          ## fitting a linear regression model of y on x
```

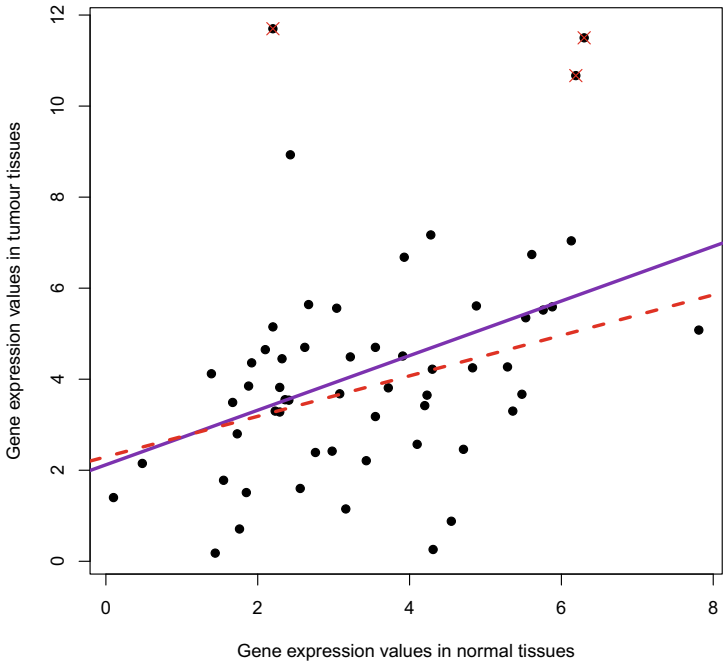



**Fig. 5.6** Least squares regression line to predict gene expression tumour using normal: purple line is for with outliers, and red line is for without outliers

The R code is very simple. However, it is better to store the entire output and extract the relevant information as and when required. A summary of output is available using R code. It is also important to see the scattered points in the diagram along with the fitted regression equation.

```
> a<- lm(y~x)          ## stores output of lm() function in 'a'
> summary(a)           ## displays summary of lm() function
> plot(x,y)            ## displays scatter diagram
> abline(a)            ## superimposes fitted linear regression equation on
                          scatter diagram
```

We shall discuss a few more issues, especially regarding testing in the context of regression analysis later in this chapter. A curious reader might ask a very relevant question at this point. If we recall the pattern given in Fig. 5.4D, it clearly indicates that $y$ (variable in vertical axis) is related as a quadratic equation in $x$ (variable in horizontal axis). A linear regression cannot be a good fit to this data. But we can fit a model that is a quadratic equation in $x$, as

$$y_i = \alpha x + \beta x_i + \gamma x_i^2 + e_i, \quad i = 1, 2, \ldots, n$$

Although $y$ is modelled as a quadratic function of $x$, this model still belongs to the class of 'linear models' because this is a linear function of parameters present in the equation that explains the relationship of $y$ using $x$.

The same 'lm()' function of R code will work. Unfortunately, R function 'abline()' will not work here as the fitted equation is not a straight line. However, using R commands, we can easily calculate the predicted value based on fitted regression equation and plot on the same scatter diagram. Note that $a\$coefficients$ gives the estimated values of the parameters $\alpha$, $\beta$, and $\gamma$ in the same order from the output of 'lm()' function.

```
> z<- x^2                          ## stores x^2 values in z
> a<- lm(y~ x+z)                   ## stores output of lm() function in 'a'
> summary(a)                       ## displays summary of lm() function
> plot(x,y)                        ## displays scatter diagram
> y.pred<- a$coefficients [1] + a$coefficients [2]*x + a$coefficients [3]*z
                                   ## calculates predicted
                                   ## y values
> lines(x,y.pred)                  ## superimposes fitted equation on scatter diagram
```

## 5.4 Some Diagnostic Testing Problems for Paired Data

To decipher the complex interplay of two genes together or study the nature of one gene in two different tissues in a dependent set-up, we first need to determine whether, at all, we should consider the two together or separately. Whatever be the situation, we can also look for whether the nature of two genes is same in terms of their expression values. Variations in observations ensure that there are two underlying random variables that occur together although each one has its own characteristics at the same time. Thus, one important technical question arises in this situation. What is the probability distribution of the two variables taken together? In other words, we want to have an idea about the joint probability distribution of two variables representing expression values of both the genes in the same tissue or same gene in two different tissues. The joint probability distribution of two variables forms the basis of data analysis. To answer the questions, more specifically, Questions 2–7 in Sect. 5.2, we need to study them in the light of hypothesis testing in statistics. Note that the gene expressions for two genes constitute two populations that might depend on each other. Similarly, we can think that two underlying distributions exist for gene expression of the same gene in two different tissues along with a correlation between them, since pairs of observations come from the same individual. There are many such examples in biological experiments where this type of data is very natural.

### 5.4.1 Test for Normal Distribution

Before undertaking a study on any interesting feature of the two populations, we need to understand the underlying probability distributions. Based on our knowledge so far, it seems that if we can ensure that they follow normal distribution, our job will be simpler. In such situations, we can easily apply many standard statistical tests while addressing many relevant questions.

However, assuming that the two populations might be related, we have to study the joint probability distribution of the expression of two genes rather than looking at the distribution of individual genes. We know that a normal distribution is the most common probability distribution for a continuous random variable. In a bivariate situation, i.e. when we have two continuous variables that seem to be related to each other, we can think of an extension of a univariate normal distribution. This extension or generalisation in the case of a bivariate set-up is known as 'bivariate normal distribution'. Let us assume that $(X, Y)$ jointly follow a bivariate normal distribution having a p.d.f. as

$$f(x,y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}\left\{\left(\frac{x-\mu_1}{\sigma_1}\right)^2 - 2\rho\left(\frac{x-\mu_1}{\sigma_1}\right)\left(\frac{y-\mu_2}{\sigma_2}\right) + \left(\frac{x-\mu_2}{\sigma_2}\right)^2\right\}}, \quad -\infty < x, y < \infty,$$

where $\mu_1 = E(X)$, $\mu_2 = E(Y)$, $\sigma_1^2 = V(X)$, $\sigma_2^2 = V(Y)$, and $\rho$ is the population correlation coefficient between $X$ and $Y$. We denote this by $(X, Y) \sim N_2(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$. A few interesting and useful properties of a bivariate normal distribution are given in the following theorem.

**Theorem 15** *If* $(X, Y) \sim N_2(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$, *then*

*(1)* $X \sim N(\mu_1, \sigma_1^2)$, *and* $Y \sim N(\mu_2, \sigma_2^2)$,
*(2)* $a + bX + cY \sim N(a + b\mu_1 + c\mu_2, b^2\sigma_1^2 + 2bc\rho\sigma_1\sigma_2 + c^2\sigma_2^2)$ *for any real constants b and c, and*
*(3)* $X$ *and* $Y$ *are independent if and only if* $\rho = 0$.

The first part of Theorem 15 implies that for a bivariate normal distribution, each of the individual variables marginally follows a univariate normal distribution. However, the converse may not be true, i.e. if each of $X$ and $Y$ follows a normal distribution with some mean and some variance, their joint distribution may not always follow a bivariate normal distribution. The second part of Theorem 15 implies that any linear function of two univariate or bivariate normals follows a univariate normal distribution. This is a very useful result and has extensive use in statistical inference.

Equipped with the above information, when we have two variables that might be related, we should check whether expressions for each of the variables follow a normal distribution. This should be the first step in the data analysis.

Let $X$ and $Y$ denote the random variables corresponding to the expression values in two populations. Although they have a joint (bivariate) distribution, we first test $H_0 : X \sim N(\mu_1, \sigma_1^2)$ for some $\mu_1$ and $\sigma_1^2$, against $H_1 : X \nsim N(\mu_1, \sigma_1^2)$. We can use Kolmogorov-Smirnov test or Shapiro-Wilk test for normality as discussed in

Sect. 4.3.9. Similarly, we can also test $H_0 : Y \sim N(\mu_2, \sigma_2^2)$ for some $\mu_2$ and $\sigma_2^2$ against $H_1 : Y \nsim N(\mu_2, \sigma_2^2)$. If either of these tests are rejected, we can conclude that the variable (corresponding to test that is rejected) does not follow a normal distribution. Hence, we can definitely conclude that the joint probability distribution of $(X, Y)$ is not normal, or rather not 'bivariate normal'.

However, acceptance of both the hypotheses does not confirm that the joint distribution is bivariate normal, because there are a few situations where the joint distribution is not normal although the marginal distribution of each of the variables is normal. Checking normality through the Kolmogorov-Smirnov test or Shapiro-Wilk test would not suffice; we have to do another test which would directly check whether $X$ and $Y$ jointly follow a bivariate normal distribution, i.e. $H_0 : (X, Y) \sim N_2(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$ for some $\mu_1, \mu_2, \sigma_1^2, \sigma_2^2$ and $\rho$.

Literature survey reveals that there are quite a few statistical tests that can check whether a bivariate data conforms to bivariate normal distribution. We briefly discuss three tests for checking bivariate normality. Let $\{(X_i, Y_i),\ i = 1, \ldots, n\}$ be $n$ pairs of random observations from $(X, Y)$. Our test of hypotheses are: $H_0 : (X, Y) \sim N_2(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$ against $H_1 : (X, Y)$ do not follow bivariate normal distribution.

#### 5.4.1.1 Doornik-Hansen Test

Data are given for two variables as paired data, i.e. for each unit, we have observations for both variables. We always calculate sample means and sample variances for both variables. Since each datapoint is a paired data, correlation coefficient is another feature which we must consider in order to understand the salient features of bivariate data.

We know that normal distribution is also symmetric and has moderate kurtosis; these measures should also be taken care of while constructing appropriate test statistic for checking bivariate normality. Doornik and Hansen 2008 considered all these measures and proposed a test statistic. Another interesting feature of this test is that it follows a $\chi^2$ distribution with 4 degrees of freedom asymptotically, i.e. for a large sample size, the probability distribution of the test statistic can be approximated by that for a $\chi^2$ variable with 4 degrees of freedom. The p-value is calculated using this asymptotic distribution. Based on data in Table 5.1, we see that the value of Doornik-Hansen test statistic is 7.6083 with corresponding p-value as 0.107. It is important to note that this test is very sensitive to outliers and we have removed outliers from the data before implementing the test. To apply this test, one must install a special package 'normwhn.test' that contains the R-function for Doornik-Hansen test. R function 'normality.test1()' is applied on $x$, which contains the data matrix, with columns as values for respective variables.

```
> library("normwhn.test")
> normality.test1(x)
```

#### 5.4.1.2 Generalised Shapiro-Wilk Test

Based on sample means, variances and covariance of paired observations, Alva and Estrada (2009) generalised Shapiro-Wilk test to check for multivariate normality. Since here we are interested in bivariate normality, we describe briefly the test for bivariate observations.

Let $\{(X_1, Y_1), (X_2, Y_2), \cdots, (X_n, Y_n)\}$ be the $n$ pairs of observations corresponding to $(X, Y)$. Define sample variance-covariance matrix, consisting of sample variances and covariance as

$$S = \begin{pmatrix} s_{xx} & s_{xy} \\ s_{xy} & s_{yy} \end{pmatrix}$$

where $s_{xx}=\frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2$, $s_{yy}=\frac{1}{n}\sum_{i=1}^{n}(Y_i - \bar{Y})^2$, $s_{xy} = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})$. Consider another symmetric positive definite matrix $C$ such that $CSC' = I_2$, where $I_2$ is an identity matrix of order 2.

Now we define new variables $Z_1$ and $Z_2$ such that

$$\begin{pmatrix} Z_1 \\ Z_2 \end{pmatrix} = C \begin{pmatrix} X - \bar{X} \\ Y - \bar{Y} \end{pmatrix}.$$

Generalised Shapiro-Wilk statistic is defined as

$$W^* = \frac{1}{2}(W_1 + W_2),$$

where $W_i$ is the univariate Shapiro-Wilk statistic based on $Z_i$ for $i = 1, 2$.

This test is a generalised one that tests whether bivariate normal distribution fits to a data set. It is not possible to get a compact analytical expression for the distribution of the test statistic even when $H_0$ is true. However, there is an in-built R code available for computation of p-value using $W^*$. Based on the data in Table 5.1, we obtain the value of the test statistic as 0.9783 with corresponding p-value 0.3836. This test is also very sensitive to outliers and we have applied it after removing outliers. Here also we must first install the package 'mvShapiroTest' before applying the R code and then use the command 'mvShapiro.test()' for testing bivariate normality of the data.

```
> library("mvShapiroTest")
> mvShapirto.Test(x)
```

#### 5.4.1.3 Royston Test

Royston proposed another test for bivariate normality using a modification of Shapiro-Wilk test (Royston 1982; Royston 1992). If the kurtosis of the data is greater than 3, it uses Shapiro-Francia test for (Shapiro & Francia 1972) leptokurtic distributions. On the other hand, if the kurtosis is less than 3, it uses Shapiro-Wilk test for platykurtic distributions. We apply this test to our data in Table 5.1 and obtain the value of the test statistic as 1.2722 with corresponding p-value 0.5316. This test is also sensitive to outliers. However, one must install the package 'royston' before using the R code.

```
> library("royston")
> royston.test(x)
```

There are a few other tests available in the literature. The performance of the above three test statistics is almost same. More importantly, they are sensitive to outliers present in the data. So, it is always advisable to remove the outliers from the data set before applying these tests for normality.

Sometimes, we collect data where each data point has more than two components. These are known as multivariate data. All the above tests can be used to test for multivariate normality, the analogue version of normal distribution in case of higher dimension. Thus statistics literature is very rich in providing some useful tests that should be used during any data analysis, be univariate or bivariate or multivariate. We shall discuss the appropriate statistical analyses for multivariate data later.

We have discussed three such tests using the data given in Table 5.1. But in order to demonstrate the performance of each test, we need to do extensive simulations. We calculate the empirical power and type I error rate for each of these tests assuming that data are coming from different distributions. The distributions chosen cover symmetric as well as positively skewed distribution. It would be an interesting exercise to do some more simulations covering a wider range of distributions.

We first simulate $n$ pairs of bivariate data taking $n = 50, 75, 100, 125$ from a bivariate normal distribution with means 0, 0, variances 1, 1, and covariance 0.6. Apply each test to calculate the test statistic and its corresponding p-value. Repeat this experiment for 1000 times and calculate the proportion of times the p-value is less than 0.05, the chosen type I error rate. It is expected that the type I error rate should be close to 0.05 to ensure that each test really preserves the level of significance as an upper bound.

To calculate power, we generate data from the same bivariate normal distribution and then consider the absolute values of observations generated. If $y$ is an observation, we take $|y|$. This would introduce positive skewness in the new data set. Naturally, now the data cannot be claimed as a random sample from a bivariate normal distribution. We calculate test statistic and its corresponding p-value for each test. Based on 1000 repetitions, we calculate the proportion of times p-value is smaller than 0.05, which would give the empirical power. If the data come from a null distribution, i.e. bivariate normal distribution itself, this proportion would give the type I error rate of

**Table 5.3** Simulation study to compare type I error rate and power for Shapiro-Wilk, Doornik-Hansen, and Royston tests

| | Shapiro-Wilk | | | | Doornik-Hansen | | | | Royston | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| n | 50 | 75 | 100 | 125 | 50 | 75 | 100 | 125 | 50 | 75 | 100 | 125 |
| Type I error | 0.049 | 0.044 | 0.053 | 0.04 | 0.046 | 0.046 | 0.054 | 0.051 | 0.07 | 0.067 | 0.066 | 0.062 |
| Power | 0.762 | 0.962 | 0.994 | 0.999 | 0.668 | 0.914 | 0.981 | 0.997 | 0.845 | 0.981 | 1 | 1 |

the test. Table 5.3 gives a comparative study of these three tests in respect of power and type I error rate.

It is observed that all three tests are very powerful even for a moderate sample size. However, Royston test has slightly inflated type I error rate, but at the same time, it is more powerful. So any of these three tests can be applied in practice. The main reason for the more or less same performance of all three tests is probably that here we are dealing with bivariate data, i.e. each data point has two dimensions only. This might be more prominent if we consider multivariate data with higher dimensions. In any case, we can use any of these tests to check normality in data having more than one dimension.

### *5.4.2 Are Genes Correlated?*

In order to start our investigation about the behaviour of genes, we have to first understand whether the two genes, in some sense, are acting together or one gene might influence the other, or the same gene in two different tissues forms bivariate data with some correlation between them. As we see that correlation coefficient is a nice measure that can determine the degree or extent of linear relationship between two variables, we start with this measure. Note that use of correlation coefficient only ensures the existence of linear relationship between two variables.

To see whether the two variables are related, we can do a statistical test to check any absence of correlation. So, we can propose a null hypothesis $H_0 : \rho = 0$ against $H_1 : \rho \neq 0$. Here, $\rho$ is the population correlation coefficient between two variables of interest. Naturally, we can think that correlation coefficient value calculated on the basis of sample observations would provide information about $\rho$, since the samples are drawn at random from the population. Thus, we can consider sample correlation coefficient $r$ as a test statistic and reject $H_0$ if the observed $r$ is too large or too small compared to 0, the value of $\rho$ specified by $H_0$. Although assuming bivariate normal distribution, we can find the probability distribution of $r$ when $H_0$ is true, it is slightly difficult to calculate the p-value. However, a small trick makes our life simple!

First, let's assume that $(X, Y)$ jointly follow a bivariate normal distribution. Then, we can take another function

$$\psi(r) = r\sqrt{n-2}/\sqrt{1-r^2},$$

which is a non-decreasing function of $r$. Naturally, the fact that "$r$ is too large or too small" is equivalent to "$\psi(r)$ is too large or too small". Moreover, it can be shown that $\psi(r)$ follows a $t$-distribution with $n-2$ degrees of freedom, when $H_0$ is true.

Thus, the p-value can be easily calculated using $t$-distribution with $n-2$ degrees of freedom. In our data set of normal-tumour gene expression, $r = 0.4227$ and hence $\psi(r) = 3.6126$ and the corresponding p-value is 0.00062. Since p-value is extremely small compared to the chosen level of significance 0.05, we can immediately say that on the basis of the given data set (Table 5.1), it seems that there is a strong correlation between the expression values of the two genes.

Based on our discussion in the previous two subsections, we can safely assume the expression values $(X, Y)$ follow a bivariate normal distribution with means $\mu_1$, $\mu_2$, variances $\sigma_1^2$, $\sigma_2^2$, and correlation coefficient $\rho$. Note that all calculations are done without removing outliers present in the data, as we see from boxplots previously. However, even if remove outliers, the p-value for testing $H_0 : \rho = 0$, is 0.00143, which is still very small and hence there is no change in the conclusion. It is always recommended to understand the presence and importance of outliers before doing any test.

R code for the above testing problem is given below, where $x$ and $y$ are two series of observations for which we want to check $H_0 : \rho = 0$. Note that here the alternative hypothesis is $H_1 : \rho \neq 0$ and hence we have to do two-sided test. Since $t$-distribution is symmetric about 0, we can calculate the p-value based on the right tail probability and then multiply by 2.

```
> r<-cor(x,y)                    ## calculates sample correlation coefficient
> test.statistic<-r*sqrt(length(x)-2)/sqrt(1-r*r)        ## calculates test
                                                          statistic ψ(r)
> p.value<- 2*pt(abs(test.statistic),length(x)-2,lower.tail=F)    ## calculates
                                                                    p-value

> cor.test(x,y)          ## test for zero correlation coefficient using
                            in-built R function
```

This test is developed under the assumption that the two variables jointly follow a bivariate normal distribution. However, when this assumption is invalid, we should not apply this test. In fact, to check whether the two variables are related, we cannot even do a test for the absence of correlation because that may not ensure the independence of the variables. Only in the case of bivariate normal distributions, it is necessary and sufficient that zero value of correlation coefficient indicates that there is no relation between the two variables. However, if the test for bivariate normality fails, we cannot use the above $t$-test to ensure independence between two variables. We have to look for another statistical test suitable for a general set-up.

### 5.4.3 Test of Independence

Based on the discussion in the previous section, it is clear that we have to consider a general test of independence in order to see whether two variables are related or independent. But before that, we first explain the notion of independence mathematically and then propose an adequate test.

#### 5.4.3.1 What is Independence?

The joint probability distribution of two variables $X$ and $Y$ is denoted by $F_{X,Y}(x, y)$ where

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y) \text{ for any } x \text{ and } y.$$

This is known as "joint probability distribution function" of $(X, Y)$. On the other hand, we can always think of the marginal probability distribution of the variables $X$ and $Y$ separately as

$$F_X(x) = P(X \leq x) \text{ for any } x \text{ and } F_Y(y) = P(Y \leq y) \text{ for any } y.$$

These are called "probability distribution function" of respective variables. Naturally, there should be a relation between the joint distribution function and the marginal distribution functions.

From the basic probability theory, we know that two events $A$ and $B$ are said to be *independent* if $P(A \cap B) = P(A).P(B)$. Now take $A = \{X \leq x\}$ and $B = \{Y \leq y\}$. Then from the notion of independence of events, we can immediately say that two variables $X$ and $Y$ are *independent* or *independently distributed* if

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y) = P(A \cap B) = P(A)P(B) = F_X(x)F_Y(y) \text{ for all } (x, y) \in \mathscr{R}^2.$$

Here, $\mathscr{R}^2$ denotes the two-dimensional Euclidean space or simply a two-dimensional plane that we know from high school level.

As a consequence of the above definition of independence, we can simplify it when both $X$ and $Y$ are discrete random variables with p.m.f.s $f_X(x) = P(X = x)$ and $f_Y(y) = P(Y = y)$ respectively or both are continuous variables with p.d.f.s $f_X(x)$ and $f_Y(y)$ respectively. Let the joint p.m.f. or p.d.f of $(X, Y)$ is $f(x, y)$. Then the equivalent definition of independence is

$$X \text{ and } Y \text{ are independent } if \text{ and only } if \ f(x, y) = f_X(x).f_Y(y) \text{ for all } x, y.$$

So given a joint distribution it is important to know whether the two variables are independently distributed. If not, we have no way but to consider them jointly in all subsequent analyses for a better understanding of phenomenon. In most analyses related to genetic or genomic data, two variables occurring from the same experiment

or sometimes otherwise, are related because of several reasons. One such reason might be the synergistic behaviour of joint actions of two variables that is intrinsic to biological mechanism.

**Ex-5.1:** Let $X$ and $Y$ denote the numbers appearing respectively on die 1 and die 2 when both are rolled once. Clearly, $X$ and $Y$ are independent because outcome on die 2 has no way dependent on the outcome of die 1. Here, $P(X = 2, Y = 3) = P(X = 2).P(Y = 3) = \frac{1}{6}.\frac{1}{6} = \frac{1}{36}$. In fact this is true for any $x$ and $y$, i.e.

$$P(X = x, Y = y) = P(X = x).P(Y = y) = \frac{1}{36} \text{ for any } x, y = 1, 2, \ldots, 6.$$

**Ex-5.2:** Suppose $(X, Y) \sim N_2(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$, where $\rho = 0$. Then, for any $(x, y) \in \mathscr{R}^2$, we have putting $\rho = 0$ in the p.d.f. of bivariate normal distribution,

$$f(x, y) = \frac{1}{2\pi\sigma_1\sigma_2} e^{-\frac{1}{2}\{(\frac{x-\mu_1}{\sigma_1})^2 + (\frac{x-\mu_2}{\sigma_2})^2\}} = \frac{1}{\sqrt{2\pi}\sigma_1} e^{-\frac{1}{2}(\frac{x-\mu_1}{\sigma_1})^2} . \frac{1}{\sqrt{2\pi}\sigma_2} e^{-\frac{1}{2}(\frac{x-\mu_2}{\sigma_2})^2} = f_X(x).f_Y(y)$$

where $f_X(x)$ and $f_Y(y)$ are p.d.f.s of $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$ variables respectively.

Hence, if $\rho = 0$, $X$ and $Y$ are independently distributed in the case of bivariate normal distribution.

**Ex-5.3:** When we throw a die, a number appears on the upper face of the die. Let $A$ be an event denoting that the number is greater than 4 and $B$ denote the number 6. Here, $P(A) = \frac{2}{6} = \frac{1}{3}$ and $P(B) = \frac{1}{6}$. Now, we need to see whether these two events are independent,

$$P(A \cap B) = P(\text{number appearing is greater than 4 and it is 6}) = P(\text{number is 6}) = \frac{1}{6} \neq P(A)P(B).$$

Hence, $A$ and $B$ are not independent.

### 5.4.3.2 Nonparametric Test of Independence

If two variables $X$ and $Y$ are independently distributed, correlation coefficient ($\rho$) between them must be equal to zero. But $\rho = 0$ does not necessarily imply that the two variables are independent. However, it can be easily shown that if the correlation coefficient between two variables is zero, the variables are independent provided they follow a bivariate normal distribution (Ex-5.2). If they have other joint distribution, they may not be independent even though the correlation coefficient is zero.

By now it is clear that if the two variables do not jointly follow a bivariate normal distribution, we cannot do a test for the absence of correlation as this may not necessarily indicate the independence between them. So, now we think in terms of a more general framework. Instead of considering a test for correlation coefficient, we can think a test of independence if accepted, would imply that there is no relation between the two variables.

But why do we do any test for independence? If somehow we can prove that there is enough evidence that underlying variables $X$ and $Y$ (say) have no relation or dependence among them, we can treat them as completely unrelated variables. Hence our problem boils down to considering them individually. All relevant univariate studies would be enough to throw some light on the given scenario. Thus, knowing whether variables are independent, will control the entire downstream analysis and hence the conclusion.

Note that the test for independence is slightly less powerful since we do not assume any particular form of distribution. Naturally, we lose some information that is conveyed by this assumption, but it is better to sacrifice this information rather than wrongly assuming a distribution which would give more adverse effect on the entire statistical analysis and subsequent decision-making process. So, here we concentrate on describing some nonparametric tests that do not depend on normality assumption.

Puri and Sen (1971) developed a few tests to see whether two variables are independent. This test is based on ranks of the observations of $X$ and $Y$, i.e. based on ranks of $(x_i, y_i)$, $i = 1, \ldots, n$. Two test statistics are available in R code. Thus, in our example, if $X$ and $Y$ are two variables indicating gene expression values for normal and tumour samples from the same group of individuals, we want to test whether $X$ and $Y$ are independent, assuming no particular distribution. Thus, our hypotheses of interest are

$$H_0 : F_{X,Y}(x, y) = F_X(x)F_Y(y) \text{ for all } x, y \text{ against } H_0 : F_{X,Y}(x, y) \neq F_X(x)F_Y(y) \text{ for at least some } x \text{ and } y.$$

The p-value corresponding to the test based on normal score is 0.00036, whereas that for rank test is 0.00098. Since in each case the p-value is very small, we can conclude that there is enough reason to believe that the two variables are not independent.

If $x$ is the data matrix for which the first two columns contain the values for $x$ and $y$, we can use in-built function in R; but, first, we must install the package 'ICSNP'.

```
> b<-ind.ctest(x,1,2,scores="normal")
> b$p.value
> b<-ind.ctest(x,1,2,scores="rank")
> b$p.value
```

Although this type of nonparametric test for independence is not widely used in practice, but this is extremely important. Usually, people are reluctant in checking the distributional assumption behind every test that they employ in analysis of data; however, it should be remembered that standard tests are valid only under some particular assumptions about the parent probability distribution. The evidence against or in favour of this must be explored very carefully, as the rest of the analysis will completely depend on this study. We emphasise that the Puri-Sen test should be employed to test for independence in case tests for bivariate normality ensures that the distribution is other than normal.

One reason for the reluctance of using these tests frequently by the scientific community is probably that it involves lots of calculation and understanding of the statistical tests. Needless to mention that these tests are complicated compared to standard tests. So one must write computer programs in order to use them in practice, which in turn requires substantial knowledge of programming languages. However, this obstacle has now been overcome completely because of the availability of in-built functions in R.

## 5.5 Some Standard Paired Sample Testing Problems

### *5.5.1 Are Two Means Equal?*

Once we determine that two variables are correlated and they follow bivariate normal distribution, we can proceed one step further to see whether they are the same (at least approximately) with respect to their central tendencies. Here we assume that two variables $(X, Y) \sim N_2(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$ and we have a random sample $\{(x_1, y_1, ), \ldots, (x_n, y_n)\}$. Our hypotheses of interest for testing equality of two means are

$$H_0 : \mu_1 = \mu_2 \text{ against } H_1 : \mu_1 \neq \mu_2.$$

The reason for taking both-sided alternative in $H_1$ is the complete lack of prior knowledge about the nature of means if they were not equal. We naturally assume the stringent and natural situation where the population variances and correlation coefficient are all unknown.

Note that a simple idea can make development of a test very easy. If we define a variable as $U = X - Y$, then, by the second part of Theorem 15, we can easily show that $U \sim N(\mu, \sigma^2)$, where $\mu = \mu_1 - \mu_2$ and $\sigma^2 = \sigma_1^2 - 2\rho\sigma_1\sigma_2 + \sigma_2^2$. Now since $\mu = 0$ under $H_0$, we can look at the testing problem as testing $H_0 : \mu = 0$ against $H_1 : \mu \neq 0$ when $\sigma^2$ is unknown and $U \sim N(\mu, \sigma^2)$. Naturally, for the $i$-th individual, we have an observation $u_i = x_i - y_i$ for the random variable $U$. So, the sample observations are now modified as $\{u_1, u_2, \ldots, u_n\}$. This test boils down to a standard test for mean in case of univariate normal distribution, which is nothing but one sample $t$-test (discussed in Sect. 4.3.3). Hence, the test statistic would be:

$$t = \frac{\sqrt{n}\bar{U}}{s}$$

and $t \sim t_{n-1}$ under $H_0$. We reject $H_0$ if its observed value is too large or too small or the calculated p-value is extremely small. This test is known as 'paired $t$-test'. Note that although we can express $\bar{U}$ and $s^2$ in terms of measures based on $X$ and $Y$

as $\bar{U} = \bar{X} - \bar{Y}$ and $s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(U_i - \bar{U})^2 = s_1^2 - 2rs_1s_2 + s_2^2$, there is no need to evaluate them; we can directly use $\bar{U}$ and $s^2$ calculated on the basis of $\{u_1, u_2, \ldots, u_n\}$ or use the in-built function in R.

In reference to data in Table 5.1, we denote that $x$ and $y$ represent gene expression values for normal and tumour tissues respectively. Based on our data set, we see that the p-value is 0.0139 for paired $t$-test. Sometimes, prior knowledge indicates that tumour gene expression may be greater than that for normal tissue. This knowldge, if available with reasonable confidence, may be used to modify the alternative hypothesis. In this case, we may consider alternative hypothesis as $H_1 : \mu_1 < \mu_2$. P-value is calculated based on left tail probability of the null distribution of the paired $t$ statistic. Since p-value is small (although not extremely small) compared to the chosen level of significance 0.05, we conclude there is some evidence in the data set that tumour gene expression is slightly larger than that in normal tissue. Further study is required to see more functionality of this gene in the context of the disease. R code for this paired sample t-test is given below. If we use $H_1 : \mu_1 \neq \mu_2$, we should write "two.sided" instead of "less" inside 't.test' function.

```
> t.test(x, y, paired = T, alternative = "less")       ## summary output of t-test
> t.test(x, y, paired = T, alternative = "less")$p.value       ## p-value of t-test
```

### 5.5.2 *Test for Locations for Non-Normal Distribution*

The above-discussed paired $t$-test is a nice way to check whether the two location parameters, as described by means, for a bivariate normal distribution, are equal. However, like the two-sample problem (Sect. 4.3.8), this test may not be valid or may produce an unrealistic result when the random variables jointly do not follow a bivariate normal distribution. We now describe a nonparametric test for location parameters in such a scenario. This does not require knowledge of the bivariate distribution.

As mentioned earlier, it is natural to test whether two medians are equal, as mean may not exist always. Thus, if $\tilde{\mu}_X$ and $\tilde{\mu}_Y$ are two medians for $X$ and $Y$ respectively, our null hypothesis of interest is: $H_0 : \tilde{\mu}_X = \tilde{\mu}_Y$. A very simple distribution-free test can be obtained if we apply a little tweaking to the null hypothesis and instead consider $H_0 : \tilde{\mu}_{X-Y} = 0$ against $H_1 : \tilde{\mu}_{X-Y} \neq 0$, where $\tilde{\mu}_{X-Y}$ is the population median of $X - Y$. Here, we implicitly assume that median of differences is approximately equal to the difference of medians, which is not always exactly true.

For this set of hypotheses, we define $D_i = X_i - Y_i$ for $i = 1, \ldots, n$, i.e. we define the difference of each paired observation as a new variable. Count the number of positive $d_i$'s and consider it as our test statistic; denote it by $T$. Naturally, if $H_0$ is

true, there would be more or less an equal number of positive and negative values, whereas, if $H_1$ is true, the observed value of $T$ is either too large or too small. So, we calculate the p-value as $P(T > t|H_0) + P(T < n - t|H_0)$, where under $H_0$, $T \sim Binomial(n, 0.5)$. This is known as the sign test for paired sample.

Assuming that our data set does not conform to a bivariate normal distribution, we do this sign test for paired data and obtain the p-value as 0.2619. So, we can conclude that based on the given data there is no reason to believe that their locations are significantly different. Note that this test is less powerful than an exact test based on the actual distribution of the random variables.

It is important to remember that this test is basically a test for proportion for a binomial random experiment and we can use the in-built R code for binomial test for single sample proportion.

```
> d<- x-y
> binom.test(sum(d>0), n=length(d), p=0.5, alternative = “two.sided")
```

This test is carried out without removing outliers. Since we are using medians, this is not seriously affected by their presence. If we do the same test removing the outliers, the conclusion remains same as the p-value in such a case is 0.2976.

If there is reason to believe that the difference of medians is significantly different from the median of differences, we cannot apply this test. However, we can use bootstrap technique for testing equality of two medians in such a scenario.

### *5.5.3 Regression-Based Testing*

When two variables are related, more specifically when we see that there exists a significant correlation coefficient between them, we can always think of fitting an appropriate regression of one variable (response variable) on the other (independent or explanatory variable). Domain knowledge should be used to identify the response and explanatory variables. Interpretation of the results would depend on this choice.

Suppose $Y$ and $X$ are response and explanatory variables respectively in a regression framework. Using least square technique as discussed in Sect. 5.3.3, we can easily find an optimum regression equation that explains the effect of $X$ on $Y$. But the major question is whether it is worthwhile to find such a regression line. To answer this, we would need to test whether the regression coefficient is significantly different from zero. Naturally, zero regression coefficient or a value very close to 0 would indicate a very weak (or no) linear relation making linear regression study not worthwhile.

Therefore, our testing problem would be

$$H_0 : \beta = \beta_0 \text{ against } H_1 : \beta \neq \beta_0,$$

where $\beta$ is the regression coefficient and $\beta_0$ is its value as specified by null hypothesis. $\beta_0 = 0$ indicates the non existence of linear regression of $Y$ on $X$. In this context, it is worth mentioning that $\beta$ and $\rho$ are related through the relation $\beta = \rho\sigma_y/\sigma_x$. Thus testing $H_0 : \beta = 0$ is equivalent to testing $H_0 : \rho = 0$, indicating the absence of linear relationship.

To develop an appropriate test, a natural test statistic would be $\hat{\beta}$, the least square estimate based on paired observations $\{(x_i, y_i),\ i = 1, \ldots, n\}$, which is given by:

$$\hat{\beta} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}.$$

Since $\hat{\beta}$ is a linear function of $y_i,\ i = 1, \ldots, n$, it can be shown that $\hat{\beta}$ follows a normal distribution, under the assumption of normality of random error components, i.e. $\hat{\beta} \sim N(\beta, \frac{\sigma^2}{S_{xx}})$, where $S_{xx} = \sum_{i=1}^{n}(x_i - \bar{x})^2$. Following the logic explained in Sect. 4.3.3, we should take the standardised version of $\hat{\beta}$. But the presence of error variance $\sigma^2$ in the denominator makes the calculation impossible as we don't know the exact value of $\sigma^2$. So, we replace it with a good (unbiased) estimator $\hat{\sigma}^2$, where is $\hat{\sigma}^2 = \frac{1}{n-2}\sum_{i=1}^{n}(y_i - \hat{\alpha} - \hat{\beta}x_i)^2$ and $n$ is the sample size. Thus, our final test statistic would be:

$$T = \frac{\hat{\beta} - \beta_0}{\sqrt{\hat{\sigma}^2/S_{xx}}}.$$

It can be shown that the probability distributions of $\hat{\sigma}^2$ and $\hat{\beta}$ are independent and hence under $H_0$, $T$ follows a $t$-distribution with $n - 2$ degrees of freedom.

Suppose we want to predict the gene expression in tumour tissue based on that in normal tissue. We can use the regression equation of $Y$ on $x$ and test whether this regression equation can predict the value of $Y$ reasonably well. For the data given in Table 5.1, after removing outliers, the p-value associated with this test is seen to be 0.00166. So, we can conclude that the regression coefficient is significantly different from 0 meaning that the prediction is possible. The outcome of R code of fitting regression equation contains lot of information. In fact, it always provides p-value for testing $H_0 : \beta = 0$.

```
> lin.reg<- lm(y~x)      ## fits linear regression and stores output in 'lin.reg'
> summary(lin.reg)       ## provides summary of output including p-values
```

This would give a lot of information about the fitting of regression equation of $y$ on $x$. The p-value corresponding to the variable $x$ (or explanatory variable) will be indicative of the existence of linear regression and we can take decision accordingly.

## 5.6 Points to Remember

Bivariate data occur frequently in the context of genetics. Study of individual variables would be good but it does not take into account the correlation between them. Thus, we might lose information if we avoid analysis considering the two variables at the same time. Statistics is very rich in this area. There are a myriad of statistical methods and tools that can be used in many scenarios of bivariate data. We have seen in Table 5.1 that gene expression of same gene in normal as well as tumour tissue is expected to be correlated as both observations come from the same individual. Similarly, data in Table 5.2 also reveal the same feature because gene expression of two genes comes from the same individual. In this case, correlation might be more pronounced and meaningful if the two genes belong to the same pathway.

This chapter is devoted to analysing gene expression data or data on two variables when they may be dependent. Thus, given any such bivariate data set, we should first check whether the two variables follow bivariate normal distribution. Life would be relatively easy if we get an affirmative answer based on test for normality in bivariate samples. In such a case, a simple $t$-test for the absence of correlation would ensure the dependence status between two variables. Otherwise, we have to do nonparametric tests like Puri-Sen tests to check whether they are independent.

It is most likely that data generated for two seemingly related variables from the same set-up or experiment, follow a bivariate distribution. However, after doing a test for independence, we have to proceed accordingly. If assumption for bivariate normal distribution is valid based on the given data, standard available tests based on bivariate normal distribution can be performed to have more knowledge about central tendency or dispersion of two random variables. Otherwise, we have to do appropriate nonparametric tests. As mentioned earlier, it is better to do nonparametric test when bivariate normality does not hold rather than wrongly applying tests that are only suitable for bivariate normal distribution.

In this chapter, we have discussed a few statistical tests that are not very commonly used. The reason behind this is probably our reluctance to go for a distribution other than bivariate normal distribution, mainly for computational issues and looking at some slightly complicated mathematical formula. Computational challenges, that one faced earlier, have largely been addressed by the use of R, or other software/programming language. Now suitable methods for analysis of bivariate data are available with computational ease. The only thing is that we have to understand the problem and we should have a clear idea on what we really want to analyse. Hence, it is advisable to use statistical treatment of the data that is deemed most appropriate.

**Exercise**

5.1 For data in Table 5.2, draw histograms and boxplots for each gene and on a single plot. Write a short report on your observations.

5.2 Consider the following data: (12.97, 35.5), (6.99, 52.36), (7.47, 37.93), (7.93, 5.13), (10.33, 7.96), (12.76, 14.87), (15, 48.35), (13.14, 54.91), (6.18, 47.5), (9.65, 0.96), (9.51, 10.05), (13.33, 36.25), (11.14, 5.4), (9.85, 13.17), (12.04, 16.96), (6.8, 56.8), (12.51, 33.51), (13.49, 30.2), (7.54, 29), (8.42, 19.76), (8.6, 15.29), (11.34, 8.98), (7.28, 46.34), (7.3, 42.98), (11.67, 23.7).

(a) Calculate the correlation coefficient between two variables.
(b) Draw a scatter diagram.
(c) Fit a linear regression equation to this data set and plot it on the scatter diagram.
(d) Do you think fitting this regression equation is appropriate? If so, why? If not, modify your answer with justification.
(e) After fitting the above linear regression equation of $y$ on $x$, plot the residuals against $x$ variable. What can you say from the diagram? Assuming these residuals are independent, perform a test to see whether they follow normal distribution with mean 0.
(f) The data set contains 25 pairs of observations, corresponding to 25 randomly chosen individuals. Split the data set randomly into two data sets: data set 1 contains 12 pairs whereas data set 2 contains remaining 13 pairs. Fit separate linear regression lines of $y$ on $x$ for each of two data sets. Now plot these regression lines and also the one that has been obtained using all observations as in (c) on the same graph. Comment on your findings with justification. Do you understand why it is recommended to draw random samples?
(g) Based on this bivariate data set, check whether the underlying variables follow a bivariate normal distribution?

5.3 For the data in 5.2, assume that they come from bivariate normal distribution. Apply a statistical test to check whether the two variables are independent. If you use Puri-Sen test to the same data set, what would be your conclusion? Comment on the basis of two conclusions based on the two tests.

5.4 If the correlation coefficient between two variable is $+1$, what would the scatter diagram look like?

5.5 Draw a scattered diagram using R where correlation coefficient is close to zero, but there exists a relationship between the two variables.

5.6 For Table 5.1, calculate $r$, the correlation coefficient, fit an appropriate linear regression line, and test for bivariate normality. Interpret your results.

5.7 In Table 5.2, suppose we want to know that gene 1 can be explained through gene 2 (at least partially) through a linear regression equation. Write this clearly as a problem of testing of hypothesis. Do the testing and calculate the p-value for this test and interpret the result. Develop a test to see whether, on an average, gene expression for gene 2 is more the that for gene 1 by 2.4 and interpret your result.

5.8 Generate a bivariate non-normal data using R and check for normality.

5.9 Consider two data sets on gene expression values (may be transformed):
Data set-1: 2.802, 5.324, 6.363, 6.176, 4.976, 6.261, 5.068, 4.483, 5.324, 6.576, 5.599, 7.377, 5.217, 6.072, 5.490, 6.118, 7.523, 7.702, 6.977, 5.293, 6.309, 6.047, 6.653, 6.527, 6.372, 7.019, 4.337, 3.995, 7.669, 3.773, 6.350, 7.188, 6.752, 7.085, 6.479, 7.423, 4.264, 5.362, 8.094, 8.410, 8.555, 9.605
Data set-2: 4.078, 5.287, 2.818, 5.755, 5.376, 4.116, 5.009, 4.107, 3.888, 6.835, 4.625, 6.448, 2.803, 5.185, 4.483, 4.838, 3.489, 4.463, 4.803, 3.645, 3.726, 5.938, 6.507, 4.771, 5.706, 4.936, 6.650, 4.328, 6.096, 4.095, 1.931, 4.091, 3.789, 4.774, 5.493, 5.429, 6.285, 5.810, 5.409, 6.533, 1.661, 4.776

(a) Test whether the variances are same in the two sets.
(b) Perform an appropriate test to see whether location parameters for the two distributions are same?
(c) Can you use a Kolmogorov-Smirnov test to check whether the probability distributions are same in two data sets?
(d) After you perform the above two testing procedures, suppose the experimenter tells you that the values in the two data sets actually form a single data set from the same samples for tumour and normal tissues (order of samples collected being same for tumour and normal). How would you modify your test in (b)?

5.10 Take logarithm transformation of the data given in 5.2 above. Fit a linear regression line of the second component on first component for pairs of data points. Do a statistical test with $H_0 : \beta = 2.3$ against $H_1 : \beta \neq 2.3$ and comment on your findings.

5.11 For a bivariate data set on two variables $x$ and $y$, $\{(x_i, y_i),\ i = 1, 2, \ldots, n\}$ values are not available. However, we have data on sum of the variables and their differences, i.e. we have data as $\{(u_i, v_i),\ i = 1, 2, \ldots, n\}$ where $u_i = x_i + y_i$ and $v_i = x_i - y_i$ for each $i = 1, 2, \ldots, n$. We can easily calculate the correlation coefficient between $u$ and $v$, say $r_{uv}$. Can we retrieve the correlation coefficient between $x$ and $y$ from $r_{uv}$? If so, how? Justify your answer. After fitting linear regression equation of $v$ on $u$,we get the regression coefficient of $\beta$ as a function of data on $(u_i, v_i), i = 1, 2, \ldots, n$. What would be the regression coefficient of $y$ on $x$ in terms of $\beta$ already obtained?

5.12 A study of 100 patients is performed to determine if cholesterol levels are lowered after 3 months of taking a new drug. Cholesterol levels are measured on each individual at the beginning of the study and 3 months later. The cholesterol change is calculated which is the value at 3 months minus the value at the beginning of the study. On an average the cholesterol levels among these 100 patients decreased by 15.0 and the standard deviation of the changes in cholesterol was 40. What can be said about the two-sided p-value for testing the null hypothesis of no change in cholesterol levels? Provide justifications for your answer.

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# Chapter 6
# Tying Genomes with Disease




Genes, genome, etc. are common words that feature in any genetic analysis. It is now believed that a disease, be it monogenic or complex, has manifested due to the genetic architecture of the organism. Humans are no different. There are around 15000–20000 genes in the human body. These genes are scattered over the human genome which consists of 22 homologous pairs of chromosomes (autosomes) and one pair of sex chromosomes. Quite naturally, we assume that one set of chromosomes that is transmitted from the father and the other set that is transmitted from the mother are completely independent. The holy grail of any genetic study is to first identify the gene(s) responsible to cause a disease or has some role in at least increasing the disease risk. It was with this question in mind, that a genetic association study was launched with the primary focus of identifying a gene or a set of genes that are associated with the disease or phenotype in general. Note that we have no idea where the disease gene is located on the human genome and this is what we try to find out. Eventually, this identification of genes that are responsible for disease might lead to specific drugs targeting to cure the disease. In this quest, we need data. Our hope is that appropriate data can answer, or at least throw some light, in deciphering the genetic architecture. Data consist not only of genetic data, but also phenotype data along with information on covariates, environment, etc., that are deemed relevant to our study of interest.

## 6.1 Characteristics of Genomic Data

The basic and very naive idea about the genetic structure is that genes are located in different regions of the chromosomes. However, one gene might present itself in different forms. These forms are known as ‘alleles’, which may be looked at as competing genes. Depending on the allele type and/or the number of a particular allele, disease status or risk might change. Change in alleles is the main source that adds variation in genetic data, making it amenable to statistical analysis. Thus we may be interested to see whether any particular allele is more (or less) observed in affected

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I. Mukhopadhyay and P. P. Majumder, *Statistical Methods in Human Genetics*,
Indian Statistical Institute Series, https://doi.org/10.1007/978-981-99-3220-7_6

individuals compared with unaffected individuals. There are other types of data in this context, like gene expression data, methylation data, etc. We already have had some exposure to gene expression data and its analysis methods. On the other hand, data collection mechanism and study design are also important to fully understand the data. In this paradigm, sometimes we collect families and collect genetic data from all available individuals in a family. Sometimes we collect independent or unrelated individuals, among them a group affected by the disease while others are normal or free from the disease. Thus several characteristics of data prevail and we need to understand them. In order to proceed further with the study of genetic data, it is essential to represent them mathematically. This mathematical representation emerges very naturally.

## 6.2 Representing Mathematically

The human genome map tells us the precise position of a gene with other information like number of its alleles. In general this known information at a position or locus on genome helps us in identifying a gene that may be associated with the disease. This polymorphic entity with known locus is known as marker. A marker is the building block of genetic analysis. Usually it remains unchanged during transmission from parents to offsprings, i.e. from one generation to another.

Nowadays we usually consider Single Nucleotide Polymorphisms (SNP). SNP is defined as a marker having exactly two alleles at a particular nucleotide position on a chromosome. We know that DNA consists of four nucleotides, Adenine (A), Guanine (G), Cytosine (C), and Thymine (T), organised like beads on a string; two strings are twisted to form a double helix structure. Suppose a SNP has two alleles $A$ or $C$. It indicates that in the population we can observe only $A$ and $C$ as alleles for this SNP marker and no other allele is possible. This information is known to us beforehand, from the human genome map or from other sources. Although at a nucleotide position we might expect to see any one of four nucleotides, i.e. $A$, $T$, $G$, or $C$, we can only observe two bases (or nucleotides) for a SNP. Thus a SNP always has two alleles. In all subsequent discussions, we use SNP and a gene interchangeably, i.e. our treatment for these two entities would be the same, unless otherwise mentioned. In fact, a gene is a more general term, having deep biological interpretation. And we can always think of different forms in which a gene might exist in the population. So 'a SNP has two alleles' and 'a gene has two alleles' have the same meaning and will be treated similarly in all subsequent discussions, be it mathematical or intuitive.

Suppose that a gene (or a marker) has two alleles, '$A$' and '$a$' (say). This 'A' is not necessarily adenine of DNA; it is just a generic notation for an allele. Since we observe two alleles on two chromosomes of the same pair, there are in all three possible combinations of two alleles. This combination is known as 'genotype'. Thus with two alleles '$A$' and '$a$', three genotypes are possible, viz, $AA$, $Aa$, and $aa$. The probability of observing an allele in the population is called *allele frequency* and similarly we can define *genotype frequency* as the probability of a particular

genotype. The allele with smaller allele frequency is termed as *minor allele* and its corresponding frequency is called *minor allele frequency (MAF)*.

Given that a marker or a gene has two alleles, we can define a random variable $X$ as the number of minor alleles of the genotype of an individual at a particular locus. Clearly $X$ is a random variable as the individual is selected randomly from the population without any prior knowledge of her/his genetic profile.

Let the minor allele be '$a$' and the other allele be '$A$', at a bi-allelic locus. So we can write the genotype frequency as

$$P(AA) = P(X = 0), \quad P(Aa) = P(X = 1) \text{ and } P(aa) = P(X = 2).$$

Thus if a marker has two alleles $A$ and $a$ with allele frequencies respectively as 0.8 and 0.2, the minor allele frequency is 0.2 ($P(a) = 0.2$).

But how does the data set look like? A small data set (Table 6.1) may provide an idea about the data. It contains case-control status (Col. 1), genotype data at four loci, each being a SNP (Col. 2–5), a covariate 'age' (Col. 6), and phenotype (Col. 7) which is cholesterol level for each individual. There are 50 cases and 50 controls in this data set. All individuals are selected randomly or independently of each other.

## 6.3 Generating Questions

Now that we have a data set with genotypes at four loci and age for two groups, cases and control, we need to ask questions on what we can or should do while analysing the data.

(1) How many people are there in the data set? How are they divided into two groups?
(2) What is the genotype distribution in controls? Do they mimic the population features?
(3) Are genotype distributions same in these two groups and for all four loci?
(4) Since there are four loci, are they related in any sense?
(5) Is there any association between disease status and genotype at any locus?
(6) Is there any effect of age on the disease status?
(7) Can we say that cholesterol level is somehow influenced by genotypes at one locus or all loci, or is it just influenced by age only?

There might be other questions; let's focus on these for the time being. The reader may proceed with other questions and find the solution once we resolve these issues.

**Table 6.1** Data for genetic association

| CC status | *G*1 | *G*2 | *G*3 | *G*4 | Age | Cholesterol | CC status | *G*1 | *G*2 | *G*3 | *G*4 | Age | Cholesterol |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | *AA* | *Aa* | *AA* | *AA* | 41 | 158.68 | 0 | *Aa* | *AA* | *aa* | *Aa* | 34 | 161.82 |
| 0 | *AA* | *AA* | *AA* | *Aa* | 44 | 160.11 | 0 | *AA* | *AA* | *AA* | *Aa* | 38 | 163.98 |
| 0 | *AA* | *AA* | *Aa* | *AA* | 38 | 159.93 | 0 | *AA* | *AA* | *aa* | *aa* | 33 | 160.49 |
| 0 | *AA* | *AA* | *aa* | *Aa* | 33 | 160.24 | 0 | *Aa* | *AA* | *aa* | *AA* | 38 | 159.56 |
| 0 | *AA* | *AA* | *Aa* | *AA* | 32 | 162.48 | 0 | *AA* | *AA* | *aa* | *Aa* | 47 | 161.11 |
| 0 | *AA* | *AA* | *aa* | *AA* | 41 | 163.04 | 0 | *Aa* | *AA* | *AA* | *Aa* | 39 | 164.20 |
| 0 | *AA* | *AA* | *Aa* | *Aa* | 39 | 158.65 | 0 | *Aa* | *AA* | *Aa* | *AA* | 36 | 159.91 |
| 0 | *Aa* | *AA* | *Aa* | *Aa* | 47 | 158.55 | 0 | *AA* | *AA* | *Aa* | *AA* | 35 | 159.99 |
| 0 | *AA* | *AA* | *aa* | *Aa* | 45 | 157.78 | 0 | *AA* | *Aa* | *Aa* | *Aa* | 44 | 159.53 |
| 0 | *AA* | *AA* | *aa* | *AA* | 36 | 158.47 | 0 | *Aa* | *AA* | *Aa* | *Aa* | 45 | 161.68 |
| 0 | *AA* | *AA* | *aa* | *aa* | 41 | 160.81 | 0 | *AA* | *aa* | *Aa* | *Aa* | 37 | 162.87 |
| 0 | *AA* | *Aa* | *AA* | *AA* | 38 | 160.56 | 0 | *aa* | *AA* | *aa* | *Aa* | 36 | 161.35 |
| 0 | *AA* | *AA* | *aa* | *AA* | 37 | 161.02 | 0 | *Aa* | *AA* | *Aa* | *Aa* | 39 | 164.08 |
| 0 | *AA* | *AA* | *aa* | *AA* | 34 | 160.15 | 0 | *AA* | *Aa* | *Aa* | *AA* | 30 | 155.44 |
| 0 | *Aa* | *Aa* | *Aa* | *Aa* | 41 | 162.33 | 0 | *AA* | *Aa* | *Aa* | *aa* | 39 | 159.62 |
| 0 | *AA* | *Aa* | *aa* | *Aa* | 41 | 159.93 | 0 | *AA* | *AA* | *AA* | *Aa* | 47 | 165.57 |
| 0 | *Aa* | *AA* | *Aa* | *AA* | 32 | 160.24 | 0 | *AA* | *AA* | *Aa* | *Aa* | 39 | 161.68 |
| 0 | *AA* | *AA* | *Aa* | *Aa* | 43 | 162.48 | 0 | *AA* | *Aa* | *aa* | *AA* | 37 | 163.42 |
| 0 | *AA* | *aa* | *AA* | *AA* | 38 | 163.04 | 0 | *AA* | *Aa* | *aa* | *AA* | 44 | 163.44 |
| 0 | *AA* | *AA* | *Aa* | *aa* | 41 | 159.53 | 0 | *AA* | *Aa* | *aa* | *Aa* | 44 | 159.24 |
| 0 | *Aa* | *Aa* | *AA* | *aa* | 36 | 159.24 | 0 | *Aa* | *AA* | *AA* | *Aa* | 46 | 158.94 |
| 0 | *AA* | *AA* | *Aa* | *AA* | 38 | 158.78 | 0 | *Aa* | *AA* | *AA* | *Aa* | 42 | 161.57 |
| 0 | *AA* | *AA* | *aa* | *AA* | 39 | 162.67 | 0 | *AA* | *AA* | *Aa* | *AA* | 41 | 162.45 |
| 0 | *AA* | *AA* | *Aa* | *AA* | 40 | 159.32 | 0 | *AA* | *AA* | *aa* | *AA* | 43 | 160.83 |
| 0 | *Aa* | *AA* | *AA* | *AA* | 44 | 161.81 | 0 | *AA* | *Aa* | *aa* | *Aa* | 43 | 157.58 |
| 1 | *Aa* | *AA* | *Aa* | *aa* | 58 | 178.94 | 1 | *Aa* | *Aa* | *Aa* | *aa* | 49 | 179.83 |
| 1 | *AA* | *AA* | *Aa* | *aa* | 53 | 182.33 | 1 | *AA* | *AA* | *aa* | *aa* | 48 | 184.18 |
| 1 | *aa* | *AA* | *AA* | *aa* | 48 | 178.40 | 1 | *Aa* | *AA* | *Aa* | *aa* | 61 | 180.31 |
| 1 | *AA* | *AA* | *aa* | *aa* | 55 | 184.32 | 1 | *AA* | *aa* | *Aa* | *aa* | 52 | 175.58 |
| 1 | *AA* | *AA* | *Aa* | *aa* | 43 | 175.68 | 1 | *aa* | *AA* | *aa* | *aa* | 64 | 181.13 |
| 1 | *aa* | *AA* | *Aa* | *aa* | 55 | 183.95 | 1 | *Aa* | *AA* | *AA* | *aa* | 50 | 180.66 |
| 1 | *aa* | *AA* | *AA* | *aa* | 61 | 186.60 | 1 | *AA* | *AA* | *aa* | *aa* | 50 | 180.10 |
| 1 | *AA* | *Aa* | *aa* | *aa* | 42 | 183.24 | 1 | *AA* | *Aa* | *AA* | *aa* | 58 | 184.03 |
| 1 | *AA* | *AA* | *aa* | *Aa* | 47 | 183.56 | 1 | *AA* | *Aa* | *AA* | *Aa* | 43 | 180.73 |
| 1 | *AA* | *Aa* | *Aa* | *Aa* | 58 | 179.33 | 1 | *aa* | *Aa* | *aa* | *Aa* | 57 | 175.91 |
| 1 | *Aa* | *AA* | *Aa* | *Aa* | 45 | 177.09 | 1 | *AA* | *AA* | *Aa* | *Aa* | 58 | 182.69 |
| 1 | *AA* | *AA* | *Aa* | *Aa* | 56 | 183.69 | 1 | *aa* | *aa* | *Aa* | *Aa* | 53 | 177.30 |
| 1 | *AA* | *AA* | *aa* | *Aa* | 51 | 175.67 | 1 | *AA* | *Aa* | *Aa* | *Aa* | 41 | 184.00 |
| 1 | *AA* | *Aa* | *Aa* | *Aa* | 60 | 175.95 | 1 | *AA* | *AA* | *aa* | *Aa* | 45 | 178.69 |
| 1 | *aa* | *AA* | *AA* | *Aa* | 48 | 182.57 | 1 | *Aa* | *AA* | *AA* | *Aa* | 60 | 182.73 |
| 1 | *Aa* | *Aa* | *AA* | *Aa* | 46 | 177.30 | 1 | *Aa* | *AA* | *aa* | *Aa* | 72 | 175.89 |
| 1 | *Aa* | *Aa* | *Aa* | *Aa* | 43 | 177.51 | 1 | *Aa* | *Aa* | *Aa* | *Aa* | 48 | 180.56 |
| 1 | *AA* | *AA* | *AA* | *Aa* | 59 | 179.89 | 1 | *Aa* | *AA* | *aa* | *Aa* | 49 | 181.70 |
| 1 | *AA* | *Aa* | *aa* | *AA* | 59 | 178.61 | 1 | *aa* | *AA* | *aa* | *AA* | 53 | 183.24 |

(continued)

**Table 6.1** (continued)

| CC status | *G*1 | *G*2 | *G*3 | *G*4 | Age | Cholesterol | CC status | *G*1 | *G*2 | *G*3 | *G*4 | Age | Cholesterol |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | *AA* | *Aa* | *aa* | *AA* | 59 | 173.33 | 1 | *aa* | *AA* | *AA* | *AA* | 58 | 179.03 |
| 1 | *Aa* | *AA* | *AA* | *AA* | 53 | 180.20 | 1 | *Aa* | *AA* | *aa* | *AA* | 71 | 184.00 |
| 1 | *aa* | *AA* | *aa* | *AA* | 67 | 180.11 | 1 | *aa* | *Aa* | *Aa* | *AA* | 62 | 178.83 |
| 1 | *AA* | *Aa* | *aa* | *AA* | 55 | 181.07 | 1 | *aa* | *AA* | *AA* | *AA* | 52 | 177.23 |
| 1 | *AA* | *Aa* | *Aa* | *AA* | 52 | 179.48 | 1 | *Aa* | *Aa* | *Aa* | *AA* | 52 | 174.78 |
| 1 | *Aa* | *Aa* | *aa* | *AA* | 39 | 179.70 | 1 | *AA* | *AA* | *Aa* | *AA* | 49 | 173.97 |

## 6.4 Relation Between Allele Frequency and Genotype Frequency

It is clear that genotype is simply a pair of alleles occurring in a pair of chromosomes for an individual. So we can expect that genotype and allele frequencies are somehow related. In this connection there is a basic theorem in population genetics which forms the basis of genetic analysis. This is known as **Hardy-Weinberg equilibrium (HWE)**. Before discussing HWE, we have to first understand the transmission probability, i.e. how to calculate the probability of a genotype of an offspring when the genotypes of her/his parents are known in a family or pedigree. To explain transmission probabilities, for the sake of simplicity, we first consider only a nuclear family. A nuclear pedigree or family consists of parents and their offspring only.

Here we assume random mating among the parents. So transmission of an allele from father to its offspring is completely independent of transmission of an allele from mother to the same offspring. In probability theory jargon, we say that the two events, transmission of allele from father and that from mother to offspring are independent of each other. Nothing surprising! This is quite simple. This simple fact is the key to evaluating the probability of observing offspring genotypes. In Fig. 6.1A,

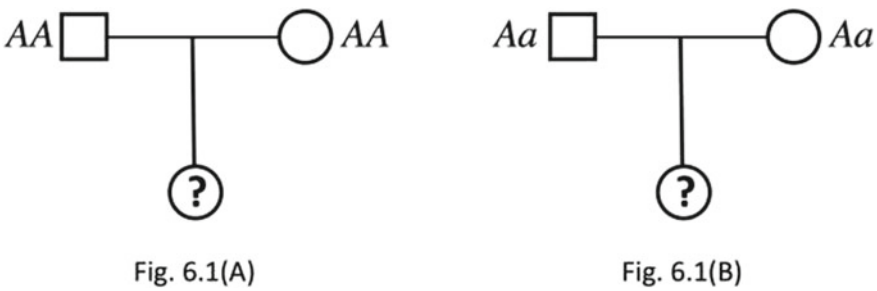



**Fig. 6.1** Two nuclear families with parental genotypes

genotypes of father is $AA$ and of mother is $AA$. Let us discuss and calculate some probability of offspring genotype for some combinations of parental genotypes.

$$\begin{aligned} P(\text{Offspring has genotype } AA) &= P(\text{Father transmits } A, \text{Mother transmits } A) \\ &= P(\text{Father transmits } A) \times P(\text{Mother transmits } A) \\ &= 1 \times 1 = 1 \end{aligned}$$

Hence we have for Fig. 6.1A,

$$P(\text{Offspring's genotype is } Aa) = P(\text{Offspring's genotype is } aa) = 0.$$

If we look at Fig. 6.1B, however, for offspring, any genotype among $\{AA, Aa, aa\}$ is possible.

$$\begin{aligned} P(\text{Offspring has genotype } AA) &= P(\{\text{Father transmits } A, \text{Mother transmits } A) \\ &= P(\{\text{Father transmits } A) \times P(\text{Mother transmits } A) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4} \end{aligned}$$

If we want calculate that offspring has genotype $Aa$, it receives $A$ allele from father (or mother) and then definitely receives $a$ allele from mother (or father). Clearly parents transmit different alleles to the offspring. Hence,

$$\begin{aligned} &P(\text{Offspring has genotype } Aa) \\ &= P(\{\text{Father transmits } A, \text{Mother transmits } a\} \textbf{or} \{\text{Father transmits } a, \text{Mother transmits } A\}) \\ &= P(\text{Father transmits } A, \text{Mother transmits } a) + P(\text{Father transmits } a, \text{Mother transmits } A) \\ &= \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2} = \frac{1}{2} \end{aligned}$$

Proceeding similarly, it can be shown that $P(\text{Offspring has genotype } aa) = \frac{1}{4}$.

### *6.4.1 Hardy-Weinberg Equilibrium for an Autosomal Locus*

Genes are transmitted from parents to offspring generation after generation. It is expected that there should be a pattern or rule that governs this natural process. However, we know that many population genetics phenomena may trigger certain and subtle changes in the genetic profiles of individuals in a population. Among them mutation, migration, selection, etc. are a few genetic forces that can change the allele and/or genotype frequency in any generation. However, it may so happen that these changes are observed only in a particular generation, i.e. it was absent in previous generations. To keep it simple, first we need to explore whether there is really any rule that can be associated with genotype frequencies at a locus when forces other

than random mating are absent. Fortunately the answer is a positive one; this leads to one of the most celebrated and simplest laws in population genetics. This was discovered by G. H. Hardy and W. Weinberg independently in 1908 and is named after them.

**Statement of Hardy-Weinberg equilibrium (HWE):** In the absence of migration, mutation and selection, and if random mating is practiced, genotype frequencies at an autosomal locus remain unchanged from one generation to another.

**Implication of HWE:** This simple law can be proved mathematically and has far reaching influence on a genetic study. It says that if random mating occurs in the present generation, genotype frequencies become stable in the next generation and will remain constant in the subsequent generations provided in every generation there will be random mating only. Thus if at any generation random mating assumption is violated or any phenomenon like mutation or migration, etc. occurs, we will have to wait for only one generation of random mating and also have to wait for mutation, migration, etc. be become absent, for genotype frequencies to become stable again. Thus at the very beginning of a genetic study involving genotypes and alleles, we have to check whether HWE exists in that population for the markers under study. We can proceed for further study with a marker only if HWE is satisfied for that locus.

Suppose in a generation, for a bi-allelic locus, the genotype frequencies are $P(AA) = Q$, $P(Aa) = H$, and $P(aa) = R$ so that $Q + R + H = 1$. Then, it can be shown that after one generation of random mating, genotype frequencies change but become stable. The new genotype frequencies are:

$$P(AA) = \left(Q + \frac{H}{2}\right)^2, \quad P(Aa) = 2\left(Q + \frac{H}{2}\right)\left(R + \frac{H}{2}\right), \text{ and } P(aa) = \left(R + \frac{H}{2}\right)^2.$$

If random mating continues in subsequent generations and there are no other external factors affecting allele frequency or genotype frequency, these frequencies will remain unchanged.

The relation between genotype frequency and allele frequency can also be established as a consequence of HWE. If we take $p = \left(Q + \frac{H}{2}\right)$ and hence $P(AA) = p^2$. Genotype frequency of a particular genotype can be thought of as a function of allele frequencies. Hence, we can interpret it in another way. We have already noted that transmission of an allele from father to offspring is completely independent of transmission from mother to the same offspring, if we assume random mating among parents. That means that an offspring genotype $AA$ may be viewed as having formed by combining two $A$ alleles chosen at random from the population of alleles. Hence,

$$P(AA) = P(Father \to A, Mother \to A) = P(Father \to A)P(Mother \to A)$$
$$= p \times p = p^2$$

Note that $1 - p = 1 - \left(Q + \frac{H}{2}\right) = R + \frac{H}{2} = q$ (say). To get the probability of $Aa$ genotype, the same argument continues. However, we have to be careful of the fact that if the father transmits $A$ allele, the mother must transmit $a$ allele whereas if the father transmits $a$ allele, the mother must transmit $A$ allele to the offspring; only in that case the genotype becomes $Aa$. Since both alleles are drawn randomly from the population with allele frequencies $P(A) = p$ and $P(a) = q$, we have

$$P(Aa) = P(Father \to A, Mother \to a) + P(Father \to a, Mother \to A)$$
$$= pq + qp = 2pq$$

Then, proceeding as above, we have the relation between genotype frequency and allele frequency under HWE, as

$$P(AA) = p^2, \quad P(Aa) = 2pq, \quad P(aa) = q^2 \text{ where } q = 1 - p, \ 0 < p < 1.$$

In the above discussion of obtaining genotype frequency from allele frequencies, we have implicitly assumed that allele frequency for males and females in the population are same.

A rigorous proof of HWE is given in the Appendix D. In the entire discussion, we have considered only bi-allelic locus on autosomal chromosomes. The same rule applies to a locus with multiple alleles.

Suppose a locus has $k$ alleles $A_1, A_2, \ldots, A_k$ where $k > 2$. Then, if the population is in HWE, the genotype frequencies would be

$$P(A_iA_i) = p_i^2, \ i = 1, \ldots, k, \text{ where } P(A_i) = p_i, i = 1, \ldots, k, \text{ and}$$
$$P(A_iA_j) = 2p_ip_j, \quad \text{for } i \neq j = 1, \ldots, k$$

Moreover, our consideration of HWE in this form is only true for locus at autosomal chromosomes, which are 22 pairs in number in humans. The same rule does not apply to locus on X chromosome. So it demands a separate discussion about HWE for X-linked locus.

### *6.4.2 HWE for X-linked Locus*

Suppose a bi-allelic locus on X chromosome has two alleles, say '$A$' and '$a$'. Clearly a female person has two copies of alleles thus giving rise to three genotypes $AA$, $Aa$, and $aa$, as in the case of autosomal chromosome. But in males, only one copy

of allele is observed because a male person has only one copy of X chromosome; his other chromosome is Y. It is thus natural to consider different allele frequencies of $A$ for males and females. Let $p_f(t)$ and $p_m(t)$ be the allele frequencies for the allele $A$ in females and males, respectively, at generation $t$. We are interested to know what would happen to these allele frequencies in the next generation under the assumption of random mating only (no other genetic force is present).

Note that a female with genotype $AA$ gets one $A$ allele from her father and another $A$ allele from her mother, because she has two X chromosomes. Since we assume random mating, the probability that a female has $AA$ combination in the $t+1$-th generation is the product of the probabilities of randomly selecting one $A$ allele from the female population at generation $t$ and that for the male population in generation $t$. Following the same arguments, we have probabilities of all genotypes in generation $t+1$ as

$$
\begin{aligned}
&P(AA \text{ genotype among females in generation } t+1) = p_f(t)p_m(t),\\
&P(Aa \text{ genotype among females in generation in generation } t+1) = p_f(t)\big(1-p_m(t)\big)\\
&\qquad\qquad + \big(1-p_f(t)\big)p_m(t), \text{ and}\\
&P(aa \text{ genotype among females in generation in generation } t+1) = \big(1-p_f(t)\big)\big(1-p_m(t)\big).
\end{aligned}
$$

On the other hand, a male person can receive his only copy of X chromosome from his mother and his other chromosome is Y which comes from his father. So he has only one allele at this locus on the X chromosome. Hence, for males, we have, following the previous arguments,

$$
\begin{aligned}
&P(A \text{ among males in generation } t+1) = p_f(t), \text{ and}\\
&P(a \text{ among males in generation } t+1) = \big(1-p_f(t)\big)
\end{aligned}
$$

Genotype frequencies are not of the same pattern in females and males. This motivates us to look at allele frequency in generation $t+1$ directly.

Now, in generation $t+1$, frequency of $A$ depends on both females and males; hence we consider $p_f(t+1)$ as the average allele frequency among females and males from generation $t$. On the other hand, for males in generation $t+1$, the frequency of $A$ allele depends only on that among females in generation $t$. Thus we have

$$
\begin{aligned}
p_f(t+1) &= \frac{1}{2}\big(p_f(t)+p_m(t)\big), \text{ and}\\
p_m(t+1) &= p_f(t) \qquad (6.1)
\end{aligned}
$$

Since allele frequency for $A$ seems different in females and males in generation $t+1$, even after random mating, in order to hold a law, we expect that some kind of average allele frequency might have been stable. We consider $\frac{1}{3}p_m(t+1)+$

$\frac{2}{3}p_f(t+1)$, which a weighted average of $p_f(t+1)$ and $p_m(t+1)$ and investigate this quantity.

$$\frac{1}{3}p_m(t+1)+\frac{2}{3}p_f(t+1)=\frac{1}{3}p_f(t)+\frac{2}{3}\Big\{\frac{1}{2}p_f(t)+\frac{1}{2}p_m(t)\Big\} \quad \text{[from (6.1)]}$$
$$=\frac{1}{3}p_m(t)+\frac{2}{3}p_f(t)=\cdots\cdots=\frac{1}{3}p_m(0)+\frac{2}{3}p_f(0)$$

Thus it is clear that although allele frequencies are different among females and males in every generation, however, their average (weighted average) remains stable over generations after random mating occurred in $t$-th generation.

We might be interested to know the nature of difference of allele frequencies among females and males and how it varies over generations. A little bit of calculation would reveal a startling feature! From (6.1), we have,

$$p_m(t+1)-p_f(t+1)=p_f(t)-\frac{1}{2}\big(p_f(t)+p_m(t)\big)=\frac{1}{2}\big(p_f(t)-p_m(t)\big)$$
$$=\Big(-\frac{1}{2}\Big)\big(p_m(t)-p_f(t)\big)=\Big(-\frac{1}{2}\Big)^2\big(p_m(t-1)-p_f(t-1)\big)$$
$$\cdots\cdots=\Big(-\frac{1}{2}\Big)^n\big(p_m(t-n+1)-p_f(t-n+1)\big)\to 0 \text{ as } n\to\infty$$

Thus in the long run, allele frequency of $A$ at a locus on X chromosome will be the same for females and males. Note that although we have worked with a particular weighted average of female and male allele frequencies, these weights can be interpreted easily. Each male carries one X chromosome whereas each female carries two X chromosomes. Hence, we have chosen the weights according to the contribution of $A$ alleles by males and females. These weights can also be obtained by mathematically assuming a general weighted average with unknown weights $w_1$ and $w_2$ (say), subject to $w_1+w_2=1, 0<w_1, w_2<1$; we omit this detailed mathematical derivation for simplicity and empasise on the intuitive justification.

We discuss a few problems on HWE below.

**Ex-6.1.** Four alleles in a randomly mating population are 0.10, 0.25, 0.35 and 0.30. In a population of size 20000, what is the expected genotype distribution in that population?

Let the four alleles be $A_1$, $A_2$, $A_3$ and $A_4$ with frequencies 0.10, 0.25, 0.35 and 0.30 respectively. To get the expected genotype frequency for a particular genotype in the population, we find the respective genotype frequency assuming HWE and then multiply it by 20000. Let $n_{ij}$ be the expeceted genotype frequency of genotype $A_iA_j$ for $i, j=1,\ldots,4$. Hence, we have,

$n_{11} = 20000 \times (0.1)^2 = 200,\ n_{22} = 20000 \times (0.25)^2 = 1250,\ n_{33} = 20000 \times (0.35)^2 = 2450,$
$n_{44} = 20000 \times (0.3)^2 = 1800,\ n_{12} = 20000 \times 2(0.1)(0.25) = 1000,\ n_{13} = 20000 \times 2(0.1)(0.35) = 1400,$
$n_{14} = 20000 \times 2(0.1)(0.3) = 1200,\ n_{23} = 20000 \times 2(0.25)(0.35) = 3500,\ n_{24} = 20000 \times 2(0.25)(0.3) = 3000,$
$n_{34} = 20000 \times 2(0.35)(0.3) = 4200.$

**Ex-6.2.** At a bi-allelic autosomal locus, the allele frequency for two alleles $A$ and $a$ are respectively 0.22 and 0.78. Assuming Hardey-Weinberg equilibrium,

(a) what is the percentage of heterozygous people in the population?
(b) what is the percentage of homozygous recessive people in the population?

It is given that $q = P(A) = 0.22$ and $p = P(a) = 0.78$. Assuming HWE, $P(AA) = (0.22)^2 = 0.0482$, $P(Aa) = 2 \times 0.22 \times 0.78 = 0.3432$, and $P(aa) = (0.78)^2 = 0.6084$.

(a) Percentage of heterozygous people in the population $= 100 \times P(Aa) = 34.32\%$.

(b) Since $A$ is the minor allele in this case, we can assume $A$ is the disease allele. Hence, percentage of homozygous recessive people in the population $= 100 \times P(AA) = 4.82\%$.

**Ex-6.3.** Suppose a population in Hardy-Weinberg equilibrium contains 11% recessive homozygote individuals for a certain trait. In a population of 30000,

(a) what is the percentage of homozygous dominant individuals in the population?
(b) what is the percentage of heterozygous individuals in the population?

Let $q$ be the recessive allele frequency. Since HWE is maintained in the population, we have $q^2 = 0.11$ and hence $q = 0.3317$ and hence the frequency of the dominant allele is $p = 1 - q = 0.6683$.

(a) Percentage of homozygous dominant individuals in the population $= 100 \times p^2 = 100 \times (0.6683)^2 = 44.66\%$.

(b) Percentage of heterozygous individuals in the population $= 100 \times 2pq = 44.33\%$.

### *6.4.3 Estimation of Allele Frequency*

It is essential to see whether HWE exists for a marker in a given population under study. Since we deal with genotype data, a statistical test is welcome. HWE explains relationship between genotype frequency and allele frequency. We have to exploit this information in order to check HWE. Technology provides us only genotype data

and not allele level data. Genotype proportions in the sample are available and hence we need to devise a method to estimate allele frequency from genotype data.

Suppose we select $n$ individuals at random from a population and genotype them at a number of markers. Let us consider one marker having two alleles $A$ and $a$ and also let $n_{AA}$, $n_{Aa}$ and $n_{aa}$ be the number of individuals having genotypes $AA$, $Aa$, and $aa$ respectively. Clearly $n_{AA} + n_{Aa} + n_{aa} = n$. However, since we have no idea about the allele frequency beforehand, which is usually the case, we have to estimate allele frequency from the given sample on genotypes.

### Gene Counting Method

Allele frequency of an allele in the population indicates the probability of observing a particular allele when one allele is selected at random. It is simply the population proportion. Thus allele frequency can be estimated easily by just counting the number of alleles present in the sample. Note that each individual having a homozygous genotype, say $AA$, contributes two $A$ alleles whereas a person with $Aa$ genotype contributes only one $A$ allele. Thus the total number of $A$ alleles in the sample is $2n_{AA} + n_{Aa}$. For a bi-allelic locus, a sample of $n$ individuals contributes a total of $2n$ alleles. Since we know a sample proportion is usually a good estimate of the corresponding population proportion, we have the estimates of allele frequencies as

$$\hat{p} = \hat{P}(A) = \frac{2n_{AA} + n_{Aa}}{2n} \text{ and } \hat{q} = \hat{P}(a) = 1 - \hat{p} \tag{6.2}$$

Thus genotype frequencies under HWE may be obtained as

$$\hat{P}(AA) = \hat{P}^2, \quad \hat{P}(Aa) = 2\hat{p}\hat{q} \text{ and } \hat{P}(aa) = \hat{q}^2 \tag{6.3}$$

There is no restriction on sample size. Sometimes for a rare allele, recessive genotype with two copies of rare allele may not be observed if the sample size is small. However, as long as at least two types of genotypes are observed in the sample, we can estimate the allele frequencies. Of course, a larger sample would increase the efficacy of the estimate.

### Maximum Likelihood Method

A more statistically sound and intuitively appealing way to estimate allele frequency from genotype frequency is to use likelihood function for the data and then maximise it to get the maximum likelihood estimator (MLE) for allele frequency. Given the data as described in the previous section and noting the genotype probabilities under HWE, the likelihood function is given by

$$L(p|n_{AA}, n_{Aa}, n_{aa}) = \frac{n!}{n_{AA}!n_{Aa}!n_{aa}!}(p^2)^{n_{AA}}(2pq)^{n_{Aa}}(q^2)^{n_{aa}} = \frac{2^{n_{Aa}}n!}{n_{AA}!n_{Aa}!n_{aa}!}p^{2n_{AA}+n_{Aa}}q^{n_{Aa}+2n_{aa}} \tag{6.4}$$

$$\text{or } \log(L(p|n_{AA}, n_{Aa}, n_{aa})) = \log\left(\frac{n!}{n_{AA}!n_{Aa}!n_{aa}!}\right) + (2n_{AA} + n_{Aa})\log(p) + (2n_{aa} + n_{Aa})\log(q) + n_{Aa}\log 2 \tag{6.5}$$

Maximising likelihood (or log-likelihood) function with respect to $p$, we get the estimate of allele frequency exactly same as that obtained by gene counting method!

### 6.4.4 Mean and Variance of Allele Frequency Estimator

In statistics, we are always curious to know whether an estimator is good (in some sense). Naturally an estimator performs well if its expectation is same as the population parameter which we want to estimate. This indicates, ideally, that if we draw many samples and calculate this estimate based on each sample, we can expect these values to cluster around the parameter value, although we don't know the value of the parameter. This property, known as 'unbiasedness' is a very popular measure to judge whether an estimator is good. On the other hand, we also hope that the variance of the estimator should be small, or become smaller and smaller as the sample size gets larger. Naturally this leads to evaluating the expectation and variance of any estimator that might take an important role in further downstream analysis. If we somehow find that an estimator is not so good, we should explore for a better estimator.

In the above context, we now try to evaluate mean, i.e. expectation and variance of $\hat{p}$, the estimator obtained by gene counting or MLE method. Since this requires some rigorous calculation using a few results, we provide the detail proof in the Appendix D. It can be shown that

$$E(\hat{p}) = p \text{ for all } p \text{ and } V(\hat{p}) = \frac{pq}{n} \text{ for all } p.$$

**Ex-6.4.** Genotype frequencies at a bi-allelic locus for 100 individuals are as follows:

$$DD: \ 50, \quad Dd: \ 40, \quad dd: \ 10$$

Estimate the frequency of allele $D$ and calculate its standard error.

Let $p$ be the allele frequency for allele $D$. Then using the maximum likelihood method, as in Sect. 6.4.3, we have

$$\hat{p} = \frac{2 \times 50 + 40}{2 \times 100} = \frac{140}{200} = 0.7$$

$$\hat{V}(\hat{p}) = \frac{\hat{p}\hat{q}}{n} = \frac{0.7 \times 0.3}{200} = 0.00105 \text{ implies standard error}(\hat{p}) = \sqrt{0.00105} = 0.032$$

### 6.4.5 Test for HWE

Now we are in a position to check whether HWE exists in a population of interest. This can be framed as a statistical testing problem. The null and alternative hypotheses of interest are

$$H_0: \text{ HWE is satisfied against } H_1: \text{ HWE is not satisfied.}$$

Since we know genotype frequencies in terms of allele frequencies when HWE is satisfied, we can reformulate the hypotheses as

$$H_0: P(AA) = p^2, P(Aa) = 2pq, P(aa) = q^2 \text{ against } H_1: \text{at least one inequality in } H_0.$$

As we have seen earlier, in any testing problem the main component is the data at hand; it contains information about the phenomenon on which our testing problem depends. In this situation, our data consists of genotypes of $n$ randomly chosen individuals at a particular locus. Hence using this data we can easily count the number of individuals having a particular genotype (Table 6.2).

If HWE holds for the marker under study, we expect that the expected frequencies should be very close to the observed frequencies. The expected genotype proportions under HWE, which are $p^2$, $2pq$, and $q^2$ for $AA$, $Aa$, and $aa$ genotypes respectively. Hence, we can easily get the expected genotype frequencies as $np^2$, $2npq$ and $nq^2$ respectively. However, there is only one problem: we don't know the value of $p$, i.e. allele frequency of allele $A$. Again we can do a simple trick; just replace the allele frequency by its corresponding estimate based on the sample that we have. The whole scenario can be presented nicely as in Table 6.2.

Now we are interested to see how close the observed and expected frequencies are for each genotype. To check that we should consider the squared difference between the two. However, since this is in regard to the expected (or observed) frequency, we should divide the squared difference by the expected frequency and define a measure of closeness as

$$\frac{(n_{AA} - n\hat{p}^2)^2}{n\hat{p}^2} + \frac{(n_{Aa} - n2\hat{p}\hat{q})^2}{n2\hat{p}\hat{q}} + \frac{(n_{aa} - n\hat{q}^2)^2}{n\hat{q}^2} = T \text{ (say)} \tag{6.6}$$

**Table 6.2** Observed and expected frequency table for testing HWE

| | $AA$ | $Aa$ | $aa$ |
|---|---|---|---|
| *Observed* | $n_{AA}$ | $n_{Aa}$ | $n_{aa}$ |
| *Expected* | $n\hat{p}^2$ | $2n\hat{p}\hat{q}$ | $n\hat{q}^2$ |

It is clear that if HWE is satisfied, i.e. if $H_0$ is true, the value of $T$ should be small; otherwise it would be large. Thus we reject $H_0$ if the observed value of $T$ is large. Using statistical theory it can be shown easily that $T$ follows a $\chi^2$ distribution with 1 (one) degree of freedom approximately (asymptotically) when the sample size is large. So we calculate p-value as $P(\chi_1^2 > t)$ where $t$ is the observed value of $T$ and reject $H_0$ if the p-value is very small compared to the chosen level of significance. On the other hand if the p-value is large, we do not reject $H_0$ concluding that HWE is satisfied.

For the genotype data for $G1$ in Table 6.1, estimate of MAF is $\hat{p} = 0.27$. Now to test whether HWE is satisfied, we consider only control individuals. Thus within controls, the value of the $\chi^2$-statistic that is used for testing purpose is 0.0192, using (6.6). Hence the p-value for this test is 0.8897. Since this value is quite large compared to the level of significance 0.05, we can conclude that based on the given data, it seems that HWE is satisfied for the locus $G1$. We have to check at all loci whether HWE exist. Repeating the same procedure, we see that the p-values for $G2$, $G3$, and $G4$ are 0.447, 0.0001, and 0.7756 respectively. So we can say that HWE fails to exist for the locus $G3$ whereas for other loci HWE is satisfied. So we should look for possible issues on why at locus $G3$ HWE is violated. However, for further study we can proceed without the genotype data on $G3$ or resolve this issue by relooking the entire process.

Note that testing for HWE is important and we should work with those markers for which HWE is satisfied. It is not advisable to consider markers where HWE is not satisfied based on the above testing method. In such situations, we should look for other reasons like problems during data generation, technical issues, etc. and start further downstream analysis only after resolving these issues.

There are a few important points to note. In testing for HWE, we should work only with genotypes of control individuals although case data are available. If a locus is associated with the disease or phenotype of interest, frequency of a particular allele and hence a particular genotype may be more frequent in cases than that in controls. This may show that occurrence of this allele in cases is different than that in the population and hence case individuals may not constitute a random sample from the population. The basic requirement of HWE testing is that the sample should be random which may be violated if case individuals are included. Hence sacrificing a few samples is recommended rather than inducing non-randomness in the data. This gives a nice but important information that increasing sample size is not always welcome; it depends on the nature of sample and the related problem. However, we may assume that case individuals in our sample constitute a random sample from the population of affected people. This subtlety of randomness property needs to be understood correctly; otherwise some corrections are necessary during the analysis if this assumption is violated.

Another important issue is the level of significance. Note that we only know the distribution of the test statistic $T$ asymptotically, i.e. only when the sample size is large, $T$ follows a $\chi^2$ distribution. So the entire test is carried out at level of significance $\alpha$ *approximately*.

### 6.4.6 Study Design

Study designs are normally conceptualised to estimate effect sizes of various exposure factors on disease outcomes. In the context of genetic factors, the exposure is to the genes of the parents for a simple Mendelian disease, such as sickle-cell disease. However, for a complex disease, e.g. diabetes or cardiovascular disease, in addition to genetic factors, exposures to various environmental factors modulate disease outcome. Thus, some cardinal features of study design are to unravel effects of factors on disease outcome that are common between a simple Mendelian disease and a complex disease. However, for a complex disease study designs may need to be altered to gain statistical efficiency for estimating effect sizes.

The study designs that are commonly used to identify factors that are associated with diseases are observational. This contrasts with experimental study designs that underlie estimation of the impact of a drug for treating a disease. For a simple Mendelian disorder, we wish to identify a genomic alteration that is associated with a disease outcome. A study design that was popular for this purpose was a family study design. Observations on genotypes at several loci along with disease status were obtained on members of multi-generational families. Co-segregation of alleles at a locus with the disease was estimated using 'linkage analysis'. A co-segregating locus would be declared to be associated or 'close' to the disease locus because suppression of recombination resulting from physical contiguity of the loci would result in co-segregation. Even though this is a powerful study design, unless data are available on large, multi-generation families, the efficiency of detection of co-segregation is severely compromised. Thus, 'linkage analysis' and family study design have made way for the case-control study design. It may be noted that for many diseases, notably cystic fibrosis, disease genes have been mapped using family study designs and linkage analysis.

The case-control study design is also observational, in which association of an exposing factor (genotype or allele) is inferred by contrasting two groups of individuals—those with disease (cases) and those without (controls). In this study design, therefore, a set of individuals with disease are identified. If the disease is common, then individuals with disease can be recruited by surveying members of a population. However, if the disease is rare, then recruitment through a population survey becomes inefficient; in such a situation, since those afflicted with the disease are likely to seek advice from doctors, recruitment of cases can be made from hospitals and clinics. Generally, controls can be easily recruited from the general population. If we were to try to identify a gene for sickle-cell disease today, using the case-control study design we would first recruit a set of individuals with sickle-cell disease and then a set of normal individuals without any haemoglobin disorder. In these two sets of individuals, we would collect data on genotypes at a large number of loci and test whether there is significantly higher preponderance of a genotype or an allele among those with sickle-cell disease compared to those who are normal.

Even if a disease is a single-gene defect, there may be many factors that need to be understood and taken into account in formulating a study design. For example,

there are some mental-health disorders that are single gene, but do not manifest at birth. They are late-onset diseases and the age at onset is variable. Great care needs to be taken in the recruitment of controls. An individual may be much younger than the mean or the median age at onset, and hence will not exhibit characteristics of the disease even if the person carries the disease-predisposing allele. Such an unaffected person may be recruited as a control in a case-control study. This will adversely impact on the ability to identify the genetic factor associated with the disease. For a late-onset disease, it is better to recruit controls who are much above the mean or median onset-age of the disease. Huntington disease is a rare genetic disorder in which parts of the brain degenerate as the person carrying the gene mutation gains age. The disease causes rapid, jerky body movements and the loss of mental skills (dementia). Symptoms of Huntington's disease usually develop between ages 30 and 50 years. An individual who is 18 years old may be free of the disease even if the person is a carrier of the disease gene. In a case-control study design, such a person will be misclassified as a control, when in theory the person should be classified as a case. This misclassification will result in many controls carrying the disease gene, thereby reducing the power of detecting the disease gene using the case-control study design that relies on detecting the significance of the difference in the proportion of the disease gene among cases and controls.

For a complex disease for which there are multiple genetic predisposing factors and possibly some known environmental exposures also modulating the disease outcome, a case-control study has to be designed with great care. For example, for heart disease, it is known that smoking significantly enhances disease risk. For such a disease, matching is the key to identify genetic risk factors over and above the risk imposed by smoking. A patient needs to be carefully assessed for exposures to known environmental factors. An unaffected control needs to be chosen in such a way that the environmental exposures of this person match those of the patient. Thus, if we have recruited a patient who has suffered a heart attack and has been a smoker, we need to recruit another smoker of the same age and gender as the patient who has not suffered from heart disease. By matching, we are effectively ruling out the possibility that environmental factors (here, smoking) could have resulted in the disease outcome in the case. By effectively eliminating this possibility, we can now explore for genomic factors that are significantly associated with the disease outcome using statistically valid analytical procedures that are described elsewhere in this book.

'Matching' may be practical in designing a study if the number of environmental factors that are known to modulate disease outcome is small. If this number is beyond four or five, then it may become impossible to find controls that match with all the exposure characteristics of cases. Under such circumstances, elimination of the effects of differences in environmental exposures between cases and controls has to be done statistically by using regression analysis, either prior to or simultaneously while conducting association analysis of disease outcome with genomic factors.

The case-control study design may be severely compromised if there are many unknown environmental factors, possibly in addition to known factors, exposures to which can modulate disease outcome. An example is liver disease. It is known that infection with hepatitis viruses causes liver disease. Also, exposure to alcohol.

However, there are possibly many dietary factors or water pollutants that can also cause liver disease. If this is the reality, then the study design of choice is a longitudinal, cohort study design. In this design, a large number of individuals are selected at an early age and are longitudinally followed up. At each time-point of follow-up, assessments of exposures to relevant environmental factors (hepatitis virus infection, water pollutants, dietary chemicals, etc.) and of outcome of the disease under study are made. When adequate numbers of individuals with and without disease are noted, then the collected data are analysed to identify significant environmental factors that impact on disease outcome and also to identify significant genetic factors after elimination of the effects of identified environmental factors that are significant.

## 6.5 Genetic Association

The phrase 'genetic association' itself is self explanatory. In simple language, we need to explore the phenomenon of a gene being somehow associated or related or connected to a disease or a phenotype of interest. In other words, genetic association means whether a gene is associated with a disease or any phenotype of interest. The phenotype may be quantitative or qualitative in nature. For example, we may be interested to know whether a gene is associated with cholesterol level. Here cholesterol is the phenotype which is quantitative in nature and can take any value within a certain interval. Thus the phenotype cholesterol level is a continuous variable. On the hand, we may want to know whether any gene is associated with a disease, e.g. cardiovascular disease. Here the disease status, i.e. whether a person is affected or unaffected is the phenotype of interest. This is an example of qualitative phenotype. The phenotype disease status has two categories, viz 'affected' or 'unaffected'. We can write '0' for unaffected and '1' for affected individual. Note that '0' and '1' represent the two categories and not their exact numerical values.

A detailed study of genetic association helps us in identifying, or at least having some idea about possible gene(s) that may somehow be related with the phenotype or disease. It basically narrows down search for disease associated gene(s) so that a more focused downstream study may be undertaken to understand and decipher the genetic architecture in more detail.

Since phenotype data may be categorical as well as continuous, study of genetic association requires different statistical tools depending on the nature of data, and of course, the problem concerned. Thus we need to discuss various analytical methods.

### *6.5.1 Genetic Association for Qualitative Phenotype*

We first discuss genetic association for qualitative phenotype. To study genetic association, we first select a random sample from the population. We usually that the disease is rare and not a very common phenomenon. Hence the population mostly

**Table 6.3** Data structure for genetic association at a marker

| | $AA$ | $Aa$ | $aa$ | Total |
|---|---|---|---|---|
| *Control* | $n_{11}$ | $n_{12}$ | $n_{13}$ | $n_{10}$ |
| *Case* | $n_{21}$ | $n_{22}$ | $n_{23}$ | $n_{20}$ |
| Total | $n_{01}$ | $n_{02}$ | $n_{03}$ | $n$ |

consists of normal or unaffected individuals; however, it contains a few affected individuals as well. We have to select random samples from two specific groups, affecteds (cases) and unaffecteds (controls). A random sample from the population can be regarded as controls. However, we need to select a group of cases, supposedly as a random sample among the case individuals in the population. Thus individuals belonging to the same group or different groups are independent of each other. Once we select an individual, we can find the genotypes of multiple markers across her/his genome. For the sake of simplicity, we now concentrate only on one marker. Our main research question is whether a marker is associated with the disease.

After genotyping at a marker for individuals, we have information (data) about their genotypes. If the marker under study has nothing to do with the affection status, genotypes in cases and those in controls are distributed randomly or exhibit the same pattern. We assume that genotypes are in HWE among controls. If there is no association between the marker and the disease, we would expect the same among case individuals in that particular marker. This means that when the marker is not associated with the disease, we observe same genotype distribution in cases and controls. However, if this marker has some influence in increasing or decreasing the disease risk, genotype distributions in the two groups should be markedly different. For example, for a recessive disease, suppose $a$ is the minor allele that increases the disease risk. So we expect to find more $aa$ genotypes in cases than in controls.

Thus our main objective is to see whether a marker is associated with the disease, i.e. whether genotype distributions are different between cases and controls. This basic problem in genetics can be thought or reformulated as a statistical testing problem. Here the hypotheses of interest are

$H_0$ : marker is not associated with the disease against $H_1$ : marker is associated with the disease

In Table 6.3, $n_{i0} = n_{i1} + n_{i2} + n_{i3}$ for $i = 1, 2$ and similarly $n_{0j} = n_{1j} + n_{2j}$ for $j = 1, 2, 3$. So the number of cases is $n_{20}$ whereas number of controls is $n_{10}$ and the total number of individuals is $n$. Now the arguments follow exactly the same way as in case of testing HWE. Here it is natural to say that when $H_0$ is true, frequencies of respective genotypes are same in the two groups. The test statistic that would be used for testing the above hypothesis, should reflect different scenarios under $H_0$ and $H_1$ appropriately. Keeping the idea of testing HWE, we can consider a test statistic, which is based of the idea of comparing observed and expected frequency in each

cell in Table 6.3, as

$$T = \sum_{i=1}^{2} \sum_{j=1}^{3} \frac{(n_{ij} - E_{ij})^2}{E_{ij}} \tag{6.7}$$

where $n_{ij}$ and $E_{ij}$ respectively denote observed and expected frequency for the $(i, j)$th cell in Table 6.3.

To calculate $E_{ij}$, the expected frequency of a cell, first assume that $H_0$ is true, i.e. there is no genetic association. So we can expect that the proportion of genotype $AA$ in controls is the same as that in the entire sample. Hence we have, under $H_0$,

$$\frac{n_{11}}{n_{10}} = \frac{n_{01}}{n} \implies n_{11} = \frac{n_{10} n_{01}}{n} \tag{6.8}$$

Similarly, for each $(i, j)$th cell, assuming $H_0$ to be true, we have $n_{ij} = \frac{n_{i0} n_{0j}}{n}$, $i = 1, 2$, $j = 1, 2, 3$. Substituting (6.8) in (6.7), we get after elementary algebraic simplification,

$$T = \sum_{i=1}^{2} \sum_{j=1}^{3} \frac{(n_{ij} - \frac{n_{i0} n_{0j}}{n})^2}{\frac{n_{i0} n_{0j}}{n}} = n\Big( \sum_{i=1}^{2} \sum_{j=1}^{3} \frac{n_{ij}^2}{n_{i0} n_{0j}} - 1 \Big) \tag{6.9}$$

It can be shown that $T \sim \chi^2_2$ under $H_0$, approximately (asymptotically). Based on this null distribution of the statistic $T$ we can calculate the p-value approximately and take a decision. Note that this method provides a very simple test for genetic association of a marker with the disease (qualitative phenotype).

Based on the data given in Table 6.1, we can easily test whether a locus is associated with the disease. Here the first column indicates the status of individual '0' and '1' indicating control and case respectively. R code is available for this $\chi^2$ test. First we read the data, store it and then apply the test between two columns; one being case-control (CC) status and the other the genotype column (locus) of interest. Suppose the data are stored in a file named 'data.csv' with column headings indicating the contents of each column.

```
> a<-read.csv("data.csv",header=T)          ## reading the datafile
> names(a)          ## heading of columns
> head(a)          ## first six rows
> test.1<-chisq.test(a$CC.status,a$G1)          ## store results of the test
> test.1$p.value          ##displays p-value on screen
```

The p-value obtained is 0.00211 which is very small compared to the level of significance 0.05. So we reject $H_0$ of no association at level 0.05 approximately and conclude that on the basis of the given data, it seems that the locus G1 is associated with the disease.

We should note a few important things here. The level of significance for the test is *approximately* $\alpha = 0.05$, because the test statistic follows a $\chi^2$ approximately (asymptotically). Hence, the p-value calculated on the basis of $\chi^2$ distribution is only approximate. However, if the sample size is large, the approximation is good and can be used for all practical purposes.

The data set contains other loci and we can repeat the same procedure to other loci also. But a word of caution! We should not test for association at G3 because Hardy-Weinberg equilibrium is not satisfied for this locus (Sect. 6.4.5) and hence we should remove this locus from our analysis.

Now that we have a nice test for basic genetic association using qualitative phenotype (disease status), this triggers plenty of questions such as:

- This is a test of genetic association at only one marker. What to do when we have more than one marker? Can we just repeat this test for each marker and take decision at a fixed level, say $\alpha = 0.05$?
- If the markers are close enough, can we incorporate the closeness of markers in the test?
- When we collect data, in this case usually blood (or other tissues) for generating genotype data, we may also collect data on other covariates like age, gender, other environmental variables (smoking habit, food habit, etc.). How do we incorporate the information on these covariates while developing a genetic association test using a qualitative phenotype?
- However, there may exist other phenotypes that are not qualitative in nature. Can we develop a test for association for such trait? If so, can we ask the same questions as above, especially while dealing with a quantitative phenotype?

### 6.5.2 *Genetic Association in Presence of Covariates*

As discussed earlier, data for genetic association may contain variables other than genotypes and disease status. These variables might be important. Suppose a marker is associated with the disease but depends on the age of onset. For example, although blood sugar issues may be seen at a young age, it is, in general, more prevalent among the elderly population. A random sample includes individuals across different age groups. If the disease of interest is seen more frequently among persons of older ages, it might so happen that a genetic locus is actually not associated but our statistical test shows strong association. This is definitely a false positive; this occurs because we fail to incorporate or eliminate the effect of age while testing for the association. So in order to identify the presence of any genetic association with the disease, we should eliminate the effect of the covariate(s). A detailed and dedicated discussion and methods are also given in Chap. 7.

Naturally we cannot extend our previous discussion of using $\chi^2$ test as it depends on cell frequency alone. On the other hand, regression technique is a nice approach to eliminate the effect of other variables. Thus keeping all these in mind, we have

to explore an adequate method of testing whether a marker locus is associated with the qualitative phenotype or disease status. However, we have studied regression analysis only when the response variable is continuous in nature. We need to develop an analogous idea and subsequent statistical methods in case of categorical data. But first let's introduce a new concept based on probability theory.

### *6.5.3 Odds Ratio*

The new concept that we introduce is called 'odds' in favour or against an event. For instance, the odds that 'I can work out an algebra problem is 1 to 3' means out of 4 problems, I can do 1 and I cannot do 3. If we were to say "There is a 75% chance that it will rain tomorrow", then 3 out of 4 times we can say that it will rain. This means that it will rain three times out of 4 probable chances of raining. This can also be restated as "the odds of raining tomorrow is 3 to 1". This can also be understood in terms of probabilities of events associated with the problem. In the above example,

$$\text{Odds of raining tomorrow} = \frac{P(\text{raining})}{P(\text{not raining})} = \frac{3/4}{1/4} = \frac{3}{1} = 3.$$

Similarly the odds against raining tomorrow $= \frac{1/4}{3/4} = \frac{1}{3}$. We also interpret this in another way. The chances of raining tomorrow is three times more than the chance of not raining. Thus unlike probability, odds of any event may take any positive real number as its value, since it is the ratio of two probabilities. It is clear that 'odds' is more welcome to common people than 'probability' as it carries a very easy interpretation in connection with the real world.

Now in the context of disease prevalence, we can define the odds of a person being affected as the ratio of the probability that a person is affected to that of the person who is not affected. However, we can now easily extend this concept to incorporate another covariate that is available during the data collection and might occur naturally.

Suppose the probability that a young person is affected by a disease is 0.3 whereas the same probability is 0.8 for an elderly person. Thus we have two odds, one for the young and another for the old person, which are respectively $\frac{0.3}{0.7}$ and $\frac{0.8}{0.2}$. Now we define another measure, 'odds ratio', as

$$\text{odds ratio} = \frac{\text{odds for elderly person having disease}}{\text{odds for young having disease}} = \frac{0.8/0.2}{0.3/0.7} = 9.33 \quad (6.10)$$

This indicates that the chance of having the disease in an elderly person are 9.33 times more than that for a younger person. Note that each odds is defined in terms of conditional probability, i.e.

**Table 6.4** Smoking status and affectedness

| | $Y = 0$ | $Y = 1$ | $P(X = x)$ |
|---|---|---|---|
| $X = 0$ | $p_{11}$ | $p_{12}$ | $p_{10}$ |
| $X = 1$ | $p_{21}$ | $p_{22}$ | $p_{20}$ |
| $P(Y = y)$ | $p_{01}$ | $p_{02}$ | 1 |

$$\text{odds ratio} = \frac{P(\text{person is affected}|\text{person is old})/P(\text{person is unaffected}|\text{person is old})}{P(\text{person is affected}|\text{person is young})/P(\text{person is unaffected}|\text{person is young})}$$

In order to define odds ratio in a general manner, we define the probability distribution of two jointly distributed random variables. Let $X$ and $Y$ be two random variables taking only two values 0 and 1 (say). As an example we can think that $Y$ denotes the disease status (0 means healthy and 1 means affected) whereas $X$ denotes the smoking status (0 for non-smoking and 1 for smoking). To study any relationship between disease and smoking habit, we have to consider both variables jointly (Table 6.4).
where $p_{01} = p_{11} + p_{21}$, $p_{02} = p_{12} + p_{22}$, $p_{10} = p_{11} + p_{12}$, $p_{20} = p_{21} + p_{22}$ and $p_{01} + p_{02} = p_{10} + p_{20} = p_{11} + p_{12} + p_{21} + p_{22} = 1$. As explained earlier, odds ratio can be deduced in terms of probability, this eventually leads to a definition of odds ratio. We need to recall the conditional probabilities with reference to Table 6.4 to express odds ratio in terms of $p_{ij}$s. First note that

$$P(Y = 1|X = 1) = \frac{P(Y = 1, X = 1)}{P(X = 1)} = \frac{p_{22}}{p_{20}}.$$

Similarly, we have

$$P(Y = 0|X = 1) = \frac{p_{21}}{p_{20}}, P(Y = 1|X = 0) = \frac{p_{12}}{p_{10}}, P(Y = 0|X = 0) = \frac{p_{11}}{p_{10}}$$

$$\text{Thus, } OR = \text{odds ratio} = \frac{\text{odds}|X = 1}{\text{odds}|X = 0} = \frac{P(Y = 1|X = 1)/P(Y = 0|X = 1)}{P(Y = 1|X = 0)/P(Y = 0|X = 0)} = \frac{\frac{p_{22}}{p_{20}}/\frac{p_{21}}{p_{20}}}{\frac{p_{12}}{p_{10}}/\frac{p_{11}}{p_{10}}} = \frac{p_{11}p_{22}}{p_{12}p_{21}} \quad (6.11)$$

Naturally based on the data structure, we can easily get an estimate of odds ratio just by replacing the population parameters by their corresponding estimates. Since the estimate of $p_{ij}$ is $\hat{p}_{ij} = n_{ij}/n$ for $i, j = 0, 1$, an estimate of odds ratio would be

$$\widehat{OR} = \frac{n_{11}n_{22}}{n_{12}n_{21}} \quad (6.12)$$

'Odds' and 'odds ratio' play important roles in dealing with categorical variables along with other continuous and/or categorical variables. Especially, it is of utmost

importance, when the response is a categorical variable in a regression set up. Let us look at this afresh.

Note that $0 < \text{odds}(Y|x) < \infty$ and hence $-\infty < \log(\text{odds}(Y|x)) < \infty$. Sometimes it is advantageous to work with $\log(OR)$ instead of simple $OR$. It can be shown that for large sample size,

$$\log(\widehat{OR}) \sim N(\log OR, \sigma^2) \text{ approximately where } \sigma^2 \approx \frac{1}{n_{11}} + \frac{1}{n_{12}} + \frac{1}{n_{21}} + \frac{1}{n_{22}} \tag{6.13}$$

Although our study variable $Y$ is categorical, in many real situations it may depend on other variables or covariates. Since we are expecting that the response variable depends on covariate $(X)$ and hence the odds, we can think of a regression model of $y$ on $x$. However, $Y$ is a categorical variable whereas log(odds) can take any real value; it would be easy to consider a regression equation of log(odds) on $x$. Naturally the easiest way is to consider a linear regression model of log(odds) on $x$ as

$$\log(\text{odds}) = \alpha + \beta x \tag{6.14}$$

A little mathematical simplification gives more insight into this model. Noting that $P(Y = 1|x) + P(Y = 0|x) = 1$, after simplification we get

$$\log\left(\frac{P(Y=1|x)}{P(Y=0|x)}\right) = \alpha + \beta x \implies \frac{P(Y=1|x)}{P(Y=0|x)} = e^{\alpha+\beta x}$$
$$\implies P(Y=1|x) = \frac{e^{\alpha+\beta x}}{1+e^{\alpha+\beta x}} \text{ and } P(Y=0|x) = \frac{1}{1+e^{\alpha+\beta x}} \tag{6.15}$$

Thus we can say that given $x$, $Y \sim Binomial(1, p(x))$ where $p(x) = \frac{e^{\alpha+\beta x}}{1+e^{\alpha+\beta x}}$

Here, $X$ takes only two values, 0 and 1. So putting values of $x$ as 0 and 1 in (6.15), we have

$$\text{Odds ratio} = \frac{P(Y=1|X=1)/P(Y=0|X=1)}{P(Y=0|X=0)/P(Y=0|X=0)} = \frac{e^{\alpha+\beta}}{e^{\alpha}} = e^{\beta}.$$

Hence, $e^{\beta}$ is nothing but the odds ratio $(OR)$. We also have

$$\beta >=< 0 \implies e^{\beta} >=< 1. \tag{6.16}$$

Thus expression in (6.16) may generate a testing method related to the odds ratio. Note that (6.14) looks similar to a linear regression equation except that here the response variable is the log(odds) that is linearly dependent on the explanatory variable (covariate) $x$. This is known as 'logistic regression', which should be used when the response variable is categorical instead of a continuous variable.

**Ex-6.5** In a group of 120 smokers (exposed) and 280 non-smokers (not exposed), we look at lung cancer (event) status. If 39 smokers have lung cancer and 6 non-smokers have lung cancer, what is the odds ratio for lung cancer?

$$\text{odds for lung cancer among smokers} = \frac{\hat{P}(\text{lung cancer}|\text{smokers})}{\hat{P}(\text{no lung cancer}|\text{smokers})} = \frac{39/120}{81/120} = \frac{13}{27},$$

$$\text{odds for lung cancer among non-smokers} = \frac{\hat{P}(\text{lung cancer}|\text{non-smokers})}{\hat{P}(\text{no lung cancer}|\text{non-smokers})} = \frac{6/280}{274/280} = \frac{3}{137},$$

$$\text{Hence, odds ratio} = \frac{\text{odds for lung cancer among smokers}}{\text{odds for lung cancer among non-smokers}} = \frac{13}{27}.\frac{137}{3} = 21.988.$$

### 6.5.4 *Statistical Test Relating to Odds Ratio*

To test whether a qualitative phenotype (e.g. disease status) is associated with genotypes in the presence of other covariates like age, gender, etc., we can calculate odds ratio and perform a statistical test for odds ratio. The asymptotic distribution of sample OR as given in (6.13) provides a method for such testing. Thus the hypotheses of interest are

$$H_0 : OR = 1 \text{ against } H_1 : OR \neq 1$$

We can use a test statistic

$$T = \frac{\log(\widehat{OR}) - 0}{\sigma} \text{ which follows } N(0, 1) \text{ under } H_0, \text{ approximately (asymptotically),} \tag{6.17}$$

where the expression for $\sigma^2$ is given in (6.13). Based on a data set, we can estimate the OR, calculate p-value, and take a decision accordingly. Note that here the level of significance is approximately $\alpha$ as the entire method depends on the asymptotic distribution of $\log(\widehat{OR})$, i.e. assuming that the sample size is large.

We can extend this idea of testing for association in the presence of other covariates. Let us discuss this with an example. Based on the data in Table 6.1, let $Y$ denote the disease status and $X$ be a random variable taking values 0, 1 or 2 for genotypes $AA$, $Aa$ or $aa$ respectively. Also, let $Z$ be another variable denoting age. Note that although we can think of age as a continuous variable, we can include more variables, be it categorical or continuous or both. In this situation we should be interested in the association of phenotype with genotype after adjusting the effects of other variables. Hence a regression type model is essential in this case; we can easily consider a logistic regression of $Y$ on $X$ and $Z$. Thus our model would be

$$\log(odds(y|x, z)) = \beta_0 + \beta_1 x + \beta_2 z \tag{6.18}$$

Note that $\log(odds)$ is also known as 'logit'. It looks like a linear regression of $\log(odds)$ on other variables and we can use the least square method. However, given the values of explanatory variables, especially when the auxiliary variable is continuous, we may get only one value of the explanatory variable. Thus it is always better to estimate the parameters using the likelihood method. Noting that $Y_i$ can take values 0 or 1 for $i = 1, \ldots, n$, the likelihood function for (6.18) is given as

$$P(y_1, \ldots, y_n | x_1, \ldots, x_n, z_1, \ldots, z_n) = \prod_{i=1}^{n} P(Y_i = 1 | x_i, z_i) = \prod_{i=1}^{n} \left( \frac{e^{\beta_0 + \beta_1 x_i + \beta_2 z_i}}{1 + e^{\beta_0 + \beta_1 x_i + \beta_2 z_i}} \right)^{y_i} \times \left( \frac{1}{1 + e^{\beta_0 + \beta_1 x_i + \beta_2 z_i}} \right)^{1 - y_i}$$

Maximising with respect to the parameters $\beta_0$, $\beta_1$, and $\beta_2$ we get their estimates. However, it involves solving simultaneous equations that are not even linear. Hence iterative or other procedures are required to get the final result. However, a simple R command fits the logistic regression directly and provides the estimates of all parameters along with their p-values.

Clearly $\beta_1$ is the logarithm of odds ratio. Hence p-value associated with $\hat{\beta}_1$ can be used directly to test $H_0 : \beta_1 = 0$ against $H_1 : \beta_1 \neq 0$. This is a test for genotypic association eliminating the effect of other variables.

In the data set as in Table 6.1, we see that age is another covariate available for individuals. So it is of interest to see whether there is any effect of this covariate, i.e. age while investigating the genetic association of any genetic locus. Naturally we cannot directly apply $\chi^2$ test as it does not have any provision to include the effect of age. However, we can use logistic regression incorporating age as another covariate and test whether the odds ratio is one for a particular genetic locus. Hence, to see genetic association of G1 locus with the disease, our model becomes

$$\log(odds(y | G1, \text{age}) = \beta_0 + \beta_1 G1 + \beta_2 \text{age}$$

Using in-built R code we can easily test $H_0 : \beta_1 = 0$ against $H_1 : \beta_1 \neq 0$.

```
> assoc<-glm(a$CC.status ~ a$G1+ a$Age, family="binomial")      ## fitting
                                                                logistic regression
> summary(assoc)                    ## displays summary of association
                                      testing using logistic model
```

It shows that the p-value corresponding to age is very small whereas that for genotypes p-value is very large compared to 0.05, the chosen level of significance. Hence, although we previously found an indication of association of G1 with disease, it is probably not the actual association; the signal is mainly due to the effect of age.

### 6.5.5 Genetic Association for Quantitative Phenotype

Quantitative phenotype (or variable) usually carries more information than qualitative variable. Instead of labelling a person as having 'high', 'low', or 'normal' blood pressure, it is always better to have the exact value of her/his blood pressure as measured by a sphygmomanometer. Categorising a continuous variable into a few categories would lead to loss of information. Thus, if we have genotype data at a locus along with a quantitative phenotype of interest, we may explore genetic association of the quantitative phenotype with the genetic locus.

Here no concept of case or control exists; we only have numerical, mostly continuous values of the quantitative phenotype. Genotype data are distributed over three categories viz, with genotypes '$AA$', '$Aa$', or '$aa$' for a bi-allelic locus. All phenotype values are distributed in these three categories.

#### 6.5.5.1 Analysis of Variance Method

Assuming that genotype at a marker locus has three categories (e.g. $AA$, $Aa$, and $aa$), our interest would be to see whether the quantitative phenotype values are distributed homogeneously in these three categories for a sample of individuals. That means the phenotype values show strong homogeneity within each class corresponding to genotypes. We can very easily represent this scenario using some mathematical notations.

Suppose individuals having a particular genotype form a class or group. Let $i = 1, 2, 3$ respectively denote genotypes $AA$, $Aa$ and $aa$. Also let $y_{ij}$ be the quantitative phenotype value for $j$-th person having genotype $i$. Then we write a model as

$$y_{ij} = \mu_i + e_{ij}, \qquad j = 1, \ldots, n_i, \ \ i = 1, 2, 3 \tag{6.19}$$

where $\mu_i$ is the effect genotype class $i$. $e_{ij}$s are assumed to be random errors associated with $(i, j)$-th observation. For statistical analysis we assume that $e_{ij} \sim N(0, \sigma^2)$.

A few important points to note:

- Number of individuals belonging to different genotype classes may not the be same. So $n_i$ values are not same for $i = 1, 2, 3$.
- Population genotype frequencies must depend on allele frequencies. This fact should also be reflected in the sample.
- If the effect of genotype is present, $\mu_i$s would be different leading to different means for individuals in different genotype classes.
- It is clear that if the phenotype is not associated with the genotype at the locus of interest, different genotypes have no special effects on phenotype. Thus $\mu_1 = \mu_2 = \mu_3$ in this situation, where no effect of genotype is present on the quantitative phenotype.
- However, each $y_{ij}$ has the same (constant) variance, i.e. $V(y_{ij}) = \sigma^2$. This is a very important assumption under which we develop the analysis method.

Based on the points discussed just now, we can frame the hypotheses of testing genetic association as

$$H_0 : \mu_1 = \mu_2 = \mu_3 \text{ against } H_1 : \text{ at least one inequality in } H_0. \tag{6.20}$$

In other words here we are testing equality of means in different genotype classes assuming variances are constant. This is standard Analysis of Variance (ANOVA) method.

A little mathematical deduction would give the test statistic as

$$\mathscr{F} = \frac{MSB}{MSE} = \frac{SSB/2}{SSE/(n-3)} \tag{6.21}$$

where $SSB = \sum_{i=1}^{3}\sum_{j=1}^{n_i}\left(\bar{y}_{i0} - \bar{y}\right)^2$, $SSE = \sum_{i=1}^{3}\sum_{j=1}^{n_i}\left(y_{ij} - \bar{y}_{i0}\right)^2$ with $\bar{y}_{i0} = \frac{1}{n_i}\sum_{j=1}^{n_i} y_{ij}$, $\bar{y} = \frac{1}{n}\sum_{i=1}^{3}\sum_{j=1}^{n_i} y_{ij}$, and $n = \sum_{i=1}^{3} n_i$.

This statistic although looks complicated at the first sight, may be understood from an intuitive point of view. If $H_0$ is true, there is no effect of genotype on phenotype and hence we can expect that all observations come from one class, i.e. they are homogeneous. So the class means, i.e. $\bar{y}_{10}$, $\bar{y}_{20}$ and $\bar{y}_{30}$ should be very close to each other and hence they should not differ much from the grand mean, i.e. $\bar{y}$. On the other hand, if $H_0$ is not true, class means vary and might be different from the grand mean. A measure of such variation may be given by (assuming there are k classes)

$$SSB = \sum_{i=1}^{k}\sum_{=1}^{n_{ij}}(\bar{y}_{i0} - \bar{y})^2.$$

Clearly $SSB$ would be small if $H_0$ is true and large if $H_0$ is not true. However, as in testing of hypothesis for other problems, the crucial question is: how large is really large? To understand this, we need to look at the scenario from another perspective.

Note that whatever be the differences, it is due to random sample only, provided $H_0$ is true. For each class, amount of total variation in that class is measured by $\sum_{j=1}^{n_i}(y_{ij} - \bar{y}_{i0})^2$ and combining all classes, we have $SSE = \sum_{i=1}^{k}\sum_{j=1}^{n_i}(y_{ij} - \bar{y}_{i0})^2$ as a measure of variation. For the $i$-th class, $\bar{y}_{i0}$ may be taken as a representative value for that class and so $\sum_{j=1}^{n_{ij}}(\bar{y}_{i0} - \bar{y})^2$ may be taken as a measure of total variation present in the $i$-th class. Hence, we define a measure of such variation based on the entire data, as

$$SSE = \sum_{i=1}^{k}\sum_{j=1}^{n_{ij}}(\bar{y}_{i0} - \bar{y})^2.$$

Clearly we can get an estimate of variation $\sigma^2$ from SSE. In fact, it can be shown that:

$$E\left(\frac{SSE}{n-k}\right) = \sigma^2.$$

Note that if $H_0$ is not true, this $SSB$ should be much larger compared to the variation due to random sampling and they should be close to each other if $H_0$ is true. Thus the ratio $SSB/SSE$ or its scaled variation may be taken as a test statistic. Hence our final test statistic becomes $\mathscr{F} = \frac{MSB}{MSE}$ as defined in (6.21).

Now it is clear that if there is no association, i.e. $H_0$ is true, we can expect that class means ($\bar{y}_{i0}$) would be close to each other and hence MSB would be small with respect to MSE. On the other hand, if there is any association, class means would differ substantially from each other and hence their squared deviation from grand mean, i.e. MSB would be much larger than MSE. Thus if the ratio $MSB/MSE$ is very large, we reject $H_0$ claiming that there is enough evidence of significant genetic association with the phenotype.

It can be shown that when $H_0$ is true, the test statistic $\mathscr{F}$ as given in (6.21) follows an $F$ distribution with degrees of freedom 2 and $n-3$. Thus in practice we calculate the p-value using $F_{2,n-3}$ distribution and take the decision accordingly.

For the data set given in Table 6.1, we want to examine any possible existence of associate of the quantitative phenotype, cholesterol level in this case, with genotype locus G1 (say). First we group the observations with respect to the genotypes and then apply ANOVA technique as discussed.

The very small p-value 0.000639 is indicative of the presence of association of G1 with cholesterol level. We can repeat the same for all other loci. Note that we completely ignore the case-control status here because when we have the quantitative phenotype, it is advisable not to search for any qualitative characteristic like affectedness of an individual. We use R code for the calculation using the ANOVA method.

```
> y1<-a$Cholesterol[which(a$G1=="AA")]      ## extracting data for genotype AA
> y2<-a$Cholesterol[which(a$G1=="Aa")]      ## extracting data for genotype Aa
> y3<-a$Cholesterol[which(a$G1=="aa")]      ## extracting data for genotype aa
> y<-c(y1,y2,y3)                            ## creating phenotype data for analysis
> label<-c(rep(0,length(y1)),rep(1,length(y2)),rep(2,length(y3)))   ## creating labels for classes
> anova<-aov(y~label)                       ## applying R function to do ANOVA
> summary(anova)                            ## displays the output
```

#### 6.5.5.2 Regression Based Test

Sometimes we use a slightly different test to check whether there is any association at a locus. If a locus is associated with the disease, usually it is the minor allele that seems to be the disease allele or responsible for increasing the disease risk. It is natural to think that increased number of minor alleles in a genotype induces more adverse effects. If we use the number of minor alleles as indicators of genotypes, i.e. if we consider a variable $X$ that takes values 0, 1, or 2 for genotypes $AA$, $Aa$, or $aa$ respectively where '$a$' is the minor allele, increase in the value of $X$ increases the disease risk, if associated. Thus, we can use regression approach to test for genetic association. Thus we consider a linear regression model as

$$y_i = \alpha + \beta x_i + e_i, \qquad i = 1, \ldots, n \tag{6.22}$$

where $e_i \sim N(0, \sigma^2)$ for all $i = 1 \ldots, n$.

Using this model we can test for $H_0 : \beta = 0$ against $H_1 : \beta \neq 0$. We can calculate the p-value based on $t$ distribution and take decision accordingly. However, this implicitly assumes that minor alleles, if associated, have a linear effect of the phenotype which, may not always be the case.

#### 6.5.5.3 Kruskal-Wallis Test

There is one issue about which we must be very cautious. This method is applicable only when the underlying assumptions are valid. Unless the variances are same for all classes, we cannot apply this ANOVA method nor when the variables do not follow a normal distribution. However, in such situations we can perform a nonparametric test, called Kruskal-Wallis test. This is nonparametric analogue of ANOVA. Although a little less powerful than ANOVA, we have no option but to use this test when assumptions of ANOVA are not valid.

A test for equality of variances of three groups and a test for normal distribution using Kolmogorov-Smirnov test or Shapiro-Wilk test is advised before applying ANOVA. Violation of any one goes against the use of ANOVA in this case. In our given data set, normality test for each group of genotypes fails and hence we should not use ANOVA; rather we use Kruskal-Wallis test, the nonparametric version of ANOVA. Here we just mention how to do Kruskal-Wallis test in R; we shall discuss this test in detail later.

```
> kruskal.test(y,label)$pvalue        ## provides p-value for Kruskal-Wallis test
```

The p-value 0.006773 is very small compared to 0.05. So we reject $H_0$ of equality of medians in three genotype groups concluding that G1 might be associated with cholesterol level.

## 6.6 Problems in Association Testing

The above discussion guides us into the nice world of genetic association study. Especially it converts some very simple idea to testing the existence of genetic association of a locus with either qualitative or quantitative phenotype. But that's not all. Let's consider one real situation. Suppose there are one million markers across the genome. We have genotype data for all loci and also phenotype data. Now apply an appropriate test that we learned. It is expected that p-values for a few tests are small compared to the level of significance (fixed before the test) and for others, p-values are large enough. This operation, in its simplest form can be described in four steps, as follows.

- Collect a group of cases and controls.
- Genotype at each marker.
- Perform case-control association test at each marker.
- Get the p-values and take decision at chosen level of significance, say 0.05 for each locus. If we see that the value is very small compared to 0.05, we feel excited because we find a locus associated with the phenotype or disease!

But, unfortunately, we are missing a few things here. There are several issues related to the inherent population structure from where the sample is taken, choice of level of significance, power of the association test, sample size required to achieve a high power, etc. Unless we focus on these issues, we might end up with too many false positives or fail to identify some true signals. Thus, we need to understand the problems and their eventual solutions so that our results may be replicated in future and hold promise for identification of disease associated loci.

### *6.6.1 Multiple Testing*

Recall the meaning of type I error rate (or level of significance). If the type I error rate is 0.05, it indicates that the probability that our test identifies a marker to be significant when, in fact, it has no effect at all. This is known as 'false positive'. Then we may think that about 5% significant results may be false positives. Let's imagine a scenario. Assuming that type I error rate is 0.05, out of 1 million markers, our testing method can identify around $1000000 \times 0.05 = 50000$ markers to be false positives and we don't know which are the false positives really! That's too much. We can't afford to waste our time and money and other resources by searching too many false positives. Don't worry. There are some nice and easy statistical ways that have been developed specifically to deal with this problem.

The problem, just discussed, is known as multiple testing problem. Because of a very large number of tests, our initial choice of 'small' level of significance does not remain small enough to control the number of false positives. Note that this entire problem occurred because of our choice of $\alpha$, known as level of significance for the

test. Type I error rate cannot not exceed the level of significance. Playing with this $\alpha$ might provide a solution. We discuss two most popular and effective approaches on how to control the number of false positives.

**Bonferroni correction:** Bonferroni conceived this problem about a century ago and suggested a very simple way to tackle this. Let's assume for each test, the level of significance is denoted by $\alpha$. He suggested that instead of taking $\alpha$ for each testing, just take $\alpha/k$ where $k$ is the number of tests. So if there is 1 million markers giving 1 million tests, take $\alpha = 0.05/1000000 = 5 \times 10^{-8}$ as the level of significance for each test. Naturally this value of $\alpha$ ensures that the approximate number of false positives among one million markers would be only 0.05. This is a simple and good solution.

However, there are some drawbacks to this method. In fact, this problem persists for all multiple testing procedures, to a varying degree. We know that if type I error rate decreases, type II error rate increases and consequently the power of the test decreases. Thus, Bonferroni correction might lead to serious compromise of the ability of the test to capture the associated loci, especially those having low or moderate effects. So, although we control the number of false positives, while doing so, we miss a few truly associated signals (markers). Although Bonferroni correction is a very simple and easy solution to the problem of multiple testing, it is too conservative giving too much penalty in finding true associations. Moreover, in order to ensure a reasonably high power, we require an extremely large sample, which is sometimes very difficult to collect. Addressing this issue, a few methods have been developed that are less conservative compared to Bonferroni correction.

**Benjamini-Hochberg (BH) correction:** This is probably the most popular method that is supposed to control the number of false positives in a much less conservative way than Bonferroni's method. While Bonferroni correction keeps the level of significance for each test at a fixed level, BH correction deviates from this idea. The type I error rate should change with the nature of p-value keeping the global type I error rate fixed at level $\alpha$. The procedure of BH correction is as follows.

- Calculate the p-values for $k$ markers, say $P_1, P_2, \ldots, P_k$.
- Arrange them in increasing order of magnitude, i.e. $P_{(1)}, P_{(2)}, \ldots, P_{(k)}$, where $P_{(1)}$ is the minimum of $P_1, P_2, \ldots, P_k$, $P_{(2)}$ is the next higher minimum, and so on, $P_{(k)}$ being highest one.
- Reject the $i$-th test if $P_{(i)} < \frac{i}{k}\alpha$.

It is seen that this method is less conservative in the sense that power is not too low compared to the Bonferroni method. However, we need a much larger sample to achieve the power that can be obtained in single marker analysis. Although there are some more methods that are improvements over the BH method, this is very popular mainly due to its simplicity and wide applicability in many scenarios.

We now apply these multiple testing corrections to see whether any of the genetic loci G1, G3, and G4 survives the correction. R code is very much useful as it has an in-built function to calculate corrected p-values.

```
> test.1<-chisq.test(a$CC.status,a$G1)
> test.2<-chisq.test(a$CC.status,a$G2)
> test.3<-chisq.test(a$CC.status,a$G3)
> pvalue<-c(test.1$p.value,test.2$p.value,test.4$p.value)
> p.adjust(pvalue,"bonferroni")
[1] 0.006352761 1.000000000 0.062294624
> p.adjust(pvalue,"BH")
[1] 0.006352761 0.417810511 0.031147312
```

Adjusted p-value by BH method seems less conservative as the corrected p-value for G3 is less than that obtained by Bonferroni method. If we fix level of significance as $\alpha = 0.05$, based on corrected p-values, it seems that both G1 and G3 are associated with the disease according to BH method. Note that the decision should be taken based on only one method of multiple correction chosen by the experimenter and we recommend the use of the BH correction especially when the number of tests is large.

### *6.6.2 Population Stratification*

Solution to controlling Type I error rate in order to avoid too many false positives leads to the discovery of a wide range of multiple testing correction methods. However, there is no denying that we require a larger sample to achieve a reasonably high power when we have multiple markers to be tested. However collecting larger samples and generating genotype data at each marker for all individuals, is not so easy. The reason is not only cost and limited resources, but there are also some intrinsic characteristics of the data that need to be taken care of while collecting samples. Nowadays with multi-centric research facilities sometimes resources are available. But we need to understand and consider the population characteristics while collecting data. One such problem is due to 'population stratification'. Let's consider an example.

Suppose two different ethnic groups are mixed in a population. We don't have any knowledge of that. Consider case-control genotype data at a bi-allelic locus, where the number of control individuals with genotypes $AA$, $Aa$, and $aa$ are 25, 68, and 58 respectively, whereas that among case individuals are 28, 70, and 50 respectively. The p-value corresponding to $\chi^2$ using (6.9) is 0.6834. It means that at the level of significance 0.05 approximately, we have no evidence to believe that the genotype distribution among cases and controls are different, i.e. this genotype locus is not associated with the phenotype.

However, unknowingly, this is not the real situation that we see in the data. A closer look at the background information of sampled individuals reveals that there are actually two ethnic groups mixed together in the population. Assuming that the

**Table 6.5** Genotype frequencies for two ethnic groups at a bi-allelic locus with alleles $A$ and $a$

| | Group I | | | Group II | | | Combined group | | |
|---|---|---|---|---|---|---|---|---|---|
| | $AA$ | $Aa$ | $aa$ | $AA$ | $Aa$ | $aa$ | $AA$ | $Aa$ | $aa$ |
| Control | 8 | 38 | 44 | 17 | 30 | 14 | 25 | 68 | 58 |
| Case | 21 | 34 | 29 | 7 | 36 | 21 | 28 | 70 | 50 |
| Total | 29 | 72 | 73 | 24 | 66 | 35 | 53 | 138 | 108 |

ethnicity of an individual is known, Table 6.5 presents the genotype frequencies of cases and controls among the two ethnic groups.

Doing a simple association testing for each group by the $\chi^2$ statistic, we see that the p-values for Group I and Group II are 0.0115 and 0.0487 respectively indicating the presence of association (may be marginal) in each group. But there is no association signal when these groups were merged together. So sub-grouping or heterogeneity may play an important role in masking the real situation with respect to genotype association. Hence, it becomes necessary to reconsider the entire problem.

Note that this population substructure is hidden in the population and we had no idea about that during data collection. It is also difficult sometimes to get any such idea. An example may be that mixing occurred in a previous generation and the genes were inherited within each subpopulation over subsequent generations. So this hidden or unknown population structure may give a completely erroneous result. This problem is known as 'Simpson's paradox'. And we need to be very careful while doing any association studies. The same is true for quantitative phenotype data also.

Let's look at the problem a little more carefully. It is the stratification or existence of multiple subgroups within the same population that creates the fallacy. This stratification is hidden, i.e. not known. So the best way to deal with this situation is to devise a method that tries to adjust the effect of stratification or do an association testing to eliminate the effect. The most popular method encapsulated in a software that is used to address this issue is 'eigenstrat'.

Keeping in mind all these issues that put few question marks about the simple association method using $\chi^2$, standard ANOVA or regression methods, a few sophisticated methods have been developed and research is still continuing.

### *6.6.3 Polygenic Risk Score*

Genome wide association studies have paved the path in identifying a large number of genetic variants, mostly single nucleotide polymorphisms that are significantly associated with many complex diseases. These variants usually have a small effect and hence correspond to a small fraction of truly associated variants. It is expected that the genotype profile of an individual should provide an indication of the predictive power of the phenotype. To this end, polygenic risk score (PRS) has been developed

for providing an estimate of genetic liability for an individual based on his/her genetic profile (Choi et al. 2020). By genetic profile, here we mean the set of genotypes at different loci of the individual.

According to the classic PRS method (Dudbridge 2013; Purcell et al. 2009) PRS is calculated as a linear combination of genotype scores at different loci weighted by the effect sizes of those loci. Genotype score at a locus is usually taken as the number of risk alleles of that individual. The set of loci includes common, intermediate, and rare variants that are associated with the disease. Naturally the effect sizes vary. The most challenging part of PRS is the proper estimation of the effect sizes that would be included in the linear combination.

Let there be $K$ loci with effect sizes $\hat{\beta}_1, \hat{\beta}_2, \ldots, \hat{\beta}_K$ obtained through genetic association study. For continuous trait, we can use a standard regression model $y = \beta_0 + \beta_i g_i + \boldsymbol{\gamma}'\boldsymbol{x} + \epsilon$ where $g_i$ is the genotype at locus $i$, $\boldsymbol{x}$ contains other covariates, $y$ is the phenotype and $\epsilon$ is the error in the model. For qualitative trait, we use logistic regression, $logit\big(P(Y = 1)\big) = \beta_0 + \beta_i g_i + \boldsymbol{\gamma}'\boldsymbol{x} + \epsilon$, where symbols have the same meanings as before.

Based on genome-wide association study on a training set of samples, we can easily evaluate the effect sizes, i.e. $\hat{\beta}_i$ for $i = 1, 2, \ldots, K$. Now given the genotype score $g_i$ at $i$-th locus for an individual, we have the PRS score of that individual as

$$PRS = \sum_{i=1}^{K} \hat{\beta}_i g_i$$

Usually the target individual, i.e. the individual for whom we are interested to calculate the PRS, should be outside the training sample; otherwise the calculated PRS might be inflated and may not show an unbiased picture for genetic susceptibility. However, it is desirable that the target individual should match other features as that observed in the data set that is used for estimation of $\beta_i$s. PRS provides a predictive power for genetic liability. This is a score assigned to each individual based on genotype profile at a large number of loci. However, it is clear that the estimates of effect sizes play the most important role. Sometimes use of estimates from public databases is advocated. These estimates are calculated on large data sets. However, ethnicity is also an important factor. Moreover, presence of other covariates, existing LD structure (Baker et al. 2018) among the markers, etc. are some influencing factors that need to be taken care of while estimating $\beta_i$s. Several other advanced methods are also developed to fine tune these estimates. It is also seen that instead of considering only the markers that reach the genome-wide significance, we can achieve more predictive power if we use a large number of SNPs in the calculation of PRS.

## 6.7 Some Advanced Association Testing Methods

Apart from multiple testing or population stratification problems, there are some other issues. Solutions to them, even partially, might lead to a better or more powerful method of genetic association. Needless to say, these methods would also consider some other features inherent to the data and the genetic architecture. These additional features, if known, should be explored and exploited properly to come up with more sophisticated and powerful methods.

It is easy to conceive that if two loci are situated at a close proximity on the same chromosome, they might be correlated. In other words, one might affect the other and this degree of influence varies. If there is extremely high correlation between them, we may ignore one and proceed to association testing with one marker. But if the correlation is moderate, ignoring one locus might incur loss of information. Single marker test does not consider this correlation at all. It is conjectured that if we consider two or more markers together, this holistic approach might be more realistic and informative. Another issue, which is more conceptual, is that it is the gene in which we are ultimately interested. Gene is not only the unit of inheritance, but it triggers also all downstream activities that is manifested in a phenotype. So instead of single marker at a time, a gene (of course through multiple markers) is more informative in deciphering disease gene association.

Honouring these ideas we discuss two methods that are well accepted and widely used in the context of studying the possible association of a disease with a gene or a user-defined small genomic region. Information about gene or genomic region of interest comes through a set of markers, mostly SNPs, that lie in its close proximity.

### *6.7.1 Kernel-Based Association Test (KBAT)*

The goal of kernel-based association test (Mukhopadhyay et al. 2010) is to test association between a qualitative phenotype and a set of SNP markers that belong to a small genomic region or gene. In this method, no assumption is made about the direction of SNP effects. Since it considers multiple SNPs at a time, the correlation structure among the SNPs has already been taken into account implicitly. However, if two SNPs are extremely highly correlated, it is recommended to take only one marker from the two. This test, however, is not a standard test and depends on what is called 'genotype similarity score'.

The data consists of $n_1$ cases and $n_2$ controls genotyped at $K$ markers. Naturally, individuals belonging to the case group are assumed to be homogeneous with respect to their genotype distribution at each locus. The same is true for the control group. But there is heterogeneity between the two groups if there is any association; otherwise they should be homogeneous. The test is based on 'kernel' or genotype similarity score, that is defined based on genotypes at a locus for two individuals.

The most commonly used kernel is 'allele match (AM) kernel'. One advantage of AM kernel is that it does not require the knowledge of risk allele. We can also define other kernels. If $g_i$ and $g_j$ are the genotypes of $i$-th and $j$-th individuals at a locus, AM kernel takes the value 4 if $g_i$ and $g_j$ are same; 2 if one is homozygote and the other heterozygote, and 0 if they don't share any common alleles. These genotype scores are the backbone of the test. We can evaluate such scores between pairs of individuals in controls and also in cases. Moreover, we can combine single marker score to generate multi-loci score between a pair of individuals. If the overall score in the control group is close to that for the case group, we can say that there is no evidence of association for the set of loci considered together; otherwise there may exist an association. Note that through this test we try to investigate association not at a single marker level, but whether the gene or genomic region of interest as a whole is associated with the qualitative phenotype (may be disease).

Let $h_l^k(g_i, g_j)$ be a measure of genotype similarity between $i$-th and $j$-th individuals with genotypes $g_i^k$ and $g_j^k$ at $k$-th locus ($k = 1, 2, \ldots, K$) respectively in group $l$ ($l = 1, 2$). Here $l = 1$ denotes cases and $l = 2$ denotes controls. Within each group of size $n_l$, there are altogether $m_l = n_l(n_l - 1)/2$ distinct similarity scores for $m_l$ possible pairs. Keeping the idea of ANOVA in mind, we can write the above discussion in terms of a linear model as

$$y_{l(ij)} = \mu + \alpha_l + e_{l(ij)}; \;\; i < j = 1, 2, \ldots, n_l; \;\; l = 1, 2 \tag{6.23}$$

where $y_{l(ij)}$ is the genotype similarity score for $(i, j)$ pair of individuals in the $l$-th group, $\mu$ denotes the overall grand mean or the general effect for pairs of individuals, $\alpha_l$ is the group specific treatment effect for similarity scores over the general effect, and $e_{l(ij)}$ are the error components.

We assume

(i) $\alpha_1 + \alpha_2 = 0$,

(ii) $V(e_{l(ij)}) = \sigma^2$ for $l = 1, 2$,

(iii) $Cov(e_{l(ij)}, e_{l'(i'j')}) = \begin{cases} \rho\sigma^2 & \text{for } i \neq i'; \, j = j'; \, l = l' \text{ or } i = i'; \, j \neq j'; \, l = l' \\ 0 & \text{otherwise} \end{cases}$

Although it looks like an ANOVA model, it violates the basic assumption because the error variables are not always uncorrelated, and $y_{l(ij)}$s are not normally distributed. This prohibits the direct application of ANOVA. However, we use the basic philosophy behind ANOVA that the variation between groups might be more compared to that within groups if there is a significant effect of groups, on an average. Note that $\bar{U}_l^k = \sum_{i<j} h_l^k(g_i^k, g_j^k)/m_l$ is the average genotype similarity scores of all pairs of individuals in $l$-th group at $k$-th marker. Corresponding to $k$-th marker, we define measures of within and between groups variations as

$$SSW_k = \sum_{l=1}^{2} \sum_{i<j} \left(h_l^k(g_i^k, g_j^k) - \bar{U}_l^k\right)^2 \text{ and } SSB_k = \sum_{l=1}^{2} m_l \left(\bar{U}_l^k - \bar{U}_k\right)^2.$$

In ANOVA we compare between sum of squares to within sum of squares or a scaled version of them. For testing the association of $K$ markers, the proposed KBAT statistic is

$$\mathscr{F} = \frac{\sum_{k=1}^{K} SSB_k}{\sum_{k=1}^{K} SSW_k} \tag{6.24}$$

This test statistic resembles with the standard ANOVA statistic except that in ANOVA we use MS (mean squares) instead of SS (sum of squares). So a few variations of KBAT can be thought of. For example, we can consider the sum of ratios of individual statistics instead of the ratio of the sum of squares. Hence, an alternative form of KBAT may be $\mathscr{F}_A = \sum_{k=1}^{K}(SSB_k/SSW_k)$. Another important issue is involved in KBAT statistic. In standard ANOVA under normality assumption, the test statistic follows an $F$ distribution under null hypothesis of no association. However, the distribution of KBAT even under null hypothesis $H_0 : \alpha_1 = \alpha_2 = 0$ is not known although it can be shown that it is right-sided test provided $\rho > 0$. Usually for all the kernels considered in KBAT statistic, $\rho > 0$, i.e. we reject the null hypothesis of no association if observed $\mathscr{F} > F_\alpha$ where the threshold value $F_\alpha$ should be calculated either using the null distribution of the test statistic or empirically. Since the null distribution is unknown, we can obtain the value using permutation technique or using the asymptotic distribution of the test statistic under $H_0$. It is always less time consuming to use the asymptotic distribution while calculating the p-value, especially when there is an issue of multiple testing. Note that multiple testing burden also occurs here as there are many genes or gnomic regions to be tested; however, the burden is much less than that for single locus tests. For example, with 1 million markers distributed over 1000 genes, the number of individual single marker tests is 1 million whereas it is only 1000 for KBAT statistic. It is seen that the asymptotic distribution of another slightly modified version of KBAT statistic is $\chi^2$ with $K$ degrees of freedom, as given in the theorem below.

**Theorem 16** *Let* $x = \hat{p}_k^2$, $y = 2\hat{p}_k\hat{q}_k$, *and* $z = \hat{q}_k^2$. *Define* $\hat{\mu}_{12,k} = 16(x^3 + y^3 + z^3) + 8(1 - (x^3 + y^3 + z^3) - (4xz - x^2z - xz^2)) - 4(xy^2 + zy^2 + y(1-y)^2)$. *Take* $\hat{\sigma}_{1k}^2 = \hat{\mu}_{2k} - \hat{\mu}_{1k}^2$ *and* $\hat{\sigma}_k^2 = \hat{\mu}_{12,k} - \hat{\mu}_{1k}^2$. *Then* $\hat{\sigma}_{1k}^2$ *and* $\hat{\sigma}_k^2$ *are consistent estimators of* $\sigma_{1k}^2$ *and* $\hat{\sigma}_k^2$ *respectively,* $k = 1, \ldots, K$. *If* $\frac{n_2}{n_1} \to \nu$ *as* $n_1 \to \infty$, *then as* $min(n_1, n_2) \to \infty$,

$$T = \beta(n_1, n_2).\frac{K(1+\nu^2)}{2\nu(1+\nu)}.\frac{\sum_{k=1}^{K} SSB_k/\hat{\sigma}_{1k}^2}{\sum_{k=1}^{K} SSW_k/\hat{\sigma}_k^2} \xrightarrow{L} \chi_K^2, \text{ where } \beta(n_1, n_2) = \frac{\binom{n_1}{2}\binom{n_2}{2}}{n_1}$$

Theorem 16 indicates that if $t$ is the observed value of the modified KBAT statistic $T$, we can easily calculate the p-value based on the asymptotic distribution of $T$. Hence the p-value (approximately) is given by value $= P(T > t|H_0) \approx P(\chi^2_K > t)$. The notation '$\xrightarrow{L}$' indicates 'convergence in distribution,' i.e. $T$ is asymptotically distributed as a $\chi^2$ variable with $K$ degrees of freedom.

Since using $\chi^2$ distribution, the p-value can be calculated very fast, it gets rid of doing the computationally intensive permutation technique and also provides the p-value up to a very high degree of accuracy.

### *6.7.2 Sequence Kernel Association Test (SKAT)*

Proposal of KBAT (Mukhopadhyay et al. 2010) is a paradigm shift in genetic association study in the sense that it advocates the use of information from multiple markers at a time and hence is more powerful compared to single marker association tests. However, it has a disadvantage. We know that in many studies, along with genotype and phenotype data, information on other covariates like age, sex, etc. are also available. It is very difficult to adjust this KBAT statistic for other covariates. A different multi-loci association test is proposed (Wu et al. 2009) that takes care of the presence of other covariates, known as sequence kernel association test (SKAT).

Like KBAT, for a small genomic region or a gene consisting of multiple SNP markers, we can test the joint effect of these markers through SKAT adjusting the effects of other covariates like age, sex, etc. To describe this method, we first consider a linear model for quantitative trait as

$$y_i = \alpha_0 + \boldsymbol{\alpha}' \boldsymbol{X}_i + \boldsymbol{\beta}' \boldsymbol{G}_i + \epsilon_i, \quad i = 1, \ldots, n \tag{6.25}$$

where $X_i$ is the covariate and $G_i$ is the genotype vector for the $i$-th individual and $\epsilon_i$ is the error associated with the response variable $y_i$, $i = 1, \ldots, n$. For qualitative trait like case-control status, etc., we consider a logistic model as,

$$logit\, P(y_i = 1|\boldsymbol{X}_i, \boldsymbol{G}_i) = \alpha_0 + \boldsymbol{\alpha}' \boldsymbol{X}_i + \boldsymbol{\beta}' \boldsymbol{G}_i, \quad i = 1, \ldots, n \tag{6.26}$$

Here $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_K)'$ is the effect of genotype vector $\boldsymbol{G}$ on the phenotype. Therefore the vector $G_i$ contains the genotypes at $K$ loci for individual $i$, $i = 1, \ldots, n$.

Our main purpose is to test

$$H_0 : \beta_1 = \cdots = \beta_K = 0 \text{ against } H_1 : \text{ at least one } \beta_i \neq 0.$$

However to construct a more powerful test, we assume that each $\beta_j$ ($j = 1, \ldots, K$) follows an arbitrary distribution with mean 0 and variance $w_j \tau$. This in turn implies that we can modify our hypotheses as

$$H_0 : \tau = 0 \text{ against } H_1 : \tau \neq 0.$$

Here we assume a random effect for $\beta_i$s. We can now easily construct a commonly used score statistic for testing the null hypothesis. The score statistic for testing $H_0$ is

$$Q = (\boldsymbol{y} - \hat{\boldsymbol{\mu}})' H (\boldsymbol{y} - \hat{\boldsymbol{\mu}}) \tag{6.27}$$

where $H = GWG'$, $\hat{\boldsymbol{\mu}}$ is the predicted mean of $\boldsymbol{y}$ under $H_0$, i.e. $\hat{\boldsymbol{\mu}} = \hat{\alpha}_0 + X\hat{\boldsymbol{\alpha}}$ for continuous trait and $\hat{\boldsymbol{\mu}} = logit^{-1}(\hat{\alpha}_0 + X\hat{\boldsymbol{\alpha}})$ for dichotomous trait, $G$ is the matrix of $K$ genotypes for all $n$ individuals, and $H$ is the kernel matrix. We can define this $H$ in many ways. However, it is convenient to use genotype similarity score between individual $i$ and $j$ as $H_{ij} = \sum_{k=1}^{K} w_k G_{ik} G_{jk}$ and $H_{ij}$ is the $(i, j)$-th element of the matrix $H$ and $W = diag(w_1, \ldots, w_k)$ with $w_i$ being the weight of the $i^{th}$ variant. Based on prior information or intuitively we can define $H$ in many ways suitable for the specific purpose of the study.

In order to calculate the p-value for a study, we can adopt a computation intensive method like permutation of bootstrap method. However, we can calculate the p-value much faster using the asymptotic distribution of the SKAT statistic, as given below.

**Theorem 17** *Let $B = V - V\tilde{X}(\tilde{X}'V\tilde{X})^{-1}\tilde{X}'V$ where $\tilde{X} = (\mathbf{1} \;\; X)$ is a $n \times (K+1)$ matrix. Also note that for continuous phenotype, $V = \hat{\sigma}_0^2 I$ where $\hat{\sigma}_0^2$ is a consistent estimator under $H_0$, and for dichotomous trait, $V = diag\Big(\hat{\mu}_{01}(1 - \hat{\mu}_{01}), \ldots, \hat{\mu}_{0n}(1 - \hat{\mu}_{0n})\Big)$ with $\hat{\mu}_{0i} = logit^{-1}(\hat{\alpha}_0 + \hat{\alpha}' X_i)$. Then under $H_0$,*

$$Q \sim \sum_{i=1}^{n} \lambda_i \chi_{1,i}^2 \text{ asymptotically as } n \to \infty,$$

where $(\lambda_1, \lambda_2, \ldots, \lambda_n)$ are the eigenvalues of $P^{1/2} H P^{1/2}$ and $\chi_{1,i}^2$ are independent $\chi_1^2$ variables for $i = 1, \ldots, n$.

However there is no direct way to calculate the p-value based on this asymptotic distribution even under $H_0$. SKAT uses an approximate method as proposed by Davies (1980). So the p-value calculated in SKAT uses two levels of approximations; one for the asymptotic distribution of $Q$ and another for p-value calculation by Davies' method. However, it can be used in practice as the sample size is usually very large nowadays.

## 6.8 Points to Remember

We have learnt how to analyse genetic data especially genotype data along with some covariates to identify possible association with the disease or the phenotype of interest. However, some important issues emerged during our discussion.

We first identify whether the phenotype is qualitative or quantitative and appropriate tests should be chosen based on the nature of phenotype. Methods for qualitative phenotype are usually different from that for quantitative phenotype and we need to be careful about that. After the test at multiple loci, it is necessary to do multiple testing correction; otherwise our analysis might produce too many false positives, thus wasting resources. However, before doing any analysis, we must check for other factors like population stratification, HWE, etc. so that the ultimate decision would not be masked due to unwanted source(s) of influence.

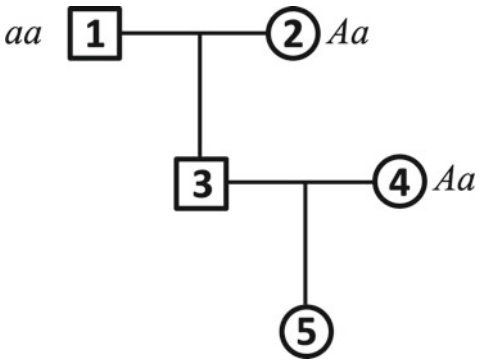
We should have a global view of the problem, data, and expected outcome. This would lead to tackling the problem at hand in the best possible way. Although single marker test is a first step for this kind of genetic analysis, after checking HWE and/or satisfying other quality control criteria, it is worthwhile to try some other sophisticated methods that might throw more light on the entire scenario. Research is still continuing in this area. Hence, search for recently developed methods of association and other related problems is welcome, especially when it has potential to strengthen our conclusion or discovering new potential biomarkers.

New methods may sometimes look complicated because it considers several issues simultaneously. However, in most cases, a new and more appropriate statistical method comes with user-friendly programming code or as an R function that can be easily used. So we should not have any problem in implementing the newly developed method if deemed most appropriate for a problem.

### Exercise

6.1 For the following pedigree,

(a) What are the possible genotypes of person 3? Find out the probability of each genotype of person 3.
(b) What is the probability that person 5 has genotype $AA$ when it is known that person 3 has genotype $Aa$?
(c) What is the probability that person 5 has genotype $AA$?
(d) Given that person 5 has genotype $Aa$, what is the probability that person 3 has genotype $Aa$?
(e) Further it is known that $A$ allele is the disease allele and two copies of $A$ are required for a person to be affected. However, not all persons with $AA$ genotype are affected. This is called incomplete penetrance. It is seen that $P(\text{person is affected} \mid \text{person has genotype } AA) = 0.83$. What is the probability that person 5 will be affected?




6.2 In a nuclear family, a child is carrying genotype $DD$ at a bi-allelic locus. Assuming random mating and allele frequency of allele $D$ in the population is $p$, what is the probability that her father has genotype (a) $dd$, and (b) $Dd$?

6.3 Suppose at a bi-allelic locus, HWE is satisfied. If the minor allele frequency at this locus is 0.12, what are the genotype frequencies at that locus?

6.4 Assuming HWE exists at a bi-allelic locus with two alleles $M$ and $m$, if the genotype frequencies of $MM$ and $Mm$ are 0.04 and 0.32 respectively, what is the minor allele frequency at this locus?

6.5 At a bi-allelic locus with alleles $A$ and $a$, if the genotype frequencies $P(AA)$, $P(Aa)$, and $P(aa)$ are respectively $a$, $ar$, and $ar^2$, for some $a$ and $r$ ($a, r > 0$), what can you say about HWE at this locus?

6.6 Suppose at a locus there are three alleles $A_1$, $A_2$ and $A_3$. Based on genotype data $n_{ij}$ for genotype $A_iA_j$, for $i = 1, 2, 3$ and $j = 1, 2, 3$, find the maximum likelihood estimates of all three allele frequencies. Can you get the same estimates if you adopt the gene counting method? Justify your answer.

6.7 Genotyping at a bi-allelic locus with alleles $D$ and $d$ was done for a sample from a small island in South Pacific Ocean. The observed frequencies for genotypes $DD$, $Dd$ and $dd$ are 243, 41 and 6 respectively. Check whether these genotype frequencies are in HWE.

6.8 In Question 6.7, suppose the given genotype frequencies are for females and $D$ locus is on X -chromosome. Let the genotype frequencies for $D$ and $d$ at this locus among males be 280 and 30 respectively. What can you say about HWE at this locus?

6.9 Suppose allele frequencies at a bi-allelic locus L are in HWE with $P(A) = p$. After two generations, due to severe epidemic, genotype frequencies of $Aa$ and $aa$ have been reduced by $2pqr$ and $pqr$ respectively, where $0 < r < 1$. How many generations will be required to stabilise the genotype frequencies? Show that genotype frequencies of $AA$, $Aa$, and $aa$ genotypes will be $p^2(1 + 2qr)^2$, $2pq(1 + 2qr)(1 - 2pr)$ and $q^2(1 - 2pr)^2$ respectively, when they again attain HWE.

6.10 Among many Indian tribes, the allele frequency of $B$ in the $ABO$ blood groups is extremely low. For example, in one sample of 600 people from one location, there were 37 $A$ and 563 $O$ blood types. What are the best estimates of the

allele frequencies of $A$, $B$, and $O$ in this population and what are the expected genotype frequencies assuming HWE?

6.11 At a particular bi-allelic locus, we have the allele frequencies as 0.575 and 0.425 respectively for alleles $A$ and $a$. The genotype counts in a sample from a particular population are 30, 55, and 15 respectively for genotypes $AA$, $Aa$, and $aa$.

(a) Define the test statistic and specify its null distribution along with the degrees of freedom to test whether the genotype frequencies in this population are significantly different than what would be expected if the population is in Hardy-Weinberg equilibrium for that locus (use 5% level of significance).
(b) What is the value of the test statistic? Draw inference by comparing the value with the value obtained from Chi-square cut off at 5% level of significance.

6.12 The colour of a person's eyes is determined by a single pair of alleles at a locus. If they are both blue-eyed alleles ($BB$), then the person will have blue eyes; if they are both brown-eyed alleles ($bb$), then the person will have brown eyes; and if one of them is blue-eyed and the other a brown-eyed allele ($Bb$), then the person will have brown eyes. (Because of the latter fact, we say that the brown-eyed gene is dominant over the blue-eyed one.) A new-born child independently receives one eye gene from each of its parents, and the gene it receives from a parent is equally likely to be either of the two eye genes of that parent. It is found that, $P(B) = 0.08$. We have 150 brown eyed and 14 blue eyed people in a sample drawn from a population. Can we test whether the locus is at Hardy-Weinberg equilibrium? Draw statistical inference from the data.

6.13 Suppose at a tri-allelic locus data are collected on the number of people having different genotypes. Assuming all possible genotypes that can be formed by three alleles, are present in the data, how do you estimate allele frequencies? Also calculate the expectation and variance of your estimators.

6.14 Suppose the p-values for testing odds ratio in three situations are calculated based on the asymptotic distribution of logarithm of sample odds ratio. Let p-values thus obtained be 0.000031, 0.053, and 0.521. How do you interpret each case, assuming that the level of significance is taken as 0.01.

6.15 In Question 6.14, suppose these p-values are calculated for three genetic loci. What do you do to preserve the maximum type I error rate at 0.01? Interpret your result.

6.16 A case-control study on smoking status and oral cancer reveals that there are 446 cases who are smokers whereas 179 controls who are also smokers. Among non-smokers, these numbers are 13 and 29 respectively. Calculate odds ratio and comment on your result.

6.17 Later some more information becomes available with respect to the problem in Question 6.16. The data can further be classified based on alcohol use, as given below. Calculate odds ratio for each of the two groups based on alcohol

use. Comment on your result combining findings with that in Question 6.16. You may calculate other statistical quantities and/or adopt some other methods to get a comprehensive view.

| | Alcohol users | | non-alcohol users | |
|---|---|---|---|---|
| | Cases | Controls | Cases | Controls |
| Smokers | 201 | 111 | 245 | 68 |
| Non-smokers | 8 | 9 | 5 | 20 |

6.18 What is a matched case-control study design and why is matching important in the context of identifying the association of a genetic marker with a disease?

6.19 For the data in Table 6.1, do the following.

(a) Assume that the disease is recessive, i.e. caused only if a person has two copies of disease alleles at a locus. Estimate the minor allele frequency at each locus based on case-control status data only.
(b) Check HWE at each locus.
(c) Test whether any locus is associated with the affectedness of people using genotype data and affection status. Do not forget to do multiple testing correction before you reach your conclusion.
(d) Using the data on cholesterol, check whether any locus is associated with cholesterol level.
(e) Suppose it is known that the cholesterol data do not follow a normal distribution. How do you test for association using cholesterol data at each locus?
(f) Divide the entire sample into two groups: those having age less than or equal to 50 years, and those above 50 years of age. Find the odds ratio for disease for the older group with respect to younger group. What is the expectation and variance of this odds ratio? Can you devise a statistical test to conclude that odds ratio is equal to 1.5? If so, please do it; otherwise, justify your answer.
(g) Address the question in 6.19(c), but in presence of covariate age. What is your conclusion?
(h) Address the same question in 6.19(d) in presence of age. What is your conclusion?
(i) Apply KBAT and SKAT tests to see whether the loci G1, G2, G3, and G4 are associated with disease, using phenotype data as case-control status.
(j) Write a detailed report on your findings obtained so far.

6.20 We are interested to see whether a quantitative trait is associated with a diallelic locus with genotypes $AA$, $Aa$, and $aa$, where '$a$' is the minor allele. Develop a statistical test for genetic association by writing down the model, the null and alternative hypotheses with justification, the test statistic, and critical region for the test, when we have no knowledge about the mode of inheritance of the alleles. If it is known, in this scenario, that more number of disease alleles

might increase the disease risk, how would you modify your analysis? Which one do you think to be more powerful and why?

6.21 An investigation carried out to study the toxic effects of mercury was described in the article "Comparative Responses of the Action of Different Mercury Compounds on Barley" (International Journal of Environmental Studies [1983]). Ten different concentrations of mercury were compared with respect to their effects on average dry weight (per 100 seven-day-old seedlings). The basic experiment was replicated 4 times for a total of 40 observations. The article reported an ANOVA $F$ statistic of 1.895. Using a significance level of 0.05, test the hypothesis that the true mean dry weight is the same for all 10 concentration levels.

6.22 We have three genotypes: $AA$, $Aa$ and $aa$. After giving a particular treatment, we measured the inflammatory response of 30 individuals divided into the above 3 genotype groups by the individual levels of a particular pro-inflammatory cytokine(pg/mL).

| | |
|---|---|
| $AA$ | 12.5, 13.2, 17.8, 15.2, 14.1, 14.9, 14.7, 13.1, 13.5, 11.9, 12.4, 13.8 |
| $Aa$ | 12.9, 14.9, 16.8, 14.1, 13.5, 14.2 |
| $aa$ | 25.8, 21.6, 19.6, 14.8, 23.7, 24.9, 21.3, 19.8, 20.2, 23.8, 26.1,20.9 |

On the basis of the data, can we say that the treatment effect differs among these three genotype groups? Which genotype is associated with a significantly higher mean pro-inflammatory response for the treatment as compared to the other two genotypes?

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# Chapter 7
# Some Extensions of Genetic Association Study




Now that we know how to identify a genetic locus associated with a phenotype, be it qualitative or quantitative, there might arise different situations, each of which demands serious attention. This requires special treatments to search for some answers to the newly arisen questions, starting from development and/or modifying existing methods, to meeting computational challenges, and sometimes, searching for prior information to handle those issues. Each issue comes with its own special characteristics. It needs to be addressed accordingly, in a specific way in tune with the issue. Thus, one unified approach to solve all issues might seem too ambitious. If any such method exists, it needs to be evaluated carefully because there might be some compromise in performance.

As far as the genomic location is concerned, location of one marker may be very close to another marker giving rise to strong correlation between them. When we study association of a locus, we usually ignore the presence of other loci although they might be correlated with the previous one, resulting in loss of information. So it raises a major issue of incorporating this information while studying genetic association with a phenotype. Naturally, the methods that we have already learnt, may not be applicable mainly because here we have to deal with a combination of alleles from two or more different markers that might or might not be correlated. Moreover, degree of this correlation varies.

Although simple inclusion of a covariate in the model might work well to find genetic association after adjustment of covariate effects like effect of age, gender, etc., it might not be enough. We need to assess whether this really gives more reliable results. Few recent studies indicate that there might be some issues that need to be considered; otherwise the power of the test might not always be higher than the test for genetic association without this covariate. More than one covariate situation requires appropriate attention while pursuing the analysis.

In case of qualitative phenotype we usually have case-control data. This kind of data set gives only two categories of phenotype, case or control, affected or unaffected, etc. However, in many diseases like cancer, information is available for more than two phenotype categories. As for cancer, while collecting data, we have the idea of the stage of cancer that the person is suffering from. This leads to more than two

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I. Mukhopadhyay and P. P. Majumder, *Statistical Methods in Human Genetics*,
Indian Statistical Institute Series, https://doi.org/10.1007/978-981-99-3220-7_7

categories, like, normal, stage I, stage II, stage II, stage IV, etc. Hence it is imperative to understand this multi-category data set for phenotype and appropriate statistical methods are required to analyse this type of data.

## 7.1 Generating Questions

In order to appropriately anlayse genetic data, we need to understand the data and its immediate features; what types of variables are there in the data and what should be their roles; how to incorporate every available information in the genetic association study, and many other related issues. These questions would add a new dimension to data analysis.

(1) How do we conceptualise the situation that one genetic marker is related to another?
(2) How do we measure such relationship and incorporate it in the genetic association study?
(3) Can we modify existing methods to incorporate such a situation? Or do we need to develop other statistical method(s) to tackle this issue? If so, what is the solution?
(4) If phenotype data correspond to more than two classes, can we check whether the gene expressions are the same, on an average, for all classes or groups?
(5) Do the data in each group follow a normal distribution?
(6) While applying a statistical method, do the data satisfy the assumptions of that method?
(7) If yes, fine; otherwise can we find a better method?
(8) What happens when data on more than one relevant covariate are collected?
(9) How do we perform genetic association study in presence of such covariates that might influence the phenotype?
(10) How do we study genetic association when there are more than one category in the qualitative phenotype under study?
(11) Should we study all covariates that are available?
(12) How do we do genetic association test with a locus adjusting the effects of covariates?
(13) Do these methods differ depending on the nature of phenotype (qualitative/quantitative)?
(14) Are there any other issues that are involved while adjusting covariate effect in association study?

## 7.2 Haplotype Association

As we have already discussed in Chap. 1, our DNA has several million nucleotide positions at which all individuals do not carry the same nucleotide. These are called DNA variants. Because chromosomes segregate from parents to offspring independently, DNA variants on different chromosomes segregate independently. It is expected that the set of DNA variants should be passed on to the offspring with the chromosomes. But this is not the case always. Even when a set of DNA variants are on the same chromosome, these may not be transmitted together from parents to their offspring. The major reason is that crossing-over events take place between a pair of homologous chromosomes that result in exchange of genetic material between the members of the pair. This exchange often leads to the fragmentation of specific arrangement of the DNA variants. Crossing-over events do not take place between two small portions of homologous chromosomes. Hence, arrangements of DNA variants within the small portions remain undisturbed and the variants get transmitted 'as a block' from a parent to offspring. The combination of DNA variants (alleles) that are transmitted together from a parent to offspring is called a haplotype. Necessarily, the variant loci underlying a haplotype must be very close on a chromosome. However, if two loci on the same chromosome are far away from each other, there is a chance that during meiosis a combination of alleles may be transmitted into offspring that are not present in parents. This can happen due to a cross-over event during meiosis (Fig. 7.1).

### *7.2.1 Linkage Disequilibrium*

When alleles at nearby loci are transmitted together from one generation to another (i.e., co-segregate) more often than is expected by chance, the loci are said to be associated non-randomly or alternatively to be in linkage disequilibrium (LD). Co-segregation is of fundamental importance in positioning a disease locus on the genome. The positions of loci, also called marker loci, that are used in genome-wide studies of diseases are usually known. Thus we know the information that these loci are located on which chromosome, which arm, which cytogenetic band, and at which nucleotide positions. If alleles at a marker locus co-segregate with a disease, we can infer that the disease locus is in proximity to the marker locus, thus enabling the determination of the approximate position of the disease locus. Figure 7.1 provides a schematic of this concept.

Variations in the human genome arise by mutation. Since the mutant allele arises on a chromosome carrying a specific set of alleles at adjacent loci, the mutant allele co-segregates with those alleles initially after it arises. If the newly arisen mutant allele causes a disease, then the disease will co-segregate with many marker loci; hence, it will be easy to position the disease locus. But as this chromosome segregates from one generation to another, cross-over events take place and the strength of association

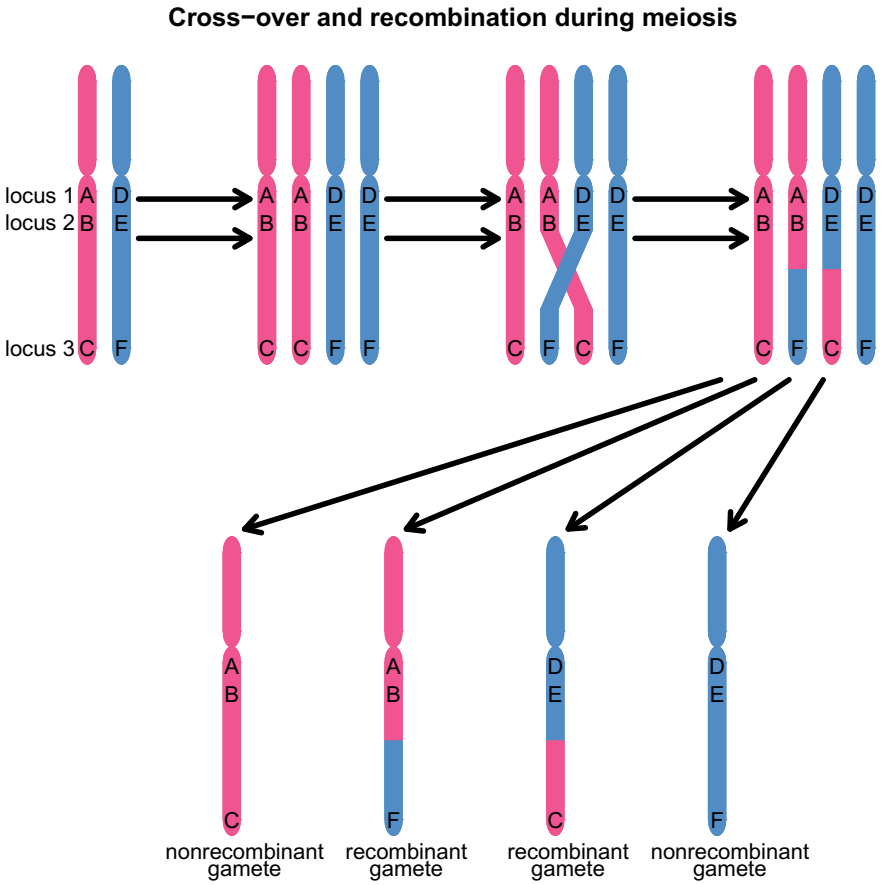



**Fig. 7.1** The loci 1 and 2 are physically close on a pair of homologous chromosomes, while locus 3 is distant from these loci. Recombination events are expected to occur at higher frequency between loci 2 and 3 than between loci 1 and 2. Recombination events produce different combinations of alleles between loci 2 and 3

of the mutant locus with pre-existing variant loci around it decreases. To exemplify, suppose the frequency of recombination between two loci (with alleles $A_1$ and $A_2$ at one locus, and $B_1$ and $B_2$ at the other locus) per generation is $r = 0.1$. Then, an individual who is a double heterozygote $A_1B_1/A_2B_2$ is expected to produce gametes in the following proportions $A_1B_1$, 0.45; $A_2B_2$, 0.45; $A_1B_2$, 0.05; $A_2B_1$, 0.05. Note that without recombination, the gametes $A_1B_2$ and $A_2B_1$ are not expected to be produced by this individual.

After many generations of evolution, the strength of linkage disequilibrium of the mutant locus with an adjacent locus diminishes to a level such that the two loci start to behave as near-independent loci (Fig. 7.2). In humans, many disease-causing variants have arisen newly. It is essential to determine the position of a locus on the

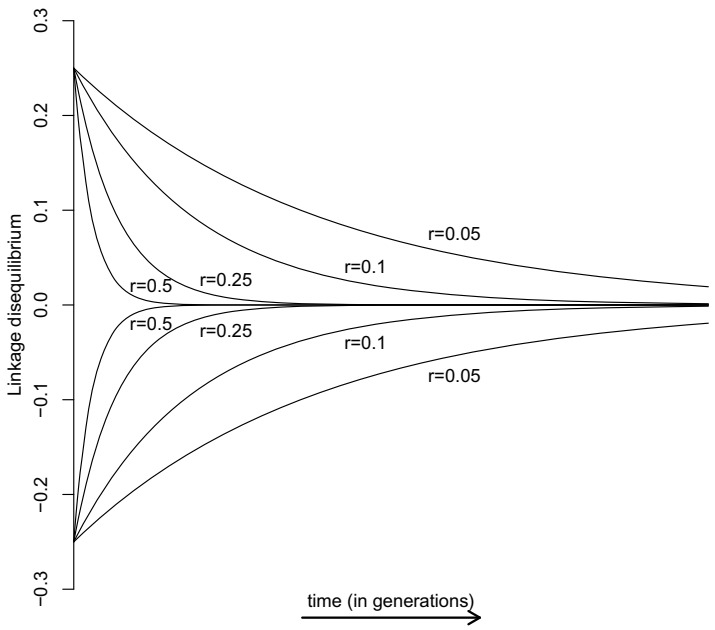



**Fig. 7.2** Behaviour of linkage disequilibrium over time (generations)

genome. This is known as 'mapping'. This mapping is usually done by exploiting the association of the disease with alleles at loci whose genomic locations are known, which helps in knowing the position of the locus that causes the disease.

In fact, the concept and the statistical expectations of the behaviour of LD over generations form the basis of case-control study of a disease for mapping the disease locus. Although we describe a case-control association study in greater detail elsewhere in this book, the basis is pictorially depicted in Fig. 7.3.

Linkage disequilibrium between two loci, or the non-random association of alleles at the two loci, can be measured in various ways. Consider two loci with alleles $(A, a)$ and $(B, b)$. In the following table (Table 7.1), $g(-+)$ denotes the relative frequency of the gamete $(-+)$.

If $g(AB) = p \times s$, $g(Ab) = p \times t$, $g(aB) = q \times s$, and $g(ab) = q \times t$, then there is no association between the alleles at the two loci. Any significant deviation implies that there is an association. This provides a natural measure $(D)$ of linkage disequilibrium (LD):

$$D = g(AB) - (p \times s) \text{ or } D = g(ab) - (q \times t).$$

If the two loci are not in LD, i.e. they are in *linkage equilibrium*, then $D = 0$. LD between two loci, as mentioned earlier, declines with the passage of generations since recombination events occur between the loci. If $r$ denotes the probability of recombination between two loci, then:

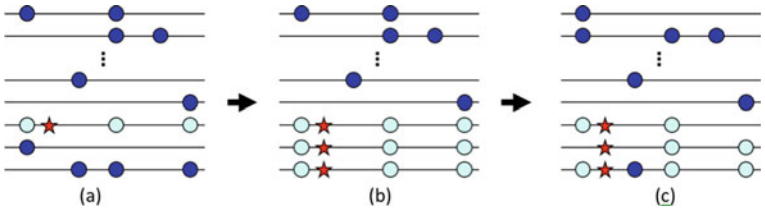



**Fig. 7.3** Each line represents a chromosome and each solid circle represents a locus on that chromosome. Different colours represent different alleles. **a** The red star is a mutation that arises on a chromosome which is a specific combination of alleles at different loci. The red star mutation causes a disease and hence is found only in individuals with the disease. **b** For some generations, all individuals with the disease (cases, in a case-control association study) will have the specific combination of alleles 'the haplotype' on which the disease-causing mutation had arisen; this combination of alleles will be absent among individuals without disease (controls, in a case-control study). **c** With the passage of generations, recombination events will start taking place between loci, and hence the disease-causing mutation will also be observed on other haplotypes. In other words, the strength of association of the disease-causing allele with the haplotype on which it had arisen will weaken

**Table 7.1** Alleles at two loci, combinations of alleles in gametes and their relative frequencies

| | $B$ | $b$ | |
|---|---|---|---|
| $A$ | $g(AB)$ | $g(Ab)$ | $p$ |
| $a$ | $g(aB)$ | $g(ab)$ | $q$ |
| | $s$ | $t$ | 1 |

$$D_t = (1-r)D_{t-1} = (1-r)^2 D_{t-2} = \cdots = (1-r)^t D_0,$$

where $D_t$ is the LD between the loci in generation $t$.

Note that recombination events are not so frequent and the probability of recombination is much less than the probability of no recombination. Thus $r < (1-r)$ implying $0 < r < 0.5$. Also from the relation $D_t = (1-r)^t D_0$, it is clear that as $t$ increases, LD decreases (Fig. 7.2). Thus after a large number of generations, LD between two loci becomes very small and hence two loci would be independent (nearly). The value of $D$ obviously depends on allele frequencies at the two loci. Therefore, a normalised measure of $D$, $D'$, based on the theoretical maximum and minimum, has been suggested.

When $D \geq 0$, $D' = D/D_{max}$, where $D_{max}$ is the smaller of $pt$ and $qs$.
When $D < 0$, $D' = D/D_{min}$, where $D_{min}$ is the larger of $-pt$ and $-qs$.

Another alternative measure that has been proposed is based on the association between the two loci and is defined as

**Table 7.2** Relative frequencies of alleles and haplotypes with two loci

| | $B$ | $b$ | |
|---|---|---|---|
| $A$ | $g(AB) = 0.6$ | $g(Ab) = 0.1$ | $p = 0.7$ |
| $a$ | $g(aB) = 0.2$ | $g(ab) = 0.1$ | $q = 0.3$ |
| | $s = 0.8$ | $t = 0.2$ | 1 |

**Table 7.3** Observed and expected frequencies when two loci are considered together

| Observed frequencies | | | | | Expected frequencies $(y_{ij})$ | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | BB | Bb | bb | Total | | BB | Bb | bb | Total |
| AA | $n_{11}$ | $n_{12}$ | $n_{13}$ | $n_{10}$ | AA | $g(AB) \times g(AB)$ | $2 \times g(AB) \times g(Ab)$ | $g(Ab) \times g(Ab)$ | $m_{10}$ |
| Aa | $n_{21}$ | $n_{22}$ | $n_{23}$ | $n_{20}$ | Aa | $2 \times g(AB) \times g(aB)$ | $2 \times g(AB) \times g(ab) + 2 \times g(aB) \times g(aB)$ | $g(Ab) \times g(ab)$ | $m_{20}$ |
| aa | $n_{31}$ | $n_{32}$ | $n_{33}$ | $n_{30}$ | aa | $g(aB) \times g(ab)$ | $2 \times g(aB) \times g(ab)$ | $g(ab) \times g(ab)$ | $m_{30}$ |
| Total | $n_{10}$ | $n_{20}$ | $n_{30}$ | $n$ | Total | $m_{01}$ | $m_{02}$ | $m_{03}$ | $n$ |

$$r^2 = D^2/(p \times q \times s \times t),$$

the value of which ranges from 0 (no LD) to 1 (complete LD).

**Ex-7.1** Consider two loci with alleles $(A, a)$ and $(B, b)$. Let the proportions of the various haplotypes and alleles be as provided in the Table 7.2.

The values of the various measures of LD from the above data (Table 7.2) will be

$$D = \big(g(AB) \times g(ab)\big) - \big(g(aB) \times g(Ab)\big) = 0.6 \times 0.1 - 0.1 \times 0.2 = 0.04$$
$$D = g(AB) - p \times s = 0.6 - (0.7)(0.8) = 0.60 - 0.56 = 0.04$$

Since $D > 0$, use $D_{max}$. Now $D_{max}$ is the smaller of $pt$ and $qs$. Noting that $pt = 0.14$ and $qs = 0.24$, we have

$$D' = D/D_{max} = 0.04/0.14 = 0.286.$$
$$\text{Finally, } r^2 = \frac{D^2}{p \times q \times s \times t} = \frac{0.04^2}{0.7 \times 0.3 \times 0.8 \times 0.2} = 0.048.$$

### 7.2.2 *Estimation of LD and Other Parameters from Genotype Counts*

When haplotypes are directly observable, i.e. linkage phase is known and hence coupling and repulsion heterozygotes can be separated, then from the equations provided in the earlier section, $D$ can easily be estimated from gamete frequencies. However, when phase is unknown, the underlying pair of haplotypes is ambiguous for the double-heterozygotes. Therefore, one cannot directly measure LD. We describe a procedure to estimate $D$, when both loci $A$ and $B$, with alleles $(A,\ a)$ and $(B,\ b)$, respectively, are codominant. Let $(p,\ q)$ and $(s,\ t)$ be the frequencies of $(A,\ a)$ and $(B,\ b)$, respectively, in a random mating population. Under this framework, there will be nine genotypes whose observed and expected frequencies are denoted, respectively, as in Table 7.3.

From Table 7.3, let $x_{11} = 2n_{11} + n_{12} + n_{21}$, $x_{12} = 2n_{13} + n_{12} + n_{22}$, $x_{21} = 2n_{31} + n_{21} + n_{31}$, $x_{22} = 2n_{33} + n_{23} + n_{32}$, $g(AB) = f_{11}$, $g(aB) = f_{12}$, $g(aB) = f_{21}$, etc.

The log-likelihood function of the observations is

$$\log(L) = \text{ constant } + \sum_{i=1}^{3}\sum_{j=1}^{3} n_{ij} \log y_{ij} \tag{7.1}$$

We use Expectation-Maximisation (EM) algorithm to obtain maximum likelihood estimates of allele frequencies and LD. We provide a brief description of the EM algorithm which is used for obtaining maximum likelihood estimates of parameters when some of the data are missing or latent/unobserved. An example is with respect to ABO blood groups. At the ABO blood group locus, genotypes of individuals of A blood group are latent. An individual with A blood group can either be of the AA genotype or the AO genotype; the actual genotype cannot be known from the blood group. Similarly, the haplotype of an individual with genotypes Aa and Bb at two loci cannot be unambiguously known; the individual may be AB/ab or Ab/aB. In such cases, EM algorithm can be used to estimate allele or haplotype frequencies.

We assume that the complete data set consists of $Z = (X, Y)$, but only $X$ is observed. The complete data log likelihood is then denoted by $L(\theta; X, Y)$ where $\theta$ is the unknown parameter vector for which we wish to find the Maximum Likelihood Estimate (MLE). Basically here we replace the missing quantity by its expectation and then maximise with respect to the parameter $\theta$. This process will continue till the parameter value converges.

E-Step: The E-step of the EM algorithm computes the expected value of $L(\theta; X, Y)$ given the observed data, $X$, and the current parameter estimate, $\theta_C$ say. In particular, we define

$$Q(\theta; \theta_C) = E[L(\theta; X, Y)|X, \theta_C] = \int L(\theta; X, y)p(y|X, \theta_C)dy \tag{7.2}$$

where $p(\cdot|X, \theta_C)$ is the conditional density of $Y$ given the observed data, $X$, and assuming $\theta = \theta_C$.

M-Step: The M-step consists of maximising over $\theta$ the expectation computed in (7.2). That is, we set $\theta_N = \max_\theta Q(\theta; \theta_C)$. Then we set $\theta_C = \theta_N$.

The two steps are repeated as necessary until the sequence of $\theta_N$ converges. Under very general circumstances convergence to a local maximum can be guaranteed (Dempster et al. 1977). It is always a good idea to run the EM algorithm many times, using a different starting value of $\theta_C$ in different runs. The ML estimate of $\theta$ is then taken to be the best (highest likelihood) of the set of local maxima obtained from the various runs of the EM algorithm.

The vast majority of the data that are normally generated in recent human genetic studies pertain to bi-allelic codominant loci, sometimes it is possible that one or both loci are dominant. Estimation in such cases has been described in Hill (1974). Further, if loci are multi-allelic, one may refer to the methodology developed by Schaid (2004).

### *7.2.3 Haplotype Block*

Two seminal papers published in 2001 (Daly et al. 2001; Jeffreys et al. 2001) showed that there were some regions of the human genome in which there was very little evidence of recombination. This evidence was gleaned from the high values of LD between pairs of polymorphic loci in that region. Restriction of recombination in that region meant that the number of haplotypes observed in that region was small. Such a region is referred to as 'haplotype block'. Between a loci on one haplotype block on a chromosome and a locus on the adjacent haplotype block, LD was observed to be high. Thus, haplotype blocks seem to be separated by short regions with many inferred recombination events. Statistical methods have been suggested to identify haplotype blocks (Patil et al. 2001; Gabriel et al. 2002). The block structure of the human genome is intriguing, because it implies that there are short regions in the human genome within which recombination rates are very high, sometimes known as 'recombination hotspots'. These short regions define the boundaries between haplotype blocks. Some experimental evidence has been generated for the existence of recombination hotspots (Badge et al. 2000). These facts do not imply that among loci within a haplotype block recombination events do not take place. Indeed they do take place, but with a low probability. Figure 7.4 provides an idealised structure of haplotype blocks in a segment of the human genome.

This implies that with the passage of evolutionary time and the occurrence of recombination events, a large haplotype block can break up into two or more smaller blocks. Therefore, it is expected that in evolutionarily older ancestral human populations, the sizes of haplotype blocks will be smaller than in populations that have evolved more recent. Indeed, this has been observed in Africa, where humankind arose. Fewer and smaller multisite haplotype blocks are usually observed in African populations compared to other populations (Gu et al. 2007).

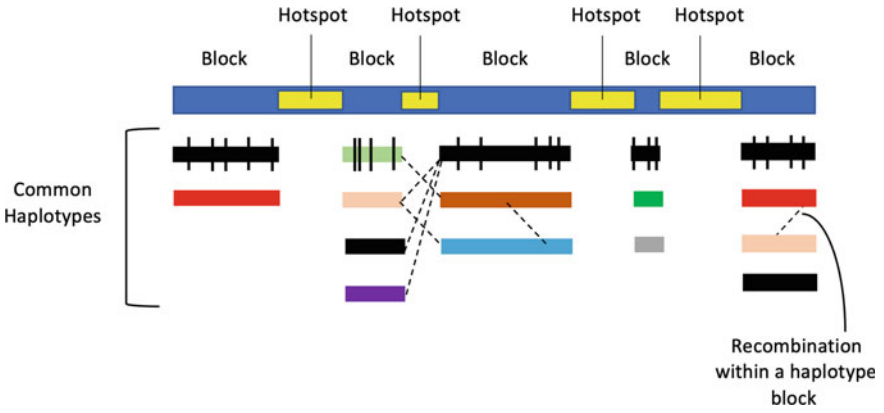



**Fig. 7.4** An ideogram of block structure of a region of the human genome is shown in this diagram. Blue segments are those with little or no recombination among markers within those regions. Segments in yellow are with significantly high recombination; designated as hotspots. Physical sizes of the blocks and hotspots can be variable. The number of SNPs in a block (location of each SNP denoted by a short vertical line) can be variable and may not be correlated with block size. Within each block, there may be variable numbers of common haplotypes and a larger number of rare haplotypes. Diversity of haplotypes within each block is depicted by differences in colour. Crossovers between and within blocks are shown as dotted lines

### *7.2.4 Haplotype Determination*

Determination of haplotypes from diploid genotype data poses a challenge. For example, consider three bi-allelic loci on a chromosome. Suppose an individual at these three loci has genotypes CG, TC, and GA. Then, the possible haplotypes can be CTG/GCA or CCG/GTA or CCA/GTG, or CTA/GCG. Thus, haplotypes may not be unique given information on genotypes at a set of loci. The number of possible haplotype combinations increases rapidly with the increase in the number of loci under consideration.

If parental genotypes are available, then the problem of haplotype determination is sometimes simplified. Suppose the genotypes of an individual are AT, AA, and CG at three loci. If the parental genotypes at these loci are available, then it may be possible to uniquely determine the haplotypes of this individual. For example, if the paternal and maternal genotypes are, respectively, (TT, AA, CG) and (TA, AA, CC), then the haplotypes of the individual are AAC/TAG. If, however, the paternal and maternal genotypes were, respectively, (AT, AA, CG) and (AT, AA, CG), then the haplotypes of the individual under consideration cannot be determined.

In general, haplotype determination is impossible. Considerable additional data on extended pedigrees can sometimes help determine haplotypes of an individual, but even that is not guaranteed. Some algorithms for haplotype determination have been proposed (Tapadar et al. 2000).

### 7.2.5 *Haplpotype Phasing*

Haplotype phasing is the resolution of a set of genotypes at multiple loci into the two haplotypes. There are various statistical methods for phasing. We shall outline only the maximum likelihood method via the Expectation-Maximisation (EM) algorithm (Excoffier and Slatkin, 1995).

The data are genotypes of $n$ individuals at a set of $m$ loci. Let the multi-loci genotypes be denoted as $1, 2, \ldots, m$ with unknown probabilities $P_1, P_2, \ldots, P_m$, and whose frequencies among the $n$ individuals as $n_1, n_2, \ldots, n_m$, respectively. Then, the probability of observations on the sample of $n$ individuals is

$$P(\text{sample} \mid P_1, P_2, \ldots, P_m) = \frac{n!}{n_1!n_2!\cdots n_m!} \times P_1^{n_1} \times P_2^{n_2} \times P_m^{n_m},$$

Let $c_j$ be the number of haplotype pairings that will lead to the multi-loci genotype $j$. Obviously, $c_j$ depends on the number of heterozygous loci in the multi-loci genotype $j$. If we assume that the observations are from individuals belonging to a random mating population, then we can write:

$$P_j = \sum_{i=1}^{c_j} P(\text{genotype } i) = \sum_{i=1}^{c_j} P(h_k h_l),$$

where $P(h_k h_l)$ is the probability that the $i$-th multi-loci genotype is made up of haplotypes $k$ and $l$. If we denote $p_k$ and $p_l$ as the frequencies of the haplotypes $k$ and $l$ in the randomly mating population, then $P(h_k h_k) = p_k^2$ if $k = l$, and $P(h_k h_l) = 2p_k p_l$, if $k$ and $l$ are different. We can now write the likelihood of the observations on the multi-loci genotype counts as

$$L(p_1, p_2, \ldots, p_h) = a_1 \prod_{j=1}^{m} \left( \sum_{i=1}^{c_j} P(h_k h_l) \right)^{n_j},$$

where $p_h = 1 - p_1 - p_2 - \cdots - p_{h-1}$, and $a_1$ is the multinomial coefficient which is a constant.

We now use the EM algorithm to estimate the haplotype frequencies. We start with some arbitrary initial values $p_1^{(0)}, p_2^{(0)}, \ldots, p_k^{(0)}$ to estimate the genotype frequencies $P(h_k h_l)^{(0)}$. This is the E (expectation step). We use,

$$P(h_k h_l \text{ in multi-loci genotype } j)^{(0)} = P_j(h_k h_l)^{(0)} = c_j^{-1}, \text{ for all } j.$$

In the $g$-th iteration, we use

$$P(h_k h_l)^{(g)} = \frac{n_j}{n} . \frac{P_j(h_k h_l)^{(g)}}{P_j^{(g)}}.$$

**Table 7.4** Haplotype frequency distribution among cases and controls. $H_1, \ldots, H_k$ denote the $k$ possible haplotypes. Frequency of haplotype $H_i$ in cases is $n_{1j}$ and that in controls is $n_{2j}$, $j = 1, \ldots, k$

| | $H_1$ | $H_2$ | $\cdots$ | $H_k$ | Total |
|---|---|---|---|---|---|
| Cases | $n_{11}$ | $n_{12}$ | $\cdots$ | $n_{1k}$ | $2n_1$ |
| Controls | $n_{21}$ | $n_{22}$ | $\cdots$ | $n_{2k}$ | $2n_2$ |

Haplotype frequencies are computed at each $M$ (maximisation) step using,

$$\hat{P}_i^{(g+1)} = \frac{1}{2} \sum_{j=1}^{m} \sum_{i=1}^{c_j} \delta_{it} P_j (h_k h_l)^{(g)},$$

where $\delta_{it}$ is an indicator variable equal to the number of times haplotype $t$ is present in multi-loci genotype $i$ $(= 0, 1, 2)$. The iterative procedure is terminated when haplotype frequencies attain stable values.

There are many limitations of this EM algorithm (Excoffier and Slatkin, 1995), the most significant of which is that the performance of the algorithm rapidly diminishes with the increase in the number of polymorphic loci. The assumption of random mating, i.e. the assumption that relative frequencies of haplotypes are those expected under Hardy-Weinberg equilibrium is another serious limitation. This haplotype phasing algorithm and estimation of frequencies of inferred haplotypes have been implemented in the Arlequin software (http://cmpg.unibe.ch/software/arlequin35/).

There are many other methods that have been developed for haplotyping from genotype data on unrelated individuals. It is beyond our scope to discuss these methods and the software packages in which these methods have been implemented. A review of these methods and packages is given in Salem et al. (2005). In recent times, various computational, rather than statistical, methods are being developed for haplotyping (e.g. Tangherloni et al. 2019; and references therein). Additionally, methods have been developed for whole genome haplotyping (Kuleshov et al. 2014).

### *7.2.6 Haplotype Association*

In a case-control study, when the cases and controls are unrelated to each other, a test for association is based on differences in estimated frequencies of haplotypes among cases and controls. This is performed using the chi-squared test of independence in a $2 \times k$ table (Olson & Wijsman 1994) as given in Table 7.4.

The test statistic is

$$\chi^2 = \sum_{i=1}^{k} \frac{(n_{1i} - n_{2i})^2}{(n_{1i} + n_{2i})},$$

which follows a $\chi^2$ distribution (approximately) with $(k-1)$ degrees of freedom.

Here, the genomic region on which the polymorphic loci are located is considered as a single multi-allelic locus, with haplotypes being considered as the alleles. While conceptually this method may be acceptable, when the number of loci increases, the number of haplotypes also rapidly increases resulting in loss of power of the chi-squared test because of a large increase in the degrees of freedom. Further, since in practice the haplotypes are inferred and not actually observed, the uncertainty in phase inference impacts on the assessment of the overall significance of the association test. When the loci are in strong LD, the impact is expected to be small.

Other approaches for measuring the association of haplotypes with a disease have been proposed. In one of these (Tzeng et al. 2003), the approach was based on the tenet that if haplotypes differ between cases and controls then it is evident that the disease-liability mutation lies in the genomic region of the haplotypes. Tzeng et al. (2003) have proposed tests based on the similarity of haplotypes among cases and controls. Score test and other more sophisticated tests have also been proposed, including simultaneous haplotype phasing and estimation of haplotype association with a disease (Schaid et al. 2002; Lin & Zeng 2006).

## 7.3 Studying Levels of Gene Expression at Various Stages of Cancer

Difference in gene expression levels in two independent populations, like case and control, can be studied or tested through appropriate statistical techniques. These include two-sample $t$-test or Wilcoxon rank sum test depending on whether the gene expression follows normal distribution or not. We can think of a generalisation to this problem.

In many diseases, especially in cancer, the severity of the disease depends on the stage of the disease. For example, stage I cancer patients usually have more survival chances after treatment compared to stage IV patients. It would be of clinical and moral importance if we can identify a few genes that are more prevalent in stage IV than that in stage I patients, in general. An overall idea about these genetic markers might be helpful during screening at the first stage. It could lead to different and more effective treatment protocols that may result in increasing the survival time of a patient.

Here data sets on gene expression consist of independent samples from patients belonging to two or more different groups. With three or more number of groups the analysis methods that we learnt, viz., $t$-test, Wilcoxon test, etc., cannot be directly applied. However, more groups or classes come with new information but not without other types of data analytic challenges. We discuss this issue with the help of a data set that has four groups. We assume that the individuals in a group are a random sample from the population of individuals having the same characteristics as that group. Naturally it is rational to assume that the individuals in different groups are

**Table 7.5** Data of gene expression for a gene for person at different stages of cancer

| Normal | Stage I | Stage II | Stage IV |
|---|---|---|---|
| 1.22 | −1.9 | 3.82 | 2.71 |
| 1.76 | −0.49 | 0.83 | 3.34 |
| 0 | −0.39 | 2.93 | 5.51 |
| 0.49 | 1.83 | 5.93 | 4.46 |
| 2.06 | −0.1 | 4.15 | 2.47 |
| 0.22 | 3.41 | 4.85 | −0.22 |
| 1.56 | 1.53 | 3.72 | 1.7 |
| 2.52 | 1.05 | 1.59 | 4.5 |
| 1.97 | 3.61 | 4.91 | 1.18 |
| −0.46 | −1.65 | 2.47 | 5.45 |
| 1.03 | 2.04 | 3.28 | 4.74 |
| −0.18 | 1.14 | 2.34 | 9.13 |
| 1.61 | | 1.97 | 5.07 |
| 4.01 | | | 1.28 |
| | | | 0.82 |

completely independent; features of an individual in a group do not in any way influence the features of another person in another group. Moreover, since we work with a random sample, there is also within group independence as far as sampled individuals are concerned. We start with data on gene expression of individuals distributed over four groups (Table 7.5). Here we try to mainly address the questions (4)–(7) as described in Sect. 7.1.

### *7.3.1 Testing Means of Multiple Samples*

To begin with, first recall two-sample t-test (Sect. 4.3.6). Here we test whether the population means are equal in two groups when observations in each of the two groups follow a normal distribution with additional assumption that their variances are same. Naturally our current problem is an extension of this standard two-sample t-test, but there is a problem. Taking difference of sample means, as in case of $t$-test, will not work here as we have more than two means, because the number of classes is more than two. This leads to a famous technique, known as 'Analysis of Variance', abbreviated as ANOVA. This compares the means of two or more different groups of independent observations. As a passing remark, this ANOVA technique comes under a more general statistical topic called 'Linear Model'. The name 'linear model' will give an indication of the function used in computation using R.

Taking a cue from two-sample t-test, we assume that the variances of each group are the same and each observation corresponds to a random variable that follows a

normal distribution. Hence we can mathematically represent a model that describes the gene expression of an individual in a particular group. This mathematical representation is not very hard; but once we understand this, the entire downstream analytic method would be very easy. We can think that observations within a group are homogeneous. However, there are some fluctuations in data because of the presence of random error or noise that is inherent in any data set. Thus our model becomes:

$$y_{ij} = \mu_i + e_{ij};\ j = 1, \ldots, n_i;\ i = 1, \ldots, k; \tag{7.3}$$

where $y_{ij}$ is the observation corresponding to $j$-th individual in the $i$-th group, $\mu_i$ is the average or mean expression of $i$-th group, and $e_{ij}$ is the random error component associated with $(i, j)$-th observation, assumed to follow a normal distribution with mean 0 and variance $\sigma^2$. Needless to say that here $y_{ij}$s and hence, $e_{ij}$s are all independent.

In our example, $k = 4$ indicating that there is altogether four groups, the number of observations in $i$-th group being $n_i$, i.e. the number of observations in 1st, 2nd, 3rd, and 4th groups are $n_1$, $n_2$, $n_3$, and $n_4$ respectively.

We want to check, based on the given data, whether the group means are equal to each other, i.e. our hypotheses of interest would be

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k \text{ against } H_1 : \text{ at least one inequality in } H_0$$

Now to test the above hypothesis, we need a test statistic. The mathematical derivation of the test statistic is slightly complicated; we try to develop it using our intuition. Just focus on what happens when $H_0$ is true and what happens when $H_0$ is not true. We can then compare these two situations that would automatically lead to a nice statistic. However, we also need to see whether the statistic developed this way would have a proper known probability distribution so that we can calculate the p-value quickly. Fortunately, this method provides a standard distribution of the test statistic.

Recall interpretations of simple mean and variance. Mean is a measure that represents the data, more precisely a central value around which the data points are clustered. If there were no fluctuations, the mean is the value that would be taken by each data point. Variance is the square of standard deviation, that gives a measure of dispersion. In other words, variance gives an idea about how the data are scattered around the mean. Higher variance indicates that the data are very much scattered from its central value whereas smaller variance ensures that observations are closely located around its mean. We just apply these simple ideas to develop a statistical testing procedure to compare the means of different groups.

First suppose that $H_0$ is true. That means all group means are same. We have already assumed (in our model) that variances are also same. So, we can think that all observations are supposed to come from the same distribution. Hence it is expected that the sample variances for each group are same more or less (except for sampling fluctuations) and this should be same if all groups are merged and considered as one single group. We can combine the total variance of each group to get an idea of total

variance present in the data, under the assumption that $H_0$ is true. This is known as 'Within Sum of Squares', as given by

$$SSW = \sum_{i=1}^{k} \sum_{j=1}^{n_i} \left(y_{ij} - \bar{y}_i\right)^2 \tag{7.4}$$

where $\bar{y}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} y_{ij}$ for $i = 1, \ldots, k$. Since there are $n$ observations but we are estimating $k$ means, the number of independent observations involved, i.e. the degrees of freedom associated with $SSW$ is $n - k$. We may divide $SSW$ by $n - k$ to have an idea of average of such total variation. This measure, known as 'Mean Square Error', is defined as $MSW = SSW/(n - k)$. Sometimes it is also denoted as 'MSE' which stands for 'Mean Square Error'.

Again look at the problem from a different view point. If $H_0$ is true, then each of the group means would be very close to the global mean, i.e. mean of all observations in the data set. So the difference between $\bar{y}_i$ and $\bar{y}$ is very close to zero. Now this difference may be positive or negative so it is better to take the squared difference. Since there $n_i$ observations in $i$-th group, we consider the total such squared difference as

$$SSB = \sum_{i=1}^{k} n_i \left(\bar{y}_i - \bar{y}\right)^2. \tag{7.5}$$

This is known as 'Between Sum of Squares'. Similar to the case of $SSW$, here also we define 'Mean Square due to Between groups' as $MSB = SSB/(k - 1)$, since the number of independent observations involved, i.e. degrees of freedom associated with $SSB$ is $k - 1$.

Now let us look at $SSW$ and $SSB$ more carefully. Clearly, if $H_0$ is true, $MSB$ should be and close to $MSW$. On the other hand, if $H_1$ is true, i.e. if $H_0$ is false, then the groups means are not equal to each other. So the total variation present in each group is large and hence $MSB$ is substantially larger compared to $MSW$. Hence we can propose a test statistic as

$$\mathscr{F} = \frac{MSB}{MSW}.$$

It is expected that $\mathscr{F}$ is small when $H_0$ is true, i.e. group means are equal; but when $H_0$ is false, $\mathscr{F}$ is likely to be large.

Fortunately, under the assumptions of the model, it can be shown that this test statistic $\mathscr{F} \sim F_{k-1,n-k}$ under $H_0$. So we can easily calculate the p-value using an $F$ distribution with degrees of freedom $k - 1$ and $n - k$. Naturally we reject $H_0$ if the p-value is very small compared to the chosen level of significance $\alpha$. Note that we emphasise that the level of significance for the test should be chosen before data collection, keeping in mind the degree of seriousness of type I error.

#### 7.3.1.1 Validating Assumption Underlying ANOVA

Although we have described ANOVA for testing whether average gene expressions are same, at least approximately, for all the groups, we need to see whether the data set under consideration satisfies the assumptions underlying ANOVA. We know how to check whether observations in a data set comes from a normal distribution using Kolmogorov-Smirnov test or Shapiro-Wilk test. So we can easily apply any of these tests to each group to examine whether it follows a normal distribution.

But another important assumption of ANOVA is that the variances of all groups are same. To test this, our hypotheses of interest are

$$H_0 : \sigma_1^2 = \cdots = \sigma_k^2 \text{ against } H_1 : \text{ at least one inequality in } H_0.$$

We are not going into the details of the appropriate test as it is slightly technically difficult. The test statistic for the above $H_0$ is

$$T = \frac{(n-k)\log(S^2) - \sum_{i=1}^{k}(n_i - 1)\log(S_i^2)}{1 + \frac{1}{3(k-1)}\left(\sum_{i=1}^{k}\frac{1}{n_1 - 1} - \frac{1}{n-k}\right)}$$

where $n = \sum_{i=1}^{k} n_i$, $S_i^2 = \frac{1}{n_i - 1}\sum_{j=1}^{n_i}\left(y_{ij} - \bar{y}_i\right)^2$, and $S^2 = \frac{1}{n-k}\sum_{i=1}^{k}(n_i - 1)S_i^2$.

$S^2$ is known as 'pooled variance' based on $k$ groups. This test is known as Bartlett's test for homogeneity of variances. It can be shown that under $H_0$, $T$ follows approximately (asymptotically) $\chi_{k-1}^2$. Hence, for a chosen level of significance $\alpha$, we can calculate the p-value corresponding to the test statistic $T$ using a $\chi^2$ distribution with degrees of freedom $k - 1$. If the p-value is large compared to $\alpha$, we can conclude that there is no reason to believe that the variances are unequal at level of significance $\alpha$ approximately. Note that the level of significance is approximately $\alpha$ as this is based on an approximate test, which in this case is $\chi_{k-1}^2$. Of course, here the total sample size $n$ should be large enough to apply this test.

### *7.3.2 Kruskal-Wallis Test*

Discussion in Sect. 7.3.1.1 raises an important point. If any of the assumptions of ANOVA are violated, what should we do? Is it still okay to go with ANOVA? Strictly speaking, the answer is 'NO'. Instead it would be better if we can find another statistical test that does not require these assumptions. Fortunately one such test exists; however, it is less powerful than ANOVA provided the assumptions of ANOVA are valid. So when they are not valid, we have only one choice; use a nonparametric test instead of standard ANOVA. The most commonly used test is

known as Kruskal-Wallis test. Although we mentioned this test earlier, only for three classes, however, this can be generalised to any number of classes, as discussed in detail below.

Kruskal-Wallis test requires only one assumption: observations are continuous in nature, i.e. they are a random sample from an absolutely continuous random variable. In our case, gene expression values are continuous and hence there is no problem in using this nonparametric test. This test is based on ranks of observations instead of their actual values.

Let $R_{ij}$ be the rank of $y_{ij}$ in the entire data set, where $y_{ij}$ is the $j^{th}$ observation in $i^{th}$ class; $j = 1, \ldots, n_i$; $i = 1, \ldots, k$. First we arrange the data irrespective of the groups they belong to, in increasing order with respect to their values. Define $R_{ij}$ as rank of $y_{ij}$ as the number of observation that are less than or equal to $y_{ij}$ among $n$ observations. So minimum of all observations has rank 1 whereas maximum has rank $n$. The test statistic compares the differences due to ranks in different groups with respect to the total variance of all ranks when the data are taken without considering any groups. Define $\bar{R}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} R_{ij}$ for $i = 1, \ldots, k$ and $\bar{R} = \frac{1}{n} \sum_{i=1}^{k} \sum_{j=1}^{n_i} R_{ij}$ where $n = \sum_{i=1}^{k} n_i$. Kruskal-Wallis test statistic is given by

$$T_{KW} = (n-1) \frac{\sum_{i=1}^{k} n_i \left(\bar{R}_i - \bar{R}\right)^2}{\sum_{i=1}^{k} \sum_{j=1}^{n_i} \left(R_{ij} - \bar{R}\right)^2} \tag{7.6}$$

Although observations are continuous, in a real data set few observations might be same giving rise to confusion in ranking them. Usually if two observations are same, we replace their ranks by the average of their individual ranks if they were unequal. Suppose the ranks of two individuals with respect to their phenotype values (or variable of interest) are both 5. It indicates a tie in ranks. However if there is no tie, their ranks would have been 5 and 6. Hence, we assign an average rank $(5 + 6)/2 = 5.5$ to each individual so the next rank would start from 7. Thus we use tied ranks in case a few observations are equal to each other. However, if there is no tie, simplified expression of the test statistic can be obtained as

$$T_{KW} = \frac{12}{n(n+1)} \sum_{i=1}^{k} n_i \left(\bar{R}_i - \frac{n+1}{2}\right)^2 = \frac{12}{n(n+1)} \sum_{i=1}^{k} n_i \bar{R}_i^2 - 3(n+1) \tag{7.7}$$

It can be shown that when the sample size is large, $T_{KW}$ follows approximately (asymptotically) a $\chi^2$ with $k - 1$ degrees of freedom. We can easily calculate the p-value and take decision accordingly comparing it with the level of significance $\alpha$ approximately.

## 7.4 Covariate Adjustment in a Genetic Association Study

Genetic association study involves genetic data as well as phenotype data with which we seek association of genetic variants. We are usually interested in disease phenotype, although any phenotype can be studied in the same paradigm. Phenotype might be of two types: (1) qualitative and (2) quantitative. For example, a qualitative phenotype might be the status of a disease for a particular individual under study. It indicates only whether the person is affected or unaffected. In most cases if $Y$ denotes the disease status, we designate 0 or 1 value for $Y$ if the person is unaffected or affected respectively. On the other hand, for traits like HDL, blood pressure, etc., it is easy to get the actual value for an individual. If we assume that 120 as a normal value for diastolic blood pressure, 140 and 160 are both indicative of high blood pressure. However, a qualitative phenotype, in this respect, can put both individuals in the same group, i.e. cases or affected individuals. Thus it is clear that quantitative phenotype is usually more informative than qualitative phenotype provided it is available. In many situations, sometimes it is difficult to get the values of quantitative phenotype or in fact, it is difficult or sometimes impossible to define an appropriate quantitative phenotype. In such cases, we have to work with the available qualitative phenotype, where phenotypic discrimination can be done unambiguously (for example, case or control).

Whatever be the situation, utmost care should be taken while collecting data, both genetic and phenotype. Genetic data are collected with a certain amount of precision that depends on the machine that generates data from the genetic material, i.e. DNA or mRNA, which is extracted from blood or other tissue from the person. However, phenotype data are external. Collection of such data does not depend on genetic material or blood or tissue. Often it is collected by external measurement or by interview. For example, we can measure blood pressure using a sphygmomanometer which is a completely non-invasive device. Cardiovascular disease can be identified, at least at a preliminary level, using ECG or other external non-invasive tests. HDL and other cholesterol levels can be measured by a simple blood test. Moreover, when a person is identified with a high possibility of cardiovascular disease, the doctor usually wants to know whether the person is a smoker, her/his food habits, age, gender, etc. These variables that might have some relation with the phenotype are known as ‘covariates’. Note that in most scenarios, data on covariates are easy to collect and these are usually collected, especially when a person with some health issues meets the doctor. Few diseases are only observed when a person reaches a certain age. Thus age of onset is sometimes worth studying while studying the progression of a disease. Thus, it is clear that along with genetic and phenotype data, another set of variables can provide data that might have some influence on phenotype, in fact, on the whole analysis regarding genetic association and subsequent inference. In this section, we shall try to address questions (8)–(14) as described in Sect. 7.1.

### 7.4.1 Example Data Set with Covariates

In Table 7.6, we present a data set with both qualitative and quantitative phenotypes, genotype data at a few genetic loci, and three covariates, viz., age, gender, and smoking status for each individual. For gender, 'M' and 'F' denote male and female respectively where in the smoking status column, '1' means smoker and '0' means non-smoker. Age is given in years although it can be exact age. We consider age as a continuous variable whereas gender and smoking status are categorical variables. The disease status column gives information on whether a person is affected or unaffected with respect to the disease of interest. Here we use '1' to indicate that a person is affected while '0' is indicative of a normal (control) individual. For example, in the context of cardiovascular disease, '1' means that the person is suffering from cardiovascular disease and '0' means the person does not have cardiovascular disease. Quantitative phenotype provides a measurement of a variable that is in some way phenotypic marker of the disease, like low level of HDL might increase the risk of cardiovascular disease. In Table 7.6, we generate a representative data set containing these types of variables. We include genotype data at three genetic loci.

In Table 7.6 G1, G2, and G3 denote the genotype data for all individuals at three loci. If $A$ and $a$ are two alleles at a locus, usually genotypes are denoted by $AA$, $Aa$, $aa$. However, we can recode it as 0, 1, and 2 respectively. This indicates the number of minor alleles in a given genotype. Here, we observe that $a$ is the minor allele and hence $AA$, $Aa$, $aa$ genotypes are coded as 0, 1, and 2 respectively. The same is true for all loci under consideration. 'CC status' presents the case-control status, i.e. whether a person is affected or unaffected and 'Quant Phenotype' column gives the value of a quantitative variable of interest for each individual. Each row represents data for an individual. We assume that each individual is independent of the other because the samples are drawn randomly from the population.

### 7.4.2 Covariate Adjustment for Quantitative Phenotype

We first consider a quantitative phenotype. If we are interested to see any genetic association with the phenotype, we can adopt a regression model or an ANOVA model for analysing the data. Both these models belong to the class of 'linear models' that has enough flexibility to extend itself in various directions. We first explain the use of a linear regression model that can adjust for covariates effects.

Let $y_i$ be the quantitative phenotype value for the $i$-th individual. Let $x_i$ be the value of the covariate, like age for the $i$-th individual. Since we want to adjust the effect of this covariate, we need to include it in the model and develop an analysis protocol so that keeping or isolating the effect of $x$, we can search for association of genetic locus with the phenotype, which is, in this case, quantitative in nature.

Thus we propose our model as

**Table 7.6** Data for case-control status, genotypes at three loci, a quantitative phenotype and age and gender as two covariates

| G1 | G2 | G3 | Age | Smoking status | Gender | CC status | Quant phenotype | G1 | G2 | G3 | Age | Smoking status | Gender | CC status | Quant phenotype |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 2 | 1 | 36 | 1 | 0 | 0 | 49.52 | 0 | 0 | 2 | 41 | 0 | 0 | 0 | 51.63 |
| 0 | 2 | 1 | 50 | 0 | 0 | 1 | 50.72 | 0 | 1 | 2 | 38 | 0 | 0 | 0 | 53.37 |
| 0 | 1 | 2 | 39 | 0 | 0 | 0 | 48.95 | 0 | 1 | 0 | 38 | 0 | 0 | 0 | 54.57 |
| 0 | 2 | 2 | 39 | 1 | 0 | 0 | 48.75 | 0 | 0 | 0 | 46 | 0 | 0 | 1 | 53.87 |
| 0 | 2 | 1 | 41 | 1 | 0 | 0 | 50.68 | 0 | 0 | 0 | 37 | 1 | 0 | 0 | 48.3 |
| 0 | 1 | 2 | 40 | 1 | 0 | 0 | 48.14 | 0 | 2 | 1 | 34 | 1 | 0 | 0 | 52.85 |
| 0 | 1 | 2 | 47 | 1 | 0 | 1 | 52.37 | 0 | 0 | 2 | 35 | 1 | 0 | 0 | 50.4 |
| 1 | 0 | 2 | 46 | 1 | 1 | 1 | 57.57 | 0 | 2 | 0 | 50 | 0 | 0 | 1 | 54.37 |
| 0 | 2 | 0 | 52 | 1 | 0 | 1 | 47.58 | 2 | 0 | 0 | 64 | 1 | 0 | 1 | 70.66 |
| 0 | 1 | 2 | 41 | 0 | 0 | 0 | 52.54 | 0 | 1 | 2 | 39 | 0 | 1 | 0 | 49.05 |
| 0 | 1 | 1 | 38 | 0 | 0 | 0 | 49.45 | 0 | 0 | 1 | 40 | 0 | 0 | 0 | 51.99 |
| 0 | 0 | 1 | 32 | 0 | 0 | 0 | 48.43 | 0 | 1 | 0 | 40 | 1 | 0 | 0 | 55.18 |
| 1 | 1 | 0 | 56 | 1 | 0 | 1 | 53.91 | 0 | 0 | 2 | 32 | 1 | 0 | 0 | 43.27 |
| 1 | 0 | 1 | 51 | 1 | 0 | 1 | 54.39 | 0 | 0 | 1 | 48 | 1 | 0 | 1 | 49.17 |
| 0 | 1 | 0 | 42 | 1 | 0 | 0 | 47.76 | 1 | 0 | 0 | 37 | 0 | 1 | 1 | 52.48 |
| 1 | 1 | 0 | 47 | 1 | 0 | 1 | 54.77 | 0 | 1 | 0 | 34 | 1 | 1 | 0 | 49.48 |
| 0 | 2 | 0 | 41 | 0 | 0 | 0 | 53.38 | 0 | 2 | 0 | 42 | 1 | 0 | 0 | 50.5 |
| 0 | 1 | 2 | 47 | 1 | 0 | 1 | 48.1 | 0 | 0 | 1 | 39 | 0 | 0 | 0 | 51.22 |
| 0 | 2 | 0 | 35 | 0 | 0 | 0 | 45.95 | 0 | 0 | 1 | 38 | 0 | 0 | 0 | 46.53 |

(continued)

**Table 7.6** (continued)

| G1 | G2 | G3 | Age | Smoking status | Gender | CC status | Quant phenotype | G1 | G2 | G3 | Age | Smoking status | Gender | CC status | Quant phenotype |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 2 | 1 | 46 | 0 | 1 | 1 | 51.41 | 0 | 0 | 1 | 37 | 1 | 0 | 0 | 50.12 |
| 0 | 1 | 1 | 41 | 1 | 1 | 0 | 52.57 | 0 | 0 | 2 | 43 | 1 | 0 | 1 | 54.24 |
| 0 | 0 | 2 | 49 | 1 | 0 | 1 | 52.65 | 1 | 1 | 0 | 41 | 0 | 0 | 1 | 53.07 |
| 1 | 0 | 0 | 54 | 0 | 0 | 1 | 54.48 | 0 | 0 | 2 | 32 | 1 | 0 | 0 | 47.67 |
| 0 | 1 | 2 | 39 | 1 | 0 | 0 | 50.25 | 0 | 0 | 2 | 40 | 1 | 0 | 0 | 50.26 |
| 0 | 0 | 2 | 40 | 0 | 0 | 0 | 50.23 | 1 | 0 | 2 | 42 | 1 | 0 | 1 | 57.13 |
| 0 | 0 | 2 | 48 | 0 | 0 | 1 | 47.94 | 0 | 2 | 1 | 42 | 0 | 0 | 0 | 53.39 |
| 0 | 1 | 2 | 50 | 1 | 0 | 1 | 49.32 | 0 | 2 | 1 | 41 | 0 | 0 | 0 | 55.1 |
| 0 | 0 | 0 | 40 | 1 | 0 | 0 | 49.29 | 0 | 0 | 1 | 57 | 1 | 0 | 1 | 53.27 |
| 0 | 0 | 0 | 34 | 1 | 0 | 0 | 51.42 | 0 | 0 | 2 | 37 | 1 | 0 | 0 | 49.1 |
| 0 | 0 | 0 | 53 | 1 | 1 | 1 | 51.01 | 0 | 2 | 0 | 44 | 0 | 0 | 1 | 45.24 |
| 0 | 2 | 0 | 49 | 1 | 0 | 1 | 46.92 | 0 | 0 | 0 | 44 | 0 | 0 | 1 | 52.8 |
| 0 | 0 | 1 | 54 | 1 | 0 | 1 | 48.82 | 0 | 1 | 2 | 40 | 1 | 0 | 0 | 47.68 |
| 0 | 1 | 2 | 30 | 1 | 0 | 0 | 50.42 | 2 | 0 | 0 | 52 | 1 | 1 | 1 | 70.81 |
| 0 | 0 | 0 | 41 | 1 | 0 | 0 | 50.18 | 0 | 1 | 2 | 40 | 0 | 0 | 0 | 51.78 |
| 1 | 1 | 0 | 46 | 1 | 1 | 1 | 52.16 | 0 | 1 | 0 | 53 | 1 | 0 | 1 | 49.08 |
| 1 | 2 | 1 | 42 | 1 | 0 | 1 | 53.82 | 0 | 1 | 1 | 43 | 0 | 0 | 1 | 47.6 |
| 0 | 1 | 1 | 41 | 0 | 0 | 0 | 51.16 | 0 | 1 | 2 | 53 | 1 | 0 | 1 | 48.5 |
| 0 | 0 | 1 | 38 | 1 | 0 | 0 | 52.96 | 0 | 1 | 0 | 54 | 1 | 0 | 1 | 45.72 |
| 0 | 1 | 0 | 41 | 0 | 0 | 0 | 53.41 | 1 | 1 | 0 | 45 | 1 | 0 | 1 | 55.49 |
| 0 | 1 | 1 | 52 | 0 | 0 | 1 | 49.27 | 0 | 0 | 2 | 50 | 1 | 1 | 1 | 48.55 |
| 0 | 2 | 1 | 53 | 0 | 0 | 1 | 50.42 | 0 | 1 | 0 | 52 | 0 | 1 | 1 | 51.29 |
| 0 | 1 | 1 | 39 | 1 | 0 | 0 | 48.08 | 0 | 2 | 2 | 41 | 1 | 0 | 0 | 49.16 |
| 0 | 0 | 0 | 41 | 0 | 0 | 0 | 52.2 | 0 | 1 | 0 | 56 | 1 | 0 | 1 | 51.65 |

(continued)

**Table 7.6** (continued)

| G1 | G2 | G3 | Age | Smoking status | Gender | CC status | Quant phenotype | G1 | G2 | G3 | Age | Smoking status | Gender | CC status | Quant phenotype |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 2 | 55 | 0 | 0 | 1 | 58.24 | 1 | 0 | 0 | 46 | 0 | 0 | 1 | 53.21 |
| 0 | 0 | 0 | 58 | 0 | 0 | 1 | 56.34 | 0 | 1 | 1 | 40 | 0 | 0 | 0 | 51.93 |
| 0 | 1 | 2 | 41 | 0 | 0 | 0 | 49.38 | 0 | 1 | 1 | 37 | 1 | 0 | 0 | 50.16 |
| 0 | 0 | 1 | 47 | 0 | 0 | 1 | 51.51 | 0 | 1 | 2 | 38 | 0 | 0 | 0 | 49.77 |
| 0 | 1 | 2 | 38 | 1 | 0 | 0 | 46.32 | 0 | 1 | 0 | 47 | 0 | 0 | 1 | 48.94 |
| 0 | 2 | 1 | 45 | 1 | 0 | 1 | 48.98 | 0 | 0 | 0 | 55 | 1 | 0 | 1 | 53.02 |
| 0 | 1 | 1 | 62 | 1 | 0 | 1 | 48.66 | 0 | 1 | 2 | 42 | 0 | 0 | 0 | 50.21 |
| 0 | 1 | 0 | 58 | 1 | 0 | 1 | 52.65 | 2 | 0 | 1 | 54 | 1 | 1 | 1 | 70.72 |
| 0 | 1 | 0 | 51 | 1 | 0 | 1 | 52.4 | 0 | 1 | 0 | 55 | 0 | 0 | 1 | 53.49 |
| 0 | 2 | 0 | 54 | 1 | 0 | 1 | 48 | 1 | 0 | 0 | 51 | 0 | 1 | 1 | 55.73 |
| 1 | 0 | 2 | 37 | 1 | 1 | 1 | 55.43 | 0 | 1 | 0 | 41 | 0 | 0 | 0 | 48.84 |
| 0 | 1 | 0 | 42 | 0 | 0 | 1 | 52.76 | 1 | 2 | 2 | 42 | 0 | 0 | 1 | 52.58 |
| 0 | 1 | 2 | 57 | 1 | 0 | 1 | 46.95 | 0 | 0 | 1 | 45 | 1 | 0 | 1 | 48.19 |
| 0 | 0 | 0 | 51 | 0 | 0 | 1 | 48.6 | 0 | 2 | 0 | 40 | 1 | 0 | 0 | 51.13 |
| 0 | 0 | 0 | 40 | 1 | 0 | 0 | 46.91 | 0 | 0 | 1 | 58 | 1 | 0 | 1 | 49.75 |
| 0 | 0 | 2 | 42 | 1 | 0 | 0 | 51.46 | 0 | 0 | 2 | 52 | 1 | 0 | 1 | 47.76 |
| 0 | 1 | 2 | 40 | 0 | 0 | 0 | 49.93 | 0 | 1 | 0 | 42 | 0 | 0 | 0 | 52.99 |

$$y_{ij} = \mu_i + \beta(x_{ij} - \bar{x}) + e_{ij}, \quad j = 1, \ldots, n_i; i = 1, \ldots, k. \tag{7.8}$$

Here $\mu_i$ is the effect of $i$-th genotype and $\beta$ is the effect of the covariate. We assume that the random error components, i.e. $e_{ij} \sim N(0, \sigma^2)$ for all $i$ and $j$.

Note that this model can also estimate the effect of the covariate along with the association of genetic locus after adjusting for the covariate. To see whether there exists any effect of covariate, we can frame this as a hypothesis testing problem. The relevant hypotheses are

$$H_{01} : \beta = 0 \text{ against } H_{11} : \beta \neq 0.$$

Similarly, to identify any genetic association after adjusting the effect of the covariate, we can also frame another set of hypotheses as

$$H_{02} : \mu_1 = \mu_2 = \mu_3 = 0 \text{ against } H_{12} : \mu_i \neq 0 \text{ for at least one } i = 1, 2, 3..$$

We should first find the estimates of the parameters involved in the model and then only should we proceed to carry out testing. Using our knowledge of least squares method, we can derive the least squares estimates of the parameters as

$$\hat{\mu}_i = \bar{y}_i - \bar{y} - \hat{\beta}(\bar{x}_i - \bar{x}), \ i = 1, 2, 3; \ \hat{\beta} = \frac{\sum_{i=1}^{3}\sum_{j=1}^{n_i}(y_{ij} - \bar{y}_i)(x_{ij} - \bar{x}_i)}{\sum_{i=1}^{3}\sum_{j=1}^{n_i}(x_{ij} - \bar{x}_i)^2}. \tag{7.9}$$

It can be shown that

$$\hat{\beta} \sim N\big(\beta, \sigma^2_{\beta}\big) \text{ where } \sigma^2_{\beta} = \frac{\sigma^2}{\sum_{i=1}^{3}\sum_{j=1}^{n_i}(x_{ij} - \bar{x}_i)^2}$$

Hence a test statistic for testing $H_{0i1}$ can be given as

$$T = \frac{\hat{\beta} - 0}{\sqrt{MSE / \sum_{i=1}^{3}\sum_{j=1}^{n_i}(x_{ij} - \bar{x}_i)^2}} \sim t_{n-k-1} \text{ under } H_{01} \text{ with } k = 3. \tag{7.10}$$

where $MSE = \frac{1}{n-4}\sum_{i=1}^{3}\sum_{j=1}^{n_i}\big(y_{ij} - \hat{\mu}_i - \hat{\beta}(x_{ij} - \bar{x})\big)^2$.

Given a fixed sample size, we can easily calculate the p-value. If this p-value is significantly small compared to the level of significance, we may reject $H_{01}$ concluding that there is a significant effect of the covariate and it is of utmost importance

that we should adjust its effect in order to study the genetic association of the locus of interest with the quantitative phenotype.

We now propose a method for testing $H_{02}$, i.e. for the association of genotype, following the idea of ANOVA but in presence of a covariate. This is known as Analysis of Covariance (ANOCOVA). In this situation, the test statistic proposed is

$$\mathscr{F} = \frac{SSE^* - SSE}{SSE} . \frac{n-k-1}{k-1} \sim F_{k-1,n-k-1} \text{ under } H_{02} \text{ with } k = 3, \qquad (7.11)$$

where $n = n_1 + n_2 + n_3$, $SSE = \sum_{i=1}^{3} \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_i)^2 - \hat{\beta}^2 \sum_{i=1}^{3} \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)^2$, $SSE^* = \sum_{i=1}^{3} \sum_{j=1}^{n_i} (y_{ij} - \bar{y})^2 - \hat{\beta^*}^2 \sum_{i=1}^{3} \sum_{j=1}^{n_i} (x_{ij} - \bar{x})^2$,

$\bar{y}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} y_{ij}$, $\bar{x}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} x_{ij}$, $\bar{y} = \frac{1}{n} \sum_{i=1}^{3} \sum_{j=1}^{n_i} y_{ij}$, $\bar{x} = \frac{1}{n} \sum_{i=1}^{3} \sum_{j=1}^{n_i} x_{ij}$,

$\hat{\beta} = \frac{\sum_{i=1}^{3} \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_i)(x_{ij} - \bar{x}_i)}{\sum_{i=1}^{3} \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)^2}$, $\hat{\beta^*} = \frac{\sum_{i=1}^{3} \sum_{j=1}^{n_i} (y_{ij} - \bar{y})(x_{ij} - \bar{x})}{\sum_{i=1}^{3} \sum_{j=1}^{n_i} (x_{ij} - \bar{x})^2}$.

A very small p-value calculated using the $F$ distribution with 2 and $n-4$ degrees of freedom would indicate the genetic association with the quantitative in presence of a covariate $x$. We can also test whether this covariate needs to be included in the model at all. In such a case we test $H_{01}$ using the statistic as given in (7.10) and a small p-value rejects this null hypothesis in favour of including the covariate in the model. However, if the p-value is much larger than the level of significance, we can conclude that it is not necessary to include the covariate in the model and we can proceed safely with an ANOVA model as discussed in Sect. 6.5.5.

We take this as an example where we write a code that is meant for this analysis without using a given function for ANOCOVA in R. This helps in knowing how to write a code if we require a code that is not available in R. We demonstrate this using the data in Table 7.6. We assume that the data in Table 7.6 are stored in 'a' in R workspace. An user can import the data by using 'read.table()' command.

It is seen that the p-value is extremely small compared to the chosen level of significance 0.05. Hence, we can conclude that even after adjustment of the effect of age, G1 seems to be associated with the phenotype, as revealed from the given data.

```
> head(a)       ## prints first six rows of the data set
> x1<- a$Age[which(a$G1==0)]       ## extracts age values for people with G1=0
> x2<- a$Age[which(a$G1==1)]       ## extracts age values for people with G1=1
> x3<- a$Age[which(a$G1==2)]       ## extracts age values for people with G1=2
> y1<- a$Phenotype[which(a$G1==0)]       ## extracts phenotype values for
                                              people with G1=0
> y2<- a$Phenotype[which(a$G1==1)]       ## extracts phenotype values for
                                              people with G1=1
> y3<- a$Phenotype[which(a$G1==2)]       ## extracts phenotype values for
                                              people with G1=2

> n1<- length(x1)            ## number of individuals with G1=0
> n2<- length(x2)            ## number of individuals with G1=1
> n3<- length(x3)            ## number of individuals with G1=2

> x<- c(x1,x2,x3)       ## merge age values for all individuals in one variable 'x'
> y<- c(y1,y2,y3)       ## merge phenotype values for all individuals in one
                              variable 'y'

>              beta.hat<              -(sum((y1-mean(y1))*(x1-mean(x1)))+sum((y2-
mean(y2))*(x2-mean(x2)))+sum((y3-mean(y3))*(x3-mean(x3))))/(sum((x1-
mean(x1))^2)+sum((x2-mean(x2))^2)+sum((x3-mean(x3))^2))
> beta.1<- cov(x,y)/var(x)

>  SSE<-  sum((y1-mean(y1))^2)+sum((y2-mean(y2))^2)+sum((y3-mean(y3))^2)-
beta.hat*beta.hat*(sum((x1-mean(x1))^2)+sum((x2-mean(x2))^2)+sum((x3-
mean(x3))^2))
> SSE.1<- sum((y-mean(y))^2) - beta.1*beta.1*(sum((x-mean(x))^2))

> F.statistic<- ((SSE-SSE.1)/SSE)*((n-3-1)/(3-1))
## calculates F statistic which is used as test statistic

> pvalue<- pf(F.statistic,2,n1+n2+n3-3,lower.tail=F)          ## calculates p-value
```

Suppose in the same setup it is known that disease risk increases with an increase in the number of disease alleles. If the disease associated allele is not known beforehand, we may assume that it would be the minor allele at the locus, if it is associated with the disease. We have seen that a simple linear regression model would suffice to explore genetic association in such a case but without any covariate. However, this model can be easily extended even when covariates are present. We consider only age as a covariate. Inclusion of multiple covariates would be straightforward.

In this situation, our model becomes

$$y_i = \alpha + \beta g_i + \gamma x_i + e_i, i = 1, \ldots, n$$

where $y_i$, $g_i$ and $x_i$ denote respectively, the quantitative phenotype, genotype value (0,1, or 2 depending on the number of minor alleles at the locus), and covariate value

corresponding to $i$-th individual; total number of individuals in the sample being $n$. $e_i$s are random error component assumed to follow $N(0, \sigma^2)$ independently.

Simple 'lm( )' function in R provides all information and p-value for tesing genetic association $H_0 : \beta = 0$ in presence of a covariate. However, we might be interested to know whether there is any effect of a covariate at all, i.e. how to test $H_0 : \gamma = 0$. The p-value for this test is also available from summary of 'lm( )' function, based on which we can take our decision.

### *7.4.3 Covariate Adjustment for Qualitative Phenotype*

Adjusting for a single covariate in a genetic association study is usually done by including the covariate in the right-hand side of the model that explains how phenotype is dependent on genotype. In case of quantitative phenotype, this leads to a nice statistical model in the framework of regression analysis. This is nothing but an extension of standard ANOVA technique incorporating the effect of covariate, known as ANOCOVA, or simple inclusion of covariates in a linear regression model, as discussed in the previous section. The same idea can be thought of as an appropriate one, even in case of a qualitative phenotype. We have used a logistic regression model (Sect. 6.5.4) while exploring genetic association using a qualitative phenotype. We see that although a linear model with qualitative or categorical variable as response variable is meaningless, a transformation, in this case a'logit' transformation is linear in explanatory variables. We can easily include covariates on the right-hand side as explanatory variables. Thus our model becomes

$$logit(y_i) = \alpha + \beta g_i + \gamma x_i \tag{7.12}$$

i.e. here we consider the conditional probability of $y_i$ given values of $x_i$ and $g_i$, as

$$P(Y_i = y_i | x_i, g_i) = \left(\frac{e^{\alpha+\beta g_i+\gamma x_i}}{1+e^{\alpha+\beta g_i+\gamma x_i}}\right)^{y_i} \left(\frac{1}{1+e^{\alpha+\beta g_i+\gamma x_i}}\right)^{1-y_i}, \quad i = 1, \ldots, n$$

where $g_i$ and $x_i$ are genotype and covariate value respectively for $i$-th individual, $y_i$ being the affection status taking values 0 or 1 depending on whether $i$-th person is unaffected or affected respectively.

To test the genetic association, our hypothesis of interest would be

$$H_0 : \beta = 0 \text{ against } H_1 : \beta > 0.$$

Note that here we are interested to see whether the genetic locus under consideration increases the disease risk or somehow associated with disease adversely, i.e. whether increase in the number of alleles would increase the disease risk. However, if we are

interested to see whether the genetic locus is in any way associated with disease, be protective or causal, we should consider the alternative hypothesis as $H_1 : \beta \neq 0$.

Smaller p-value in comparison to a chosen level of significance would ensure the existence of genetic association in the presence of a covariate. Using this approach we can also check whether the inclusion of the covariate is at all required, i.e. whether there is any possibility that genetic association gets affected by the effect of a covariate. To do this, we can frame this as another test of the hypothesis problem with the following hypotheses.

$$H_0 : \gamma = 0 \text{ against } H_1 : \gamma \neq 0.$$

The R code for applying this logistic regression includes the p-values for both the hypotheses described above. Hence a look at these p-values would answer our query very easily. It is to be noted that if $H_0 : \gamma = 0$ is accepted, it indicates that the covariate has no major role to play in genetic association with the qualitative phenotype under study. In that case, we need not use any covariate in our model; rather we should use the statistical tests as described in Sect. 6.5.4 without any covariate. It is advisable that we should first check whether there is any significant effect of the covariate on the phenotype. Depending on our conclusion, we can proceed with the appropriate logistic regression model with or without the covariate effect.

Here the covariate $x$ can be categorical (e.g. gender. etc.) or continuous (e.g. age, etc.) in nature. Logistic regression can easily accommodate any type of variable as covariate. Moreover, we can include more than one covariate. However, inclusion of each covariate would increase the number of parameters making estimation of parameters less efficient. Hence, we should not blindly increase the number of covariates; covariates should be chosen judiciously and carefully considering the problem of association. We must have a clear idea about which variables might have an effect on the qualitative phenotype and we should only include them in the model.

```
> head(a)            ## prints first six rows of the data set
> x<- a$Age          ## extracts age for all individuals
> g<- a$G2           ## extracts genotype data at locus G2 for all individuals
> y<- a$CC.Status    ## extracts case-control status for all individuals
> b<- glm(y~ x+g, family="binomial")      ## fits a logistic regression model
> summary(b)

Coefficients:
             Estimate  Std. Error  z value  Pr(> |z|)
(Intercept) -38.0500     9.0292    -4.214   2.51e-05 ***
x             0.9078     0.2189     4.148   3.36e-05 ***
g            -0.4861     0.5000    -0.972      0.331
```

With reference to the data in Table 7.6, suppose we are interested to see whether there exists any association between G2 and the qualitative phenotype, i.e. case-control status in this case, we can adopt a logistic regression model with additional

variable age. R code for this analysis is given. As before, we assume that the entire data is stored in the variable 'a' in R workspace.

The output clearly reveals that age has a significant effect as the p-value corresponding to it is very small, i.e. 0.000034. Hence, we cannot ignore the covariate effect on the phenotype and would perform a genetic association test after adjusting for the covariate effect. However, this result shows a lack of any association signal as the p-value is 0.331, which is very large compared to the chosen level of significance of 0.05. Thus we conclude that after adjusting the effect of age, it seems that there isn't enough evidence in favour of any association signal. Interested readers can do an exercise to see what happens if we use a logistic regression model without including age as a covariate.

### *7.4.4 A Few Issues on Covariate Adjustment*

By now we understand that we can include the effect of a covariate in our linear regression model or logistic regression model while studying genetic association using quantitative or qualitative phenotype, as the case may be. We have just discussed in the previous two sections, how one can modify the model to adjust the effect of covariates and isolate the genetic effect eliminating covariate effect. So the question naturally arises whether we can include more than one covariate in the same model. Of course, we can do, but there is a limitation. We know that in a regression setup we cannot estimate more parameters than the number of observations. Usually, in our genetic association study, the number of sample individuals is much larger than the number of covariates. So it is expected that in this context this kind of problem should not arise.

This approach of including the covariate in the model might also lead to some other problems. Conditioning on covariates does not include information on disease prevalence and non-random ascertainment. This might induce a correlation between candidate variants and known associated variants. There are different sampling schemes to collect data, especially for affected individuals. Sometimes, failure to adjust the mechanism of sampling and some other issues can cause reduction in power of the genetic association test (Kuo & Feingold 2010). In the context of logistic regression, Neuhaus and Kalbfleisch (1998) showed that the power of the test gets decreased if the covariate is correlated to the random variable that is being tested. This standard conditioning on known variants, might significantly reduce power because it induces a dependence structure between associated variants in a case-control data set (Zaitlen et al. 2012). Thus, retrospective sampling, non-availability of covariate specific prevalence information, especially for rare phenotypes, are among the main problems that occur during adjustment of covariates in genetic association using regression or logistic regression models. Moreover, it is really an interesting idea to know whether the known risk variants can be exploited, at least to some extent, in the discovery of new variants that are expected to increase disease risk. With vast amount of information available, appropriate use of prior knowledge in a relevant context

sounds promising. However, the context and relevance need to be carefully justified before using such prior information; otherwise it might induce some confounding effect and give misleading results. We discuss two main approaches to overcome these issues. This involves little more sophisticated statistical techniques and may be omitted during the first reading.

#### 7.4.4.1 Liability Thresholding Model

Zaitlen et al. (2012) proposed a liability thresholding model in the context of logistic regression-based association study when case-control phenotypes are ascertained non-randomly. We assume that there exists a latent random variable, i.e. liability $Z$ (say) that ascertains whether an individual will be termed as a case or a control. We label those individuals as cases if $Z \geq 0$ and controls for those $Z < 0$. This $Z$ remains unobserved. Assuming that $Z$ follows a normal distribution and $F$ is the prevalence of the disease, we can consider our phenotype model as

$$z = -\mu + \varepsilon, \quad \varepsilon \sim N(0, 1), \tag{7.13}$$

where $\mu = \Phi^{-1}(1 - F)$, $\Phi(x)$ being the cumulative distribution function of a normal variable. For the sake of notational simplicity, we omit the use of suffix '$i$' in each variable for $i$-th individual.

Let $s$ be a SNP marker for which we are interested to study the genetic association with the phenotype, i.e. case/control status in this case. The mean adjusted genotypes, i.e. $g$ can take values $-2p$, $1 - 2p$, and $2 - 2p$. Introducing the genotypes, our model becomes

$$z = -\mu + \beta g + \varepsilon, \text{ where } \varepsilon \sim N(0, \sqrt{1 - Var(\beta g)}). \tag{7.14}$$

This leads to our basic assumption, i.e. $Var(Z) = 1$, as assumed in the model.

Note that here the discussion is not confined to adjusting covariates like age, gender, etc., but rather an important issue is addressed. There might exist a covariate or a latent variable that gives rise to the case-control status ascertainment. Thus the main focus is to control such latent variable that happens mainly due to retrospective sampling.

Now our main job is to estimate $\beta$, which might give an idea about the possible association of the variant with the phenotype. However, we need to know the value of $F$, the disease prevalence. Fortunately the value of $F$ is known from prior knowledge and available from literature. At this point, we should be careful that this prior knowledge about $F$ would be applicable for the population under study.

Let $p^+$ and $p^-$ be the minor allele frequencies for $g$ in cases and controls respectively. Assume that we know the ascertainment probability $F$ apriori. Under this circumstance, a better estimate of minor allele frequency (MAF) is given by

$$\hat{p} = p^+ F + p^-(1 - F).$$

Let $\sigma_{\varepsilon}^2 = 1 - Var(\beta g)$. If $\hat{\beta}$ is an estimate of $\beta$, the effect size for the marker under consideration, we have

$$P(Y = 1|g = 0) = \left(1 - \Phi(\mu, \hat{\beta}(-2\hat{p}), \sigma_{\varepsilon}^2)\right) \tag{7.15}$$

$$P(Y = 1|g = 1) = \left(1 - \Phi(\mu, \hat{\beta}(1 - 2\hat{p}), \sigma_{\varepsilon}^2)\right) \tag{7.16}$$

$$P(Y = 1|g = 2) = \left(1 - \Phi(\mu, \hat{\beta}(2 - 2\hat{p}), \sigma_{\varepsilon}^2)\right) \tag{7.17}$$

where $\Phi(x, \mu, \sigma^2)$ is the cumulative distribution function of a normal variable with mean $\mu$ and variance $\sigma^2$. Then given the phenotype values, i.e. considering the fact that $Y = 1$ for case and $Y = 0$ for control, and assuming Hardy-Weinberg equilibrium for genotype frequencies, we have the posterior probability of a genotype when the phenotype status is known, as

$$P(g = 0|Y = 1) = \frac{P(Y = 1|g = 0)P(g = 0)}{P(Y = 1)} = \frac{P(Y = 1|g = 0)(1 - \hat{p})^2}{F} \tag{7.18}$$

$$P(g = 1|Y = 1) = \frac{P(Y = 1|g = 1)2\hat{p}(1 - \hat{p})}{F} \tag{7.19}$$

$$P(g = 2|Y = 1) = \frac{P(Y = 1|g = 2)\hat{p}^2}{F} \tag{7.20}$$

Hence, the MAF among cases is given by

$$\hat{p}^+ = P(g = 1|Y = 1) + 2P(g = 2|Y = 1). \tag{7.21}$$

Proceeding similarly and considering data for controls, i.e. putting $Y = 0$, we can easily calculate the MAF at $g$ for controls. Note that the calculation up to this point depends entirely on the initial estimate $\hat{p}$. However, there is no guarantee that it is good, in some sense of statistical theory. In order to provide a reasonably good estimate, we should use some optimisation method and ensure the goodness of our estimate. We adopt a computational approach coupled by statistical argument.

Consider the squared error between observed and expected frequencies in cases and controls. Thus we get a quantity,

$$S_e^2 = (p^+ - \hat{p}^+)^2 + (p^- - \hat{p}^-)^2. \tag{7.22}$$

Clearly $S_e^2$ is a function of $\beta$. Now we adopt a grid search technique and identify $\hat{\beta}$ that minimises $S_e^2$. This estimate of $\beta$ would provide a better idea about the existing association, if any. For a detailed study, please read the paper by Zaitlen et al. (2012).

#### 7.4.4.2 Using Profile Likelihood Function

Although the approach by Zaitlen et al. (2012) is a nice one, however, it fails to accommodate some other problems while adjusting for covariates. If the prevalence information specific to the covariate is not available, this method might not be efficient. To this end, Zhang et al. (2018) proposed a profile likelihood based approach that might be more suitable for matched case-control design. However, using profile likelihood method in a straightforward way leads to certain theoretical as well as numerical problems regarding stability of the estimates of the parameters. This problem is solved by another simple modification of the profile likelihood approach. This method also takes care of different strata present in the population.

Let the variable $S$ denote the stratum from which frequency-matched case-control samples have been drawn. Suppose we randomly draw $n_s$ number of cases and controls from the stratum $S = s, s = 1, \ldots, I$. Assume that $X$ is a non-confounding covariate and $G$ is the genotype at a bi-allelic locus coded as 0, 1, and 2, based on the number of minor alleles. Let $Y$ denote the affection status or disease status with 0 and 1 representing control and case respectively. Then using a logistic model we can write,

$$P(Y_{si} = y_{si}|S = s, X = x_{si}, G = g_{si}) = \left(\frac{e^{\alpha+\beta g_i+\gamma_1 x_{si}+\gamma_2 s_i}}{1+e^{\alpha+\beta g_{si}+\gamma_1 x_{si}+\gamma_2 s_i}}\right)^{y_{si}} \left(\frac{1}{1+e^{\alpha+\beta g_{si}+\gamma_1 x_{si}+\gamma_2 s_i}}\right)^{1-y_{si}} \tag{7.23}$$

where $y_{si}, x_{si}$, and $g_{si}$ are affection status, covariate value and genotype respectively for $i$-the individual in $s$-th stratum ($i = 1, \ldots, n_s; s = 1, \ldots, I$).

Let $\pi_{si} = P(X = x_{si}|S = s)$ be the empirical distribution of $X$ in stratum $s$, and $r_g(p) = P(G = g)$ be genotype frequency of genotype $g$. So we have $r_0(p) = (1 - p)^2, r_1(p) = 2p(1 - p)$ and $r_2(p) = p^2$ assuming that Hardy-Weinberg equilibrium distribution holds in the population of interest. We assume that

$$P(X = x_{si}, G = g|S = s) = \pi_{si} r_g(p),$$

which implies that neither $X$ nor $S$ confounds the association between $Y$ and $G$ indicating that $G$ is independent of $(X, S)$; however, $X$ and $S$ may be correlated. Moreover, stratum specific prevalence is assumed to be known, i.e. we know $f_s = P(Y = 1|S = s)$ apriori, for $s = 1, \ldots, I$.

Assuming HWE holds and independence of gene and covariate, the retrospective likelihood function is given by

$$\prod_{s=1}^{I}\prod_{i=1}^{n_s} P(X_{si} = x_{si}, G_{si} = g_{si}|S = s, Y_{si} = y_{si}) = \prod_{s=1}^{I}\prod_{i=1}^{n_s} \frac{P(X_{si} = x_{si}, G_{si} = g_{si}, Y_{si} = y_{si}|S = s)}{P(Y_{si} = y_{si}|S = s)}$$
$$= \prod_{s=1}^{I}\prod_{i=1}^{n_s} \frac{P(Y_{si} = y_{si}|X_{si} = x_{si}, G_{si} = g_{si})P(X_{si} = x_{si}, G_{si} = g_{si}|S = s)}{P(Y_{si} = y_{si}|S = s)} \tag{7.24}$$

Taking logarithm and simplifying, we have the retrospective likelihood function as

$$l(\theta) = \sum_{s=1}^{I}\sum_{i=1}^{n_s} y_{si} \log\left(\frac{e^{\alpha+\beta g_{si}+\gamma_1 x_{si}+\gamma_2 s_i}}{1+e^{\alpha+\beta g_{si}+\gamma_1 x_{si}+\gamma_2 s_i}}\right) + \sum_{s=1}^{I}\sum_{i=1}^{n_s}(1-y_{si})\left(\frac{1}{1+e^{\alpha+\beta g_{si}+\gamma_1 x_{si}+\gamma_2 s_i}}\right)$$
$$+ \sum_{s=1}^{I}\sum_{i=1}^{n_s} \log \pi_{si} - \sum_{s=1}^{I}\sum_{i=1}^{n_s} \log\left\{f_s^{y_{si}}(1-f_s)^{1-y_{si}}\right\} \tag{7.25}$$

Although Eq. (7.25) looks like a simple log-likelihood function, it is subject to several restrictions making it difficult to handle mathematically. Another way to bypass this issue is to use profile likelihood function; however this leads to saddle point issue that makes it further difficult to maximise. Thus, Zhang et al. (2018) proposed a score function that can provide consistent estimates of the parameters of the likelihood function. We avoid the detailed discussion as this is mathematically more difficult; interested readers may refer to the paper by Zhang et al. (2018).

## 7.5 Points to Remember

So far this chapter contained some interesting discussion and statistical methods on how to tackle some issues that arise during genetic association study. Of course, when multiple loci are present, one can do multi-loci test like KBAT or SKAT. However, if the loci are close to each other, there is a chance that alleles at these loci may be transmitted together as a block from one generation to another. In such a case, along with genetic association, one should also look whether a block of DNA is being transmitted from parent to offspring that causes the disease already present in the previous generation. So not only genetic association at a single locus is important, it is worth investigating haplotype association also. However, if the number of haplotypes is large, we have to find some way to group them together or a very large sample would be required. Depending on the practical situation, we have to take a decision.

When more than one class is available, for example, in studying gene expression data for different cancer stages, we need to do ANOVA or ANOCOVA provided any covariate is absent or present, respectively. However, after the analysis, if the null hypothesis is rejected indicating that gene expressions are different over stages of cancer, we need to investigate further. It is extremely important to know which stage is actually responsible for rejection of null hypothesis. Hence, after the analysis using ANOVA or ANOCOVA technique, we have to do pairwise comparison using a two-sample $t$-test. To compare the gene expression between Stage II and Stage III, this is just a two-sample problem as the samples are collected independent of each other. Similarly for any pair of stages, we can proceed exactly in the same way. Care should be taken at the very beginning; we have to first test the assumptions underlying the ANOVA technique and then only do we decide which test should be adopted. If normality assumption or more strictly, equal variance assumption is

violated, we cannot do ANOVA but can do Kruskal-Wallis test easily and then do a pairwise comparison using Wilcoxon test. Blindly doing ANOVA in comparing more than two means sometimes leads to wrong conclusion.

Covariate adjustment is very important in the context of genetic association analysis. Very often we collect data on covariates without putting too much effort or resources. Hence, it would be wise to use this data when studying genetic association. As discussed, we should be careful about the pros and cons of the method used for such analysis and accordingly choose the appropriate method. Note that situation might be different depending on how we collect data, sample identification and collection mechanism, which population is under study, availability of prior knowledge about certain parameters, etc. So, there is nothing called 'best method' that would be applicable in all scenarios. This is still a fertile area of research and a few more new genetic association methods addressing several issues might come up in future.

**Exercise**

7.1 Suppose a genomic region has four loci, each bi-allelic in nature. Assuming Hardy-Weinberg equilibrium at each of the four loci, find out all possible haplotypes and their corresponding frequencies. Now suppose that haplotypes with at least three minor alleles are extremely rare in the population and hence can be ignored. How do you modify your answer?

7.2 Using the following table of haplotype frequencies for two loci A (with alleles A1 and A2) and B (with alleles B1 and B2), determine whether the two loci are in linkage disequilibrium.

| Haplotype | Frequency |
|---|---|
| A1B1 | 756 |
| A2B2 | 723 |
| A2B1 | 123 |
| A1B2 | 127 |

7.3 Generate Fig. 7.2 for different values of $r$ with different starting values of $D_0$, where symbols have their usual meanings. To draw this figure, write an R code. This code would generate each line with a different colour that will change automatically if we want to incorporate more values of $r$ and/or $D_0$. [hint: you can use appropriate loop in R code]

7.4 Data in Table 6.1 represents genotypes at four loci and case-control status.

(a) Construct the possible haplotypes from the data.
(b) Do an association test at level of significance 0.01 using these haplotype data.
(c) Write a short report on your findings.

7.5 Calculate the linkage disequilibrium for the following genotype data at two codominant loci.

| | $BB$ | $Bb$ | $bb$ | Total |
|---|---|---|---|---|
| $AA$ | 57 | 140 | 101 | 298 |
| $Aa$ | 39 | 224 | 226 | 489 |
| $aa$ | 3 | 54 | 156 | 213 |
| Total | 99 | 418 | 483 | 1000 |

7.6 Using the data in Table 7.5, perform a statistical test to see whether the variances in different stages are same? Based on your conclusion, suggest your next step to study whether this gene has anything to do with cancer stages. Suppose it is a general idea that variance of gene expression in Stage IV patients is more than that in normal people. How do you do a statistical test to validate this conjecture using data in Table 7.5?

7.7 In Table 7.6, assume that on a chromosomal region, the order of three loci is ($G1$, $G2$,G3). Assume that '1' codes for genotypes $Aa$ and $aA$ with probabilities 0.5 and 0.5. That means if we observe a '1', it would be $Aa$ with probability 0.5 and $aA$ with probability 0.5. Incorporating this uncertainty, construct a set of haplotypes and evaluate their frequencies.

7.8 Based on the result obtained in 7.7, calculate linkage disequilibrium between (i) G1 and G2, and (ii) G2 and G3. We can think of a haplotype association test as well as a multi-loci association test. So,

(a) do a haplotype association test at the level of significance 0.01, and
(b) do a multi-loci association test at the same level of significance.
(c) Interpret your results.

7.9 A research study was conducted to examine the clinical efficacy of a new antidepressant drug. Fifteen depressed patients were randomly assigned to one of three groups: a placebo group, a group that received a low dose of the drug, and a group that received a moderate dose of the drug. After four weeks of treatment, the patients completed a questionnaire. The higher the score, the more depressed the patient. The data are presented below.

| Placebo | Low Dose | Moderate Dose |
|---|---|---|
| 38 | 22 | 14 |
| 47 | 19 | 26 |
| 39 | 8 | 11 |
| 25 | 23 | 18 |
| 42 | 31 | 5 |

(a) What would be the null hypothesis in this study?
(b) What would be the alternate hypothesis?

(c) Is there a significant difference between the groups? [You may assume that the scores are Normally distributed, with the same variance among the three groups.]
(d) If there is a significant difference, where specifically are the differences? Interpret your answer.

7.10 Using the data in Table 7.6, do the following.

(a) Test whether gender and disease status are independent?
(b) Using case-control status as phenotype, do a statistical test to see whether a particular genetic locus is associated in presence of (i) age only, (ii) gender only, and (iii) both covariates age and gender. Do the same for all loci and give your conclusion.
(c) Using quantitative phenotype (Quant Phenotype in Table 7.6), do a statistical test to see whether a particular genetic locus is associated in presence of age and gender. You can assume that the quantitative phenotype follows a normal distribution. Do the same for all loci and give your conclusion.
(d) Based on your conclusion in (b) and (c), can you suggest any modification of your tests? Justify your answer.
(e) Suppose we are interested in genetic locus G2. What is the odds ratio in favour of disease for male and age group above 50 years compared to female and age group below 50 years? Evaluate the algebraic expression first and then estimate its value.

7.11 For the data in Table 7.6, using liability threshold model, estimate the minor allele frequency assuming (a) $F = 0.1$ and (b) $F = 0.4$. Compare these estimates with the situation when there is no ascertainment bias in sampling. Write an R code to do this.

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# Chapter 8
# Exploring Multivariate Data




Moving from one gene to two genes naturally leads to the idea of existence and consideration of multiple genes together. Not only genes, this may be true for phenotypes also. If we suspect that more than two phenotypes need to be jointly considered while studying any genetic association, the same issue arises. If we have expression data for more than two genes, we might study them separately. We can use a similar idea for two or more phenotypes occurring together. Dealing with a single gene or one phenotype at a time seems okay, but there is a high possibility that we might lose the inherent correlation structure among multiple genes or multivariate phenotypes. On the other hand, a statistical way of treating them together seems more realistic; this might retain the correlation structures among genes and is more informative to decipher the genetic architecture. But how do we look at such types of data? Depending on our outlook, questions, and the problem under consideration, there are different ways and various statistical concepts and methods to deal with it.

The idea of correlation among the variables is probably the major reason to consider variables together, i.e. from a multivariate perspective. While discussing correlation, we have studied correlation and regression in the context of two variables occurring together. Similarly in the multivariate set up we might ask the question as how and to what extent a variable is related to other variables. If the variables are related, we can easily calculate pairwise correlation coefficient that gives an idea about the extent of linear relationship between any two variables. However it might so happen that two variables might seem to have a high correlation, but this correlation is only due to their relation with another third variable. This typical situation is very common in multivariate set up but does not naturally arise in case of two correlated variables. Moreover, we might also explore whether we can propose a model that explains how one variable is related to all other variables. In other words, we should explore the possibility of developing predictive models to have an idea on the value of one variable for an individual or item based on available information on other variables. This generates a plethora of questions regarding the interrelationships among variables.

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I. Mukhopadhyay and P. P. Majumder, *Statistical Methods in Human Genetics*,
Indian Statistical Institute Series, https://doi.org/10.1007/978-981-99-3220-7_8

Any data analysis exercise should start with a visual presentation to get an overall idea about patterns in the data. Visualisation of two genes is possible. We can plot the expression values of two genes in two axes in a scatter plot and visualise the relationship between them. For three genes, we can get some visualisation idea as it can be plotted on a three-dimensional plane, although slightly more complicated than a scatter diagram. But what happens when there are more than three genes? There is no way we can visualise them through a diagrammatic presentation. Moreover we know how to analyse two genes together; using bivariate normal distribution or paired t-test depending on the question. But more genes trigger more questions that we try to explore here. What we said about more than two genes, same holds true for phenotypes. Although two or more genes (phenotypes) occur together, each can be considered as a single gene (phenotype). Each component is a random variable. This type of data, with two or usually more than two components (variables), is known as multivariate data. Most important feature of multivariate data in contrast to univariate data is the existence of correlations among its components. These correlations, directly or indirectly, play important roles in multivariate analysis that should not be ignored.

## 8.1 Generating Questions

More genes, more dimensions, more questions. We need to understand the data before planning for analysis.

(1) How do we capture the various possible correlation structure in multivariate data?
(2) How do we develop a model, may be a linear model in the line of linear regression, of one variable that might depend on other variables? If we can develop such a model, how do we justify whether the model is 'good' (in some sense)?
(3) In such a situation, where linear relation (at least to some extent) seems meaningful, can we measure such a relationship?
(4) If two variables are dependent only through a third variable, how do we identify that relation and the degree of this relationship, if it exists?
(5) Data might consist of four or more components and we have only three dimensions in our real life! Can we visualise multivariate data with more than three dimensions? If so, how?
(6) Can we apply the methods or their extensions that we learnt for univariate or bivariate data, especially in the context of genetic association study?
(7) Is there any pattern hidden in the data that we need to understand and extract even if immediate raw data visualisation is not possible?
(8) Data with four components (dimensions) are difficult to handle and impossible to visualise directly. Can we make our life simple by reducing the data to a lower dimension?

We try to explain and address these questions through our intuition, statistical knowledge, and of course, common sense. In multivariate data, we usually have $p\,(>2)$ variables denoted by a vector $\boldsymbol{X}$ with $p$ components or variables $\{X_1, X_2, \ldots, X_p\}$. We write it using a vector notation as

$$\boldsymbol{X} = \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_p \end{pmatrix}.$$

If $r_{ij}$ is the correlation coefficient between $X_i$ and $X_j$ for $i \neq j = 1, \ldots, p$, we can represent the correlation coefficients in the form of a $p \times p$ matrix as

$$R = \begin{pmatrix} r_{11} & r_{12} & r_{13} & \cdots & \cdots & r_{1p} \\ r_{21} & r_{22} & r_{23} & \cdots & \cdots & r_{2p} \\ r_{31} & r_{32} & r_{33} & \cdots & \cdots & r_{3p} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ r_{p1} & r_{p2} & r_{p3} & \cdots & \cdots & r_{pp} \end{pmatrix} = \begin{pmatrix} 1 & r_{12} & r_{13} & \cdots & \cdots & r_{1p} \\ r_{21} & 1 & r_{23} & \cdots & \cdots & r_{2p} \\ r_{31} & r_{32} & 1 & \cdots & \cdots & r_{3p} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ r_{p1} & r_{p2} & r_{p3} & \cdots & \cdots & 1 \end{pmatrix} \text{ since } r_{ii} = 1 \text{ for all } i = 1, \ldots, p.$$

Note that the correlation matrix $R$ is a symmetric matrix as $r_{ij} = r_{ji}$ for all $i, j = 1, \ldots, p$. Naturally $\boldsymbol{x}$ is the observed realisation of $\boldsymbol{X}$ where

$$\boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{pmatrix}.$$

## 8.2 A Multivariate Data Set

We first present a data set (Table 8.1) that contains four variables viz. HDL, LDL, VLDL, and TGL, measures of cholesterol levels in a person. We explain different topics in multivariate analysis using this data set. Although this is a simulated data set, we have kept our attention to standard values that one might observe in a sample from a population. In other words, we have generated these observations mimicking real-life situations. We have mentioned that more than two genes may be coexpressed, i.e. we need to study them together. The same is true for multivariate phenotypes. Here we consider only a multivariate data set that has four-variate phenotype data and a single variable for genotype. Our main focus is to understand different multivariate techniques that might be required while doing analysis when we encounter any such multivariate data. In Table 8.1, although each cholesterol indicator shows integer value, without any loss of generality, we can consider each of them as a continuous

**Table 8.1** Data on different parameters measuring cholesterol level in a few individuals

| HDL | LDL | VLDL | TGL | Genotype | HDL | LDL | VLDL | TGL | Genotype |
|---|---|---|---|---|---|---|---|---|---|
| 49 | 95 | 23 | 146 | 0 | 46 | 107 | 24 | 157 | 0 |
| 45 | 103 | 22 | 150 | 0 | 44 | 107 | 26 | 168 | 1 |
| 44 | 100 | 21 | 155 | 0 | 66 | 58 | 19 | 100 | 0 |
| 63 | 72 | 20 | 120 | 0 | 68 | 65 | 18 | 104 | 1 |
| 50 | 90 | 23 | 148 | 1 | 64 | 66 | 19 | 119 | 0 |
| 47 | 106 | 23 | 148 | 0 | 43 | 105 | 25 | 153 | 1 |
| 66 | 62 | 19 | 110 | 0 | 44 | 106 | 26 | 148 | 1 |
| 63 | 73 | 21 | 109 | 1 | 60 | 75 | 18 | 110 | 2 |
| 62 | 82 | 16 | 117 | 0 | 48 | 106 | 26 | 158 | 0 |
| 68 | 64 | 17 | 107 | 1 | 64 | 60 | 16 | 106 | 0 |
| 65 | 75 | 18 | 102 | 0 | 50 | 113 | 22 | 147 | 0 |
| 68 | 60 | 20 | 117 | 0 | 45 | 92 | 26 | 153 | 0 |
| 45 | 114 | 28 | 162 | 0 | 53 | 98 | 25 | 156 | 1 |
| 64 | 69 | 16 | 100 | 0 | 53 | 96 | 23 | 146 | 1 |
| 52 | 98 | 17 | 151 | 1 | 63 | 79 | 23 | 120 | 0 |
| 61 | 78 | 20 | 130 | 1 | 55 | 91 | 21 | 144 | 0 |
| 65 | 55 | 14 | 93 | 1 | 68 | 55 | 18 | 105 | 1 |
| 47 | 109 | 22 | 161 | 2 | 63 | 68 | 20 | 112 | 1 |
| 61 | 68 | 20 | 115 | 0 | 44 | 101 | 25 | 161 | 1 |
| 65 | 74 | 14 | 92 | 1 | 53 | 96 | 18 | 135 | 0 |

variable, due to the limitation of measuring instruments. However, genotype is a discrete variable indicating the number of minor alleles at a locus of interest.

## 8.3 Multiple Regression

Like linear regression in the case of bivariate data, we might be interested to know how one variable depends on the other variables. Here we mainly focus our attention only to linear regression model. The idea behind such a model is to predict the value of one variable using the values of other variables for an item or individual.

Suppose we want to fit a linear regression model to explain how $x_1$ depends on other variables $x_2, x_3, \ldots, x_p$. Let such a linear regression model be

$$x_{1l} = \beta_0 + \beta_2 x_{2l} + \beta_3 x_{3l} + \cdots + \beta_p x_{pl} + e_l, \quad l = 1, 2, \ldots, n \tag{8.1}$$

where $e_1, e_2, \ldots, e_n$ are errors associated with observations corresponding to $n$ individuals.

We can apply the least squares method to get estimates of the coefficients as $\hat{\beta}_0, \hat{\beta}_2, \ldots, \hat{\beta}_p$. Now using these estimated coefficients, we have the fitted value of $x_1$, denoted by $X_{1\bullet 23\ldots p}$ which is

$$X_{1\bullet 23\ldots,p;l} = \hat{\beta}_0 + \hat{\beta}_2 x_{2l} + \hat{\beta}_3 x_{3l} + \cdots + \hat{\beta}_p x_{pl}, \quad l = 1, 2, \ldots, n \tag{8.2}$$

This is known as multiple regression of $x_1$ on $x_2, x_3, \ldots, x_p$ and $X_{1\bullet 23\ldots,p;l}$ is the predicted (or fitted) value of $x_1$ for $l^{th}$ observation when $x_{2l}, \ldots, x_{pl}$ are known. Here $x_1$ is our study variable or response and $x_2, \ldots, x_p$ are regressors or explanatory variables.

We do not discuss in detail how to derive estimates of the coefficients $\beta_0, \beta_2, \ldots, \beta_p$, because it requires knowledge of matrix algebra and some rigorous mathematical calculations. However, an R code can easily evaluate them. In fact from that R function, we can get fitted values as explained by regressor variables $x_2, \ldots, x_p$. With reference to Table 8.1, if we want to fit TGL on HDL, LDL, and VLDL, i.e. we want to see how TGL depends on other cholesterol parameters through a linear regression model, we fit this directly using in-built R function.

```
> a<-lm(b$TGL~b$HDL+b$LDL+b$VLDL)   ## fitting regression model
                                      using data stored in 'b'
                                      ## output is stored in 'a'
> summary(a)                        ## gives estimates and p-value
                                      for significant coefficients
```

'summary(a)' gives the output of the multiple regression fitted to establish the linear dependence of TGL on HDL, LDL, and VLDL using the data (Table 8.1). The multiple $R^2$, which is square of the multiple correlation coefficient (to be discussed in next section) is 0.9145 indicating that there is a strong effect of HDL, LDL, and VlDL when we study the linear dependence of TGL on these variables. Moreover the extremely small p-value ($< 2.2^{-16}$) is indicative of the presence of a multiple linear relationship of $x_1$ on $x_2, x_3, \ldots, x_p$.

## 8.4 Multiple Correlation Coefficient

Multiple regression provides a linear relation through which $x_1$ is dependent on other variables. However, if we are interested to know how good the fit is, we have to find a measure that indicates the degree of this relationship, i.e. it should measure to what extent $x_1$ is being explained by other variables through the fitted linear regression equation.

$X_{1\bullet 23\ldots p}$ is the fitted value of $x_1$ through a linear regression equation of $x_1$ on $x_2, \ldots, x_p$.This quantity contains the effect of $x_2, \ldots, x_p$ on $x_1$ through the least squares linear regression equation. On the other hand, $x_1$ has the effect of all other

variables along with its own effect. Hence, we propose a measure, known as 'multiple correlation coefficient', as a product moment correlation coefficient between $x_1$ and $X_{1\bullet 23\ldots p}$ is, and is defined as

$$r_{1\bullet 23\ldots p} = \text{correlation coefficient } (x_1, X_{1\bullet 23\ldots p}) = \frac{\text{Cov}(x_1, X_{1\bullet 23\ldots p})}{\sqrt{V(x_1)V(X_{1\bullet 23\ldots p})}}. \tag{8.3}$$

We should note one very interesting phenomenon. Let $\hat{e}_l$ be the value of error that has been associated with $X_{1\bullet 23\ldots p;l}$ and $x_{1l}$ for each $l = 1, \ldots, n$. Note that $\hat{e}$ is uncorrelated with $X_{1\bullet 23\ldots p}$, i.e. $Cov(X_{1\bullet 23\ldots p}, \hat{e}) = 0$. Then, we have

$x_1 = X_{1\bullet 23\ldots p} + \hat{e}$ and hence,

$Cov(x_1, X_{1\bullet 23\ldots p}) = Cov(X_{1\bullet 23\ldots p} + \hat{e}, X_{1\bullet 23\ldots p}) = V(X_{1\bullet 23\ldots p}) + Cov(X_{1\bullet 23\ldots p}, \hat{e}) = V(X_{1\bullet 23\ldots p})$

$$\text{Thus, } r_{1\bullet 23\ldots p} = \frac{\text{Cov}(x_1, X_{1\bullet 23\ldots p})}{\sqrt{V(x_1)V(X_{1\bullet 23\ldots p})}} = \frac{V(X_{1\bullet 23\ldots p})}{\sqrt{V(x_1)V(X_{1\bullet 23\ldots p})}} = \sqrt{\frac{V(X_{1\bullet 23\ldots p})}{V(x_1)}}. \tag{8.4}$$

Thus multiple correlation coefficient, being the square root of ratio of two variances, is always positive. Moreover, being a product moment correlation coefficient, it cannot be greater than one. Hence, we have, $0 \leq r_{1\bullet 23\ldots p} \leq 1$. If $X_{1\bullet 23\ldots p}$ can really explain a substantial portion of $x_1$, this implies that $x_1$ and $X_{1\bullet 23\ldots p}$ have much in common. Hence $r_{1\bullet 23\ldots p}$ should be large. On the other hand, if $X_{1\bullet 23\ldots p}$ is unable to explain or capture a substantial portion of $x_1$, this correlation would be small. Using the correlation matrix $R$, a little bit of algebraic manipulation gives an expression for $r_{1\bullet 23\ldots p}$ as

$$r_{1\bullet 23\ldots p} = \left(1 - \frac{|R|}{R_{11}}\right)^{1/2} \tag{8.5}$$

where $R_{11}$ is the cofactor of $(1, 1)th$ element of the correlation matrix $R$ and $|R|$ is the determinant of $R$. Note that cofactor of $(i, j)th$ element of a matrix $A$ is defined as $(-1)^{i+j}|M_{ij}|$ where $M_{ij}$ is the matrix obtained by deleting $i^{th}$ row and $j^{th}$ column of the matrix $A$.

We can easily write an R code using the formula (8.5) for calculating multiple correlation coefficients. In R programming, the function 'cor( )' gives the correlation coefficient between two lists of observations from the same group of individuals. But if we input a matrix of observations on multiple variables, the same function would calculate correlations among all possible pairs of variables and arrange them in a matrix form. Thus we directly get the correlation matrix using R code.

```
> R<- cor(b[,-5]          ## correlation matrix obtained from data stored
                            in 'b'; we remove 5th column as
                          ## it contains genotype data
> R.11<- R[-1,-1]          ## matrix obtained from R deleting first row and
                            first column
> multiple.correlation<- sqrt(1 - (det(R)/det(R.11)))    ## r_{1•23...p} as
                            given in (8.5); det() calculates
                                        ## determinant of a matrix
```

The output correlation matrix is

$$R = \begin{pmatrix} 1.000 & -0.935 & -0.772 & -0.934 \\ -0.935 & 1.000 & 0.735 & 0.920 \\ -0.772 & 0.735 & 1.000 & 0.825 \\ -0.934 & 0.920 & 0.825 & 1.000 \end{pmatrix}$$

This $R$ matrix provides an idea about all possible pairwise correlation coefficients among $p$ variables. Clearly as HDL increases, LDL decreases because their correlation coefficient is $-0.935$. The multiple correlation coefficient is 0.954 indicating that there seems to exist a strong linear dependence of LDL, VLDL, and TGL on HDL as explained by the multiple linear regression equation.

Multiple correlation coefficient reduces to absolute value of Pearson correlation coefficient for bivariate data. It indicates how much of $x_1$ can be explained through a linear regression model based on other variables. If $r_{1\bullet 23\ldots p}$ is very high, i.e. close to 1, we can say that the multiple regression equation is successful in explaining the response or study variable by other variables through a linear model (equation). However, a small value is indicative of its failure for such an explanation through a linear regression model. Thus, $r_{1\bullet 23\ldots p}$ provides a measure of degree of linear relationship of response with other explanatory variables.

## 8.5 Partial Correlation Coefficient

Multiple regression provides an idea on the dependence of one variable on other variables through a linear relation whereas multiple correlation coefficient measures the extent or degree of the influence of other variables on one variable (here it is $x_1$) as explained through linear regression equation. But one might be interested to know how only two variables are related. Is there any measure that can give an idea about it? This question is in relation to bivariate data where correlation coefficient measures the degree of linear relationship between two variables. This is still a good measure but is unable to take care of the effects of other variables on these two variables. It might so happen that two variables might have a very high correlation coefficient between them, but this correlation is due to the effect of another variable.

In this case, in order to get an idea of the actual relation, we need to eliminate the effect of the third variable.

In mathematical notation, suppose we want to find any possible degree of linear relationship between $x_1$ and $x_2$ but there exist variables $x_3, \ldots, x_p$ that might have some effect on $x_1$ and $x_2$. Especially, a situation might arise where both $x_1$ and $x_2$ are actually uncorrelated but they show a high degree of correlation only because of the fact that both of them are highly influenced by $x_3$. Naturally in such cases it is advisable to first eliminate the effect of the influencing variables, i.e. $x_3, \ldots, x_p$ from each of $x_1$ and $x_2$ and then measure the correlation between them. This correlation measure thus obtained is known as 'partial correlation coefficient' between $x_1$ and $x_2$ eliminating the effects of other variables.

But how do we eliminate the effects of the other variables? Well, we know that multiple regression captures the dependence of one variable on other variables. If we subtract the fitted value as explained by multiple regression equation of $x_1$ on $x_3, \ldots, x_p$ from $x_1$, it contains the effect of $x_2$ and its own effect. Let this be denoted by $e_{1\bullet 34\ldots p} = x_1 - X_{1\bullet 34\ldots p}$ where $X_{1\bullet 34\ldots p}$ is the fitted value of $x_1$, i.e. the portion of $x_1$ that is being explained by $x_3, \ldots, x_p$. Similarly $e_{2\bullet 34\ldots p} = x_2 - X_{2\bullet 34\ldots p}$ contains the effect of $x_2$ on $x_1$ eliminating the effects of other variables from $x_2$. Note that these variables $e_{1\bullet 34\ldots p}$ and $e_{2\bullet 34\ldots p}$ are free from the effect of $x_3, x_4, \ldots, x_p$ as captured through a linear regression equation. Hence, the product moment correlation coefficient (denoted by $r_{12\bullet 3\ldots p}$) between $e_{1\bullet 34\ldots p}$ and $e_{2\bullet 34\ldots p}$ provides a measure of linear relationship between $x_1$ and $x_2$ eliminating effects of $x_3, \ldots, x_p$ as captured by respective linear regression equation and is given by

$$r_{12\bullet 3\ldots p} = \frac{Cov(e_{1\bullet 34\ldots p}, e_{2\bullet 34\ldots p})}{\sqrt{V(e_{1\bullet 34\ldots p})V(e_{2\bullet 34\ldots p})}}. \tag{8.6}$$

Since we already have a correlation matrix, i.e. $R$, we should be able to find an easy expression for partial correlation coefficient that will help us to calculate this quantity directly without evaluating fitted values through multiple regression models. It can be shown that partial correlation coefficient can be expressed as

$$r_{12\bullet 3\ldots p} = -\sqrt{\frac{R_{12}}{R_{11}R_{22}}} \tag{8.7}$$

where $R_{12}$, $R_{11}$ and $R_{22}$ are the cofactors of $(1, 2)th$, $(1, 1)th$ and $(2, 2)th$ elements respectively in the correlation matrix $R$. Partial correlation coefficient being a product moment correlation coefficient always lies between $-1$ and $+1$, i.e. $-1 \leq r_{12\bullet 3\ldots p} \leq 1$. The R code for calculating this is given below.

```
> R.11<- R[-1,-1]          ## matrix obtained from R deleting first row and
                             first column
> R.22<- R[-2,-2]          ## matrix obtained from R deleting second row
                             and second column
> R.12<- R[-1,-2]          ## matrix obtained from R deleting first row and
                             second column
> partial.correlation<- -(-det(R.12))/sqrt(det(R.11) * det(R.22))
                           ## r_{1•23...p} as given in Eq. 8.7.
```

The value of $r_{12\bullet 34\ldots p}$ is $-0.5445$ indicating that there exists a moderately strong linear relation between HDL and LDL levels eliminating the effects of VLDL and TGL based on the given data (Table 8.1).

## 8.6 Principal Component Analysis

The first typical feature of multivariate data is its dimensionality. If there are more than three dimensions, there is no way that we can see the data through simple plots, because we cannot plot anything beyond three dimensions. So it seems that a few standard and unified ideas should be built in order to visualise data and do appropriate data analysis. Two important things to remember: first, it would be great if we can somehow see the data; second, it is always easy if we can analyse uncorrelated or independent variables rather than correlated variables. But multivariate data are usually correlated and they are of higher dimensions preventing us from plotting them on a two-dimensional plane if the dimension of each data point is more than two. However, an intelligent solution exists to solve these issues.

Suppose a data set has $p$ components where $p \geq 3$. It is impossible to plot them if $p > 3$ and even if we plot 3-dimensional data, not much information can be extracted immediately. This triggered an idea: can we do 'something' so that we can reduce the data to a lower dimension, say 2-dimension? If we can, it would be very easy to plot the data immediately. Moreover, while undertaking such a process, can we make the variables free from correlations? Fortunately, in both cases the answer is 'yes'. However, there is one shadow of doubt: is it possible to retain the 'originality' of the data as it was collected from the experiment? Well, not every answer can be 'yes'; we can achieve something but may have to compromise. But first we should try to develop a method that can reduce the data to a lower dimension, at least approximately, in some sense. The most commonly used method that reduces the dimensions of a multivariate data is known as 'Principal Component Analysis' (PCA).

Let's use a few mathematical notations for the sake of simplicity and clarity. Let $X_1, \ldots, X_p$ be the multivariate data with means $\mu_i = E(X_i)$, variances $\sigma_i^2 = V(X_i)$, and covariances $\sigma_{ij} = Cov(X_i, X_j)$ for $i, j = 1, \ldots, p$. Note the correlation coefficient between $X_i$ and $X_j$ is $\rho_{ij} = \sigma_{ij}/\sigma_i\sigma_j$.

Let's represent $X_1, \ldots, X_p$ as a column vector, i.e.

$$\boldsymbol{X} = \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_p \end{pmatrix}$$

Then all possible variances and covariances of the components of $\boldsymbol{X}$ can be represented in a matrix form as

$$\Sigma = V(\boldsymbol{X}) = \begin{pmatrix} \sigma_{11} & \sigma_{12} & \cdots & \cdots & \sigma_{1p} \\ \sigma_{21} & \sigma_{22} & \cdots & \cdots & \sigma_{2p} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \sigma_{p1} & \sigma_{p2} & \cdots & \cdots & \sigma_{pp} \end{pmatrix}$$

Our objective is to reduce $X_1, \ldots, X_p$ to lower dimension variables, say, $Y_1, \ldots, Y_q$ with $q < p$. Moreover, while doing so, it would be great if we can find $Y_1, \ldots, Y_q$ such that correlation coefficient between $Y_i$ and $Y_j$ is 0, for $i \neq j$, although there exists a correlation between $X_i$ and $X_j$. Looks surprising but possible!

Since there are altogether $p$ variables, consider $p$ linear combinations (functions) of $X_1, \ldots, X_p$ as

$$\begin{aligned} Y_1 &= a_{11}X_1 + a_{12}X_2 + \cdots + a_{1p}X_p \\ Y_2 &= a_{21}X_1 + a_{22}X_2 + \cdots + a_{2p}X_p \\ &\vdots \qquad\qquad\qquad \vdots \\ Y_p &= a_{p1}X_1 + a_{p2}X_2 + \cdots + a_{pp}X_p \end{aligned} \tag{8.8}$$

where $a_{ij}$, $i, j = 1, \ldots, p$ are constants such that they have two very interesting properties:

(1) Correlation coefficient $Y_i$ and $Y_j$ is 0 for $i \neq j$, and
(2) $V(Y_1) \geq V(Y_2) \geq \cdots \geq V(Y_p)$

Note that, we can evaluate $V(Y_i)$ for $i = 1, \ldots, p$ and $Cov(Y_i, Y_j)$ for $i \neq j = 1, \ldots, p$ as

$$V(Y_i) = V\Big(a_{i1}X_1 + a_{i2}X_2 + \cdots + a_{ip}X_p\Big) = \sum_{k=1}^{p}\sum_{r=1}^{p} a_{ik}a_{ir}\sigma_{kr}, \tag{8.9}$$

$$\text{and } Cov(Y_i, Y_j) = Cov\Big(\sum_{k=1}^{p} a_{ik}X_k, \sum_{r=1}^{p} a_{jk}X_r\Big) = \sum_{k=1}^{p}\sum_{r=1}^{p} a_{ik}a_{jr}\sigma_{kr}. \tag{8.10}$$

We are not going into mathematical details of derivation and results related to PCA; we would rather prefer to discuss this entire issue more conceptually that might help in actual data analysis using any statistical package or R. First note that each $Y_j$ is a linear combination, i.e. a linear function of $X_1, \ldots, X_p$, where the coefficients $a_{ij}$s are estimated on the basis of the given data. So, although we know the meaning of each $X$ variable, after applying PCA, we might not get a clear idea about the interpretation and meaning of each $Y_j$. This definitely looks like a disadvantage! Don't worry; we have many more advantages that make this method very popular and effective.

It is clear that we have to find the coefficients $a_{11}, a_{12}, \cdots, a_{1p}$ such that variance of the first linear combination would be maximum, i.e. we have to choose $a_{11}, a_{12}, \cdots, a_{1p}$ so that $V(Y_1)$ would be maximum. It can be shown that $V(Y_1) = \lambda_1$ where $\lambda_1$ is the maximum eigen value of $\Sigma$, the dispersion matrix of $(X_1, X_2, \cdots, X_p)$.

Then, we proceed to find the coefficients $a_{21}, a_{22}, \cdots, a_{2p}$ such that $V(Y_2)$ is maximised subject to the restriction $Cov(Y_1, Y_2) = 0$. Fortunately, it can again be shown that $V(Y_2) = \lambda_2$ where $\lambda_2$ is the second largest eigen value of $\Sigma$. Thus, it is possible to derive two uncorrelated variables $Y_1$ and $Y_2$ even when there may exist some correlation between the components of $X$. Continue this way, i.e. maximise $V(Y_i)$ subject to the fact that it remains uncorrelated with all previous $Y_1, Y_2, \ldots, Y_{i-1}$ (for $i = 2, \ldots, p$). This can easily be done. So we have

$$V(Y_i) = \lambda_i, \quad i = 1, \ldots, p \text{ and } Cov(Y_i, Y_j) = 0 \text{ for } i \neq j = 1, \ldots, p.$$

Hence, the dispersion matrix or variance-covariance matrix of $\boldsymbol{Y}$, is given as

$$\Sigma^* = \mathrm{V}(\mathbf{X}) = \begin{pmatrix} \lambda_1 & 0 & \cdots & \cdots & 0 \\ 0 & \lambda_2 & \cdots & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & \cdots & \lambda_p \end{pmatrix}$$

First note that although $X_i$ and $X_j$ are correlated, the transformed variables $Y_i$ and $Y_j$ are uncorrelated for $i \neq j,\ i, j = 1, \ldots, p$. To understand what happens inside this mechanism, when we convert $X$ to $Y$, assume for the sake simplicity, that $p = 2$, i.e. we are dealing with a bivariate situation. If we plot $x_1$ and $x_2$ values in a scatter diagram, this shows the existence of correlation between them. But if we plot $y_1$ and $y_2$, any correlation is clearly absent. $y_1$ and $y_2$ are derived from the original data for each individual. So after PCA we get the transformed variables to be uncorrelated in terms of a new set of variables which are functions of $x$-variables. Here $\{Y_1, \ldots, Y_p\}$ retains the information that is contained in $\{X_1, \ldots, X_p\}$ with additional advantage that they are uncorrelated. If the $X$-variables follow normal distributions, then $Y$-variables become independent also. Thus in many cases, we would be able to work with independent variables derived from originally correlated normal variables. However, PCA works irrespective of whether $X_1, \ldots, X_p$ follow normal distribution or not.

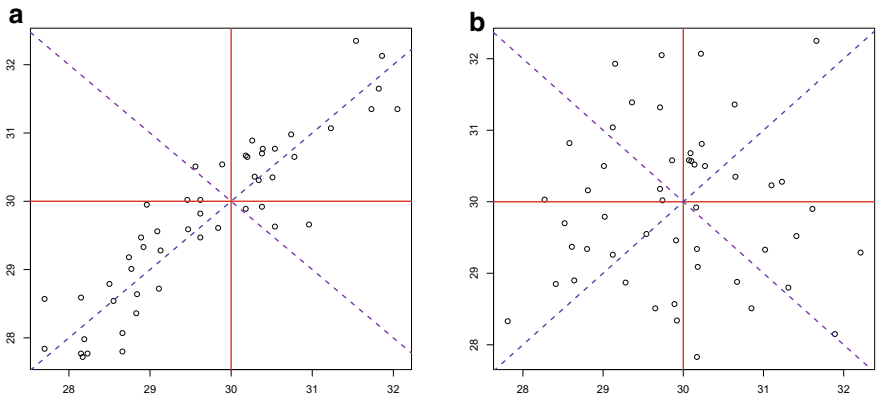



**Fig. 8.1** Geometry of principal component analysis. Red lines, the original axes for X-variables fail to explain substantial variability along each axis. But the dotted lines, the axes for principal components can very well explain variability along it axes. Figure 8.1A: Blue line, indicating PC1, explains much of the variation present in the data whereas the purple line, representing PC2 explains very small amount of variation. This happens because the variables are highly correlated and PC1 explains successfully most of the variations. Figure 8.1B: Blue line and purple lines explain almost the same amount of variation because the original variables are uncorrelated and hence PCA has no advantage in this situation

There is another way to look at the principal components. Again, for ease of understanding, we consider bivariate case and draw a scatter diagram of two variables $x_1$ and $x_2$. If we rotate the axes such that the variation along $Y_1$ axis is much more than that along $Y_2$ axis, although the total variance remains same, i.e. $V(X_1) + V(X_2) = V(Y_1) + V(Y_2)$. Now if we see that $V(Y_1)$ covers almost $80 - 90\%$ of total variance, we can easily work with only $Y_1$ instead of working with two variables without sacrificing much information (variation). This is the biggest advantage of PCA. Thus in a high dimension scenario, if the first few principal components explain a very high amount of variance, we can work with these new variables which are very small in number compared to the original set of variables but contain a very large portion of information or substantial amount of total variability. Thus PCA reduces higher dimension variables into a lower dimension. Rest of the downstream analysis may depend on these few principal components that explain majority of variance.

The geometry of PCA can be explained by a diagram using two variables. Figure 8.1 describes how the scatter looks like when we plot the original variables and the principal components, on the same graph. As explained by Fig. 8.1, PCA should be applied only when the variables provide information through their correlations, but it is not a successful technique when the variables are uncorrelated or independent. The main reason is that when $X_1, \ldots, X_p$ are independent, the derived principal components $Y_1, \ldots, Y_p$ are also independent. Hence, it is impossible to reduce dimensions, i.e. we are unable to find a lesser number of variables that contain majority of variance.

Hence, after doing PCA, we find how many of the principal components explain a large amount (percentage) of variation present in the data as given by $\sum_{i=1}^{p} V(X_i)$. By 'large' we usually take 80% or 90% or any other large value satisfying the user (but not 100%). We take first $k$ principal components such that

$$\frac{V(Y_1) + \cdots + V(Y_k)}{V(Y_1) + \cdots + V(Y_p)} \approx 0.80 \text{ (or } 0.90) \tag{8.11}$$

If we plot the variances, known as 'scree plot', it would give us an idea of the number of principal components that covers a substantially high amount of variation present in the data.

If we want to work with variables that explain the entire variation, i.e. 100% of total variability, we have no option but to work with all principal components, and hence, all actual variables. It is possible to capture the entire variance only if we work with all variables. Thus, in such a situation, PCA has no role to play.

We now present the data analysis of the multivariate data on cholesterol level for 40 individuals as given in Table 8.1. We need not write any rigorous code for doing this. There is an in-built R function that will produce all results of PCA.

```
> a<-read.csv("cholesterol.txt",header=T)          ## reading the datafile
> b<- princomp(a)          ## doing PCA on the data
> summary(b)          ## summary of analysis
Importance of components:
                          Comp.1       Comp.2       Comp.3       Comp.4
Standard
deviation            30.299694    5.80204682   2.53662405  1.940120798
Proportion of
Variance              0.954402    0.03499591   0.00668909  0.003913023
Cumulative
Proportion            0.954402    0.98939789   0.99608698  1.000000000

> screeplot(b)       ## a graphical presentation showing proportion
                     of variance explained.
```

The R command 'screeplot( )' produces Fig. 8.2. It represents vertical bar diagrams showing the contribution of each principal component. Figure 8.2 clearly shows that it is enough to work with only one, i.e. first principal component because it captures more than 95% of the total variance present in the data.

Here the proportion of variance explained by the principal component is

$$\frac{V(Y_1)}{V(Y_1) + V(Y_2) + V(Y_3) + V(Y_4)} = 0.95442 \text{ i.e more than } 95\%.$$

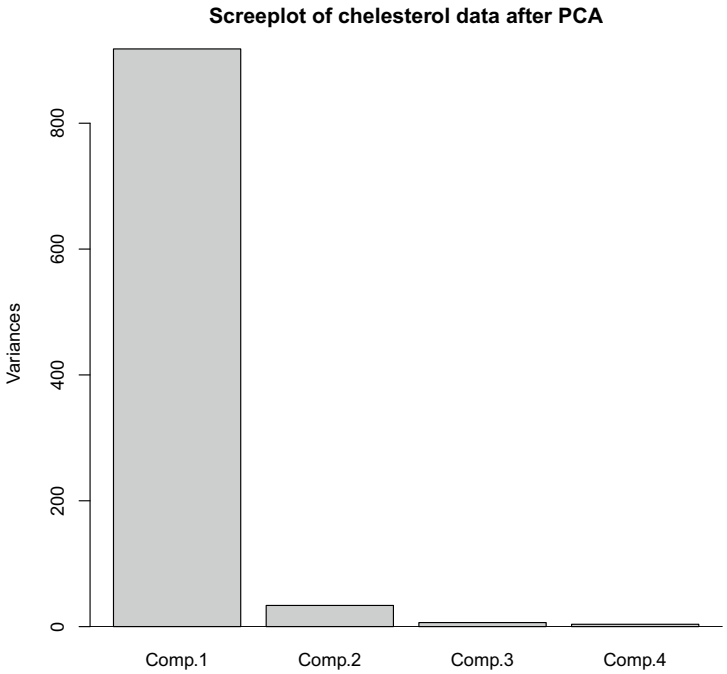



**Fig. 8.2** Scree plot showing proportion of variance explained by different principal components

Hence it is enough to work only with first the principal component for further downstream analysis. This means that any further analysis is nothing but analysis of univariate data considering that information from the data are well contained in the first principal component. In most situations, where variables in multivariate set up are really correlated, we have seen that two (or sometimes three) components can explain more that 80% of total variation. Hence downstream statistical analysis may be done using bivariate (or trivariate) distributions. However, extra care should be taken while interpreting the result obtained through analysis of principal components instead of original variables. Sometimes, it becomes difficult to interpret $Y$-variables, i.e. principal components because they are linear combinations of $X$-components.

**Example: Investigation of population stratification using PCA**

The problem of population stratification in genome-wide association analysis was introduced in Sect. 6.6.2. Existence of cryptic population stratification can adversely impact on the inferences of an association study. It is desirable that existence of population stratification be investigated prior to conducting an association analysis so that the effect of stratification can be controlled to a certain extent. A simple method to detect population stratification is to use genotype data at a large number of polymorphic loci in the set of individuals recruited into the study and extract the first two principal components. Then, based on the first two principal components,

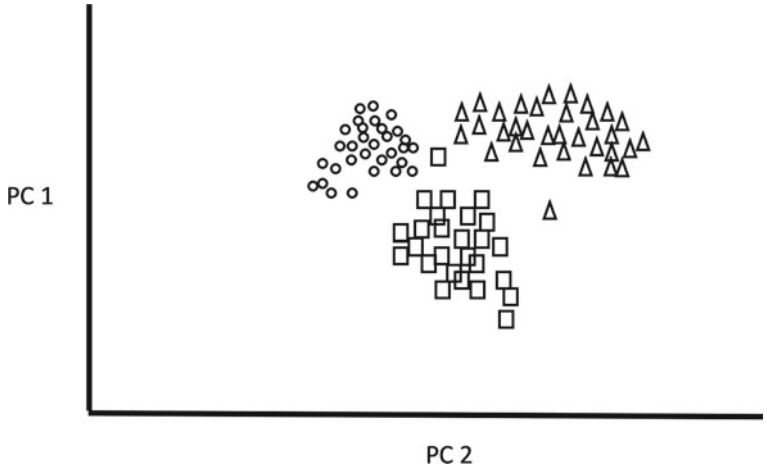



**Fig. 8.3** Scatter diagram of first two PCs for individuals based on genotype data of individuals recruited into a genome-wide association study

we plot the scatter diagram of the data on the individuals. Such a scatter diagram is provided in Fig. 8.3. This figure reveals the existence of a cryptic population stratification; individuals recruited into the study were possibly from three distinct populations.

Price et al. (2006) proposed EIGENSTRAT, which computes principal components for variants across the genome to identify population stratification. If population stratification is detected, then its effect is controlled by including the first few principal components as covariates in the regression framework that relates the trait of interest and the genomic variants (Fig. 8.3).

## 8.7 Cluster Analysis

Although the cholesterol data set has four variables, we can think of, at least to start with, a scatter diagram for a pair of variables. A scatter diagram (Fig. 8.4) of VLDL against HDL reveals some interesting feature.

It clearly reveals that the data points form two or three groups (or clusters), although they may not be very distinct. This idea of grouping or tendency to form different groups in a single data set is the key idea behind cluster analysis. Well, although it looks easy through a scatter diagram, there are some issues that make this an active area of research. We are unable to 'see' data points beyond three dimensions. We need an appropriate method or algorithm, with justification, that may provide these groups or clusters from the data set. This method should be based on sound statistical logic or intuition along with some optimised measure. At the same time, it should retain the characteristics of the data while putting them in respective groups. Note

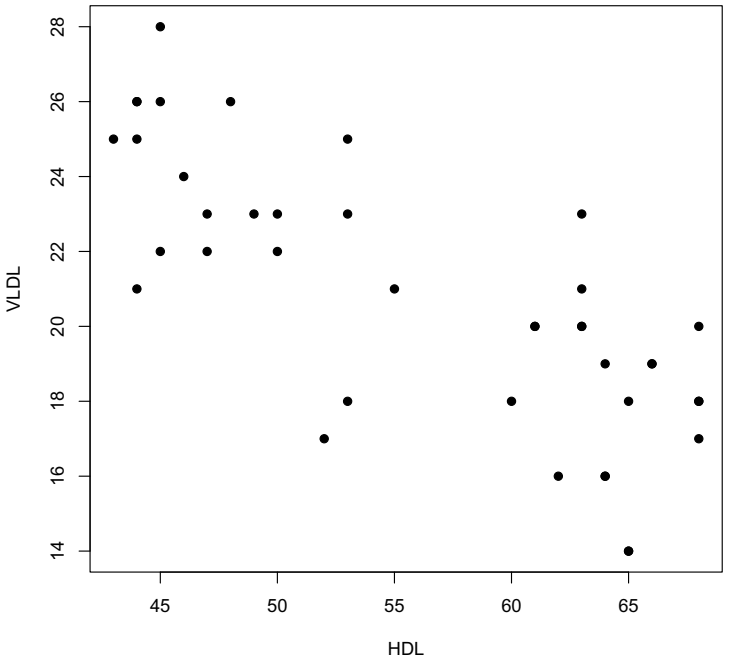



**Fig. 8.4** Scatterplot of VLDL versus HDL reveals two clusters

that this group membership is not known a priori; we have to find it from the data. We also have no idea about the number of such groups, commonly called 'clusters'. We only have multivariate data.

The basic idea is that we should start assuming something logical, but that is not final. Studying the inherent features of data through statistical ideas and methods might reveal these groups. There is no prior knowledge that is being used while identifying these latent groups. Using some logic, we could first assign cluster membership to each data point, i.e. to which group a data point belongs. We have to then continuously update the process of assigning cluster membership to a point until it gives a reasonably good solution. But what is a 'good solution'? Well, a solution is good only with respect to certain criterion. So given an optimum criterion, we have to continuously update our algorithm until it is satisfied, i.e. optimised according to our criterion. This optimum criterion, may be defined in various ways. Thus, it is natural that one technique might give different solutions depending on the choice of the optimum criterion.

Nowadays, with large data it is not possible to do clustering manually. We have to use a computer. Moreover, if the number of variables is more than three, we cannot plot them and any way of identifying clusters just by looking at the graphs is out of question. Note that at this point we have no idea about how the clusters were originally there in the population or even in a sample as we cannot see the data.

Hence we need an efficient algorithm supported by statistical theory or intuition or logic to separate the clusters from the data cloud. Thus cluster analysis is sometimes also known as unsupervised machine learning. The word 'unsupervised' indicates lack on any prior knowledge or information about the data.

Clustering method can also be looked as a multi-dimensional optimisation because some optimum criteria must be satisfied in order to cluster the data points, each point having a number of components. Multi-dimensional optimisation may not always lead to global optimisation. However, we shall never know whether it reaches a global optimum. So, the underlying principle of the clustering method is essential and it needs to be sound with respect to the problem concerned.

The basic idea of clustering is intuitively very simple. If two points are very close to each other, they are expected to belong to the same group. On the other hand, if they are not so close or far apart, they should belong to different groups or clusters. Points within a group are homogeneous whereas there is heterogeneity between groups. Thus, the concept of distance is at the heart of cluster analysis. It represents how far one data point is with respect to the other. For a given set of $n$ observations, we can have pairwise distances for all possible $n(n-1)/2$ pairs of data points. We can also think of the distance of a point from a chosen point that may not lie in the data set but might be represented in some sense. So while doing cluster analysis or choosing an appropriate clustering method, we need to be careful about our purpose or objective and choose a method wisely. Cluster analysis or unsupervised machine learning literature is very rich and research is still ongoing. Here we discuss a few of the most popular and recently developed methods that seem to be appropriate in genetic data analysis.

### *8.7.1 Hierarchical Clustering*

This is one of the oldest clustering methods but its use is ever growing. Based on a distance measure between two points or two objects, first it searches for two closest data points and connects them. Then, considering these two points as a group, it continues its search to find the next two pairs of points. Here the previously grouped two points are treated as one single point. It continues until all points are connected. We use five two-dimensional points as in Fig. 8.5 and demonstrate the algorithm of this clustering method.

Points 3 and points 4 are closest; so we connect them by a branch and consider it as a group. Next point 1 and point 5 are closest; join them similarly. However, now the distance between point 1 and point 5 as a group and point 2 is the next smallest; so we join them. At the end, we connect the group consisting of points 1, 2, 5 and the group with points 3 and 4 to get the final structure. It is clear that the length of line joining two points of groups represents the distance between them. The entire set of points are connected in a sequential manner and there is an underlying hierarchy, a tree-like structure. This is known as a hierarchical clustering method.

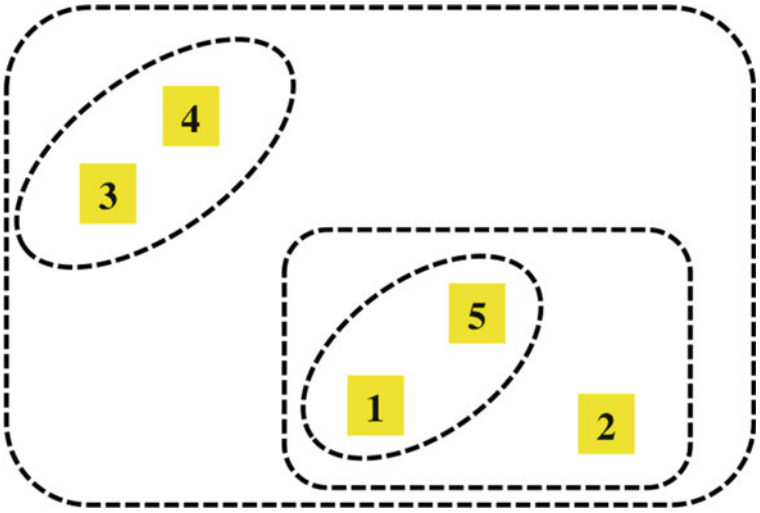



**Fig. 8.5** A schematic view of hierarchical clustering

Mathematically this can be represented easily. Let $d(\{P_1, \ldots, P_m\}, \{Q_1, \ldots, Q_n\})$ be the distance between two groups of points $\{P_1, \ldots, P_m\}$ and $\{Q_1, \ldots, Q_n\}$. If $m = 1$ or $n = 1$, the corresponding group consists of only one point. According to Fig. 8.5, we have

$$
\begin{aligned}
&d(\{P_3\}, \{P_4\}) \leq d(\{P_i\}, \{P_j\}) \text{ for all } i \neq j \\
&\qquad\qquad \Longrightarrow \text{ join } P_3 \text{ and } P_4 \\
&d(\{P_1\}, \{P_5\}) \leq d(\{P_1\}, \{P_2\}) \\
&d(\{P_1\}, \{P_5\}) \leq d(\{P_2\}, \{P_5\}) \\
&d(\{P_1\}, \{P_5\}) \leq d(\{P_3, P_4\}, \{P_j\}) j = 1, 2, 5 \\
&\qquad\qquad \Longrightarrow \text{ join } P_1 \text{ and } P_5 \\
&d(\{P_1, P_5\}, \{P_2\}) < d(\{P_1, P_5\}, \{P_3, P_4\}) \\
&d(\{P_1, P_5\}, \{P_2\}) < d(\{P_2\}, \{P_3, P_4\}) \\
&\qquad\qquad \Longrightarrow \text{ join } \{P_1, P_5\} \text{ and } \{P_2\} \\
&\text{Finally join } \{P_1, P_2, P_5\} \text{ and } \{P_3, P_4\}
\end{aligned}
$$

In this type of clustering (Fig. 8.5), it does not give an output of groups of homogeneous points. All points are connected to each other, directly or indirectly through other points. However, we can use a horizontal line to cut this tree in order to get the desired number of groups. Clearly, the number of groups thus obtained is subjective; it depends on the question and the most plausible subjective representation and interpretation.

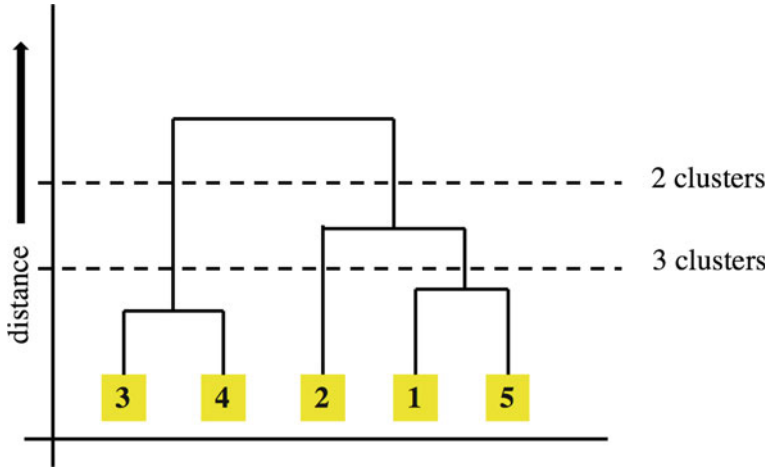



**Fig. 8.6** Hierarchical clustering

One can also start from the pair of points having maximum distance and putting them farthest apart and continue the same algorithm; only here we choose the pairs of groups of points having maximum distance. So it would be a top-down approach whereas the previous one is a bottom-to-top approach. Another interesting issue about hierarchical clustering is that, conceptually, although it identifies the closest member of a point like other clustering algorithms, however, here all points are connected to each other, either closely or remotely, unless, of course, we cut the tree to get distinct clusters (Fig. 8.6).

However, if there is any error at any stage, it will continue to propagate till all points are connected. Thus, an early stage error might lead to an erroneous output and there is no way to rectify it after we get the result. This is a major disadvantage of hierarchical clustering method. On the other hand, the output can be presented using a nice diagram, known as 'dendrogram' that gives a visual impression of connections among multi-dimensional data points. Dendrogram is very useful in interpreting results in connection to the problem.

In R, 'hclust()' function does the job of constructing a hierarchical clustering. However, it requires a dissimilarity matrix as its input. We have to use the function 'daisy()' already available in the package '{cluster}'. First we install the package '{cluster}' using the command 'install.packages("cluster")', choose any mirror that will pop up automatically if the computer is connected to the internet, and then write 'library("cluster")' to make it active in the current R workspace. So hierarchical clustering is done using the command 'daisy()' followed by 'hclust()'. The R function 'daisy()' calculates the pairwise similarity measure indicating the distance between all possible pairs of points. The function 'hclust()' acts upon the output of 'daisy()' and constructs the lineage connecting all points. Note that after the construction of hierarchical clustering, in order to plot the dendrogram, we need to use

'as.dendrogram()' function inside 'plot()' function. If we want to extract $k$ clusters, we shall use 'cutree()' command. Fig. 8.6 presents a schematic dendrogram with 2 and 3 clusters. Comprehensive R codes are given below.

```
> a<-read.txt("cholesterol.txt",header=T)      ## reading the datafile
                                                stored in "cholesterol.txt"
> d<-daisy(a,metric="euclidean")               ## calculating dissimilarity
                                                matrix using data set stored in 'a'
> b<-hclust(d)                                 ## doing hierarchical
                                                clustering
> plot(as.dendrogram(b))                       ## plotting hierarchical
                                                clustering tree
> cutree(b,k=2)                                ## provides cluster
                                                membership of each data ppoint
```

A dendrogram based on cholesterol data (Table 8.1) is given in Fig. 8.7. We did not use any 'cutree' command. This is left to the user to choose the number of clusters and accordingly create clusters.

### 8.7.2 *k-means Clustering*

One of the most popular clustering methods is 'k-means clustering'. Before executing the algorithm we assume that there are $k$ clusters. So the value of $k$ must be given as an input parameter. Then, the k-means clustering algorithm splits the data into $k$ groups so that the data points within each group are homogeneous whereas they are heterogeneous between groups. That means the distance between two points is very small if they belong to the same group, but they are far apart if they belong to different groups. This algorithm works iteratively until it satisfies a convergence criterion.

First we randomly choose $k$ data points. These are known as 'centres'. Let these centres be $c_1, c_2, \ldots, c_k$. We construct $k$ clusters, $\mathscr{C}_1, \ldots, \mathscr{C}_k$ (say), around each centre. Calculate the distance of a data point $x_i$ (say) from each of these $k$ centres and assign $x_i$ to the cluster $\mathscr{C}_j$ if its distance from $c_j$ is minimum. Let $d(x, y)$ be the distance between two points $x$ and $y$. Then we assign $x_i$ to $\mathscr{C}_j$ if

$$d(x_i, c_j) \leq \min\left\{d(x_i, c_1), \ldots, d(x_i, c_k)\right\} \tag{8.12}$$

Once we complete assigning all points to some cluster or other, we take new centres as the mean of each cluster, i.e. $c_j = \frac{1}{n_j} \sum_{i:x_i \in \mathscr{C}_j} x_i$, where $n_j$ is the number of point in cluster $\mathscr{C}_j$ after the first iteration. Note that, now the cluster centres are new, not the same as the one with which we started with. Moreover, after the first iteration, a cluster centre may not be a datapoint in the data set. Repeat the same

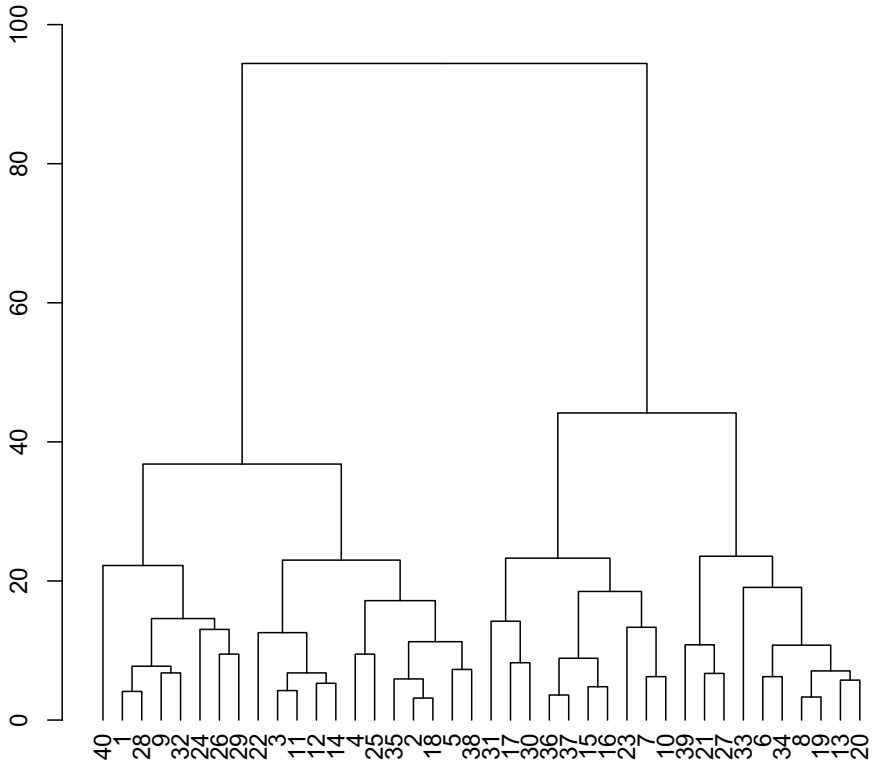



**Fig. 8.7** Hierarchical clustering of cholesterol data

procedure and assign each point to a cluster satisfying the condition (8.12). Continue until two consecutive iterations give the same set of cluster centres. We then say that the algorithm is converged and it provides the final clustering scheme for all data points.

Few important points to note. Here $k$ is known and would be given as input parameter. Each data point belongs to one and exactly one cluster. There is no data point that remains outside any cluster. Usually in k-means clustering we take Euclidean distance between two points. Let $\boldsymbol{x} = (x_1, \ldots, x_p)'$ and $\boldsymbol{y} = (y_1, \ldots, y_p)'$ be two data points each having $p$ components. Then Euclidean distance between $\boldsymbol{x}$ and $\boldsymbol{y}$ is defined as

$$d(\boldsymbol{x}, \boldsymbol{y}) = ||\boldsymbol{x} - \boldsymbol{y}|| = \sqrt{\sum_{i=1}^{p} (x_i - y_i)^2}.$$

k-means clustering is a very popular, effective, and nice method, aligning with our intuition of distance between and among the points. It is clear that output of k-means clustering algorithm depends on the random selection of $k$ centres. Theoretically, it

does not matter because we intend to get a global minimum through this algorithm. But the practical situation might be different.

One major drawback of k-means clustering is that it depends on calculation of mean, a measure of central tendency that is highly affected by extreme values or outliers. In such situations, if we somehow know the presence of outliers, this might not be a good clustering algorithm. A variation recommended to tackle such situation is to take $d(x, y) = |x - y|$, which is the absolute value of the difference between two points $x$ and $y$. This method is known as 'Partitioning Around Medoids' (PAM). PAM is expected to be more robust than k-means clustering method.

R code for k-means clustering can be applied easily to find clusters in a data set. The input parameters are the data set itself and the number of clusters or cluster centres. We apply this algorithm on the cholesterol data set. First we present the output of R code for k-means algorithm. After that we shall explain the output.

```
> kmeans(a,2)          ## k-means algorithm

K-means clustering with 2 clusters of sizes 20, 20
Cluster means:
    HDL    LDL   VLDL   TGL
1  64.35  67.90  18.3  109.40
2  47.85 101.65  23.3  152.35
Clustering vector:
[1] 2 2 2 1 2 2 1 1 1 1 1 1 2 1 2 1 1 2 1 1 2 2 1 1 1 2 2 1 2 1 2 2 2 2 1 2 1 1 2 2
Within cluster sum of squares by cluster:
[1] 3203.35  2463.85
(between_SS / total_SS = 85.3%)
```

Here it is clear that clustering scheme with two centres seems reasonable as it explains 85.3% of the total variation present in the data. Increasing the number of clusters would increase the percentage of explained variation, but we have to stop at some point where this increase is not substantial. There is no hard and fast rule for choosing the number of clusters. Few authors recommended some measures, but those suggestions may not produce a deterministic result in all data sets. Cluster means are vectors of means for four variables; each row indicating centre of the corresponding cluster.

Note that k-means algorithm is basically an optimisation problem in a multi-dimensional space. Naturally, this may not lead to global optimum always. Thus solutions of k-means algorithm is a local optimum, which may or may not be the global optimum. However, even if multiple applications of k-means algorithm with different cluster centres produce different clustering schemes, they do not differ widely in most cases. So we should be happy with the result of a single application of k-means algorithm. However, if we want to be more accurate, we have to apply k-means algorithm several times with different cluster centres and a final clustering may be obtained through consensus, although there is no guarantee that the final clustering would be the best.

### *8.7.3 Tight Clustering*

The above clustering methods are very useful and popular. Besides these, a few other clustering methods are available, among which, PAM, model-based clustering are used in many applications. However, the surge in data science research and technology has triggered the generation of different types of genomic data. These genomic data vary with respect to their characteristics and demand modification of existing clustering methods. Not only the modification of existing methods, sometimes it demands some changes in the clustering philosophy too.

Till the end of last century, clustering usually meant that each data point has a cluster membership, i.e. each data point belonged to one cluster or another. However, Tseng and Wong (2005) proposed a new clustering philosophy that seems pertinent in a few special data types like gene expression, medical data, etc.

Gene expression data are very noisy, a result of influence of many factors, some known and some unknown. Myriads of genes interact together to give the gene expression profiles. The basic philosophy or assumption behind clustering that each point must belong to a cluster seems too stringent in view of gene expression data. Tseng and Wong (2005) proposed a breakthrough philosophy that, it is not necessary for each gene to belong to a particular cluster. There might be some genes that remain scattered all over the data cloud, with their own characteristics, and do not belong to one cluster or another. Thus they have no cluster membership. This looks plausible and valid in view of the biological knowledge and background noise during data generation.

Tight clustering algorithm (Tseng & Wong, 2005) honours the above philosophy. Instead of trying to assign each data point to a cluster, the algorithm first extracts the tight-most cluster, i.e. a group of points that are homogeneous within themselves to a high degree. In order to find tight-most cluster, tight clustering algorithm uses resampling techniques. It randomly samples a substantial portion of (say 70%) of the data, applies a k-means clustering engine, and assigns membership of remaining data points. Repetition of this resampling technique for a large number times generates a consensus matrix that shows the number of times a pair of points belong to same cluster. Based on a threshold, it then extracts points that are very close to each other, i.e. it extracts the tight-most cluster at this stage or iteration. After keeping these points separate, it moves on to find a second tight-most cluster using the same approach and so on. It is expected that although subsequent iterations produce tight or homogeneous clusters, however, the degree of tightness decreases with increase in iteration number. At some point, the members within a cluster might not be really homogeneous. Thus, the sequential extraction of clusters with decreasing degree of tightness continues until the within group homogeneity, i.e. tightness of a cluster falls below a threshold. Each iteration or stage provides a cluster. When the tightness of the cluster falls below a threshold, we stop at that stage. The remaining points are left scattered over the entire space of points. These points that remained unassigned to any cluster are sometimes called noise points, or scattered genes in the context of gene expression data.

This clustering algorithm has another advantage. In k-means or PAM, the number of clusters should be given as input and the algorithm finds exactly that number of clusters. Hierarchical clustering does not require the number of clusters as input, but it connects all points and it is up to the user in extracting the number of clusters by cutting the tree representing the lineage. However, in tight clustering, it is not necessary to externally provide the number of clusters. Only an idea on the maximum number of clusters would suffice. Tight cluster algorithm will stop if the within cluster tightness does not provide sufficient evidence for being termed as a cluster. Hence it is robust to the choice of the number of clusters.

R command 'tight.clust()' in the package 'tightClust' generates output of this algorithm where the only input is the data matrix and the number of clusters that the user wants to find. Sometimes, if the number of clusters given by the user is large, tight clustering algorithm stops automatically as soon as the tightness of a group of points fall below a threshold, even if the desired number of clusters is not achieved. Thus, in some sense, this algorithm is intelligent while producing results. Thalamuthu et al. (2006) performed an extensive simulation study and real data analysis of gene expression data. It has shown the superiority of tight clustering algorithm over other common clustering methods that are applied to microarray gene expression data.

However, there is one limitation of this tight clustering algorithm. The number of genes in a standard gene expression data set is usually very large, around 20000 to 30000. Unfortunately the tight clustering algorithm proposed by Tseng and Wong (2005) is unable to handle such huge number of genes; it can run only up to $\sim$ 3000 genes. Karmakar et al. (2019) proposed another version of tight clustering that removes this drawback. This algorithm is made specifically for handling large data. It is efficient and provides robust performance while dealing with large data sets. It can even run for 100000 data points seamlessly.

The goal of this extended version is to provide a stable algorithm that allows the use of tight clustering in large data sets. The proposed algorithm follows big data analytics approach (Karmakar & Mukhopadhyay, 2020). The broader theme of this approach is splitting a problem into smaller problems and then combining the results of those smaller problems to arrive at a solution to the original problem. This algorithm takes advantage of the various nice aspects of the original tight clustering algorithm. Since the mathematical treatment of this algorithm is little difficult, we shall only describe an outline without any mathematical details.

First we randomly partition $n$ data points into a number of disjoint subsets such that each subset would be of a manageable size so as to run the original tight clustering algorithm. Let the number of such subsets of data be $l$. To reduce errors due to partitioning, this procedure is repeated $r$ times. Hence the total number of such manageable sets becomes $M_0 = l \times r$. We extract the tight-most cluster from each of these $M_0$ data sets and store them in an empty bucket. Note that elements in this bucket may not necessarily be disjoint and they may not contain all data points. An element of the bucket simply tells us that the data points therein are of a similar pattern. The next step of the algorithm performs some operations on this bucket. In this step, we redefine members of the bucket so that the data points belonging to

each element will now have a similar profile and gene expressions from different members will likely have significantly different profiles.

Now we should need to combine the elements in the bucket that will lead us towards the final clusters. Consider the first member of the bucket. Using a predefined decision rule, we combine all members of the bucket with a pattern similar to that of the first element and redefine it as a new collection. We again do this process with another element left in the bucket. Continue until all elements in the bucket are put in redefined groups.

Finally to obtain the tight-most cluster from the possible choices, i.e. the redefined members of the bucket, we use an appropriate measure of tightness of a set of data points that tells us how disperse the set of data points are, within an element in the bucket or a collection of points. Using this measure we have the first tight-most cluster. The tightness measure ensures a stable cluster.

In the next iteration, after we obtain the tight-most cluster from the data, we need to find another, i.e. the second tight-most cluster. For this, we remove all data points of the already obtained cluster from the whole data set as well as from members of the bucket. Now, consider the remaining data as a new data set and repeat the process starting from partitioning and combining to extracting another tight-most cluster and so on. Once we get $k$ desired clusters, we stop and the remaining points are treated as noise points or scattered points that might span over any space in the main data cloud. Of course, it is not necessary to extract all $k$ clusters. If the data points in a cluster do not satisfy the tightness criterion, it will not be termed as a cluster and will remain as scattered points. Thus, although an idea about the number of clusters is required as an input, it is a very weak notion; tight cluster algorithm, even for large data sets will form clusters until a tightness criterion is satisfied.

Note that the choice of decision function and threshold depends on the user. However, different choices might lead to different clusters although there should not be much difference in the clusters as this algorithm heavily depends on the substantial number of random sampling. However, the tightness of clusters will depend on the choice of measure of tightness. This algorithm removes the limitation of original tight clustering; moreover, it is scalable to time and robust in its performance. For a detailed discussion with simulated and real data sets readers are advised to read the original paper by Karmakar et al. (2019).

## 8.8 Association with Multivariate Phenotypes

We have learnt different statistical methods to detect genetic association of a locus with phenotype, be it qualitative or quantitative. Discussion with multivariate data reveals another interesting and important question. We know that any clinical test for measuring cholesterol level produces results on four phenotypes, viz, HDL, LDL, VLDL, and triglyceride (TGL) levels. Similarly status or indication of hypertension is measured by two variables, systolic and diastolic blood pressures. Thus occurrence of multivariate phenotypes is natural mainly because these variables are correlated

with each other and they usually occur together. Moreover, since all these measures are indicators of a particular broad phenotype like cholesterol level (may be a trigger for cardiovascular disease) or blood pressure, we may think that genetic variant(s) responsible for the disease might be associated with all these phenotypes. For example, four phenotypes HDL, LDL, VLDL, and TGL might all be somehow associated with the same set of genetic variants or loci.

It seems that if we study multivariate phenotypes together instead of univariate or a single variable or phenotype separately, power of the study would increase as we can exploit extra information through correlations prevailing among different variables. Hence there is a need for statistical methods for studying genetic association using multivariate phenotypes.

However, extension of univariate to multivariate phenotype in the context of association study with a genetic locus is not straightforward. For example, in a regression set up, each observation is a vector (ordered set of more than one variable). The variables in a vector may be correlated. Because of this correlation structure among the variables occurring together, it demands a fresh look at all methods of genetic association. We discuss a few methods that may be used to study genetic association using multivariate phenotypes.

Till today, there are plenty of genetic association studies that identify numerous loci that are associated with different traits. However, single marker analysis is unable to explain a substantial portion of genetic heritability. One reason might be the loss of information by using a single marker instead of multiple loci. This problem has been resolved, at least to some extent, by proposing multi-loci method (Mukhopadhyay et al. 2010; Wu et al. 2011; Thalamuthu et al. 2011). Another reason may be argued as the inability to accommodate multivariate phenotypes in a single model thus failing to exploit the pleiotropy (one genetic variant affecting multiple phenotypes simultaneously) or inter-relations among different phenotypes. However, literature survey reveals a few methods that have been developed to meet these challenges. We see that existing statistical methods addressing the analysis through multivariate phenotypes can be broadly classified into two classes: (1) use of summary statistics that result from previous genetic association studies using individual variables, and (2) direct use of multivariate phenotypes to study association with a single locus. We discuss below, a few methods for genetic association at single marker using multivariate phenotypes.

### *8.8.1 Linear Mixed Effects Model*

Linear mixed effects model looks like a regression type model but with some more extensions and assumptions. This consists of a fixed genetic effect due to the locus under study and a person specific effect that occurs due to the availability of multivariate phenotypes for each person in the sample. For the sake of simplicity, we discuss only multivariate phenotypes where all components are quantitative in nature, more specifically continuous variables.

A linear mixed effects model can be written as

$$y_{ij} = \alpha + \beta_j X(g_i) + \eta_{ij} + e_{ij},\ j = 1, \ldots, J;\ i = 1, \ldots, n \tag{8.13}$$

where $y_{ij}$ is the phenotype value for $j$-th component in $i$-th individual, $\alpha$ is the general effect, $X(g_i)$ is the genetic effect for the $i$-th individual with $\beta_j$ being the corresponding coefficient indicating the contribution of genetic effect, $\eta_{ij}$ is the person specific random effect for $j$-th component and $e_{ij}$ is the random error component. We assume that $e_{ij}$s are independently and identically distributed (i.i.d.) random variables following $N(0, \sigma_e^2)$ for all $j = 1, \ldots, J;\ i = 1, \ldots, n$. Clearly, the correlation between two univariate phenotypes is introduced through covariance among $\eta_{ij}$s, i.e.

$$Cov(y_{ij}, y_{ij'}) = Cov(\alpha + \beta_j X(g_i) + \eta_{ij} + e_{ij}, \alpha + \beta_{j'} X(g_i) + \eta_{ij'} + e_{ij'}) = \sigma_{jj'},\ \text{(say)},$$

because of independence among $e_{ij}$s and between $e_{ij}$ and $\eta_{ij}$. Multivariate phenotypes occur in the model more vividly and we write the above model using vector of phenotype observations as

$$\boldsymbol{y}_i = \alpha \mathbf{1} + X(g_i)\boldsymbol{\beta} + \boldsymbol{\eta}_i + \boldsymbol{e}_i,\ \ i = 1, \ldots, n \tag{8.14}$$

where

$$\boldsymbol{y}_i = \begin{pmatrix} y_{i1} \\ y_{i2} \\ \vdots \\ y_{iJ} \end{pmatrix}, \mathbf{1} = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}, \boldsymbol{\beta} = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_J \end{pmatrix}, \boldsymbol{\eta}_i = \begin{pmatrix} \eta_{i1} \\ \eta_{i2} \\ \vdots \\ \eta_{iJ} \end{pmatrix} \text{ and } \boldsymbol{e}_i = \begin{pmatrix} e_{i1} \\ e_{i2} \\ \vdots \\ e_{iJ} \end{pmatrix}$$

Now, if we want to test whether the genetic locus has no effect on any of the $J$ phenotypes, we can frame this as a problem of testing statistical hypothesis as

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_J = 0 \text{ against } H_1 :\ \text{at least one } \beta_j \neq 0.$$

During estimation, variance parameters in $\Sigma$ matrix and $\sigma_e^2$ can be estimated by a more sophisticated method called 'restricted maximum likelihood method' (REML). Using likelihood ratio testing method we can test the above hypothesis and find the p-value based on asymptotic distribution of the test statistic. However, one can also employ another test statistic, known as Wald's $\chi^2$ statistic, i.e.

$$\hat{\boldsymbol{\beta}}' \widehat{\Sigma_{\boldsymbol{\beta}}^{-1}} \hat{\boldsymbol{\beta}} \sim \chi_J^2 \text{ asymptotically, for large } n,$$

where $\hat{\boldsymbol{\beta}}$ and $\widehat{\Sigma_{\boldsymbol{\beta}}^{-1}}$ are the estimates of $\boldsymbol{\beta}$ and inverse of variance-covariance matrix $\Sigma$ respectively. The p-value can easily be obtained and we can take a decision accordingly.

This procedure is an extension of the standard regression model with an additional random component $\eta_{ij}$ making its analysis slightly different. This kind of model is known as 'linear mixed model', because $\beta_j$s are fixed parameters whereas $\eta_{ij}$s are unknown but they are considered as a random variable with unknown parameters. However, this provides an easy way of incorporating information from all phenotypes along with their correlations in a single model. It can also be extended to include more than one genetic locus.

### *8.8.2 Variable Reduction Method Using PCA*

To collapse information included as a whole in multivariate phenotypes, we can use some dimensional reduction method as discussed previously. Principal component analysis is probably the most popular method for such reduction of multiple variables. If the phenotype vector consists of $J$ univariate components, we can use PCA to reduce its dimension. Now only using the first principal component (PC1) we can easily do our genetic association study by adopting any method for a single phenotype variable. However, although we use PC1 as our phenotype of interest, it might have no physical meaning as it is a linear combination of the components of multivariate phenotype vector.

In this case also, we assume that all components of phenotype vector are continuous and hence PC1 would act as a quantitative univariate quantity. PC1 contains majority of information embedded in multivariate phenotype. Hence any significant genetic association with PC1 is expected to show some association with multivariate phenotype. Based on this idea, to test any possible association with genetic locus, we can adopt any regression-based approach or any other approach suitable for association using a single variable. If, along with multiple variables and genotype data, we have covariate information for all individuals, we can easily adjust the effect of covariate and identify association signal, if any, using any method discussed previously.

If the p-value of the test statistic is extremely small, we can say that this multivariate phenotype is probably associated with the genetic locus, although some difficulty may arise while interpreting the results. If the PC1 explains more than 80% of total variation, only then should we apply this method. Otherwise, a small p-value may not carry any significance.

For the data in Table 8.1, we see that principal component analysis indicates that first principal component (PC1) explains more than 85% variance. So it is enough to work with only PC1. To find association with genotype as given in Table 8.1, we adopt a linear regression method with a number of alleles as an independent or explanatory variable. Thus our model becomes

$$PC1_i = \mu + \beta g_i + e_i, \quad i = 1, \ldots, n$$

where $PC1_i$ is the value of first principal component corresponding to $i^{th}$ individual, $g_i$ is the genotype of the same individual with $e_i$ as random error component with usual assumptions as in simple regression analysis. Here, the null hypothesis, $H_0 : \beta = 0$ indicates the absence of any association of PC1 with the number of alleles at the genetic locus of interest.

Fitting a simple linear regression reveals the p-value as 0.9345 which is extremely large compared to a chosen level of significance 0.05 (say). Hence we can say that there is no evidence to declare any association of phenotype vector with the genotype of interest. Note that association of phenotype vector only implies that we assume that PC1 contains enough information to reveal the association with multivariate phenotypes.

Interpretation of result needs careful attention. If there is any association signal, it is difficult to interpret in terms of individual components, because from PC1 we may not get a clear idea about which component is really associated. Sometimes, the phenotype vector, as a whole, is associated due to synergistic effect of all components. One gene might affect multiple phenotypes with varying degrees. In such cases, it is not wise to go for interpretation in terms of individual components; rather any comment or conclusion on association with the phenotype vector as a whole seems more meaningful.

### 8.8.3 O'Brien's Method of Combining Univariate Test Statistics

Various genetic studies have made available the summary statistics from testing association between a univariate phenotype and a genetic locus of interest. So the question naturally arises, whether it is possible to combine these results obtained through univariate analysis in order to get some idea about pleiotropy, i.e. whether a single marker is associated with multiple phenotypes. O'Brien (1984) proposed a method that combines univariate statistics. However, here the basic demand is how to use the available pairwise correlations among the components of the vector of phenotypes.

Let $\boldsymbol{T} = (T_1, T_2, \ldots, T_K)'$ be a multivariate phenotype vector of order $K$, i.e. it has $K$ univariate components. We have seen in univariate genetic association study that most of the test statistics follow normal distribution either exactly or asymptotically. For example, even when the phenotype is qualitative, the test statistic, which is an odds ratio, follows an asymptotic normal distribution. Thus, without any loss of generality, we can assume that

$$\boldsymbol{T} \sim N_K(\boldsymbol{\beta}, \Sigma),$$

where $\boldsymbol{\beta} = (\beta_1, \beta_2, \ldots, \beta_K)'$ is the vector of parameters that is to be tested and $\Sigma$ is the variance-covariance matrix. Now $H_0 : \beta_1 = \cdots = \beta_K = 0$ provides a null

hypothesis which when true indicates the absence of genetic association with the entire phenotype vector, i.e. with all components of multivariate phenotype vector as a whole.

Let for the phenotype vector $\boldsymbol{T}$, $\Sigma$ and $\Sigma^{-1}$ be variance-covariance matrix and its inverse respectively, as given by

$$\Sigma = \begin{pmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1K} \\ \sigma_{21} & \sigma_{22} & \cdots & \sigma_{2K} \\ \vdots & \vdots & \cdots & \vdots \\ \sigma_{K1} & \sigma_{K2} & \cdots & \sigma_{KK} \end{pmatrix} \text{ and } \Sigma^{-1} = \begin{pmatrix} \sigma^{11} & \sigma^{12} & \cdots & \sigma^{1K} \\ \sigma^{21} & \sigma^{22} & \cdots & \sigma^{2K} \\ \vdots & \vdots & \cdots & \vdots \\ \sigma^{K1} & \sigma^{K2} & \cdots & \sigma^{KK} \end{pmatrix},$$

where $\sigma_{ij} = cov(T_i, T_j)$, for $i, j = 1, \ldots, K$.

Taking $\mathbf{1} = (1, 1, \ldots, 1)'$ as a vector with all components as unity, O'Brien proposed a test statistic as

$$V = \mathbf{1}' \hat{\Sigma}^{-1} \boldsymbol{T} = \sum_{i=1}^{K} \sum_{j=1}^{K} \hat{\sigma}^{ij} T_i, \tag{8.15}$$

where $\hat{\sigma}^{ij}$ is the $(i, j)$-th element of the estimated inverse of $\Sigma$.

It can be shown that, $V$ follows asymptotic univariate normal distribution with mean $\mathbf{0} = (0, 0, \ldots, 0)'$ and variance $\mathbf{1}' \hat{\Sigma}^{-1} \mathbf{1} = \sum_{i=1}^{k} \sum_{j=1}^{k} \hat{\sigma}^{ij}$, under $H_0$.

This test does not require the entire data on all individuals. It only requires the summary statistics or the statistics that have been used in identifying genetic association of each variable with a genetic marker. Thus in this sense it can handle missing data, if any, because missingness has already been taken care of while evaluating test with respect to univariate phenotype.

We need to be careful about some issues while applying O'Brien's test. The most important issue is the estimation of $\Sigma$. From the given data, we can have only one set of statistics for phenotypes and it is not possible to get such an estimate from the given data. Usually some prior information about possible correlation might be helpful in this regard. Summary statistics are obtained from previous studies. While doing any genetic association study, background information about the population under study, sampling schemes, etc., are very important. Thus while applying such a test, we should be very careful whether information through summary statistics is really applicable to the population under study. However, the recent richness in databases has become extremely helpful in eliminating such issues, if not fully, to a great extent.

### 8.8.4 Methods with Heterogeneous Genetic Effects

One limitation of O'Brien's method is that although this is the most powerful test among the class of tests that are linear combinations of univariate test statistics when $\beta_1 = \cdots = \beta_K > 0$, provided $\Sigma$ is known; however, there is a drop in power when $\beta_1, \ldots, \beta_K$ are not homogeneous. Another drawback of this test is that it requires the knowledge of estimates of covariances or correlations present between two univariate test statistics. In most cases, we do these kinds of tests using odds ratio. Hence, the summary statistics used are odds ratio obtained previously using single marker association test. Thus, as an approximation, we can use the pairwise correlation between two SNP markers.

To remove this limitation, Yang et al. (2010) proposed some modifications. They proposed the use of non-uniform weights to incorporate the potential heterogeneity of $\beta_1, \beta_2, \ldots, \beta_K$. The strategy is to split the entire data set into two random subsets. Use one data set to estimate the weights of $\beta_1, \ldots, \beta_K$. Then, we can construct the test statistic as given in (8.15) based on the second subset. Let the test statistic obtained using the estimation set be $\hat{\boldsymbol{T}}_w$ and that using the other set be $\boldsymbol{T}$. Then the proposed test statistic is

$$S = \hat{\boldsymbol{T}}_w' \hat{\Sigma}^{-1} \boldsymbol{T}. \tag{8.16}$$

It can be shown that $S$ follows univariate normal distribution asymptotically with variance $\hat{\boldsymbol{T}}_w' \Sigma^{-1} \hat{\boldsymbol{T}}_w$, conditionally on the splitting of data sets.

The only assumption here is that the sample size should be large so that even after randomly splitting the data set into two subsets, we should have a good estimate of weights. Also, the sample size for the second subset of data should be large enough so that asymptotic normality of the distribution will hold. This method requires the entire data set.

The only limitation is that the performance of the test statistic may depend on the actual splitting of the data. Yang et al. (2010) proposed a modification of this method using cross-validation method where p-value can be calculated using permutation-based technique. However, splitting of the original data set may sometimes lead to loss of information, because we are working with a portion of data, not the entire data set for a single operation.

Another class of test statistic is based on a quadratic form of the components of multivariate phenotypes, of course including the pairwise correlations among univariate test statistics. Xu et al. (2003) proposed a Wald's chi-square type test statistic as

$$S = \boldsymbol{T}' \hat{\Sigma}^{-1} \boldsymbol{T}. \tag{8.17}$$

Here we replace the vector $\mathbf{1} = (1, 1, \ldots, 1)'$ as in (8.15) by $\boldsymbol{T} = (T_1, T_2, \ldots, T_K)'$, the vector of univariate test statistics. It can be shown that for large sample size, the test statistic $S$ asymptotically follows a mixture of chi-square variables; however, this

distribution can be approximated by a linear function of another chi-square variable. Thus we have, in a precise form,

$$S \overset{D}{\sim} \sum_{i=1}^{K} c_i \chi_1^2 \approx a\chi_d^2 + b \text{ where } d = \frac{\sum_{i=1}^{K} c_i^3}{\sum_{i=1}^{K} c_i^2}, b = \sum_{i=1}^{K} c_i - \frac{\left(\sum_{i=1}^{K} c_i^2\right)^2}{\sum_{i=1}^{K} c_i^3}, \text{ and } a = \frac{\left(\sum_{i=1}^{K} c_i^2\right)^3}{\left(\sum_{i=1}^{K} c_i^3\right)^2}$$

where $X \overset{D}{\sim} Y$ indicates that $X$ and $Y$ have the same probability distribution for large sample size.

Let $s$ be the observed value of the test statistic as given in (8.17). Based on the above approximate distribution, we can easily calculate the p-value approximately as

$$\text{p-value} \approx P_{H_0}(\chi_d^2 > (s-b)/a),$$

and take a decision accordingly. Note that whatever decision we take, we have performed this test at level of significance $\alpha$ $(0 < \alpha < 1)$ approximately.

## Exercise

First we provide a table on measurements of diastolic and systolic blood pressure, and genotypes at three SNP loci for a number of individuals. A few problems in this exercise are based on this data set. Each genotype is coded as 0, 1, or 2 according to the number of minor alleles.

8.1 Distinguish, with an example, among simple correlation coefficient, multiple and partial correlation coefficients.

8.2 Product moment correlation coefficient may be negative; however, multiple correlation coefficient is always positive. Justify and interpret this comment.

8.3 For the correlation matrix as given in Sect. 8.4, calculate all possible multiple and partial correlation coefficients.

8.4 For four variables, the correlation matrix $R$ is given by

$$R = \begin{pmatrix} 1.000 & 0.245 & 0.765 & -0.526 \\ 0.245 & 1.000 & 0.707 & 0.384 \\ 0.765 & 0.707 & 1.000 & 0.074 \\ -0.526 & 0.384 & 0.074 & 1.000 \end{pmatrix}$$

Calculate multiple correlation coefficients $r_{1\bullet 234}$, $r_{2\bullet 134}$, and partial correlation coefficients $r_{12\bullet 34}$, $r_{13\bullet 24}$ and $r_{14\bullet 23}$. Comment on your findings.

8.5 Let there be three variables $x_1$, $x_2$, and $x_3$ such that

$$x_1 = 2x_3 + \epsilon_1$$
$$x_2 = 3x_3 + \epsilon_2$$

where $\epsilon_1, \epsilon_2 \sim N(0, \sigma_1^2)$ are independently and identically distributed, and $v(x_3) = \sigma_3^2$. Find $r_{12}$ and $r_{12\bullet 3}$. Comment on your findings.

8.6 Table 8.2 contains data on bivariate phenotype on blood pressure, i.e. systolic and diastolic measurements from the same group of individuals.

(a) Perform a principal components analysis on the bivariate phenotype.
(b) Do a scree plot and determine the percentage of variation explained by the first principal component.
(c) Check whether the total variation of two principal components is same as the total variance of the original variables. Whatever be the answer, justify.
(d) Plot systolic measurements against diastolic measurements for all individuals and also plot one principal component against the other one. Comment on your findings using these plots.

8.7 We want to know whether there exist different groups in the population with respect to cholesterol levels, using the phenotype data in Table 8.1.

(a) Perform a cluster analysis using hierarchical clustering. Identify the groups using two different cutrees, one giving two groups and another giving three groups. Which one you would prefer and why?
(b) Perform a cluster analysis using k-means clustering method and identify the group label for each person and present it in a table. Compare the groups found by using k-means clustering method with that obtained by hierarchical clustering method.

8.8 The objective of a study is to understand the factors that are associated with systolic blood pressure in infants. Systolic blood pressure (mmHg), weight (grams) and age (days) are measured in 100 infants. A multiple linear regression is performed to predict blood pressure (mm Hg) from age and weight. The following results were obtained.

| Regression coefficient | Estimate | SD(estimate) |
|---|---|---|
| Intercept ($\beta_0$) | 50.0 | 4.0 |
| Weight (gm) ($\beta_1$) | 0.1 | 0.3 |
| Age (days) ($\beta_2$) | 4.0 | 0.6 |

How much higher would you expect the blood pressure to be of an infant who weighed 3400 grams compared to an infant who weighed 2550 grams if both infants were of exactly the same age?

8.9 You can think that systolic and diastolic measurements as in Table 8.2 are related to blood pressure level of an individual as a bivariate phenotype.

**Table 8.2** Data on systolic and diastolic blood pressure for a group of individuals

| Diastolic | Systolic | G1 | G2 | G3 | Diastolic | Systolic | G1 | G2 | G3 | Diastolic | Systolic | G1 | G2 | G3 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 88 | 148 | 0 | 1 | 2 | 78 | 121 | 0 | 0 | 0 | 81 | 123 | 0 | 0 | 0 |
| 107 | 144 | 0 | 0 | 1 | 82 | 117 | 0 | 0 | 0 | 91 | 116 | 0 | 0 | 1 |
| 83 | 112 | 0 | 0 | 1 | 71 | 112 | 0 | 1 | 0 | 87 | 114 | 0 | 0 | 1 |
| 85 | 115 | 1 | 0 | 0 | 80 | 128 | 0 | 0 | 0 | 72 | 120 | 0 | 1 | 0 |
| 72 | 127 | 0 | 1 | 0 | 86 | 132 | 0 | 2 | 2 | 79 | 124 | 0 | 0 | 0 |
| 81 | 122 | 1 | 0 | 0 | 76 | 119 | 0 | 2 | 0 | 78 | 116 | 1 | 0 | 0 |
| 75 | 114 | 0 | 0 | 0 | 87 | 133 | 0 | 1 | 2 | 75 | 115 | 0 | 1 | 0 |
| 86 | 122 | 0 | 2 | 1 | 71 | 116 | 1 | 0 | 1 | 72 | 118 | 0 | 1 | 0 |
| 72 | 118 | 0 | 1 | 0 | 91 | 135 | 0 | 2 | 1 | 88 | 124 | 1 | 2 | 0 |
| 81 | 115 | 0 | 2 | 1 | 77 | 114 | 0 | 2 | 0 | 89 | 122 | 0 | 0 | 0 |
| 95 | 131 | 1 | 0 | 1 | 80 | 123 | 0 | 0 | 0 | 78 | 122 | 0 | 0 | 0 |
| 93 | 132 | 0 | 1 | 2 | 72 | 116 | 1 | 0 | 0 | 90 | 126 | 0 | 1 | 0 |
| 84 | 125 | 0 | 0 | 0 | 76 | 112 | 1 | 0 | 1 | 97 | 141 | 1 | 1 | 2 |
| 82 | 118 | 1 | 0 | 0 | 97 | 133 | 0 | 0 | 2 | 90 | 125 | 0 | 2 | 0 |
| 88 | 128 | 1 | 0 | 0 | 74 | 111 | 0 | 0 | 0 | 87 | 117 | 0 | 0 | 1 |
| 87 | 126 | 0 | 0 | 2 | 87 | 127 | 1 | 1 | 0 | 94 | 133 | 1 | 0 | 0 |
| 88 | 129 | 0 | 0 | 0 | 97 | 137 | 0 | 0 | 1 | 88 | 140 | 0 | 0 | 1 |
| 72 | 122 | 2 | 0 | 0 | 75 | 117 | 1 | 0 | 1 | 106 | 131 | 0 | 0 | 2 |
| 93 | 119 | 0 | 1 | 0 | 94 | 131 | 0 | 2 | 2 | 82 | 129 | 0 | 0 | 0 |
| 85 | 124 | 0 | 0 | 0 | 80 | 112 | 0 | 0 | 0 | 77 | 120 | 0 | 0 | 0 |
| 88 | 138 | 0 | 1 | 2 | 78 | 112 | 0 | 2 | 0 | 84 | 122 | 1 | 0 | 0 |
| 84 | 130 | 1 | 0 | 0 | 95 | 125 | 0 | 0 | 0 | 84 | 117 | 1 | 0 | 0 |
| 79 | 112 | 0 | 0 | 0 | 99 | 131 | 0 | 2 | 2 | 73 | 118 | 0 | 0 | 0 |
| 84 | 141 | 0 | 1 | 0 | 86 | 113 | 0 | 0 | 0 | 93 | 126 | 1 | 0 | 1 |

(continued)

**Table 8.2** (continued)

| Diastolic | Systolic | G1 | G2 | G3 | Diastolic | Systolic | G1 | G2 | G3 | Diastolic | Systolic | G1 | G2 | G3 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 94 | 128 | 0 | 1 | 1 | 82 | 123 | 1 | 0 | 1 | 71 | 115 | 0 | 0 | 0 |
| 71 | 118 | 0 | 2 | 0 | 71 | 117 | 0 | 0 | 0 | 85 | 121 | 1 | 0 | 0 |
| 88 | 113 | 0 | 1 | 0 | 85 | 134 | 0 | 0 | 0 | 85 | 125 | 1 | 0 | 1 |
| 83 | 120 | 0 | 1 | 0 | 78 | 115 | 0 | 0 | 0 | 73 | 124 | 0 | 0 | 0 |
| 73 | 127 | 0 | 0 | 0 | 91 | 122 | 0 | 1 | 0 | 88 | 131 | 0 | 2 | 2 |
| 90 | 116 | 1 | 0 | 0 | 94 | 118 | 1 | 0 | 0 | 79 | 125 | 0 | 0 | 1 |
| 100 | 120 | 1 | 2 | 0 | 93 | 117 | 0 | 0 | 2 | 76 | 114 | 0 | 0 | 0 |
| 83 | 128 | 0 | 0 | 0 | 87 | 126 | 0 | 1 | 2 | 79 | 119 | 0 | 0 | 0 |
| 85 | 134 | 1 | 0 | 0 | 76 | 120 | 0 | 0 | 1 | 85 | 123 | 0 | 1 | 0 |
| 74 | 128 | 0 | 0 | 0 | 82 | 129 | 2 | 1 | 0 | 80 | 126 | 0 | 0 | 0 |
| 88 | 124 | 0 | 2 | 0 | 84 | 133 | 0 | 2 | 0 | 85 | 117 | 0 | 1 | 0 |
| 80 | 116 | 0 | 0 | 0 | 74 | 118 | 1 | 1 | 0 | 98 | 124 | 0 | 0 | 2 |
| 99 | 131 | 0 | 0 | 2 | 85 | 122 | 0 | 1 | 0 | 75 | 112 | 0 | 0 | 0 |
| 77 | 117 | 0 | 0 | 0 | 78 | 132 | 1 | 1 | 0 | 71 | 119 | 0 | 0 | 0 |
| 85 | 118 | 0 | 2 | 0 | 73 | 118 | 0 | 0 | 0 | 80 | 113 | 0 | 0 | 0 |
| 91 | 120 | 0 | 2 | 0 | 79 | 119 | 0 | 1 | 0 | 93 | 123 | 0 | 0 | 0 |
| 83 | 127 | 1 | 1 | 0 | 86 | 122 | 0 | 0 | 0 | 85 | 131 | 0 | 1 | 1 |
| 91 | 138 | 0 | 2 | 2 | 88 | 123 | 0 | 2 | 0 | 82 | 128 | 0 | 0 | 0 |
| 78 | 116 | 0 | 0 | 0 | 101 | 132 | 1 | 2 | 2 | 74 | 122 | 0 | 0 | 0 |
| 78 | 114 | 0 | 1 | 0 | 71 | 113 | 0 | 1 | 0 | 89 | 122 | 0 | 0 | 0 |
| 74 | 115 | 0 | 2 | 0 | 84 | 131 | 0 | 2 | 0 | 83 | 124 | 0 | 0 | 0 |
| 86 | 112 | 0 | 0 | 0 | 86 | 136 | 0 | 2 | 2 | 81 | 122 | 0 | 2 | 1 |
| 77 | 116 | 0 | 1 | 0 | 75 | 117 | 1 | 0 | 0 | 74 | 116 | 0 | 0 | 0 |

(a) Perform an association analysis using bivariate phenotype for each locus.
(b) Also perform association analysis using each phenotype, i.e. univariate phenotypes for each locus.
(c) Compare the results obtained in (a) and (b). Comment on your results.
(d) Can you find any heterogeneity in the sample using data in Table 8.2? Justify your answer.

8.10 Consider the observations in Table 8.1.

(a) Draw histograms for each variable on a single graph.
(b) Draw scatter diagrams considering all possible pairs of variables and present them on a single graph.
(c) Find multiple correlation coefficients of TGL on HDL, LDL, and VLDL.
(d) Evaluate partial correlation coefficients of (i) HDL and (ii) LDL eliminating the effects of other variables.

8.11 For the data in Table 8.1, first divide the sample into two groups: one having at least one minor allele at the genetic locus, and the remaining persons as another group. Let these be Group 1 and Group 2 respectively.

(a) What is the multiple regression equation of HDL on LDL, VLDL, and TGL for (i) the entire group, (ii) Group 1, and (iii) Group 2?
(b) For each Group calculate multiple correlation coefficients of TGL on HDL, LDL, and VLDL. Comment on your results.
(c) For each Group calculate partial correlation coefficient between HDL and LDL eliminating the effects of TGL and VLDL. Comment on your results.

8.12 For the data in Table 8.1, do the following.

(a) Identify how many principal components explain at least (i) 70%, (ii) 80%, and (iii) 90% of total variance present in the data. Show your justification graphically and comment.
(b) Use cluster analysis to see whether there are any distinct groups in which the individuals can be grouped with respect to their phenotypes.
(c) Use an appropriate cluster analysis method to see whether any individual has any special characteristics so that s/he can be left out of any cluster.
(d) Use principal component regression to see whether there is any genetic association with any of the loci.
(e) Try to write your own R code to do association study with all cholesterol indicators using linear mixed model approach, taking one locus at a time.

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# Appendix A

## *What is R*

Before starting to describe the salient features of R, first note that we have to change our mindset in terms of using a computer. We are habituated with mouse clicking most of the times; now change that habit to write a few 'commands' in the terminal. 'Commands' are nothing but a set of instructions given to the computer so that it can understand our need and do the computing job for us. This is easy as we always type a letter or document using any word processor. Likewise, here we have to write the code in R or the 'R workspace'.

But first THREE QUESTIONS that would immediately come to mind:

(1) What is R?
(2) How do we get it?
(3) How do we learn it?

The brief but full answers to these THREE QUESTIONS are:

(1) R is a programming language, but can also be viewed as a software if we stick the objective of a software, that it is able to do lots of things using some inbuilt 'functions'. R has many functions already built-in. At the same time it is developed such that we can use R as a language in order to write our own code, using the available functions. Like all other programming language, it has the flexibility to use loops of many types, logical statements, reading and writing files, plotting high quality graphs, etc. Practically it has flexibility of a programming language and strength of a computer software. Moreover, we can also build a comprehensive code or a 'package' for doing a specific job. A package is a set of commands that can be used later for a more general purpose or for working with other data sets. Since R is freely available, one can write a package and upload it so that people from other corners of the world can use it; thus spreading scientific knowledge across the world. Note that, in R, similar types of functions or add ons for them are grouped into a 'package'. In order to do a particular

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I. Mukhopadhyay and P. P. Majumder, *Statistical Methods in Human Genetics*,
Indian Statistical Institute Series, https://doi.org/10.1007/978-981-99-3220-7

work, we first have to install the package that contains the required functions. For example, if we want to draw graphs in R, we have to install the package 'graphics' and use the 'plot' function to draw a graph. However, packages that contain commonly used functions would automatically be installed as default while installing R on the computer.

(2) R can be downloaded from the website 'http://cran.r-project.org'. It is available for many computing platforms and operating systems like Windows, Mac, Linux, and Unix. Every now and then, the R version is upgraded and so we have to upgrade it time to time in order to use the full power of R. Note that while installing R in the machine, it automatically installs some base packages, which includes thousands of commonly used functions. If we want to do some sophisticated calculation or operation, we need to find out the function and the corresponding package and install that package, again free of cost. Note that you can download basic R or R Studio, which seems to be a more user-friendly version of R. The same R will run; only in R studio, there are four windows (or terminals) each of which is reserved for a specific job. Like, if we want any help in R command, we can see it on one window.

(3) But the most important issue is how one can learn R. The best way to learn R is **self learning**. However, there are plenty of online materials, books, tutorials, etc. available in the internet. One such book is 'Introductory Statistics with R' by Peter Dalgaard. Just go through it for the first few pages at least, write the commands and codes as given in the book, understand each and every command, and one will acquire proficiency in using R. However, one may face a few problems while using R. The major problem is that, if we want do a particular work, how do we know the function that does this work? One option is to guess and in most of the cases the guess would be right. For example, if we want to calculate the arithmetic mean of a set of numbers already input in R, we can guess a command that would do this job. The most plausible guesses can be 'average', 'avg', 'mean', 'am,' etc. Surprisingly one of them is correct and for calculating arithmetic mean, 'mean' is the command in R. We have done one experiment as ask to guess R command for a job; every time the correct command emerges within first three guesses. However, one can always search in the internet using any search engine and find the command. In the above example, if we write 'calculate mean in R' in the search engine, a page will be opened that informs the actual command, the package in which it belongs, how to use that function, and all other relevant information about the command. Note that it will give the package name and we have to install it if it is not already available in the R workspace.

## *Basic operations in R*

We now describe some basic operations in R in order to give confidence to the readers, who are still hesitating to use it. First we double click the icon of R (or R studio)

and a terminal will be opened. It is always advisable to change the directory from the menu and go to the directory where the data set for the analysis is stored. Now that we are in the same folder, whatever analysis we do using R, the resulting files (we have to save them) will be available in that directory only. If we set the directory (folder) at the very beginning, we don't need to set the path while saving any result because the result file would be saved automatically in that directory.

### Variables

First note that R can be looked upon as a calculator. If we type $2 + 3$ and press 'enter', it will give 5. $sqrt()$ is the command for calculating square root of a number. Now $sqrt(49)$ gives 7, but $sqrt(\text{“}statistics\text{”})$ will give error, because 'statistics' is not a numeric value for which we can calculate square root. Note that '$<-$' or '$=$' is used to assign value to a variable. If we write '$a < -49$', press 'enter', the value 49 is stored in '$a$'.

```
> 2+3
[1] 5
> sqrt(49)
[1] 7
> a< −49
> a
[1] 49
> srqt(“statistics”)
Error in srqt(“statistics”) : could not find function “srqt”
```

The above discussion leads us to define three types of variables used in R. They are:

(1) Numeric variable: $a < -2.53$; $a$ is a numeric variable. We can do arithmetic operations using $a$.
(2) Character variable: $a < -\text{“}Knowledge\ of\ Statistics\ is\ essential\ for\ data\ analysis\text{”}$; $a$ is a character variable and we cannot do arithmetic operations using $a$, but can do other things. If we write $sub(\text{'}Knowledge\text{'}, \text{'}Knowledge\ and\ practice\text{'}, a)$, we will get '$Knowledge\ and\ practice\ of\ Statistics\ is\ essential\ for\ data\ analysis$'. Here the word '$Knowledge$' in the variable $a$ is being substituted by '$Knowledge\ and\ practice$'.
(3) Logical variable: $a < -(2 + 2 == 5)$; $a$ is a logical variable giving the output $FALSE$, because $2 + 2$ can never be 5!

### Data entry and simple operations

To input data into R, we give the data as a vector or a list of observations. After each command we have to press enter. The command is

```
x<-c(1,2,4,7,3.4)
```

Similarly if we write $x < -seq(-3, 3, 0.1)$, we input all the values starting at $-3$ and ending at 3 with an increment 0.1. Thus we enter 61 observations following a rule, i.e. they are equispaced between two known values. Now we can do some simple operations with the values in $x$.

```
y<-x*x          ## makes the square of each observation in x and store it in the same position
                ## in another variable y
```

We provide a few examples of simple R code that are very useful when we work with any data.

```
x<- c(2,5,3,7,1,9,6)          ## these values are stored as a list (vector) of values in the variable x
length(x)          ## finding the number of elements (numbers) stored in x
x[1]          ## first element of x
x[length(x)]          ## last element of x without knowing the number of elements in x
min(x)          ## finding minimum value
max(x)          ## finding maximum value
c(min(x), max(x))          ## minimum and maximum values together as a list (vector)
```

Few inbuilt functions in R help in typical data input. We shall discuss two such functions '*rep()*' and '*seq()*'. 'rep' is used when we want to repeat any number and 'seq' helps in writing consecutive numbers, not necessarily integers.

```
rep(2,4)          ## the number 2 is being repeated 4 times
c(rep(4,4),rep(3,3),rep(5,3))          ## creates a list 4, 4, 4, 4, 3, 3, 3, 5, 5, 5
rep(c(4,3,5),c(4,3,3))          ## easier way to create the previous list

seq(3,7,by=0.5)          ## all numbers starting with 3 and ending with 7 with an increment of 0.05
x<- seq(3,7,by=0.5)          ## store the above numbers
```

### Some simple statistical measures

Simple statistical measures like mean, median, standard deviation, etc. can easily be computed for a list of observations using a single command, in most cases. Sometimes we can merge a few functions to get slightly more complicated or non-standard measures.

```
x<-c(8.60,11.68,9.58,9.17,9.11,8.22,10.74,8.25, 9.84,12.14,9.58,10.80,12.17,7.16,13.32)   ## data entry
y<-c(3,2,4,1,5,7,2,9,1,4,7,3,6,1,9)              ## list of other values stored in y
n<- length(x)              ## finds the number of observations in x
sum(x)             ## finds the sum of observations in x
mean(x)            ## calculates arithmetic mean of values in x
median(x)          ## calculates median of values in x
sd(x)              ## calculates standard deviation of values in x, divisor is n − 1
cor(x,y)           ## calculates correlation coefficient between x and y
```

Using these simple functions, we can easily compute a few other measures. For example, suppose we want to calculate a measure, known as mean deviation about median for values in $y$. This is defined as

$$MD = \frac{1}{n}\sum_{i=1}^{n}|y_i - \tilde{y}| \text{ where } \tilde{y} \text{ is the median of } y.$$

An R code for mean deviation about median is written as

```
MD<- mean(abs(y-median(y)))
```

### Dataframe

Sometimes, we have data on multiple variables from the same individual. Data set given in Table 8.1 is an example of such data. We need to understand how to create or present data in a tabular format using R. Here we explain only using a small portion of the data in Table 8.1. Suppose we have data in four variables HDL, LDL, VLDL, and TGL, as follows.

```
HDL<- c(49,45,44,63,50,47,66,63,62,68)                ## data input for HDL
LDL<- c(95,103,100,72,90,106,62,73,82,64)             ## data input for LDL
VLDL<- c(23,22,21,20,23,23,19,21,16,17)               ## data input for VLDL
TGL<- c(146,150,155,120,148,148,110,109,117,107)      ## data input for TGL
```

First we should check whether each variable has the same number of values. It can be done in many ways. We use 'length( )' command in a logical mode to check this.

```
length(HDL)                    ## gives the number of values in variable 'HDL'
length(HDL)==length(LDL)       ## checks whether HDL and LDL have
                                   same number of values.;
                               ## its output is "TRUE" or "FALSE"
```

Similarly we can check whether all variables are of same length. This is essential in order to present multivariate data. Now to present them in a tabular format, we use the function 'data.frame( )'.

```
d<- data.frame(HDL, LDL, VLDL, TGL)          ## stores data in a table in this order
d                                  ## prints the output on the screen, as follows:

   HDL  LDL  VLDL  TGL
1   49   95    23  146
2   45  103    22  150
3   44  100    21  155
4   63   72    20  120
5   50   90    23  148
6   47  106    23  148
7   66   62    19  110
8   63   73    21  109
9   62   82    16  117
10  68   64    17  107
```

Given this dataframe or data set, we might want to see the values only for HDL, or calculate the mean of each column using one command or function in R. We might also add another column, say, age, on the right of this dataframe. All these can be done very easily.

```
apply(d,2,mean)          ##apply 'mean' function to each column (2 indicates column, 1 indicates row)

HDL     LDL     VLDL    TGL                    ##Output of 'apply' function applied on 'd'
55.7    84.7     20.5   131.0

age<- c(37, 41, 47, 51, 37, 44, 37, 36, 59, 53)          ## stores age data in 'age'
d<-cbind(d,age)                    ## creates another column 'age' as 5-th column in 'd'
d                        ## gives the output of 'd'

d$age<- c(37, 41, 47, 51, 37, 44, 37, 36, 59, 53) ## directly creates 5-th column as age; check output of 'd'
```

### Reading data files

Another issue without which this discussion would remain incomplete is that we need to know how to input data through a datafile. Usually most data files are available in *text* format (*.txt*) or *comma separated value* format (*.csv*). In order to directly import a datafile into the R workspace, the commands are:

(1) $read.table(\text{“}filename.txt\text{”}, header = T, sep = \text{“}\backslash t\text{”})$: to read .txt file, if there are column names, write 'header = T', otherwise 'header = F' and mention the separator, if any.
(2) $read.csv(\text{“}filename.csv\text{”}, header = T)$: to read .txt file, if there are column names, write 'header = T', otherwise 'header = F'.

We can do some operations after reading a table of data. These operations would give an idea about some features of the table. Here also we use 'apply' function of R to get accustomed to working with tables after reading them directly from the directory.

```
a<- read.table("filename.txt", header=T,sep="\ t")          ##reads a table and stores in a
dim(a)          ## dimension i.e. number of rows and columns in a
head(a)         ## first six rows of a
tail(a)         ## last six rows of a
a[1:5, 1:7]     ## first five rows and first seven columns of a; note that within square bracket, before
                ## comma (',') put the row indicators and after comma, put column indicators
a[c(1,3,4),c(2,3,7,6)]          ## values corresponding to row number 1, 3 and 4, and column
                                ## number 2, 3, 7, and 6, in this order, from the table stored in a
mean(a[,2])     ## calculates mean of elements in second column of a; note that within square bracket,
                ## before comma, there is nothing, indicating all rows have been considered
sd(a[5,])       ## calculates standard deviation of elements in fifth row of a; note that within square
                ## bracket, after comma, there is nothing, indicating all columns i.e. all elements in the
                ## entire fifth row have been considered
apply(a,1,mean)     ## a nice function, calculates means of each row of a; the function 'apply' is
                    ## applying 'mean' function on the data stored in 'a' on each row ('1' indicates
                    ## row, '2' indicates column)
apply(a,2,sd)   ## calculates standard deviations of each column of a; the function 'apply' is applying
                ## 'sd' function on the data stored in 'a' on each column ('2' indicates column)
```

## Loop in R

There are mainly two loops, as common in all programming languages, in R. One is 'for' loop and another is 'while' loop. We shall describe how to write 'for' loop in R. We leave it to the reader to explore and find how to write a loop using 'while' command.

Loops are used to do a job repeatedly. In the calculation of arithmetic mean, we have to first add second number to the first, then add third number to the previous sum, and so on. Thus in each step we have to collect one number to be added to the result obtained at the previous step. It ends as soon as we add the last number. So it seems that we need three quantities: the initial number (initiation), an increment to get the next number, and an identity of the last number. We shall write an R code for calculating arithmetic mean of a set of numbers stored in $x$.

```
x<- 11:20           ## the consecutive numbers from 11 to 20 are stored in x
n<- length(x)       ## number of numbers stored in x
s<- 0           ## initialisation of a variable where the sum of numbers will be stored
for (i in 1:n)      ## 'for' loop starts with taking the first value, it will take increment of one to get the
                    ## next number i.e. second number and so on; loop continues until it reaches the last
                    ## number (as given by 'n')
{           ## beginning of the operations to be done inside the loop
s<- s+x[i]      ## first number of x is being added to s. It continues as long i takes consecutive values up to n
}           ## end of operation inside the loop; it then searches for the next value of i; loop ends when i = n
m<- s/n         ## after the loop ends, s contains the sum of all numbers in x; mean is then just s/n
cat("mean is ", round(m,2))         ## prints as "mean is ", and then the value of mean rounded up to two
                                    ## places of decimal
```

We can do multiple jobs inside the loop, i.e. inside {...} and get all results after the loop ends. There is no need to write the loop multiple times for multiples jobs over the same set of numbers.

**Writing data files**

Similarly we can save output files, provided it is described in rows and columns. Suppose '*a*' contains the output datafile and we want to save it in file named *output.txt* with no row heading but same column heading; elements of a row are separated by *tab*. The command for that would be
$write.table(a, \text{“}output.txt\text{”}, row.names = F, col.names = T, quote = F,$
$sep = \text{“}\backslash t\text{”})$.
We write 'quote = F' because we don't want the numerical figures to be enclosed within double quotes. It is also possible to save a file which are not represented in rows and columns.

If we want to save the table, which is the output file, in .csv format, the command would be
$write.csv(d, \text{“}output.csv\text{”}, row.names = F, quote = F)$
There is no to need write 'col.names = F' as it always keeps column heads while saving in .csv format. While reading back this file, we can use either 'read.csv()' or 'read.table()' with *sep* inside the function. Check what happens if you don't use *sep* inside 'read.table()'!

**Plots**

If we want to plot $y$ against $x$, just type $plot(x, y)$, we see the graph, with all points of $y$ plotted against the corresponding value of $x$. Note that the points are 'empty circles'. If we want a smooth curve, just type $plot(x, y, ty = \text{“}l\text{”})$. Here the consecutive points are joined by straight line ('l' for line); if the points are very close to each other, the curve looks smooth. We can play around with this graph in many ways.

One important lesson: if we want to know the usage of a known function, say for the 'plot' function, in the terminal type '$?plot()$'; a full page of instructions will be opened, indicating the input variable to the function, output variables, the interpretation of each parameter that one can pass into the function along with some nice examples. Many examples of application of 'plot()' function will be available throughout this book and will serve as a good learning exercise for the reader. Moreover, there are options for other related commands that can be used for more sophistication. Thus R itself has many ways and tools that may be extremely helpful for self learning.

We shall only demonstrate scatter plot for two variables using the cholesterol indicators data stored in dataframe 'd'. We want to draw six scatter diagrams taking one pair of variables at a time. We have four variables; the number of possible pairs of variables is six, viz., (HDL, LDL), (HDL, VLDL), . . ., (VLDL, TGL). We also want to save the graph in 'pdf' format. R can do that easily!

```
pdf("Cholesterol.pdf")              ## save the output plot in file named 'Cholesterol.pdf' in the same directory
we are working. Drop this line if we don't want to save this.

par(mfrow=c(2,3))              ##splits one page into six cell, arranged in 2 rows and 3 columns
plot(d$HDL,d$LDL,xlab="HDL",ylab="LDL")              ## simple scatter plot with default setting
plot(d$HDL,d$VLDL,xlab="HDL",ylab="VLDL",pch=19)     ## scatter plot another point character
                                                                    # with default colour (black)
plot(d$HDL,d$TGL,xlab="HDL",ylab="TGL",pch=8,col="red")     ## scatter plot with another point
                                                                    #character in red colour
plot(d$LDL,d$VLDL,xlab="LDL",ylab="VLDL",pch=7,col="blue")       ## scatter plot with another point
                                                                    # character in blue colour
plot(d$LDL,d$TGL,xlab="LDL",ylab="TGL",pch=17,col=10)       ## scatter plot with another point
                                        # character in colour number 10 in R plot colour library
plot(d$VLDL,d$TGL,xlab="VLDL",ylab="TGL",pch=15,col=6)       ## scatter plot with another point
                                                                    # character in another colour
dev.off()              ## if we save the plot in a file, we have to write it; otherwise don't write.
```

Note that we can play with the type of points, colour, font size, etc. within a 'plot' function. Moreover, we can change the labels of x-axis and y-axis by changing the text inside 'xlab' and 'ylab' in a 'plot' function. We present the output of the above code, which is written in a very simple manner, for ease of understanding. However, we see that there are some spaces or gaps between plots in each column (and/or row). This can be adjusted by controlling other parameters that define the margins. Typing '?par()' at R console, a page will be opened with many options for everything that we want to change. Play with that to produce beautiful graphs. Of course, we can write the main title and subtitle in this entire plot.

In other words, think of anything that you want to put in a plot to present your data and results in the best possible way. In fact, you can draw nice figures. The entire Fig. A.1 is drawn using R code!

### Writing a new function

We know R works through functions. For example, mean, sd, cor, etc. are all functions that are inbuilt in R. However, in some cases we might require to do a job repeatedly in our program. Writing every time would make the program unnecessarily long. This can be made more precise and faster if we write a small R code for that job. This is called writing a function for a specific purpose relevant to our study or problem. Let's demonstrate this idea using a simple example.

Suppose we want to write our R code for calculating the correlation coefficient based on data stored in $x$ and $y$ variables. Careful attention to the formula for correlation coefficient reveals that we need $\sum_{i=1}^{n} x_i$, $\sum_{i=1}^{n} y_i$, $\sum_{i=1}^{n} x_i^2$, $\sum_{i=1}^{n} y_i^2$, and $\sum_{i=1}^{n} x_i y_i$. Thus, we need to write a function that calculates the sum of values in a variable. Suppose 'a' contains a list of observations. Let the name of our function be $func$, which will calculate the sum. We write the code for this function and apply it to calculate the correlation coefficient.

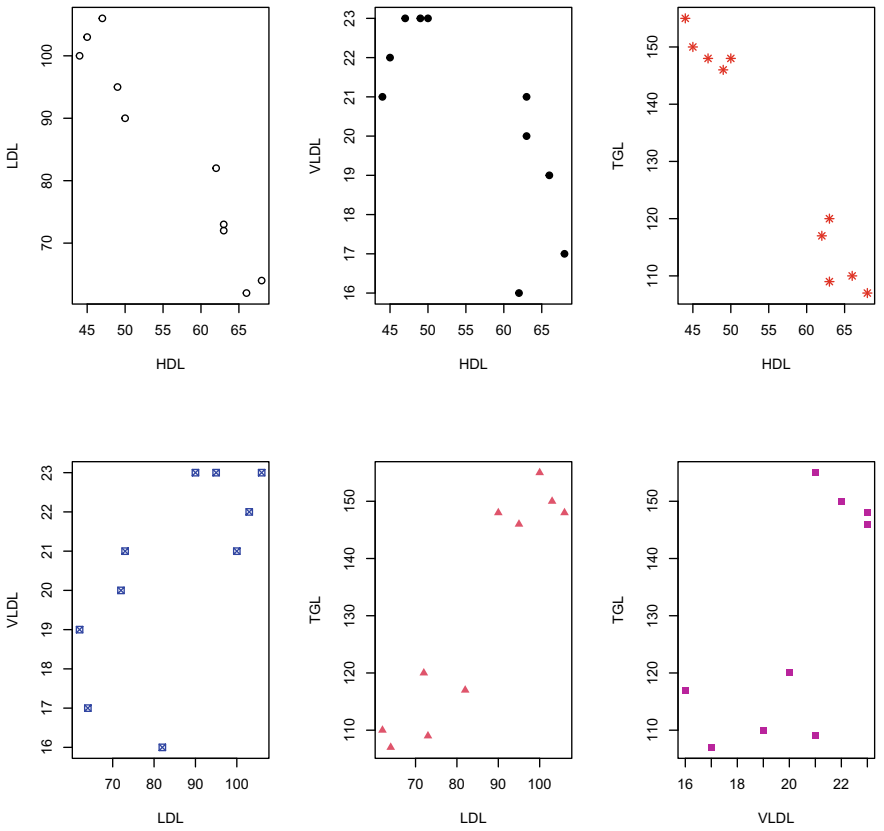



**Fig. A.1** A demonstration of R plots

```
func<- function(a)              ## declaration of function named 'func' with input 'a'
{                                  ## function starts
   out<- 0                         ## initiation of a variable where output will be stored
   n<- length(a)                ## number of observations in input variable
   for (i in 1:n)               ## 'for loop' starts
      out<- out+a[i]               ## number from 'a' is being added to previous result in 'out', one at a time
   return(out)                        ## returns the sum of numbers in 'a'
}                                  ## function ends
mean.x<- func(x)/length(x)              ## using 'func', mean calculation of numbers in 'x'
mean.y<- func(y)/length(y)              ## using 'func', mean calculation of numbers in 'y'
sd.x<- sqrt(func(x*x)/length(x)-mean.x^2)              ## using 'func', sd calculation of numbers in 'x'
sd.y<- sqrt(func((y-mean.y)^2)/length(y))              ## using 'func', sd calculation of numbers in 'y'
cov.xy<- func(x*y)/length(x)-mean.x*mean.y              ## using 'func', covariance calculation for 'x' and
'y'
corr.xy<- cov.xy/(sd.x*sd.y)   ## calculation of correlation coefficient between 'x' and 'y'
corr                                  ## print correlation coefficient on the screen
```

We demonstrate another function that takes a DNA sequence and a codon as input. It calculates the frequency of that codon in the sequence. Using this function we can prepare a frequency distribution of codons of interest. We then generate a DNA sequence of length 200 and search for a few codons.

Note that, we can of course write more than one R function inside one program, i.e. a program might require multiple user-defined functions.

```
codonSearch<-function(sequence,codon)          ## declaration of function with input 'sequence' of DNA
                                                                                and a 'codon'
  {
 x<-unlist(strsplit(sequence,split=""))          ## splits input DNA sequence into a sequence of nucleotides
 y<-unlist(strsplit(codon,split=""))             ## splits input codon sequence into a sequence of 3 nucleotides
   id1<-which(x==y[1])          ## identifies position of first nucleotide of the codon within DNA sequence
   if (length(id1>0))           ## checks whether any nucleotide matches with first codon nucleotide
      {
      id2<-which(x[id1+1]==y[2])  ## stores positions of sequence matching with second codon nucleotide
      if (length(id2>0))          ## checks whether any nucleotide matches with second codon nucleotide
         {
         id<-id1[id2]             ## stores positions from which two consecutive nucleotides match with first
                                  # two nucleotides of the codon
         id3<-which(x[id+2]==y[3])          ## stores positions from which three consecutive nucleotides
                                            # match with the codon
         k<-length(id[id3])       ## number of such positions i.e. frequency of codon in the sequence
         }
      }
return(k)                         ## returns output which frequency of codon in input DNA sequence
  }
```

Suppose we randomly generate a DNA sequence of length 10000 and try to calculate the frequency distribution of a few codons, viz., $AGC$, $ATA$, $GTT$, $CCC$, $CTC$, $CGA$,$CGT$, $TGA$, $TCG$, and $TTG$. The R code using the function 'codonSearch' can be written as

```
DNAsequence<-sample(c("A","T","G","C"),1000,replace=T)
codonChoice<-c("AGC","ATA","GTT","CCC","CTC", "CGA","CGT", "TGA", "TCG", "TTG")
n<-length(codonChoice)          ## number of codons to be studied
freq<-rep(0,n)          ## initiation of 'freq' variable to store frequency of each codon
for (i in 1:n)          ## 'for loop' starts
{
freq[i]<-codonSearch(DNAsequence,codonChoice[i])    ## frequency calculation for ith codon from
                                                                        #'codonChoice' list
}
freq.table<-data.frame(codonChoice,freq)          ## gives frequency distribution of codons
```

Definitely you can do a lot using this function. Similar other functions can be written to facilitate our computation in R.

## Exercise

A.1 Enter the data {2, 5, 3, 7, 1, 9, 6} in a variable x. Then type x and press 'enter'.

(a) Find the number of elements in $x$, i.e. in the data list.
(b) Find the last element of $x$.
(c) Find the 3rd element in the data list.
(d) Find 3rd to 5th element in the data list.
(e) Find 2nd, 5th, and 6th element from the list.
(f) Find the minimum element of $x$.
(g) Print the elements in $x$ in reverse order without entering data again.
(h) Arrange the data in increasing and decreasing order.

A.2 Type the command $rep(2, 4)$ and see what it gives.

(a) Create a data list (4, 4, 4, 4, 3, 3, 3, 5, 5, 5) using $rep$ function.
(b) Create a list (4, 6, 3, 4, 6, 3, . . . , 4, 6, 3) where there are 10 occurrences of 4, 6, and 3 in the given order.
(c) Create a list (3, 1, 5, 3, 2, 3, 4, 5, 7, 7, 7, 7, 7, 8, 7, 6, 5, 4, 3, 2, 1, 34, 21, 54) using a one-line command.
(d) First create a list (2, 1, 3, 4). Then append this list at the end with another list (5, 7, 12, 6, −8). Check whether the number of elements in the augmented list is 11.
(e) Find the numbers in odd positions.

A.3 Use of $seq$ function.

(a) Print all numbers starting with 3 and ending with 7 with an increment of 0.5. Store these numbers in $x$.
(b) Print all 17 equispaced numbers between 12 and 37, both inclusive.
(c) Print all even numbers between 2 and 14 (both inclusive).
(d) Type $2 * x$ and see what you get.
(e) Type $x * x$ and see what you get.
(f) Type $x/10$ and see what you get.
(g) If $y$ contains first $n$ positive integers where $n$ is the length of $x$, what do you get if you write $x + y$, $x * y$ and $x/y$?

A.4 Enter the data {1, 2, . . . , 19, 20} in a variable $x$.

(a) Find the sum of numbers positioned at odd numbered places.
(b) Find the sum of numbers at 2nd, 5th, 13th, and 18th positions in the list.
(c) Find mean, median, and standard deviation as a list of numbers.
(d) Find the sum of squares of the values in $x$.
(e) Calculate mean deviation about mean and mean deviation about median and print them together, as a list.
(f) Check whether both mean deviation about mean and mean deviation about median are less than the standard deviation.

A.5 R has a few in built data sets. One of them is 'iris' data set. Just type '$a < -iris$'. The entire datafile is stored in 'a'.

(a) Just type '$a$' and see what you get.
(b) How many flowers constitute this data set?
(c) How many variables are there in 'iris' data set?
(d) Print first six rows of $a$.
(e) Print ten rows of $a$.
(f) Print first six rows and first four columns of $a$.
(g) Extract the data only for 'setosa' species.
(h) Draw histogram for petal length for setosa species.
(i) For setosa species data, draw four histograms on the same graph paper for petal length, petal width, sepal length, and sepal width.
(j) Store the data for setosa species in a separate file, dropping the last column.
(k) Find the correlation coefficient between petal length and sepal length for the entire data.
(l) Find a linear regression of petal width on petal length for the entire data.
(m) Drawn a scatter diagram for petal length and petal width and also plot the regression line for the entire data on this diagram.
(n) Fit a linear regression of petal width on petal length, sepal length, and sepal width for the entire data.
(o) Find all possible pairwise correlation coefficients for sepal length, sepal width, petal length, and petal width.
(p) Extract the data only for virginica.
(q) Fit regression equations of petal width on petal length for setosa species and for virginica species separately.
(r) For each of setosa and virginica species, draw scatter diagrams of petal length and petal width separately; superimpose the respective regression lines on the same plots.
(s) Merge three plots for setosa and virginica species that you have done in (m) and (r).
(t) Calculate the column means and column standard deviations for each column except the fifth column. Augment the full data set by adding one row for mean and one row for standard deviation. Note that in each newly added row, you have to keep the fifth position as blank. Store this augmented file in a new file using 'write.table' command in R.
(u) Do a two-sample t-test to see whether means of petal lengths are same in versicolor and virginica species.
(v) Do a statistical test to see whether means of petal lengths are the same in all three species.
(w) Calculate multiple correlation coefficients of petal width on petal length, sepal length, and sepal width for (i) the entire data, and (ii) for each species separately.

(x) Calculate partial correlation coefficient between petal length and petal width, eliminating the effects of sepal length and sepal width for (i) the entire data, and (ii) for each species separately.
(y) Write a program to calculate the mean of petal length for all flowers using a loop in R. Check whether your result matches with the result obtained through 'mean' function of R.
(z) Using only one loop, write a program to calculate mean, median, and standard deviation of sepal length for all flowers in the data. Present them as a list of three values indicating the measures.
[Note: For questions (u) - (x), you can do it after reading the book.]

A.6 Using the iris data set, do the following.

(a) Draw a histogram for petal length for all flowers.
(b) Draw boxplots for sepal length for (i) all flowers, (ii) setosa only, (iii) versicolor only, and (iv) virginica only, on the same graph using 'par()' command in R.
(c) Draw a scatter diagram for petal width against petal length for all flowers. However, each species should have its own colour different from others so as to identify them properly. Also put the legend for colours used inside the graph.
(d) Draw Q-Q plots for (i) setosa only, (ii) virginica only, (iii) versicolor only, and (iv) the whole data set. Comment on your findings in comparison to a normal distribution.
(e) Do a principal component analysis with the entire data set. Note that you have to remove the last column, i.e. 'Species' from the data before doing this analysis. What is the percentage of variance explained by the first two principal components? Plot the amount of variance explained against number of principal components and comment on how you would choose the number of such components to work with.
(f) Plot first and second principal components against each other and comment on the plot.
(g) Do a cluster analysis using (i) hierarchical clustering and (ii) k-means clustering methods. Compare the results obtained by these two methods and interpret your result.

# Appendix B

We have drawn Fig. 2.1 to explain histogram and its components. The R command 'hist( )' will give only histogram. R code that produces Fig. 2.1 involves little bit of coding using different R commands. The entire code is given below. It can be made more precise; but we intentionally keep it as this, because you can try it yourself. Just copy each line one by one, paste in R terminal, and press enter; see what happens and how this entire graph is emerged at the end. Once you get Fig. 2.1, you may try to make it shorter and add different colours to different components.

Similarly we also provide below the R code for reproducing Fig. 2.3 which represents ogives, both less than and greater than types with some additional parts required for explanation. As in the previous case, use this R code to reproduce Fig. 2.3 and then try to add some changes as you wish.

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```
> br <- round(seq(0,10,length = 12),3)
> hb <- hist(x,breaks = br)
> d <- data.frame(br[−1],cumsum(hb$counts), br[−length(br)],
cumsum(hb$counts[length(hb$counts):1]))
> d <- rbind(c(0,0,0,90),d)
> d <- rbind(c(−0.909,0,−0.909,90),d)
> d <- rbind(d,c(10.909,90,10.909,0))
> d <- rbind(d,c(10.909,90,10.909,0))
> plot(d[,1],d[,2],pch=20,xaxt= 'n',xlab="Class boundary",
ylab="Cumulative frequency")
> lines(d[,1],d[,2])]
> axis(side = 1,at = d[,1],labels = d[,1],las = 2)
> points(d[,3],d[,4], pch = 20)
> lines(d[,3],d[,4])
> lines(c(2.927,2.927),c(−4,45))
> text(6,30,"Median")
> arrows(5.32,30,2.93,−3.5,length = 0.15,angle = 25,code = 2)
> text (8.2, 50, "Less-than type ogive")
> arrows(6.4,50,3.2,38,length = 0.15,angle = 25,code = 2)
> text (8.2, 70, "Greater-than type ogive")
> arrows(6.2,70,1.3,76,length = 0.15,angle = 25,code = 2)
> text(3.2,45,"O")
> lines(c(−1.4,2.927),c(45,45))
> axis(side = 1, at = 2.927, labels = "M", las = 1, line = −0.9)
> axis(side = 2, at = 45, labels = "45", las = 1, line = −0.9)
```

R code for boxplot (Fig. 2.4) is given below that contains codes for drawing arrows, inserting texts in any position of the graph, etc. You may try to make this code more precise, use other fonts and colours, etc.

```
> boxplot(b,lwd = 1.5,pch = 20)
> text(1.4,6,“Median”)
> arrows(1.34,5.9,1.1,3,length = 0.15,angle = 25,code = 2)
> text(x = 1.03, y = 9.47, ’{’, srt = 180, cex = 3.48,family = ’CenturySch’)
> text(1.3,9.5,“Outliers”)
> arrows(1.235,9.5,1.05,9.5,length = 0.15,angle = 25,code = 2)
> text(x = 1.24, y = 2.935, ’{’, srt = 180, cex = 7,family = ’CenturySch’)
> text(1.37,3.25,“Middle 50%”)
> text(1.37,2.8,“observations”)
> text(0.56,2.8,“Whiskers”)
> arrows(0.98,6,0.56,3,length = 0.15,angle = 25,code = 2)
> arrows(0.98,0.8,0.56,2.5,length = 0.15,angle = 25,code = 2)
```

# Appendix C

**Theorem 18** $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

***Proof*** From Fig. 3.2 it is clear that the set $B$ can be written as an union of two mutually exclusive sets, $A \cap B$ and $A^c \cap B$. Noting that the intersection of these two sets is a null set, we have by the definition of probability,

$$\begin{aligned} & B = (A \cap B) \cup (A^c \cap B) \\ \implies & P(B) = P\Big[(A \cap B) \cup (A^c \cap B)\Big] = P(A \cap B) + P(A^c \cap B) \\ \implies & P(A^c \cap B) = P(B) - P(A \cap B) \end{aligned}$$

Again, from Fig. 3.2, it is clear that $A \cup B$ can be written as an union of two mutually exclusive sets, $A$ and $A^c \cap B$. Hence by definition of probability, we have

$$P(A \cup B) = P\Big[A \cup (A^c \cap B)\Big] = P(A) + P(A^c \cap B) = P(A) + P(B) - P(A \cap B).$$

**Theorem 19** $P(\phi) = 0$.

***Proof*** Clearly, $\Omega = \Omega \cup \phi$. Noting that $P(\Omega) = 1$, we have,

$$1 = P(\Omega) = P(\Omega \cup \phi) = P(\Omega) + P(\phi) = 1 + P(\phi) \implies P(\phi) = 0.$$

**Theorem 20** *If $A$ and $B$ are two mutually exclusive sets, $P(A \cap B) = 0$.*

***Proof*** For two mutually exclusive events $A$ and $B$, we have $A \cap B = \phi$. Hence, by Theorem 19, $P(A \cap B) = P(\phi) = 0$.

**Theorem 21** *For any $A$ and $B$ such that $A \subset B$, $P(A) \leq P(B)$.*

***Proof*** Using the techniques in previous theorems and from first graph of Fig. 3.3, it is easy to see that $B = A \cup (A^c \cap B)$. Moreover $A$ and $(A^c \cap B)$ are mutually exclusive events so that $P\Big[A \cap (A^c \cap B)\Big] = 0$, by Theorem 20. Hence applying the definition of probability and using Theorem 18, we have

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Indian Statistical Institute Series, https://doi.org/10.1007/978-981-99-3220-7

$$P(B) = P\Big[A \cup (A^c \cap B)\Big] = P(A) + P(A^c \cap B) \geq P(A) \qquad [\text{since, } P(A^c \cap B) \geq 0\,]$$

**Theorem 22** $0 \leq P(A) \leq 1$ *for any* $A \in \mathscr{A}$.

***Proof*** Since $A \in \mathscr{A}$, we have by definition $P(A) \geq 0$. Again $\Omega$ being the largest set, $A \subset \Omega$. Using the fact that $P(\Omega) = 1$ by definition and using Theorem 21, we have, $0 \leq P(A) \leq P(\Omega) = 1$, i.e. $0 \leq P(A) \leq 1$.

**Theorem 23** $P(A^c) = 1 - P(A)$.

***Proof*** Clearly, from Fig. 3.2, we can write, $\Omega = A \cup A^c$. Moreover $A \cap A^c = \phi$. Hence, by definition and using Theorem 18, we have,

$$1 = P(\Omega) = P(A \cup A^c) = P(A) + P(A^c) \implies P(A^c) = 1 - P(A).$$

# Appendix D

## *Proof of Hardy-Weinberg equilibrium*

First we shall prove Hardy-Weinberg Equilibrium (HWE) in the context of bi-allelic locus.

***Proof*** To prove HWE, first we have to calculate offspring genotype probability conditional on parental genotypes. Suppose, a male with genotype $AA$ undergoes random mating with a female having genotype $AA$. The offspring genotype due to this random mating will always be $AA$, because female will transmit $A$ allele with probability 1 and similarly the male will transmit $A$ allele with probability 1. Moreover, transmission of an allele by mother is completely independent of the transmission of allele by the father. Hence,

$$\begin{aligned} P(\text{Offspring genotype is } AA | AA \times AA) &= 1, \\ P(\text{Offspring genotype is } Aa | AA \times AA) &= 0, \\ \text{and } P(\text{Offspring genotype is } aa | AA \times AA) &= 0. \end{aligned}$$

To write this precisely, we use a few notations. Let $O_g$, $F_g$ and $M_g$ be the genotype of offspring, female and male individuals respectively and $F_g \times M_g$ denote random mating between the male and the female. Hence, using these notations, we can write

$$P(O_g = x | M_g = AA, F_g = AA) = \begin{cases} 1 & \text{if } x = AA \\ 0 & \text{if } x = Aa \text{ or } aa \end{cases}$$

where $x$ denotes the genotype of the offspring under random mating between $F_g$ and $M_g$. Now suppose that $F_g = Aa$ and $M_g = Aa$. This male can transmit $A$ allele with probability 0.5 and the female can transmit $A$ allele with probability 0.5. These two events are also independent of each other. So,

$$P(O_g = AA | F_g = Aa, M_g = Aa) = 0.5 \times 0.5 = 0.25.$$

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I. Mukhopadhyay and P. P. Majumder, *Statistical Methods in Human Genetics*,
Indian Statistical Institute Series, https://doi.org/10.1007/978-981-99-3220-7

To calculate $P(O_g = Aa|F_g = Aa, M_g = Aa)$, note that male and female transmit $A$ allele and $a$ alleles respectively each with probability 0.5 resulting in offspring genotype as $Aa$ whereas male and female transmit $a$ allele and $A$ alleles respectively each with probability 0.5 resulting in the same offspring genotype i.e $Aa$. Also note that transmission of allele by mother (female) and transmission of allele by father (male) are independent, we have

$$\begin{aligned} P(O_g = Aa|F_g = Aa, M_g = Aa) &= P(M_g \to A, F_g \to a) + P(M_g \to a, F_g \to A) \\ &= P(M_g \to A)P(F_g \to a) + P(M_g \to a)P(F_g \to A) \\ &= 0.5 \times 0.5 + 0.5 \times 0.5 = 0.5 \end{aligned}$$

Hence, we have

$$P(O_g = x|M_g = Aa, F_g = Aa) = \begin{cases} 0.25 & \text{if } x = AA \\ 0.5 & \text{if } x = Aa \\ 0.25 & \text{if } x = aa \end{cases}$$

Proceeding similarly, we have the offspring genotype probabilities for all possible genotypes due to random mating and is given in Table A.1.

Now assume that genotype frequencies (probabilities) in the population in Generation 0 for three possible genotypes are $Q = P(AA)$, $R = P(Aa)$, and $H = P(aa)$. Table A.1 presents the conditional probability of offspring genotype given parental genotype combinations. So, in the next generation, i.e. Generation 1, after random mating at Generation 0, the offspring genotype probabilities become

**Table A.1** Offspring probability conditional on parental mating

| Mating type | Offspring genotype probability | | |
|---|---|---|---|
| $F_g \times M_g$ | $AA$ | $Aa$ | $aa$ |
| $AA \times AA$ | 1 | 0 | 0 |
| $AA \times Aa$ | 0.5 | 0.5 | 0 |
| $AA \times aa$ | 0 | 1 | 0 |
| $Aa \times AA$ | 0.5 | 0.5 | 0 |
| $Aa \times Aa$ | 0.25 | 0.5 | 0.25 |
| $Aa \times aa$ | 0 | 0.5 | 0.5 |
| $aa \times AA$ | 0 | 1 | 0 |
| $aa \times Aa$ | 0 | 0.5 | 0.5 |
| $aa \times aa$ | 0 | 0 | 1 |

$$
\begin{aligned}
P(O_g = AA) &= P(O_g = AA|AA \times AA)P(AA \times AA) + P(O_g = AA|AA \times Aa)P(AA \times Aa) + \cdots \\
&= 1 \times Q^2 + 0.5 \times QR + 0 \times QH + 0.5 \times RQ + 0.25 \times R^2 + 0 \times RH + 0 \times HQ + 0 \times HR + 0 \times H^2 \\
&= Q^2 + \frac{1}{2}QR + \frac{1}{2}QR + \frac{1}{4}R^2 = (Q + \frac{R}{2})^2 = Q_1 \text{ (say)}, \\
P(O_g = Aa) &= P(O_g = Aa|AA \times AA)P(AA \times AA) + P(O_g = Aa|AA \times Aa)P(AA \times Aa) + \cdots \\
&= 0 \times Q^2 + 0.5 \times QR + 1 \times QH + 0.5 \times RQ + 0.5 \times R^2 + 0.5 \times RH + 1 \times HQ + 0.5 \times HR + 0 \times H^2 \\
&= \frac{1}{2}QR + QH + \frac{1}{2}QR + \frac{1}{2}R^2 + \frac{1}{2}RH + QH + \frac{1}{2}RH \\
&= 2(Q + \frac{R}{2})(H + \frac{R}{2}) = R_1 \text{ (say), and} \\
P(O_g = aa) &= (H + \frac{R}{2})^2 = H_1 \text{ (say).}
\end{aligned}
$$

If HWE is true, we can claim that these genotype frequencies, i.e. $Q_1 = P(AA)$, $R_1 = P(Aa)$, and $H_1 = P(aa)$ in Generation 1 becomes stable and will continue to remain the same in subsequent generation as long as only random mating continues. To show whether these become stable, we have to consider another generation of random mating i.e. in Generation 1. Noting that $Q + R + H = 1$ and proceeding as explained above, but with genotype frequencies $Q_1$, $R_1$, and $H_1$ for genotypes $AA$, $Aa$, and $aa$ respectively, we have the genotype frequencies in Generation 2 as

$$
\begin{aligned}
P(O_g = AA) &= (Q_1 + \frac{R_1}{2})^2 = \Big\{(Q + \frac{R}{2})^2 + \frac{1}{2} \times 2(Q + \frac{R}{2})(H + \frac{R}{2})\Big\}^2 \\
&= (Q + \frac{R}{2})^2(Q + \frac{R}{2} + H + \frac{R}{2})^2 = (Q + \frac{R}{2})^2(Q + R + H)^2 = (Q + \frac{R}{2})^2 = Q_1, \\
P(O_g = Aa) &= 2(Q_1 + \frac{R_1}{2})(H_1 + \frac{R_1}{2}) \\
&= 2\Big\{(Q + \frac{R}{2})^2 + \frac{1}{2} \times 2(Q + \frac{R}{2})(H + \frac{R}{2})\Big\}\Big\{(H + \frac{R}{2})^2 + \frac{1}{2} \times 2(Q + \frac{R}{2})(H + \frac{R}{2})\Big\} \\
&= 2\Big\{(Q + \frac{R}{2})(Q + R + H)\Big\}\Big\{(H + \frac{R}{2})(Q + R + H)\Big\} \\
&= 2(Q + \frac{R}{2})(H + \frac{R}{2}) = R_1, \text{ and} \\
P(O_g = aa) &= (H_1 + \frac{R_1}{2})^2 = \Big\{(H + \frac{R}{2})^2 + \frac{1}{2} \times 2(Q + \frac{R}{2})(H + \frac{R}{2})\Big\}^2 \\
&= (H + \frac{R}{2})^2(Q + \frac{R}{2} + H + \frac{R}{2})^2 = (H + \frac{R}{2})^2(Q + R + H)^2 = (H + \frac{R}{2})^2 = H_1.
\end{aligned}
$$

Hence, the genotype frequencies remain the same in Generation 2 and subsequent generations, provided random mating continues in every generation and there is no external force acting in any generation.

## *Properties of allele frequency estimate*

We have seen that $\hat{p} = \frac{2n_{AA} + n_{Aa}}{2n}$. Now if we consider $n_{AA}$ as the number of individuals having genotype $AA$ and the rest have genotype either $Aa$ or $aa$, then we can say that, $n_{AA} \sim Bin(n, p^2)$. Similarly $n_{Aa} \sim Bin(n, 2pq)$. Note that if $X$ is a random variable

that follows a binomial distribution with parameters $n$ and $p$, we have $E(X) = np$ and $V(X) = npq$ where $q = 1 - p$. Thus we have

$$E(\hat{p}) = E\Big(\frac{2n_{AA} + n_{Aa}}{2n}\Big) = \frac{2E(n_{AA}) + E(n_{Aa})}{2n} = \frac{2n.p^2 + n.2pq}{2n} = p \qquad \text{(since, } p + q = 1\text{)}$$

To evaluate V(Y), we consider $Y = n_{AA} + n_{Aa}$ as a random variable that denotes the number of individuals having genotypes either $AA$ or $Aa$ and so the rest have genotype $aa$. Thus, $Y \sim Bin(n, p^2 + 2pq)$. So we have

$$\begin{aligned} & V(Y) = V(n_{AA} + n_{Aa}) = V(n_{AA}) + V(n_{Aa}) + 2Cov(n_{AA}, n_{Aa}) \\ \Longrightarrow\ & n(p^2 + 2pq)(1 - p^2 - 2pq) = np^2(1 - p^2) + 2npq(1 - 2npq) + 2Cov(n_{AA}, n_{Aa}) \\ \Longrightarrow\ & Cov(n_{AA}, n_{Aa}) = -2np^3q \end{aligned}$$

Hence, $V(\hat{p}) = V\Big(\dfrac{2n_{AA} + n_{Aa}}{2n}\Big) = \dfrac{1}{2n^2}.\Big[4V(n_{AA}) + V(n_{Aa}) + 4.cov(n_{AA}, n_{Aa})\Big]$

$$= \frac{1}{2n^2}\Big[4np^2(1 - p^2) + 2npq(1 - 2pq) - 8np^3q\Big] = \frac{pq}{n}.$$